WSQ COURSE SLIDES

Security Operations for Autonomous AI Agents

WSQ Course Code: TGS-2024042604

4 days · 32 hours · Intermediate

Version v1.0 · 12 September 2026

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W



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COURSE ADMINISTRATION

Welcome and course orientation

Attendance · outcomes · assessment · learner systems



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Digital attendance (mandatory)

1 **AM**

Scan the trainer-displayed SSG attendance QR code.

2 **PM**

Complete the second attendance event after lunch.

3 **Assessment**

Record attendance for the assessment session.

4 **Minimum**

At least 75% attendance is required.

5 **Evidence**

Confirm the successful attendance screen.

6 **Escalation**

Tell the trainer immediately if the scan fails.

Official course administration requirements

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About the trainer — template

1 **Name**

_____

2 **Designation**

_____

3 **Qualifications**

_____

4 **Industry experience**

_____

5 **Security expertise**

_____

6 **Contact**

_____

Tertiary Infotech trainer profile template

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About the trainer — Dr Alfred Ang

1 Role

Principal Trainer and courseware developer at Tertiary Infotech Academy.

2 Focus

Applied AI, cybersecurity, software engineering and hands-on technical learning.

3 Teaching approach

Mechanism-led explanation, reproducible activities and evidence-based verification.

4 Learner support

Clarification during class and post-course support through the academy.

5 Course role

Connect security strategy and governance to practical autonomous-agent operations.

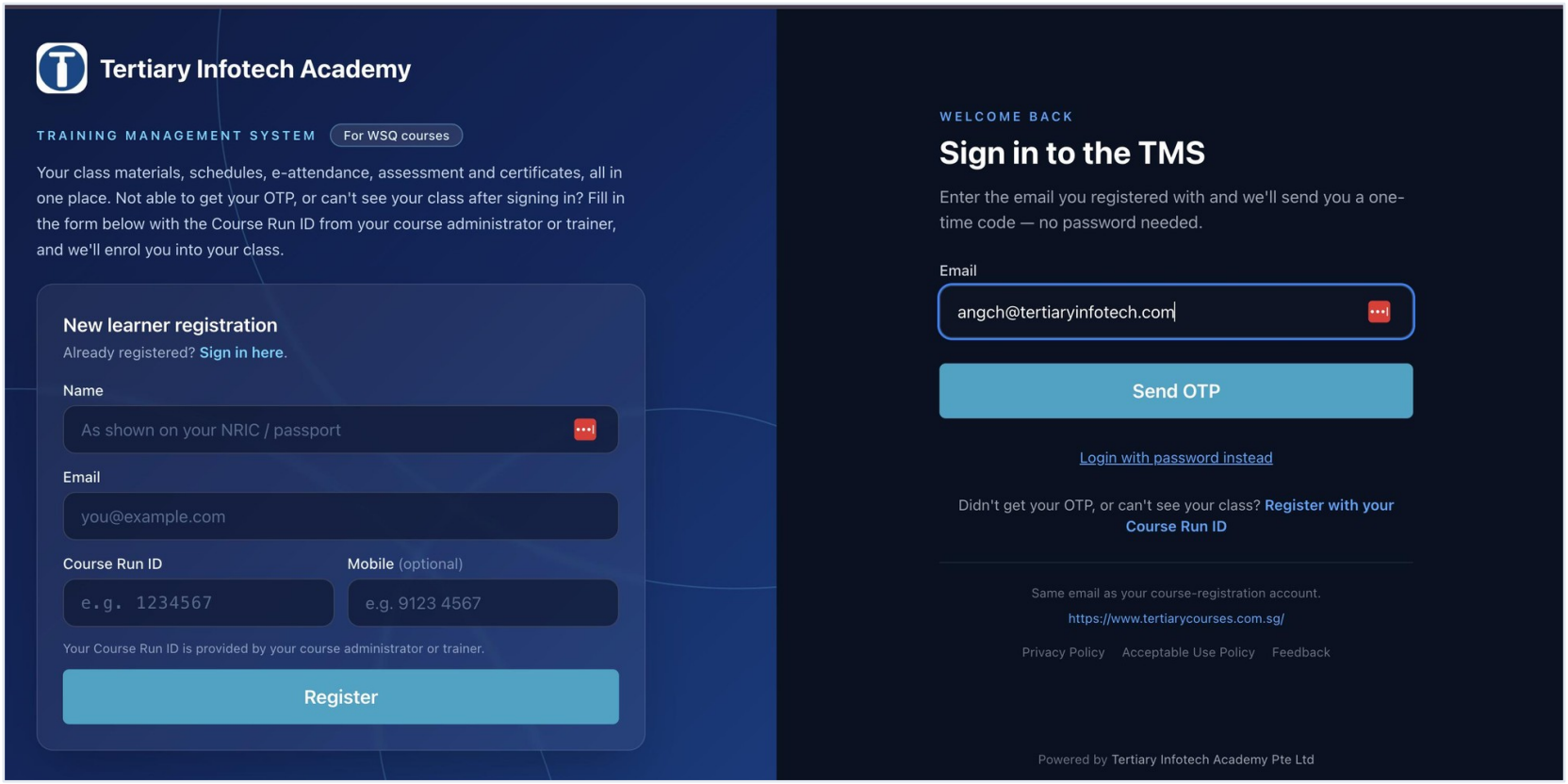
6 Contact

enquiry@tertiaryinfotech.com

Tertiary Infotech course administration

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Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

Tertiary Infotech LMS/TMS workflow

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Download your course material

1

Open

lms-tms.tertiaryinfotech.com

2

Sign in

Use your registered account

3

Select

Open this course record

4

Download

Slides, Learner Guide and activities

5

Verify

Check title, code and version

Use only files labelled Security Operations for Autonomous AI Agents · TGS-2024042604.

Tertiary Infotech LMS/TMS workflow

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Skills Framework alignment

1 **K1**

Goal setting and objectives of organisation security

2 **K2**

Information security and assurance strategy

3 **K3**

Best practices in information security policies

4 **K4**

Best practices and emerging technologies in security control

5 **K5**

Knowledge of security management benchmarks

6 **K6**

Gap analysis in organisation security

Approved course proposal · Skills Framework mapping

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Skills Framework alignment — implementation

1 **K7**

Implications and impact of security gaps

2 **A1**

Formulate security goals and objectives from business priorities, security vision, strategy directions and management benchmarks

3 **A2**

Establish standards and practices that protect information integrity, authenticity and confidentiality

4 **A3**

Manage compliance with information-security guidelines and classification or permission rules

5 **A4**

Lead communication of security goals and objectives to the organisation

6 **A5**

Review existing security controls against current and future business costs and risks

Approved course proposal · Skills Framework mapping

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Skills Framework alignment — strategy and operations

1 **A6**

Develop strategies and plans to resolve security gaps

2 **A7**

Drive organisation-wide security initiatives in line with internal and external standards

3 **Assessment**

WA: K1–K7 · PP: A1–A7

4 **Course evidence**

Ten activities produce reproducible control artifacts.

5 **Alignment gate**

Nothing assessed is absent from the slides and activities.

6 **Terminology**

Every hands-on lab is presented as an Activity.

Approved course proposal and verified legacy assessment mapping

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Learning outcomes

1

LO1

Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.

2

LO2

Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.

3

LO3

Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.

4

LO4

Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.

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FOUR TOPICS

Course outline

1 **Topic 01**

Security Strategy and Standards for Autonomous AI Agents — Agent inventory · security objectives · identity · tools · data · resilience

2 **Topic 02**

Agent Security Policies, Governance and Compliance — Policy-as-code · evidence · permissions · assurance · communication

3 **Topic 03**

AI Agent Risk Assessment and Security Control Evaluation — Threat paths · risk quantification · control effectiveness · remediation

4 **Topic 04**

Implementing and Monitoring Organisation-Wide Agent Security Operations — Telemetry · detection · incident response · containment · recovery

Official current course page

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32 HOURS

Four-day learning journey

1

Day 1

Strategy, inventory, identity and standards

2

Day 2

Policy, governance, compliance and communication

3

Day 3

Threat modelling, risk and remediation

4

Day 4

Monitoring, incident response and assessment

Each day contains 8 scheduled hours excluding lunch; detailed timing is in the Lesson Plan.

Official course duration and aligned lesson plan

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Briefing for assessment

1 **Individual**

Complete each instrument independently.

2 **Open book**

Use the course slides, Learner Guide and approved materials.

3 **No recording**

Do not photograph or copy the assessment scripts.

4 **Submission**

Upload the completed documents to the LMS/TMS.

5 **Decision**

Results are Competent or Not Yet Competent.

6 **Clarification**

Oral clarification may close minor gaps where authorised.

Verified legacy assessment instructions and house standard

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Assessment instruments

1 **WA (SAQ)**

7 open-ended questions · K1–K7 · 1 hour.

2 **PP**

6 practical tasks · A1–A7 · 3 hours as stated by the verified legacy paper.

3 **Question papers**

Learner-facing candidate documents only.

4 **Answer keys**

Trainer-only; never attached to GitHub or learner-facing LMS fields.

5 **Alignment**

Every WA answer traces to slides; every PP task traces to an activity.

6 **Evidence**

Scripts, mock data and generated artifacts support the PP tasks.

Verified legacy assessment papers downloaded from the supplied course Drive folder

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Assessment flow

1. **TRAQOM** — Complete required survey
2. **Attendance** — Record assessment attendance
3. **WA** — Answer 7 knowledge questions
4. **PP** — Complete 6 practical tasks
5. **Submit** — Upload to LMS/TMS
6. **Sign-off** — Sign Assessment Summary Record

Question papers only are learner-facing; answer keys remain trainer-only.

WSQ assessment house workflow

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Assessment and funding criteria

1 **Attendance**

At least 75% recorded attendance.

2 **Assessment**

Complete both WA and PP.

3 **Competence**

Meet every required assessment criterion.

4 **Identity**

Use the registered learner identity.

5 **Evidence**

Retain successful digital-attendance confirmations.

6 **Appeal**

Follow the documented appeal process if needed.

Official current course page and WSQ administration

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Activity environment

1 **Python**

Python 3.10 or later; standard library only.

2 **Editor**

VS Code or any text editor.

3 **Data**

Synthetic JSON, JSONL and CSV files.

4 **Execution**

Run activity.py inside each activity folder.

5 **Evidence**

Generated Markdown, CSV, JSON or HTML outputs.

6 **Safety**

No live credentials, agents, production systems or external APIs.

Aligned activity package

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Practice exam portal

1 **Portal**

https://exams.tertiaryinfotech.com

2 **Practice mode**

Use untimed practice to identify gaps.

3 **Exam mode**

Use timed mode after completing the activities.

4 **Review**

Return to the exact slide/activity mapped to each missed criterion.

5 **Scope**

K1–K7 knowledge and A1–A7 practical evidence.

6 **Reminder**

The portal supports preparation; the official assessment remains WA + PP.

Official current course page practice-exam link

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TOPIC 01

Security Strategy and Standards for Autonomous AI Agents

Agent inventory · security objectives · identity · tools · data · resilience

LO1 · K1, K2, A1, A2



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LO1 and assessment criteria

1 **K1**

Goal setting and objectives of organisation security

2 **K2**

Information security and assurance strategy

3 **A1**

Formulate security goals and objectives from business priorities, security vision, strategy directions and management benchmarks

4 **A2**

Establish standards and practices that protect information integrity, authenticity and confidentiality

5 **LO1**

LO1: Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.

Approved course proposal criteria adapted to the autonomous-agent context

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Security Strategy and Standards for Autonomous AI Agents: operating model

1. **Boundary** — Components, interfaces and ownership
2. **Runtime** — States, transitions and decision points
3. **Contract** — Fields, rules, thresholds and evidence
4. **Failure** — Attack path, diagnostic and control
5. **Verify** — Acceptance test and activity artifact

Every mechanism in this topic is taught through all five layers.

Courseware-build mechanism-level coverage contract

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Agent inventory: architecture and control boundary

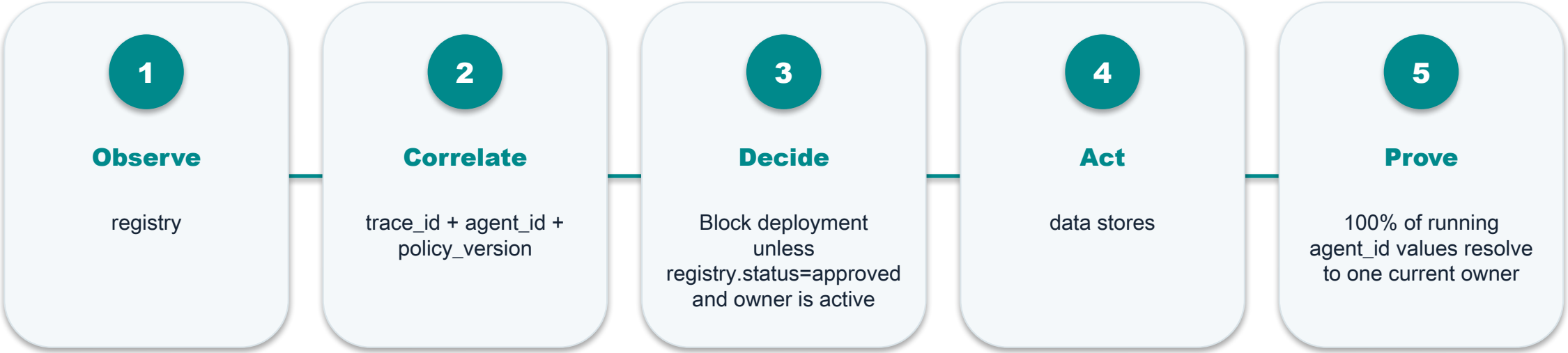


Control objective: Block deployment unless registry.status=approved and owner is active

Supplied AI-security reference corpus · NIST AI RMF · OWASP ASI

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Agent inventory: runtime decision trace



Failure to detect: Unregistered shadow agent reaches a production connector

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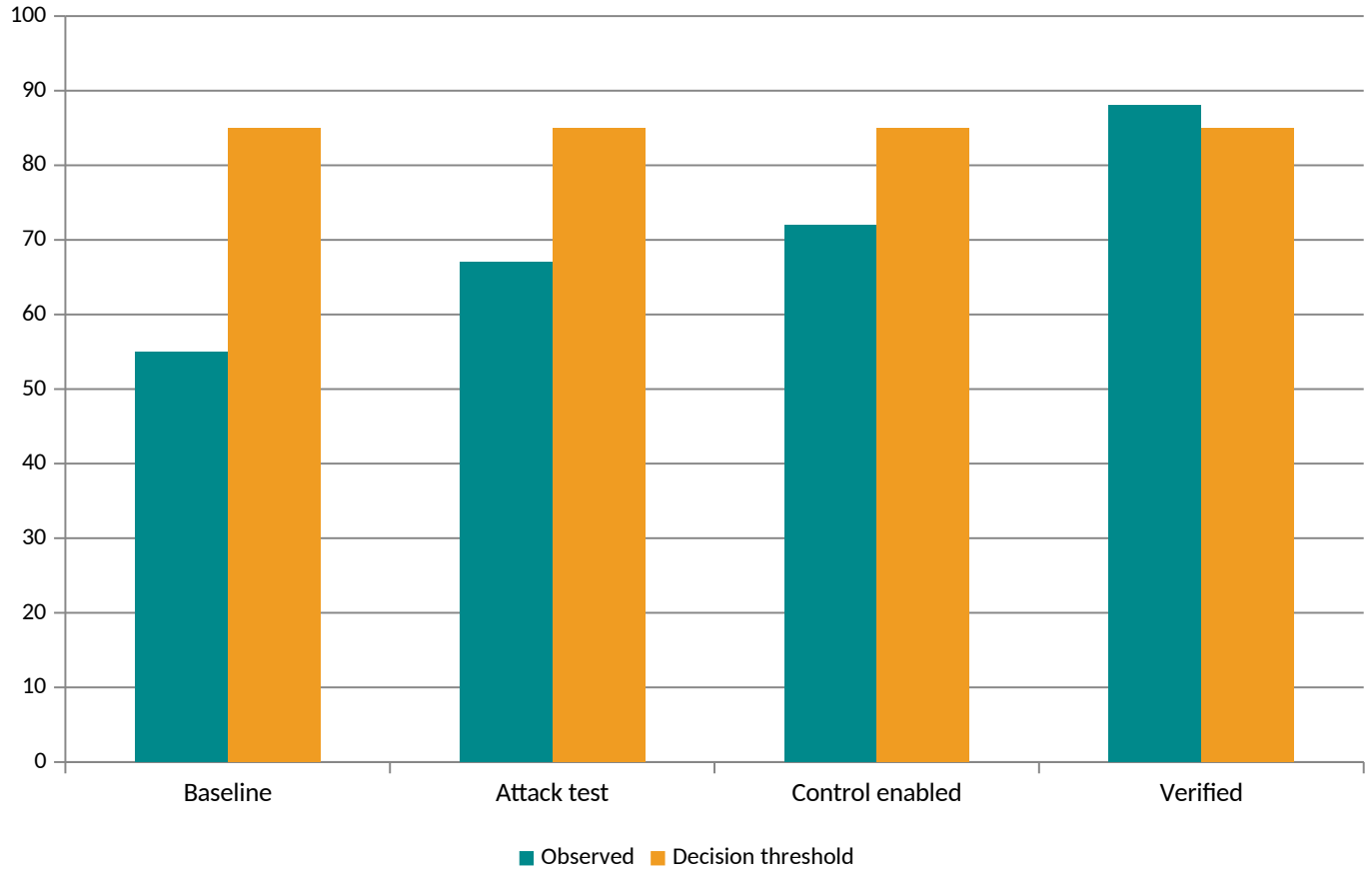
Agent inventory: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	owner	Required in decision and audit evidence
field_3	purpose	Required in decision and audit evidence
field_4	model	Required in decision and audit evidence
field_5	tools[]	Required in decision and audit evidence
field_6	data_classes[]	Required in decision and audit evidence

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Agent inventory: evidence and acceptance decision



DECISION RULE

Accept only when 100% of running agent_id values resolve to one current owner. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Business impact mapping: architecture and control boundary

1

Business Service

State and evidence for business service

2

Agent Task

State and evidence for agent task

3

Dependent Process

State and evidence for dependent process

4

Customer Impact

State and evidence for customer impact

Control objective: Assign RTO/RPO and a manual fallback before production

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Business impact mapping: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
business service	trace_id + agent_id + policy_version	Assign RTO/RPO and a manual fallback before production	customer impact	Observed recovery time ≤ approved RTO

Failure to detect: Agent outage exceeds the process recovery objective

Supplied AI-security reference corpus · MITRE ATLAS

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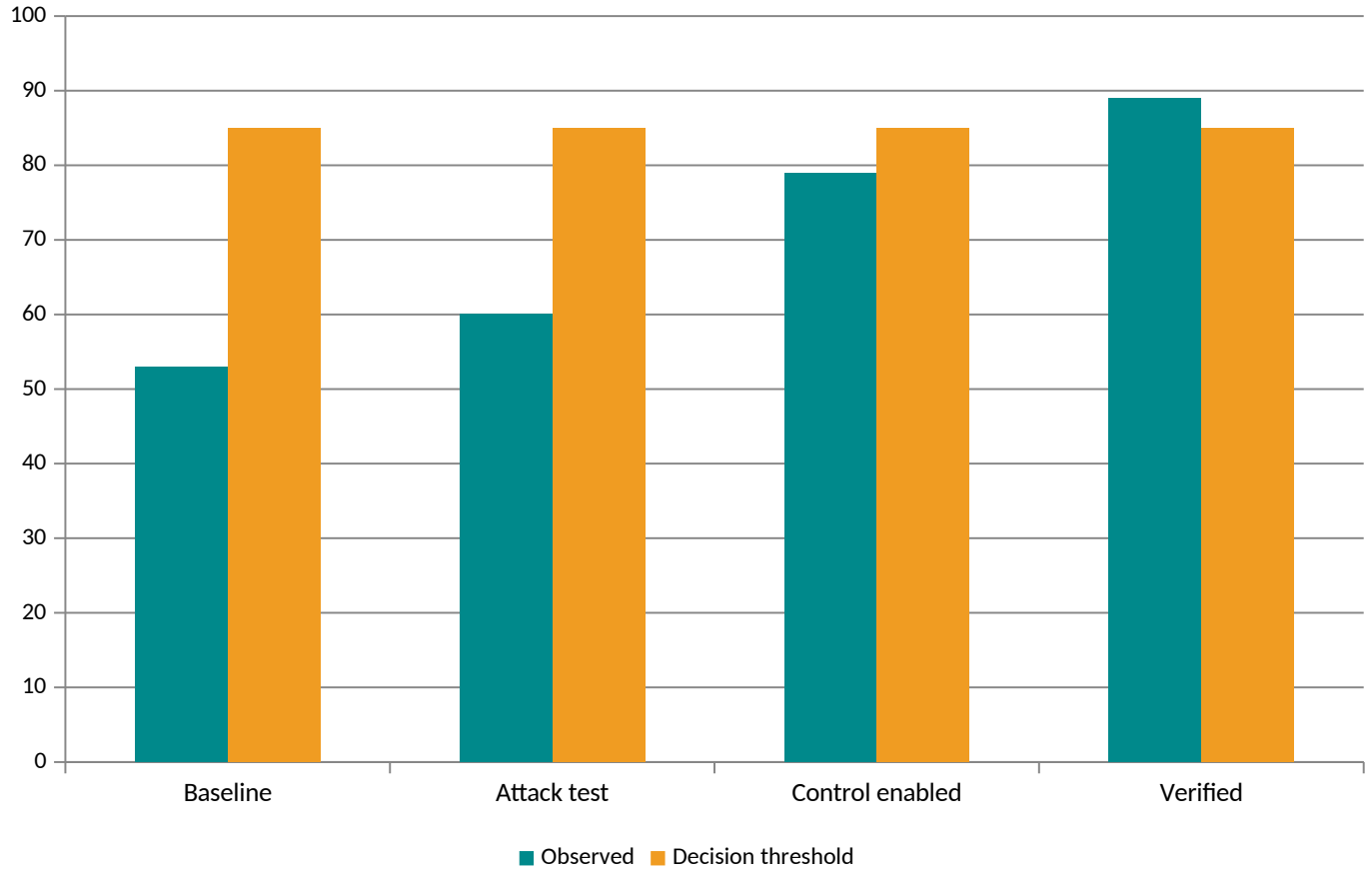
Business impact mapping: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	service_id	Required in decision and audit evidence
field_2	process_rto	Required in decision and audit evidence
field_3	data_rpo	Required in decision and audit evidence
field_4	maximum_loss_sgd	Required in decision and audit evidence
field_5	criticality	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Business impact mapping: evidence and acceptance decision



DECISION RULE

Accept only when Observed recovery time ≤ approved RTO. Otherwise record a gap, owner and due date.

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Security objective hierarchy: architecture and control boundary



Control objective: Write SMART control objectives with a named evidence source

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Security objective hierarchy: runtime decision trace

1

Observe

vision

2

Correlate

trace_id + agent_id + policy_version

3

Decide

Write SMART control objectives with a named evidence source

4

Act

metric

5

Prove

Every objective has one owner, threshold and review date

Failure to detect: A vague goal cannot be tested or owned

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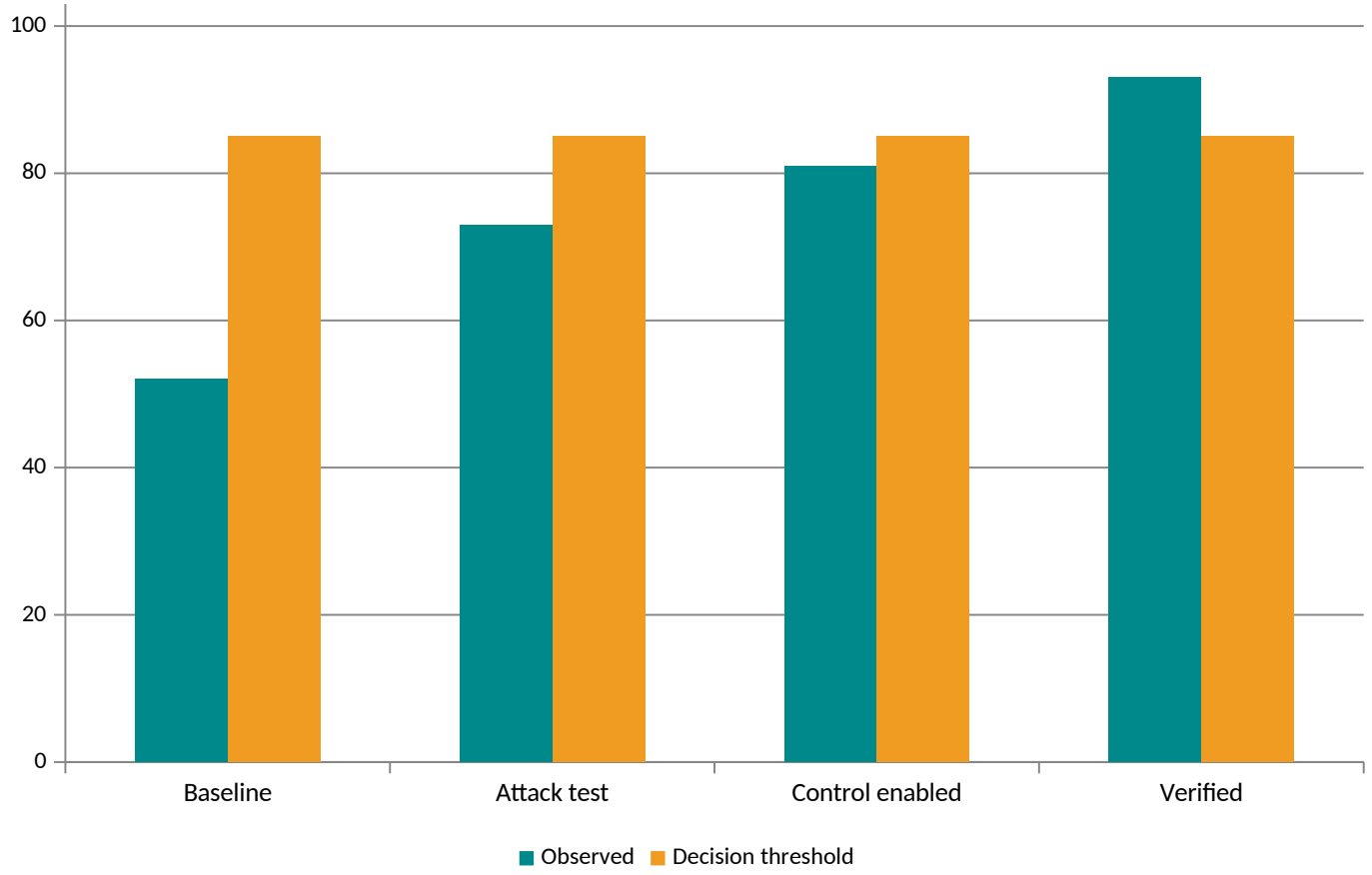
Security objective hierarchy: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	objective_id	Required in decision and audit evidence
field_2	sponsor	Required in decision and audit evidence
field_3	target	Required in decision and audit evidence
field_4	threshold	Required in decision and audit evidence
field_5	due_date	Required in decision and audit evidence
field_6	evidence_source	Required in decision and audit evidence

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Security objective hierarchy: evidence and acceptance decision



DECISION RULE

Accept only when Every objective has one owner, threshold and review date. Otherwise record a gap, owner and due date.

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Data classification: architecture and control boundary



Control objective: Enforce label-aware routing, redaction and retention

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Data classification: runtime decision trace

1 — Observe

input sources

2 — Correlate

trace_id + agent_id + policy_version

3 — Decide

Enforce label-aware routing, redaction and retention

4 — Act

retention

5 — Prove

0 restricted records leave an approved processing boundary

Failure to detect: Restricted data is inserted into an unapproved prompt

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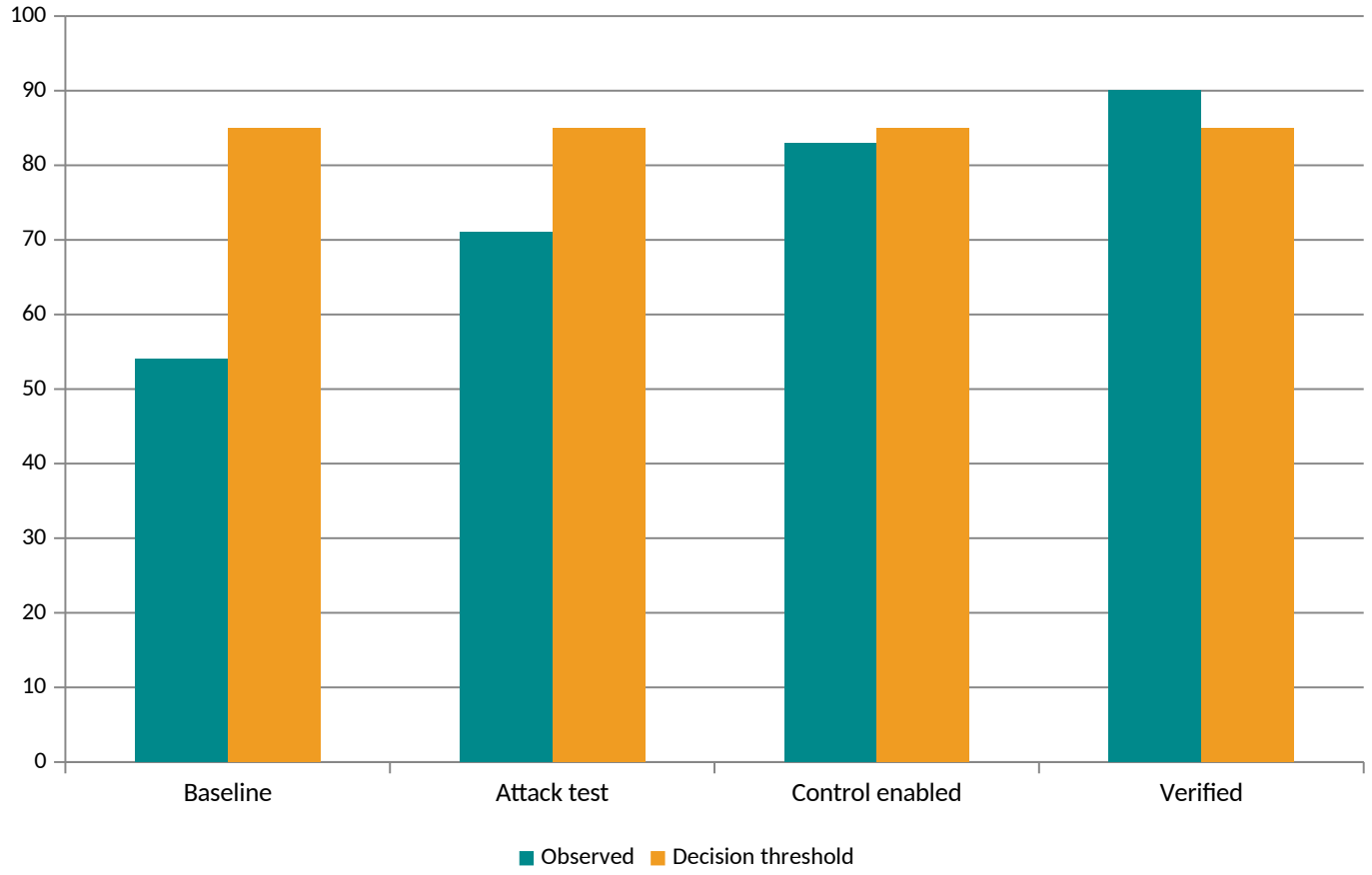
Data classification: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	data_class ∈ {public	Required in decision and audit evidence
field_2	internal	Required in decision and audit evidence
field_3	confidential	Required in decision and audit evidence
field_4	restricted}	Required in decision and audit evidence
field_5	purpose	Required in decision and audit evidence
field_6	retention_days	Required in decision and audit evidence

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Data classification: evidence and acceptance decision



DECISION RULE

Accept only when 0 restricted records leave an approved processing boundary. Otherwise record a gap, owner and due date.

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Trust-boundary map: architecture and control boundary



Control objective: Place authentication, authorisation and logging at every crossing

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Trust-boundary map: runtime decision trace

1

Observe

user zone

2

Correlate

trace_id + agent_id + policy_version

3

Decide

Place authentication, authorisation and logging at every crossing

4

Act

external service

5

Prove

Each cross-zone flow maps to a tested control

Failure to detect: A tool call crosses a boundary without an enforcement point

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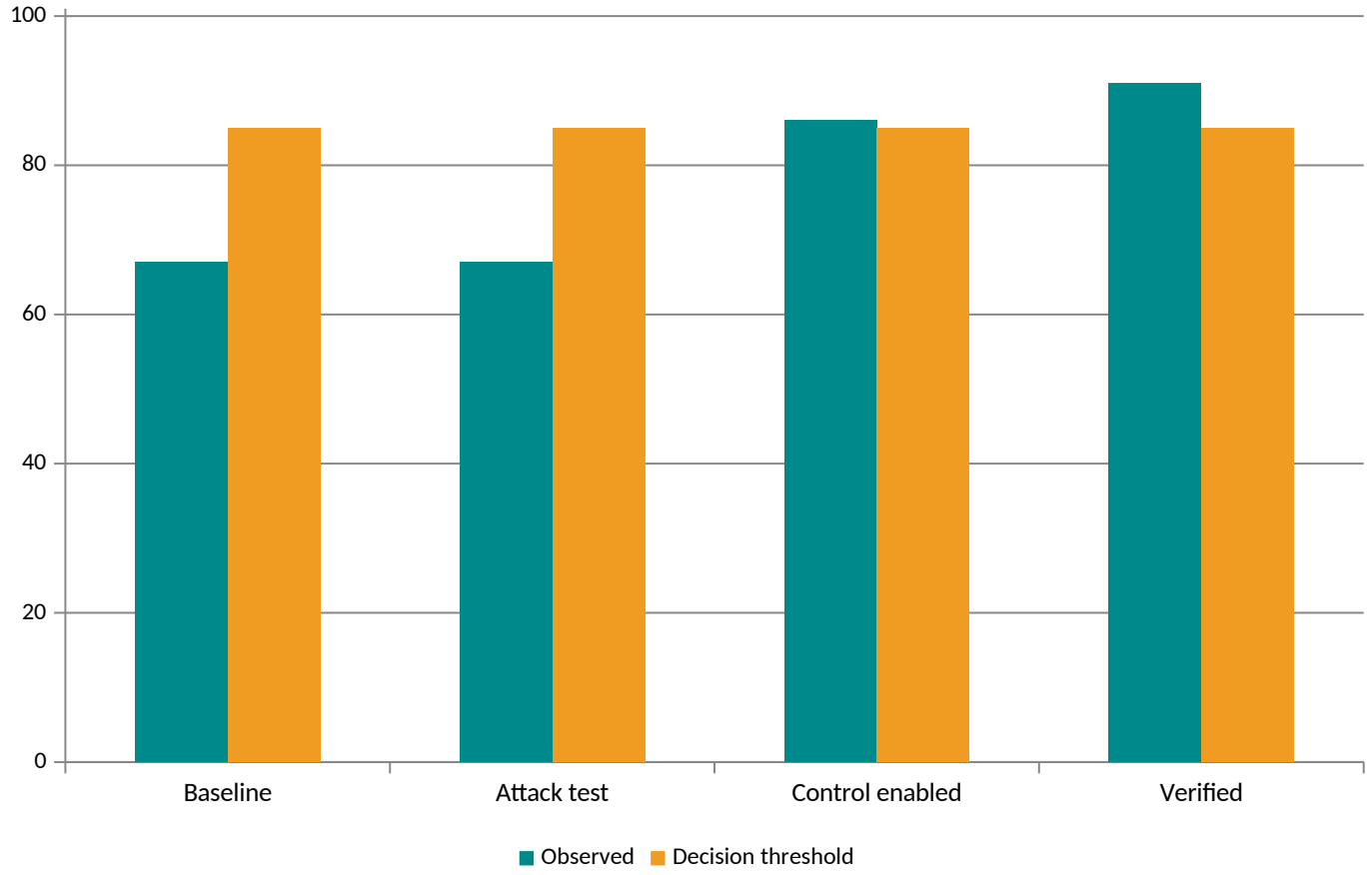
Trust-boundary map: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	boundary_id	Required in decision and audit evidence
field_2	source_zone	Required in decision and audit evidence
field_3	target_zone	Required in decision and audit evidence
field_4	protocol	Required in decision and audit evidence
field_5	authn	Required in decision and audit evidence
field_6	authz	Required in decision and audit evidence

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Trust-boundary map: evidence and acceptance decision



DECISION RULE

Accept only when Each cross-zone flow maps to a tested control. Otherwise record a gap, owner and due date.

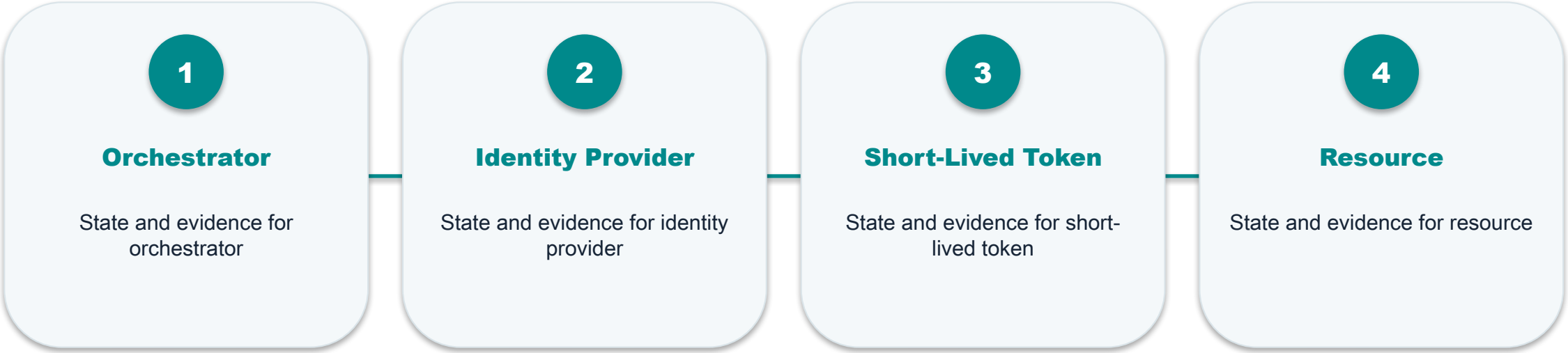
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Workload identity: architecture and control boundary



Control objective: Issue a unique workload identity and scoped token per agent

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Workload identity: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
orchestrator	trace_id + agent_id + policy_version	Issue a unique workload identity and scoped token per agent	resource	Token subject uniquely identifies the calling agent

Failure to detect: Several agents share one service-account credential

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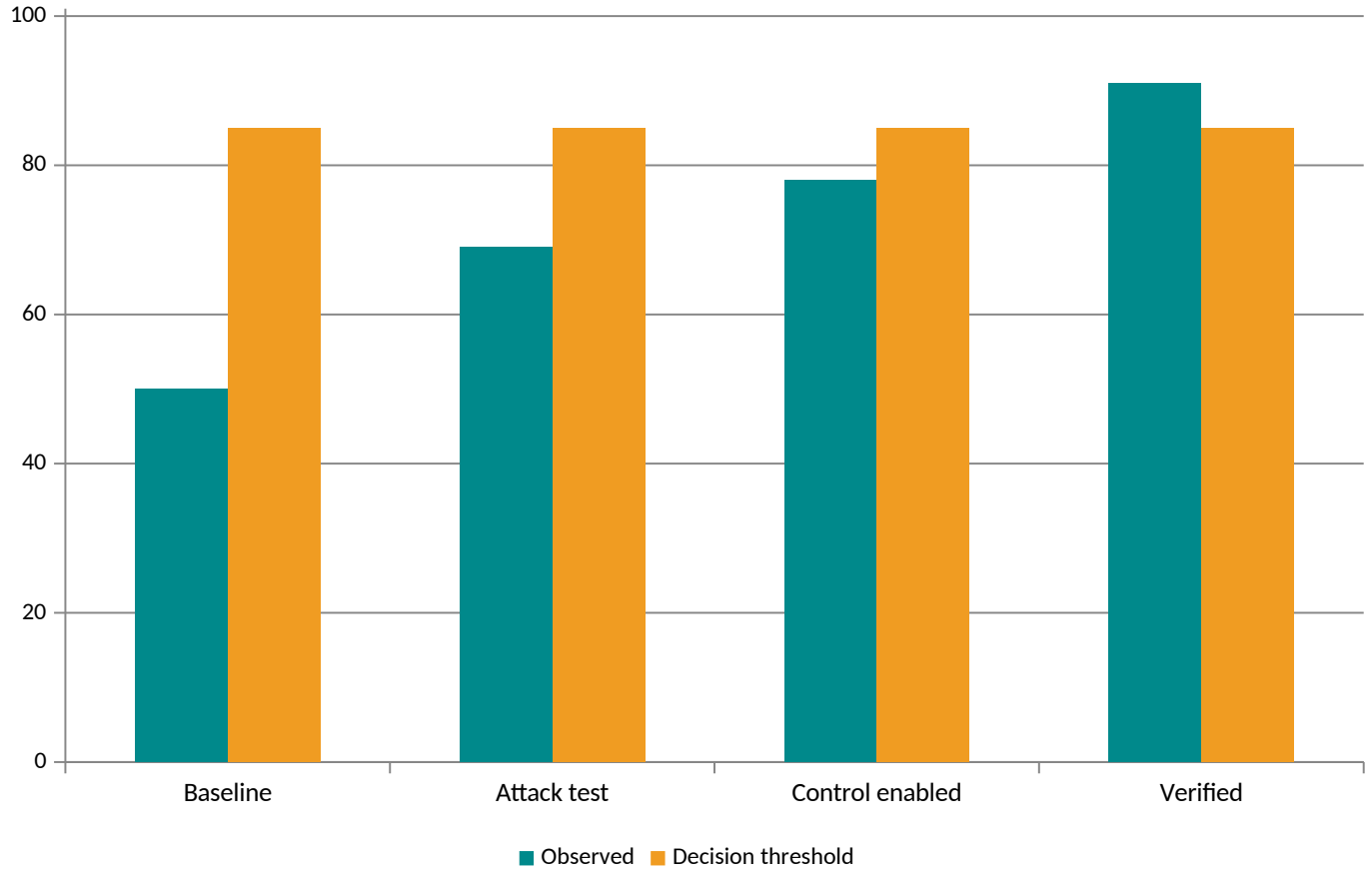
Workload identity: concrete control contract

Field / key	Expected value or type	Enforcement
subject	agent_id	Required in decision and audit evidence
audience	tool_id	Required in decision and audit evidence
field_3	ttl_seconds≤900	Required in decision and audit evidence
field_4	scopes[]	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

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Workload identity: evidence and acceptance decision



DECISION RULE

Accept only when Token subject uniquely identifies the calling agent. Otherwise record a gap, owner and due date.

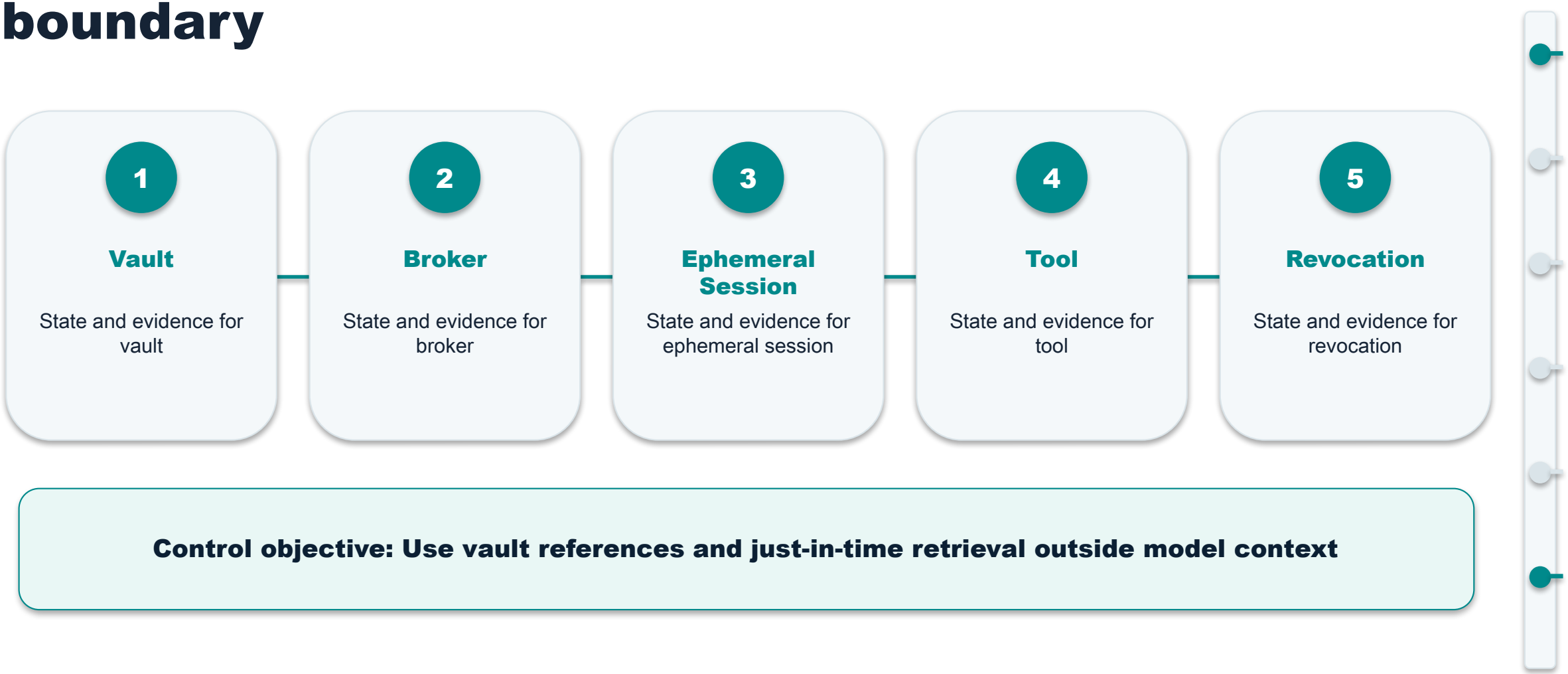
EVIDENCE STATUS

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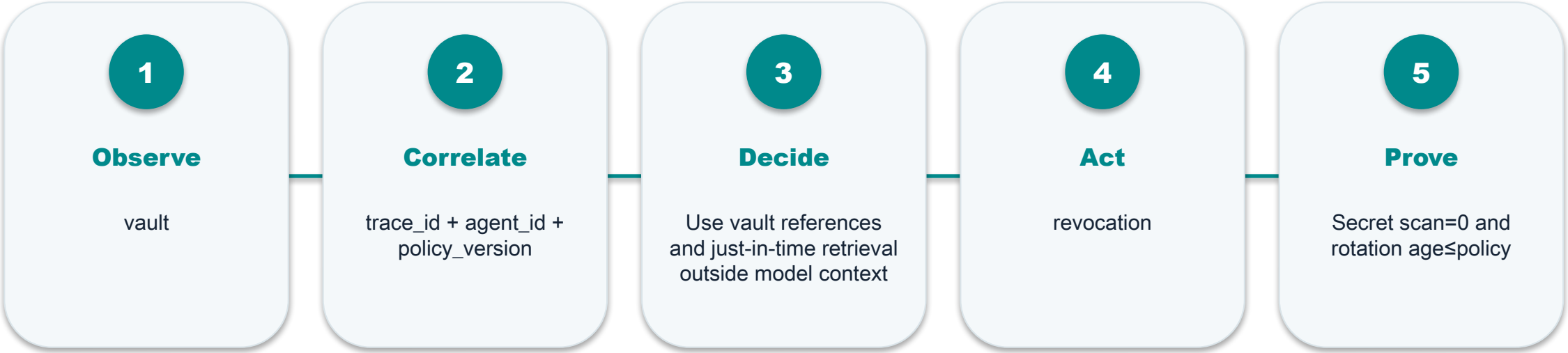
Credential lifecycle: architecture and control boundary



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Credential lifecycle: runtime decision trace



Failure to detect: A credential appears in prompt context or source code

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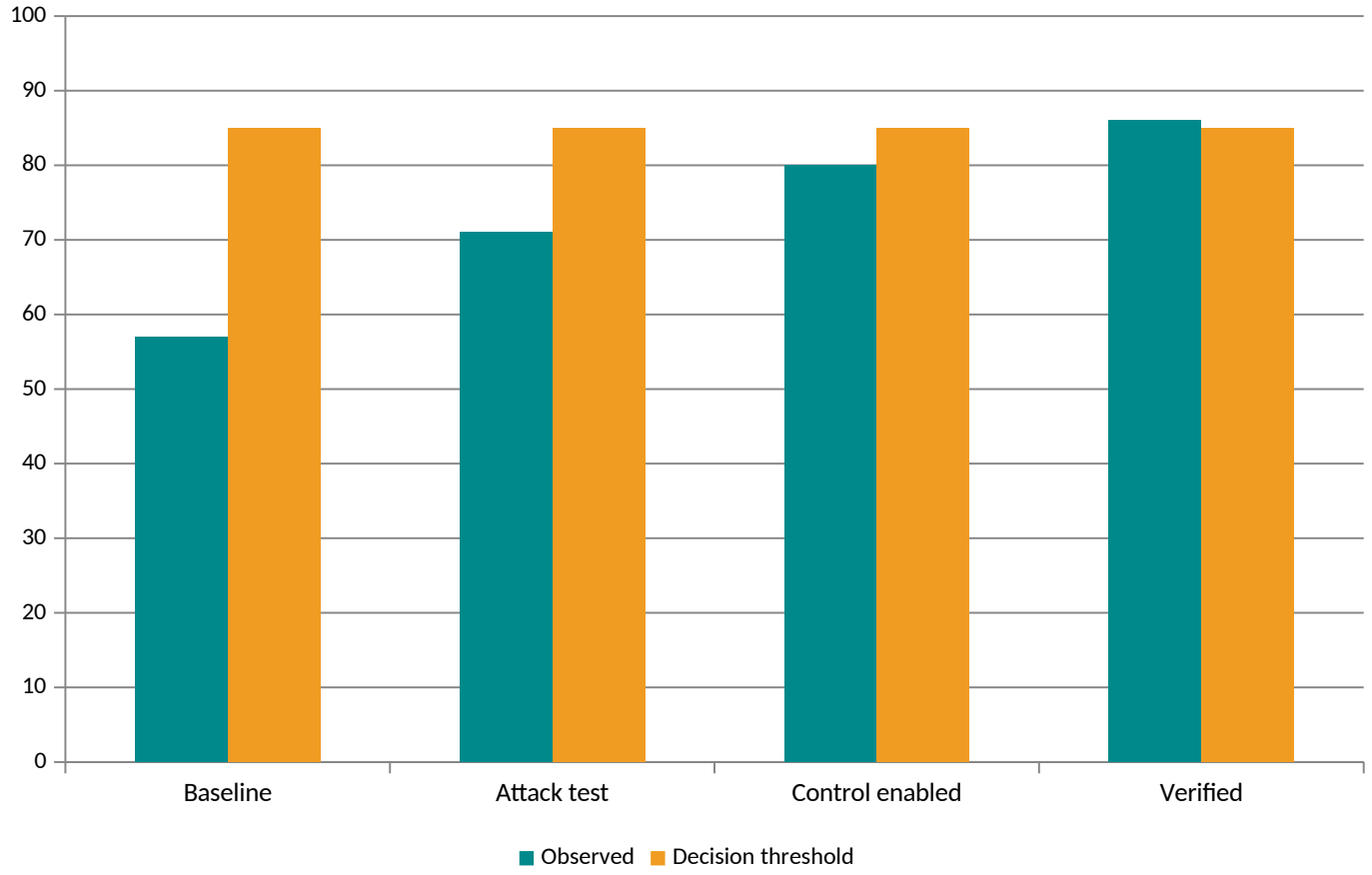
Credential lifecycle: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	secret_ref	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	issued_at	Required in decision and audit evidence
field_4	expires_at	Required in decision and audit evidence
field_5	rotation_days	Required in decision and audit evidence
field_6	last_access	Required in decision and audit evidence

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Credential lifecycle: evidence and acceptance decision



DECISION RULE

Accept only when Secret scan=0 and rotation age≤policy. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Tool registry: architecture and control boundary

1

Tool Package

State and evidence for tool package

2

Provenance Check

State and evidence for provenance check

3

Capability Record

State and evidence for capability record

4

Approval

State and evidence for approval

5

Runtime

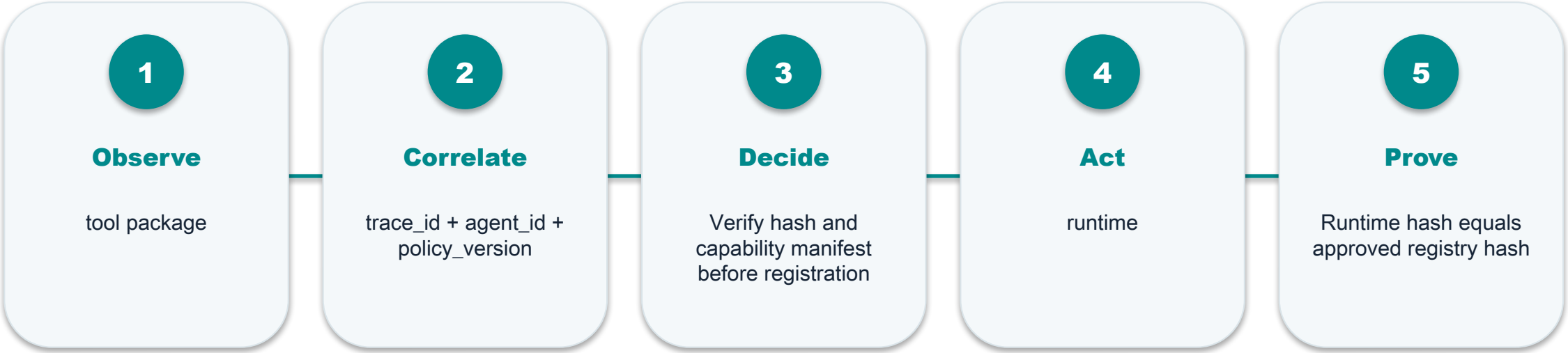
State and evidence for runtime

Control objective: Verify hash and capability manifest before registration

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Tool registry: runtime decision trace



Failure to detect: A renamed tool adds an undeclared destructive capability

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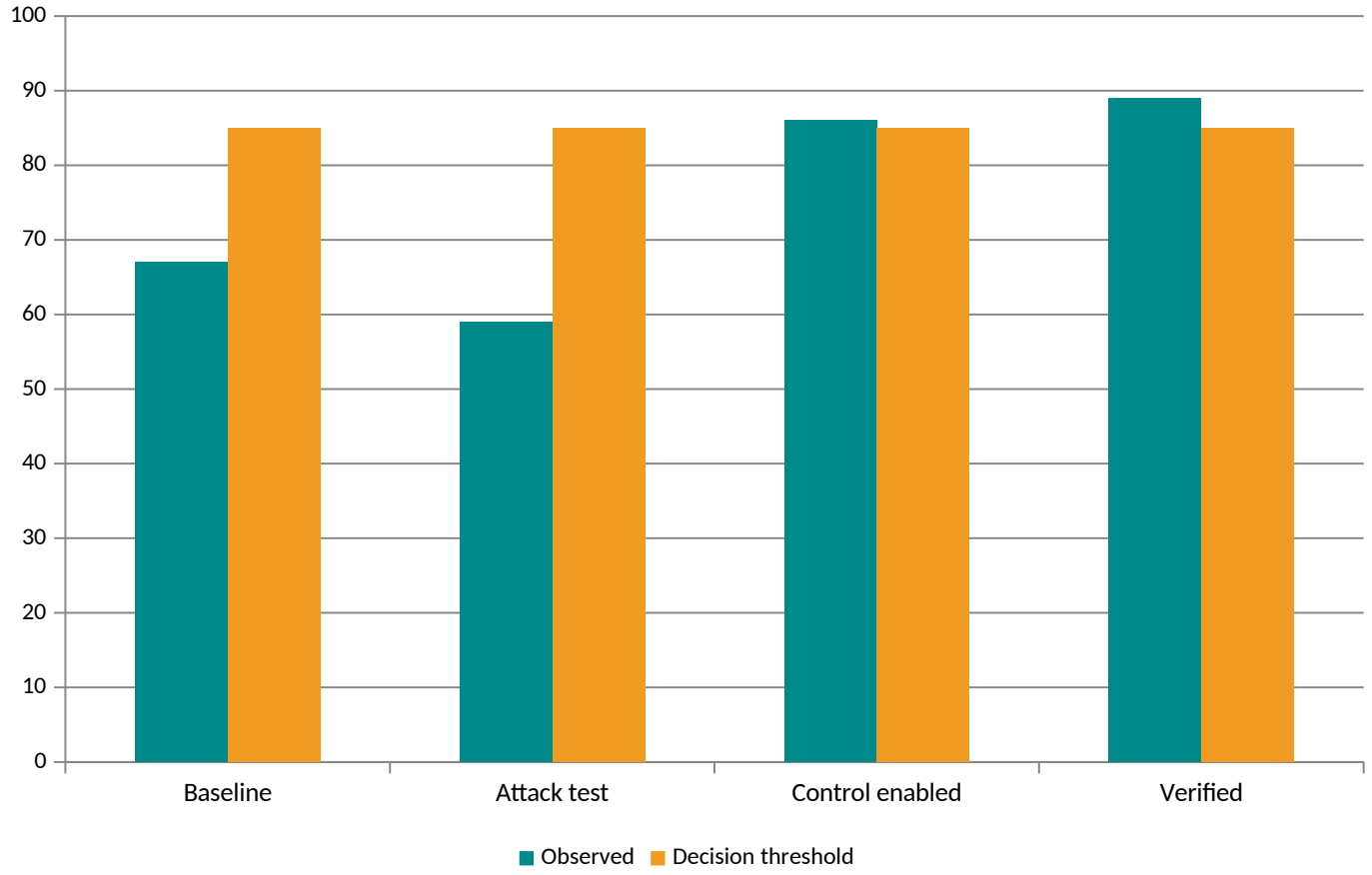
Tool registry: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tool_id	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	publisher	Required in decision and audit evidence
field_4	hash	Required in decision and audit evidence
field_5	capabilities[]	Required in decision and audit evidence
field_6	risk_tier	Required in decision and audit evidence

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Tool registry: evidence and acceptance decision



DECISION RULE

Accept only when Runtime hash equals approved registry hash. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Tool allowlist: architecture and control boundary

1

Planner Proposal

State and evidence for planner proposal

2

Policy Decision

State and evidence for policy decision

3

Argument Validation

State and evidence for argument validation

4

Tool Execution

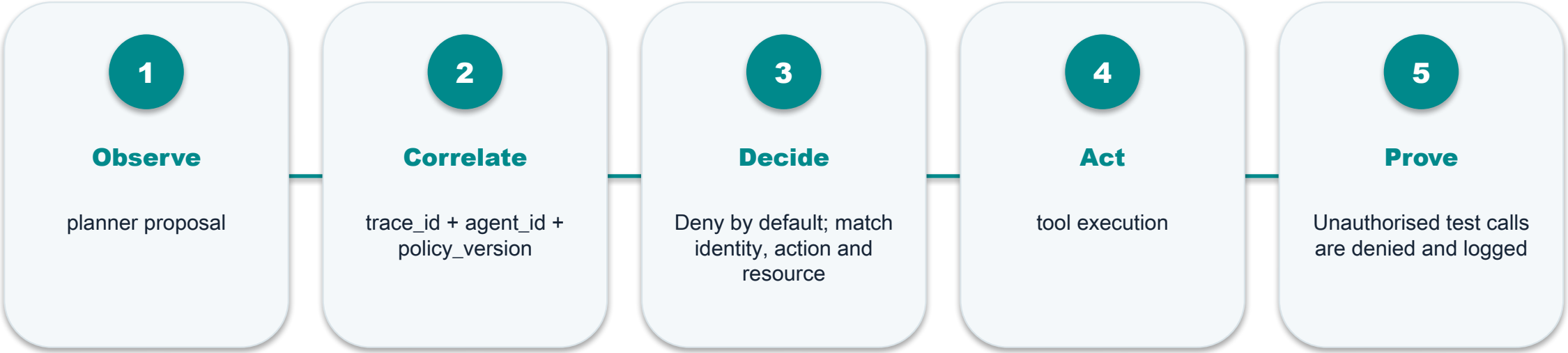
State and evidence for tool execution

Control objective: Deny by default; match identity, action and resource

Supplied AI-security reference corpus · NIST AI RMF · OWASP ASI

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Tool allowlist: runtime decision trace



Failure to detect: Model selects a valid but unauthorised administrative tool

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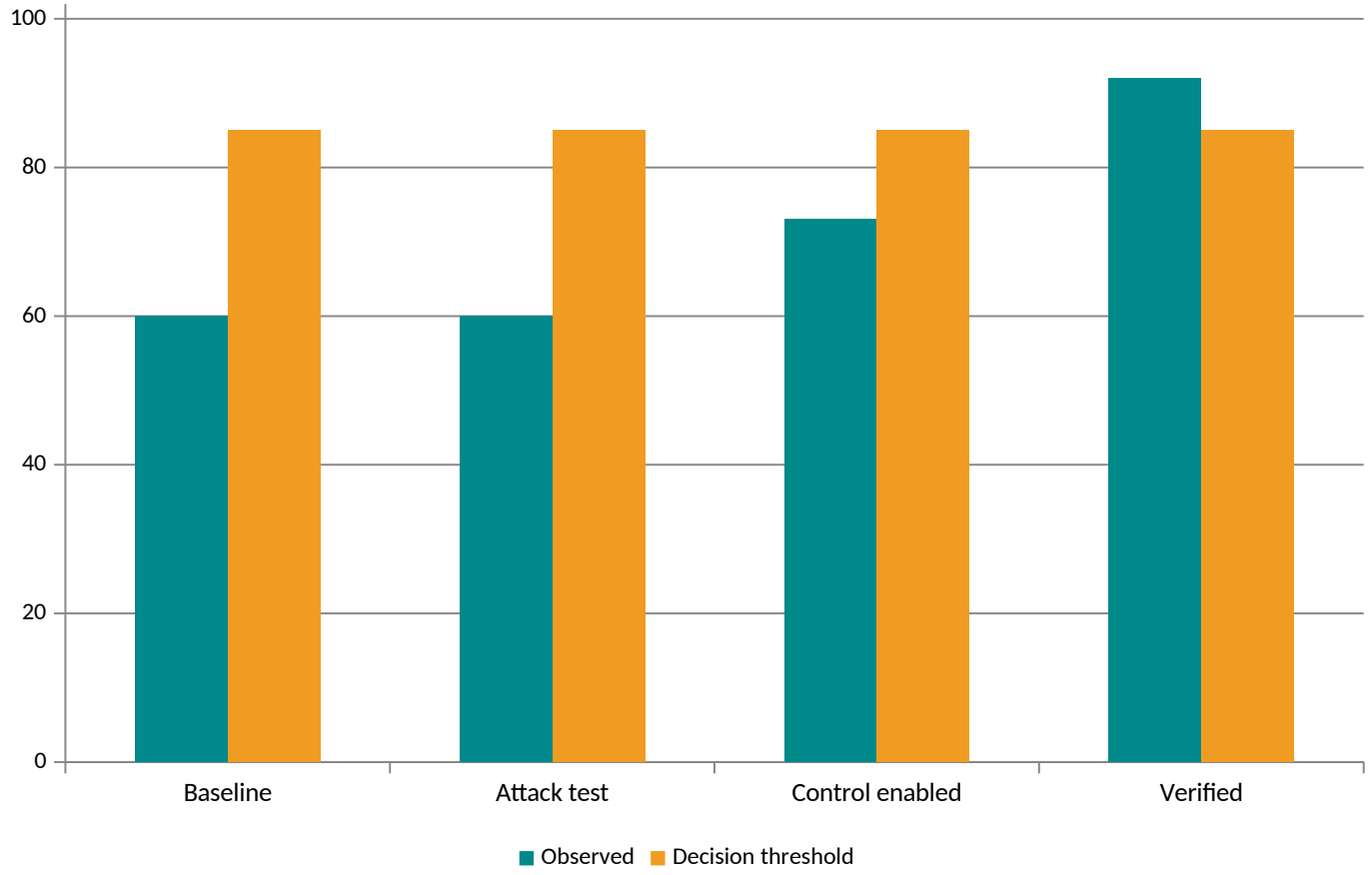
Tool allowlist: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	tool_id	Required in decision and audit evidence
field_3	action	Required in decision and audit evidence
field_4	resource	Required in decision and audit evidence
field_5	decision	Required in decision and audit evidence
field_6	policy_version	Required in decision and audit evidence

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Tool allowlist: evidence and acceptance decision



DECISION RULE

Accept only when Unauthorised test calls are denied and logged. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Approval taxonomy: architecture and control boundary



Control objective: Require signed, time-bound approval for irreversible actions

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Approval taxonomy: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
action proposal	trace_id + agent_id + policy_version	Require signed, time-bound approval for irreversible actions	execution	100% tier-3 executions carry a valid approval_id

Failure to detect: High-impact action runs without human confirmation

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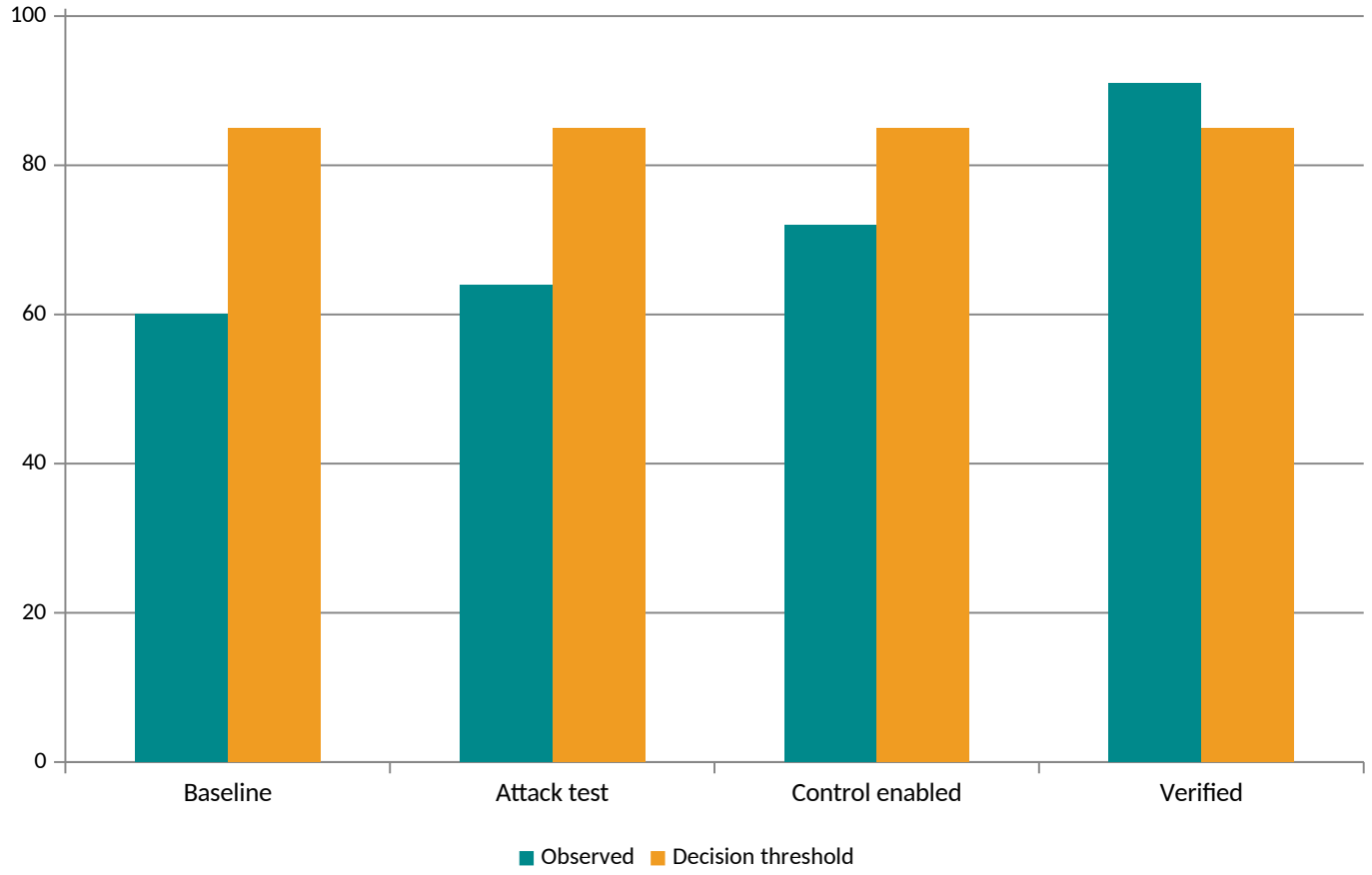
Approval taxonomy: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	risk_tier	Required in decision and audit evidence
field_2	approval_required	Required in decision and audit evidence
field_3	approver_role	Required in decision and audit evidence
field_4	timeout_seconds	Required in decision and audit evidence
field_5	reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Approval taxonomy: evidence and acceptance decision



DECISION RULE

Accept only when 100% tier-3 executions carry a valid approval_id. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Memory isolation: architecture and control boundary

1

Session Memory

State and evidence for session memory

2

Tenant Partition

State and evidence for tenant partition

3

Encrypted Store

State and evidence for encrypted store

4

Retrieval Filter

State and evidence for retrieval filter

Control objective: Partition by tenant and session; filter before retrieval

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Memory isolation: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
session memory	trace_id + agent_id + policy_version	Partition by tenant and session; filter before retrieval	retrieval filter	Cross-tenant retrieval test returns zero records

Failure to detect: One tenant retrieves another tenant's memory

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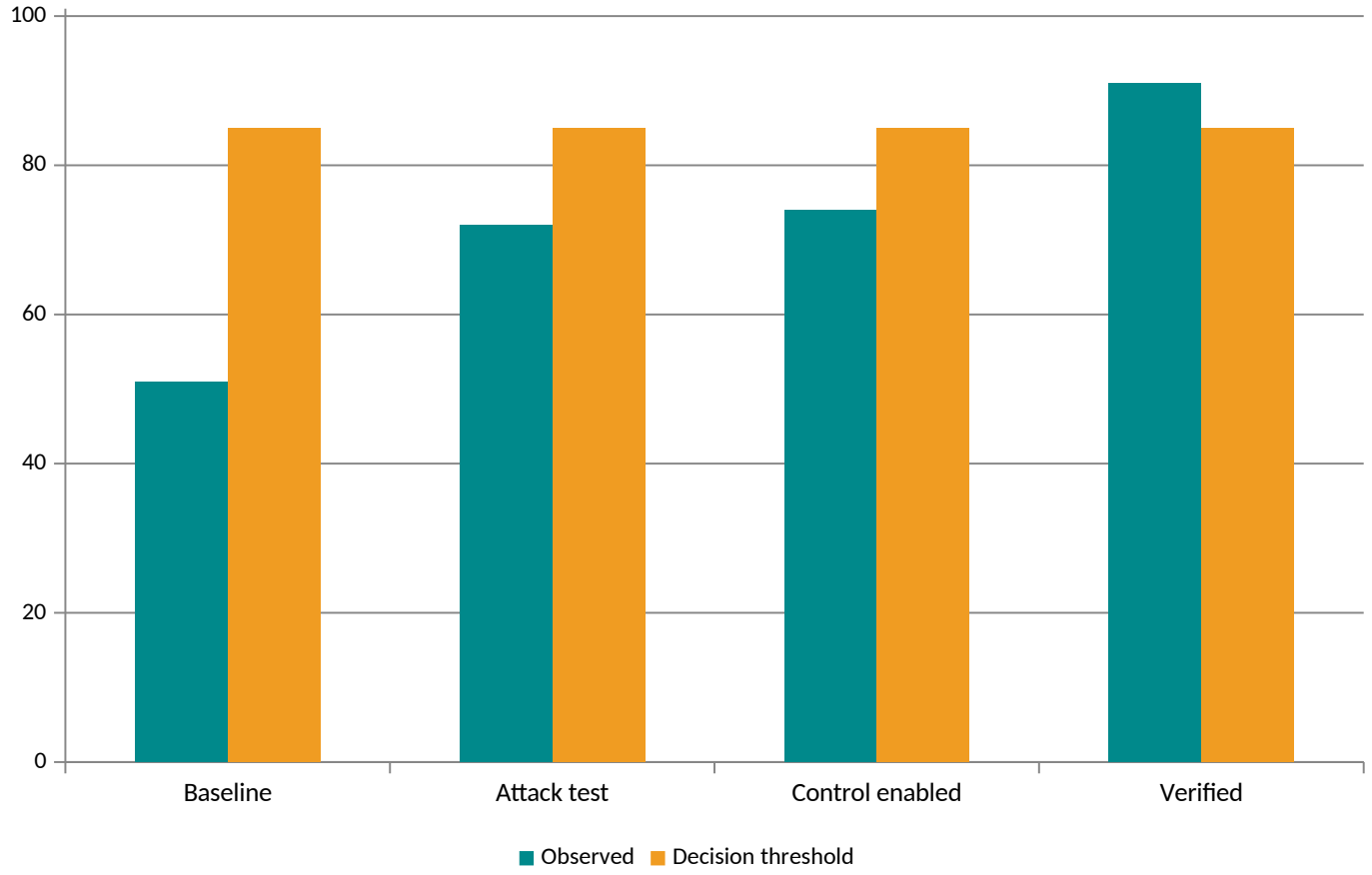
Memory isolation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tenant_id	Required in decision and audit evidence
field_2	session_id	Required in decision and audit evidence
field_3	namespace	Required in decision and audit evidence
field_4	classification	Required in decision and audit evidence
field_5	expires_at	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Memory isolation: evidence and acceptance decision



DECISION RULE

Accept only when Cross-tenant retrieval test returns zero records. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Session lifetime: architecture and control boundary



Control objective: Cap age, idle time and maximum autonomous steps

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Session lifetime: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
login	trace_id + agent_id + policy_version	Cap age, idle time and maximum autonomous steps	termination	Expired sessions reject the next tool call

Failure to detect: A forgotten session remains capable overnight

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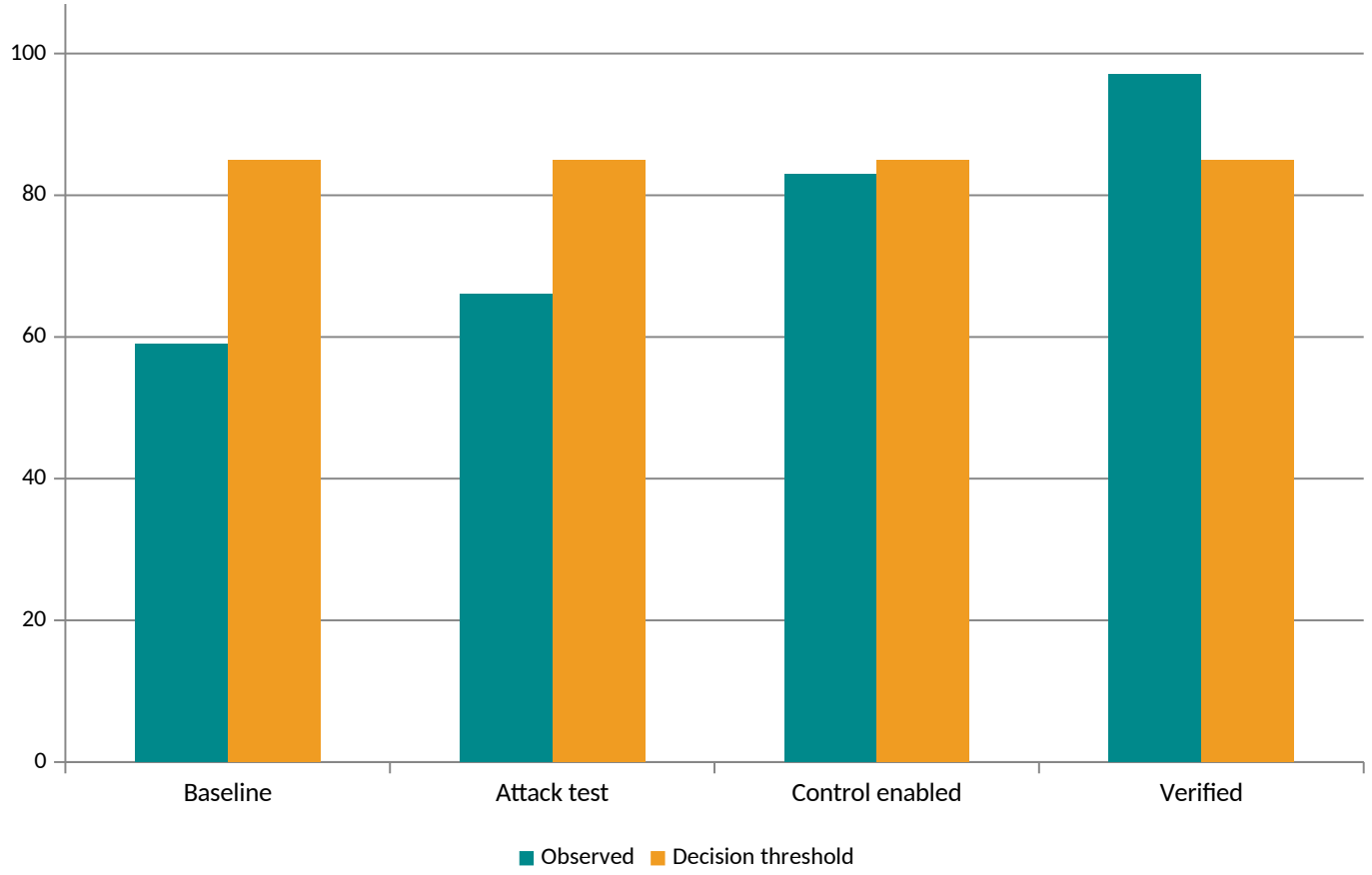
Session lifetime: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	session_id	Required in decision and audit evidence
field_2	created_at	Required in decision and audit evidence
field_3	last_active	Required in decision and audit evidence
field_4	max_age	Required in decision and audit evidence
field_5	idle_timeout	Required in decision and audit evidence
field_6	step_count	Required in decision and audit evidence

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Session lifetime: evidence and acceptance decision



DECISION RULE

Accept only when Expired sessions reject the next tool call. Otherwise record a gap, owner and due date.

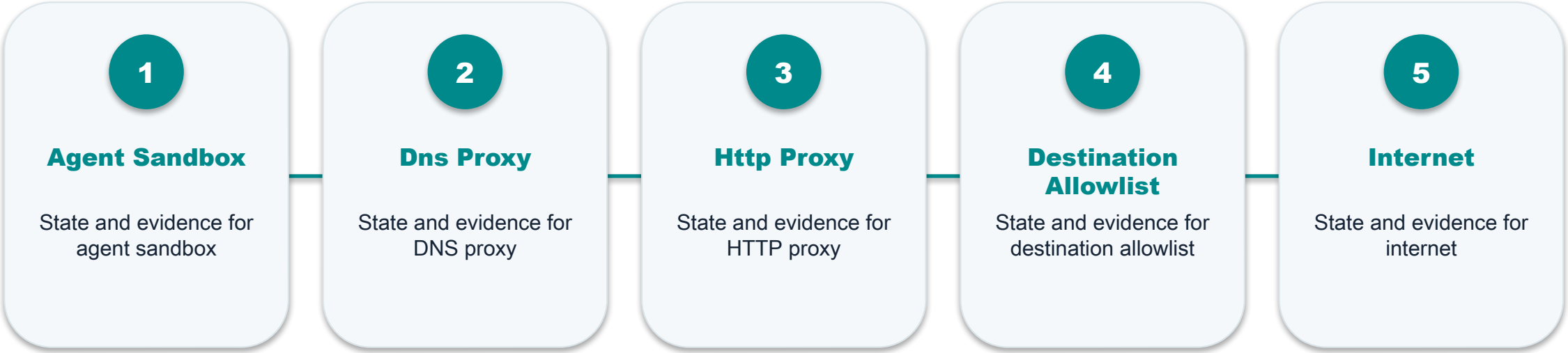
EVIDENCE STATUS

PASS

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Egress control: architecture and control boundary

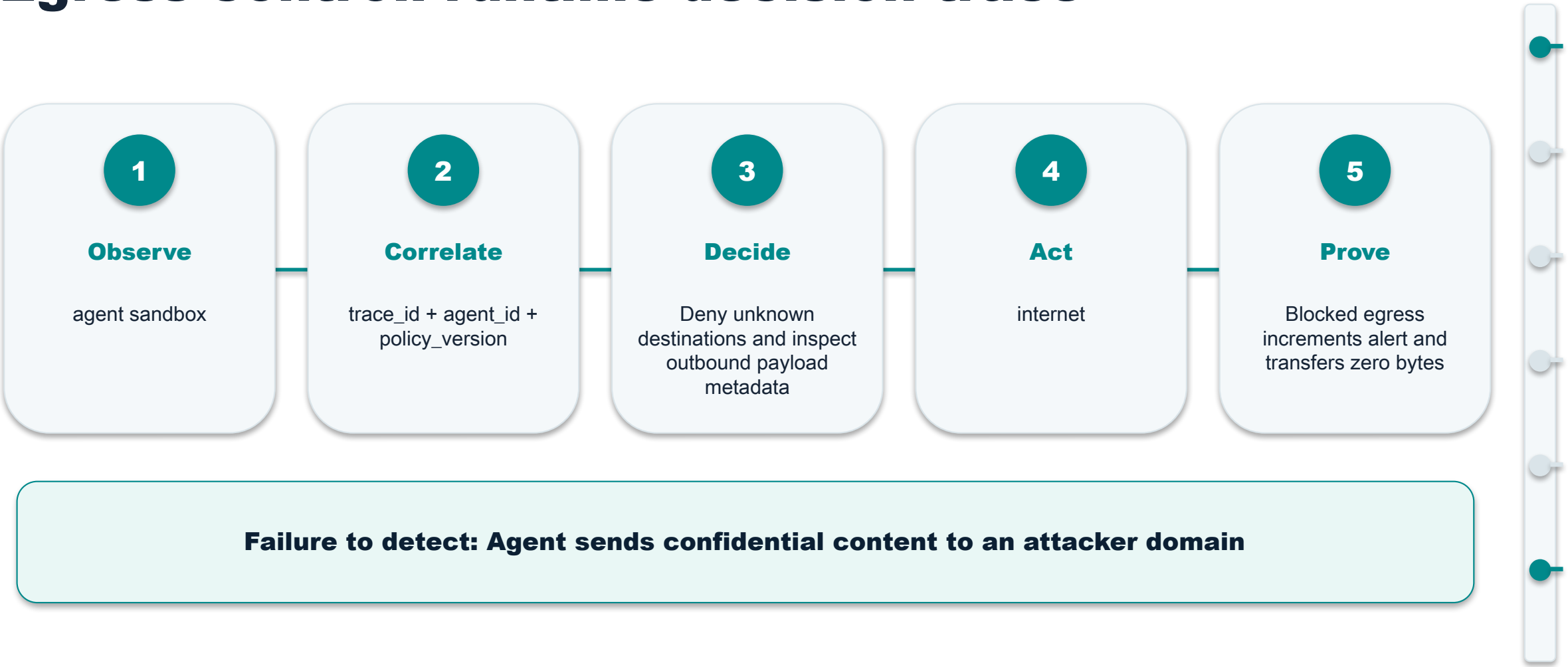


Control objective: Deny unknown destinations and inspect outbound payload metadata

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Egress control: runtime decision trace



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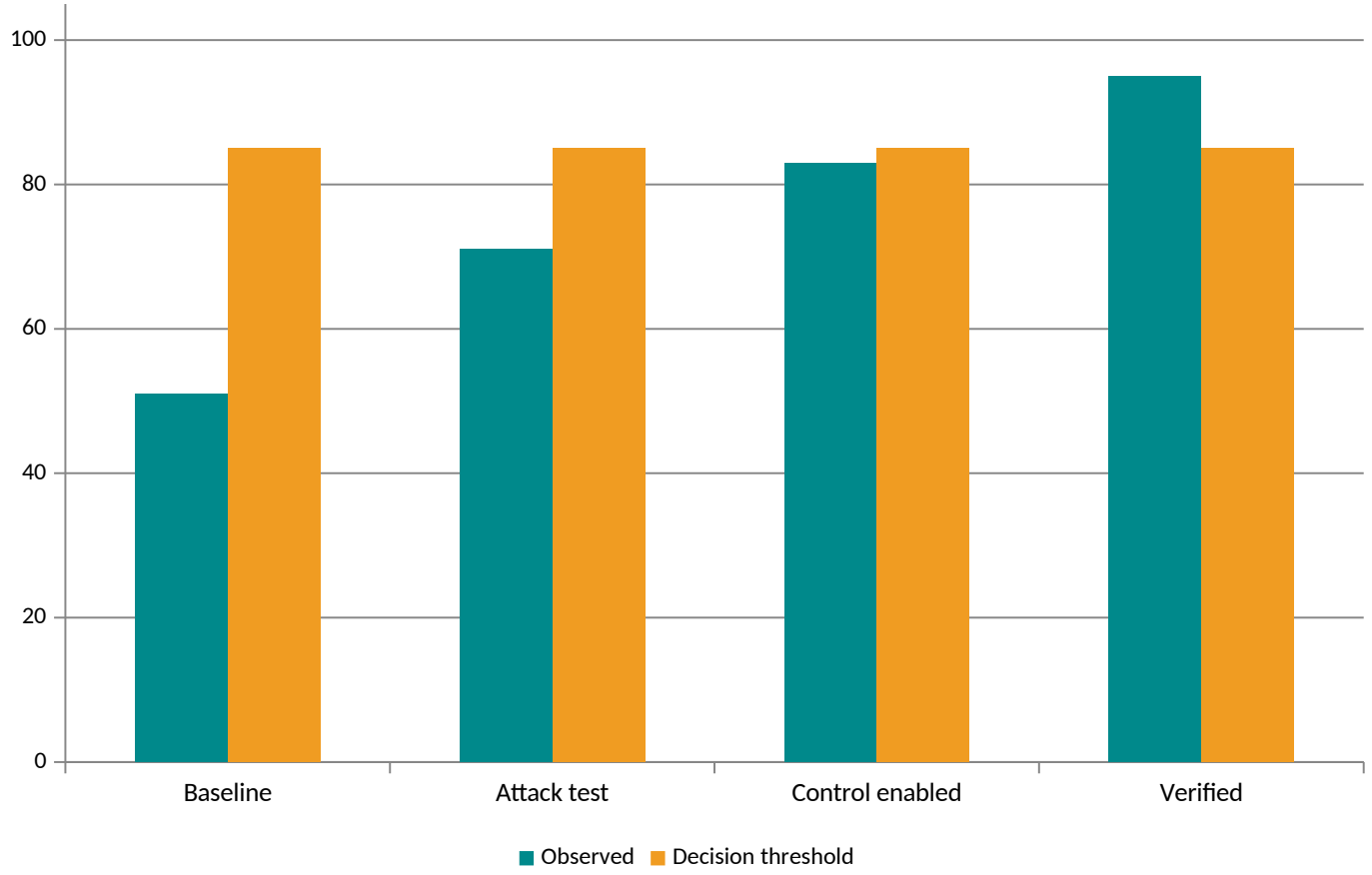
Egress control: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	destination	Required in decision and audit evidence
field_2	port	Required in decision and audit evidence
field_3	method	Required in decision and audit evidence
field_4	bytes_out	Required in decision and audit evidence
field_5	data_class	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

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Egress control: evidence and acceptance decision



DECISION RULE

Accept only when Blocked egress increments alert and transfers zero bytes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Sandbox profile: architecture and control boundary

1

Agent Process

State and evidence for agent process

2

Namespace

State and evidence for namespace

3

Filesystem Mount

State and evidence for filesystem mount

4

Network Policy

State and evidence for network policy

5

Resource Quota

State and evidence for resource quota

Control objective: Run non-root with read-only root and restricted syscalls

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Sandbox profile: runtime decision trace

1 — Observe

agent process

2 — Correlate

trace_id + agent_id + policy_version

3 — Decide

Run non-root with read-only root and restricted syscalls

4 — Act

resource quota

5 — Prove

Escape tests fail; quota breach terminates only the sandbox

Failure to detect: Generated code writes outside its workspace or escapes to host

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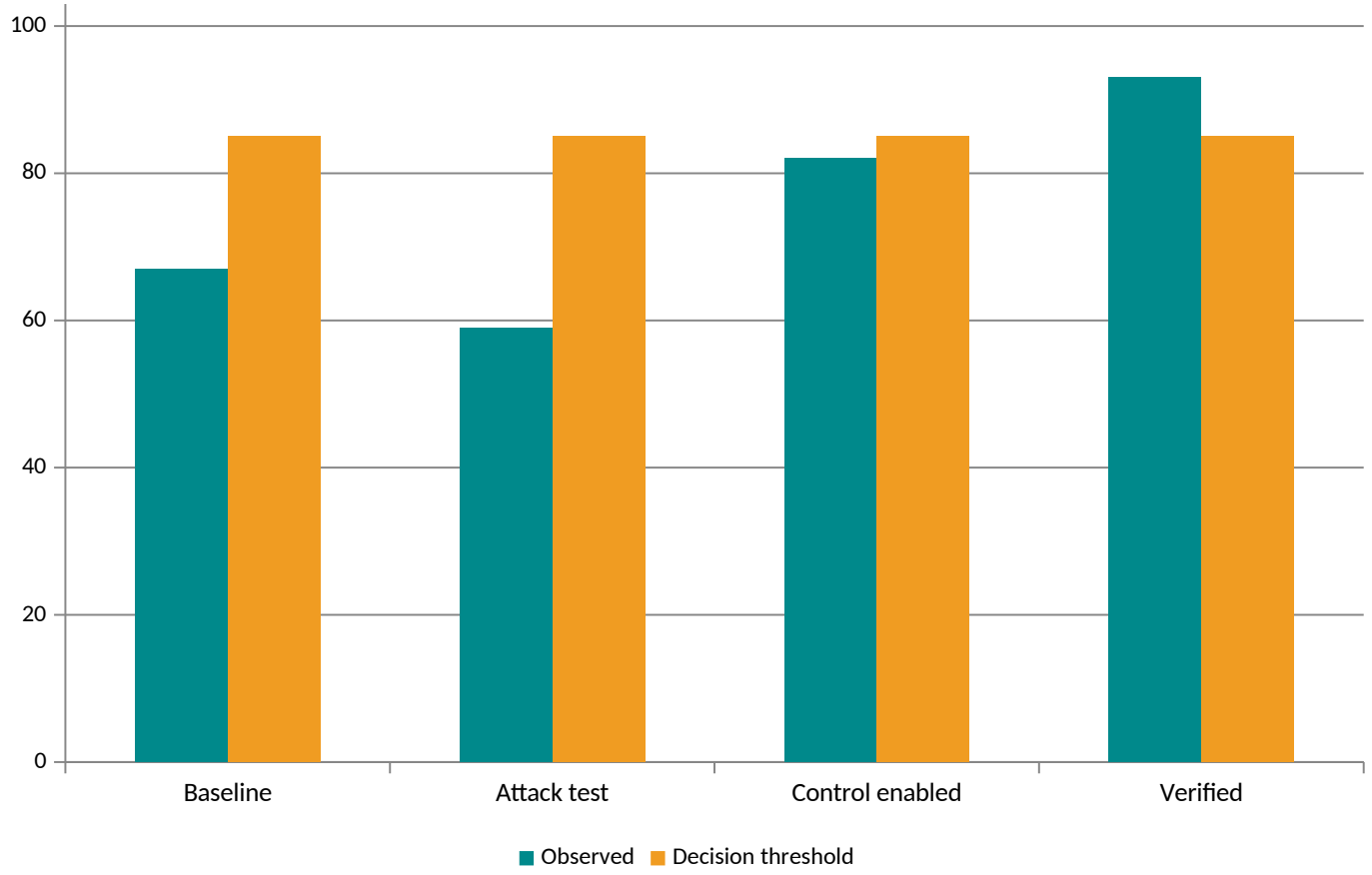
Sandbox profile: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	image_digest	Required in decision and audit evidence
field_2	uid	Required in decision and audit evidence
field_3	readonly_root	Required in decision and audit evidence
field_4	cpu_limit	Required in decision and audit evidence
field_5	memory_mb	Required in decision and audit evidence
field_6	network_profile	Required in decision and audit evidence

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Sandbox profile: evidence and acceptance decision



DECISION RULE

Accept only when Escape tests fail; quota breach terminates only the sandbox. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Artifact provenance: architecture and control boundary



Control objective: Require signed provenance tied to source and builder

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Artifact provenance: runtime decision trace



Failure to detect: An unsigned prompt/tool bundle reaches production

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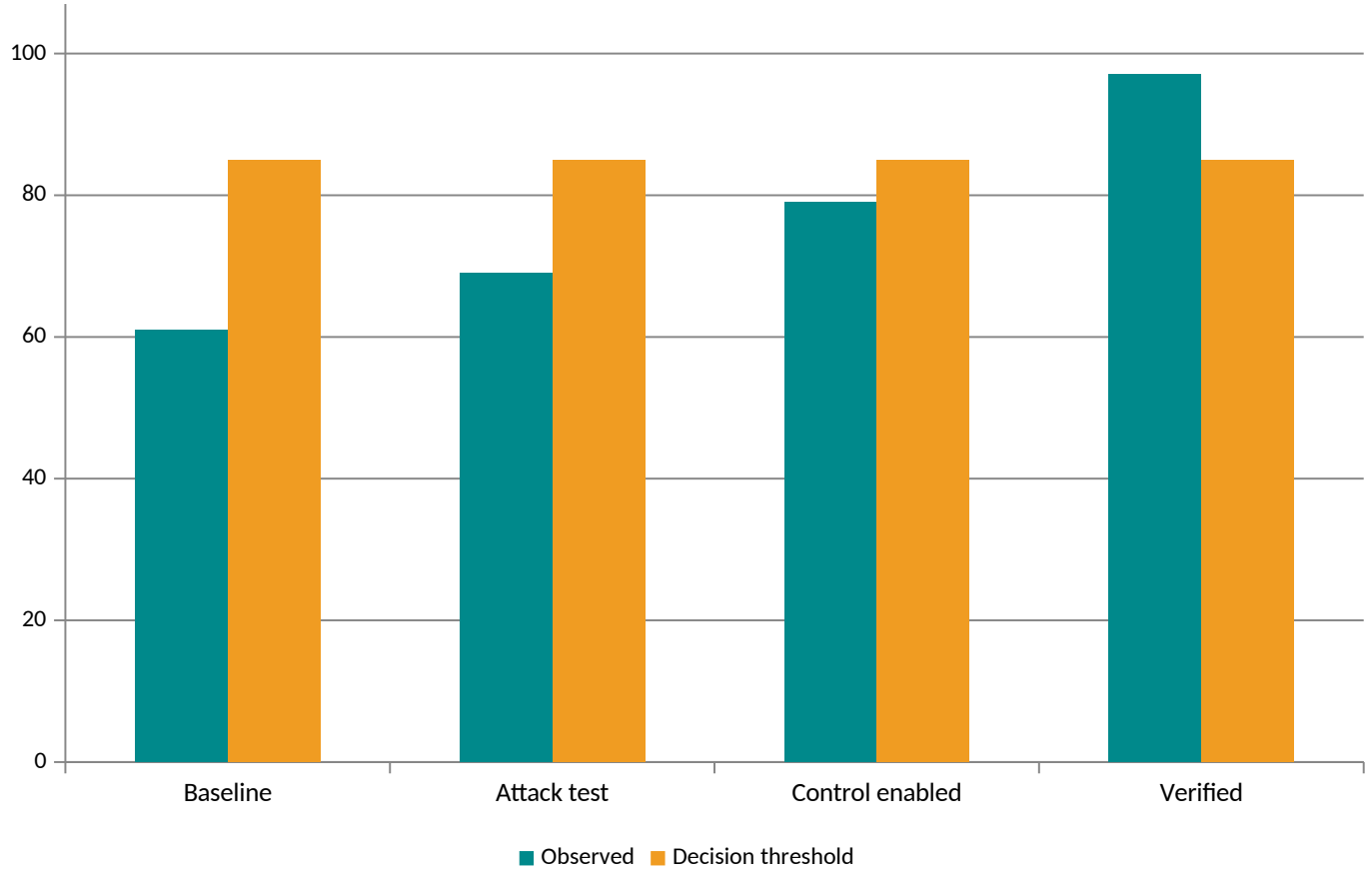
Artifact provenance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	artifact_digest	Required in decision and audit evidence
field_2	source_commit	Required in decision and audit evidence
field_3	builder_id	Required in decision and audit evidence
field_4	signature	Required in decision and audit evidence
field_5	predicate_type	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Artifact provenance: evidence and acceptance decision



DECISION RULE

Accept only when Deploy verifier rejects missing or mismatched attestations. Otherwise record a gap, owner and due date.

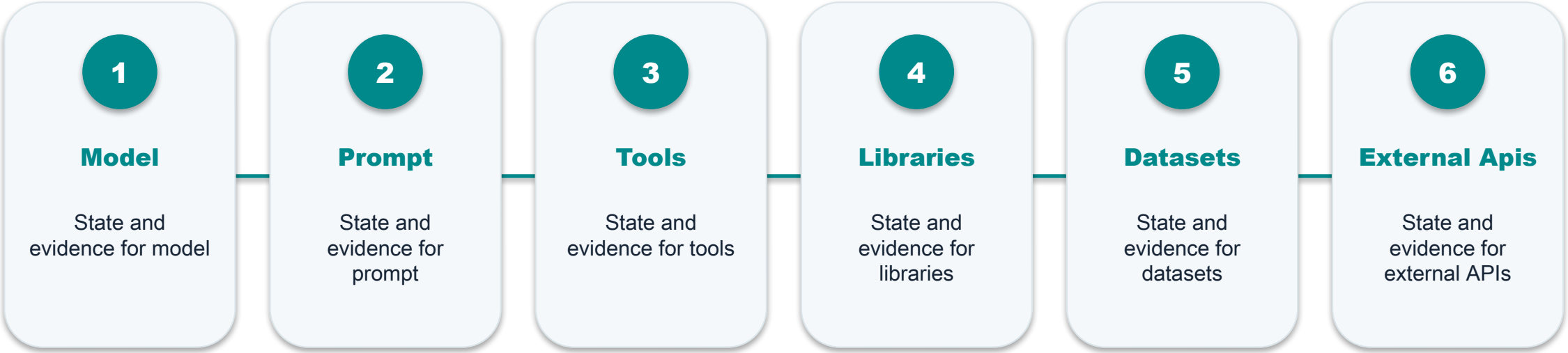
EVIDENCE STATUS

PASS

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AI bill of materials: architecture and control boundary



Control objective: Maintain an AI BOM and rescan on every release

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AI bill of materials: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
model	trace_id + agent_id + policy_version	Maintain an AI BOM and rescan on every release	external APIs	All deployed components have versions and scan timestamps

Failure to detect: Vulnerable dependency or changed model is invisible

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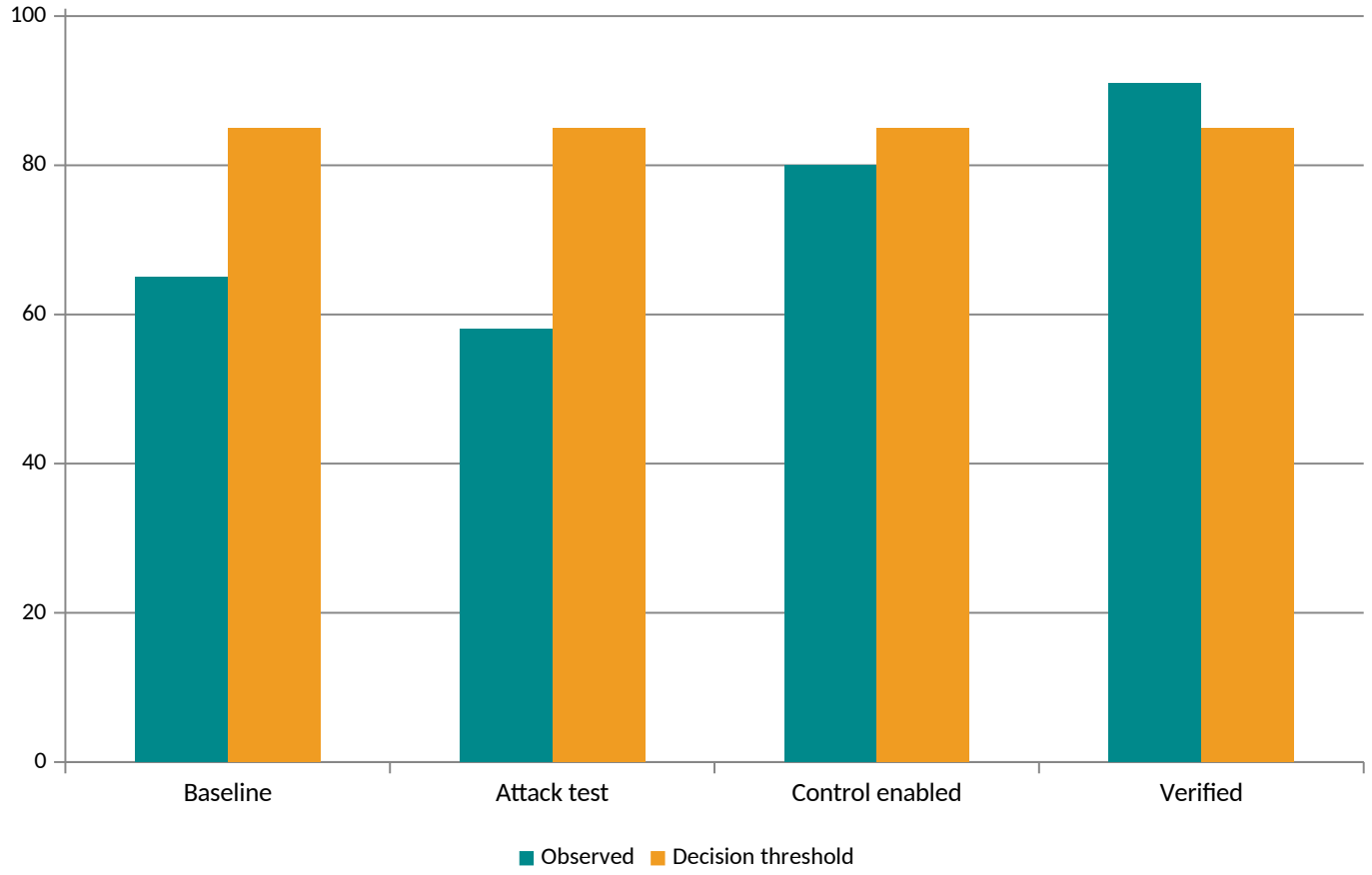
AI bill of materials: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	component	Required in decision and audit evidence
field_2	supplier	Required in decision and audit evidence
field_3	version	Required in decision and audit evidence
field_4	licence	Required in decision and audit evidence
field_5	hash	Required in decision and audit evidence
field_6	vulnerability_status	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

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AI bill of materials: evidence and acceptance decision



DECISION RULE

Accept only when All deployed components have versions and scan timestamps. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Supplier assurance: architecture and control boundary

1

Provider Questionnaire

State and evidence for provider questionnaire

2

Evidence Review

State and evidence for evidence review

3

Contract Controls

State and evidence for contract controls

4

Monitoring

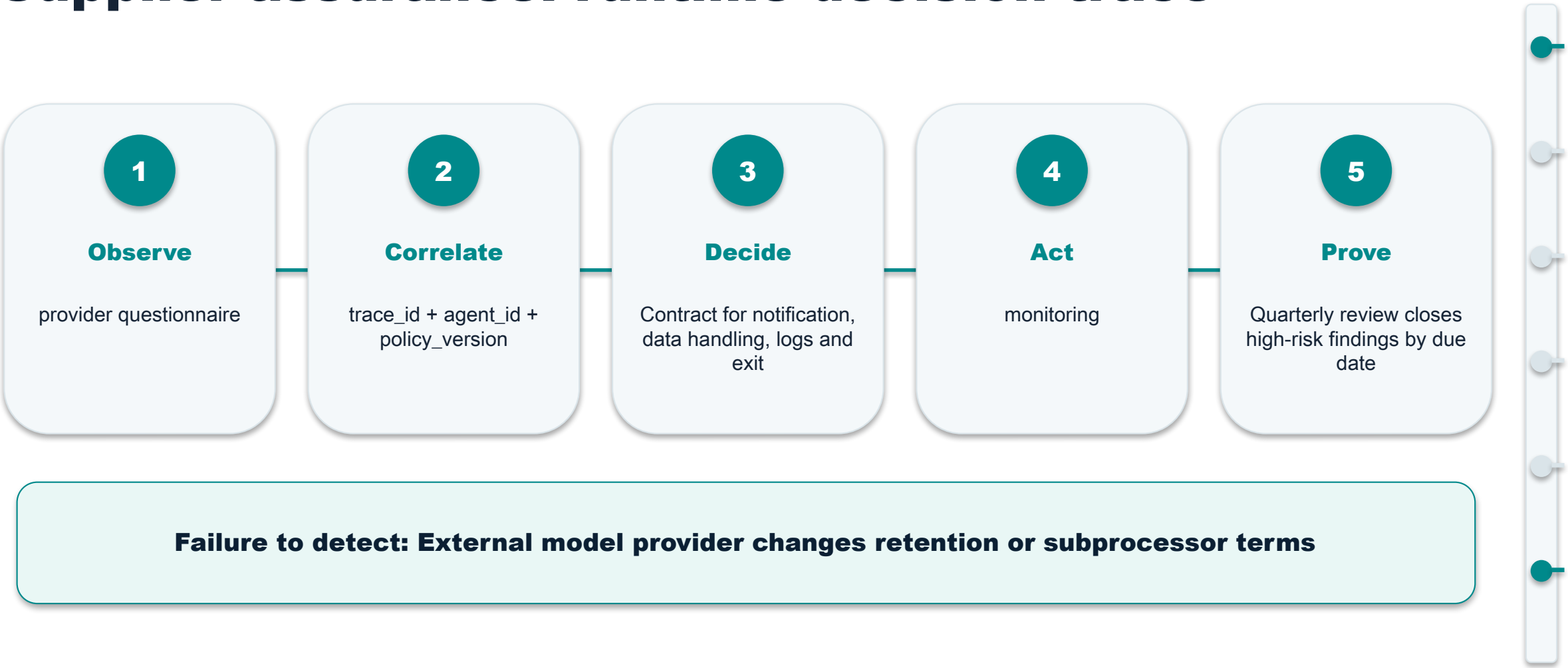
State and evidence for monitoring

Control objective: Contract for notification, data handling, logs and exit

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Supplier assurance: runtime decision trace



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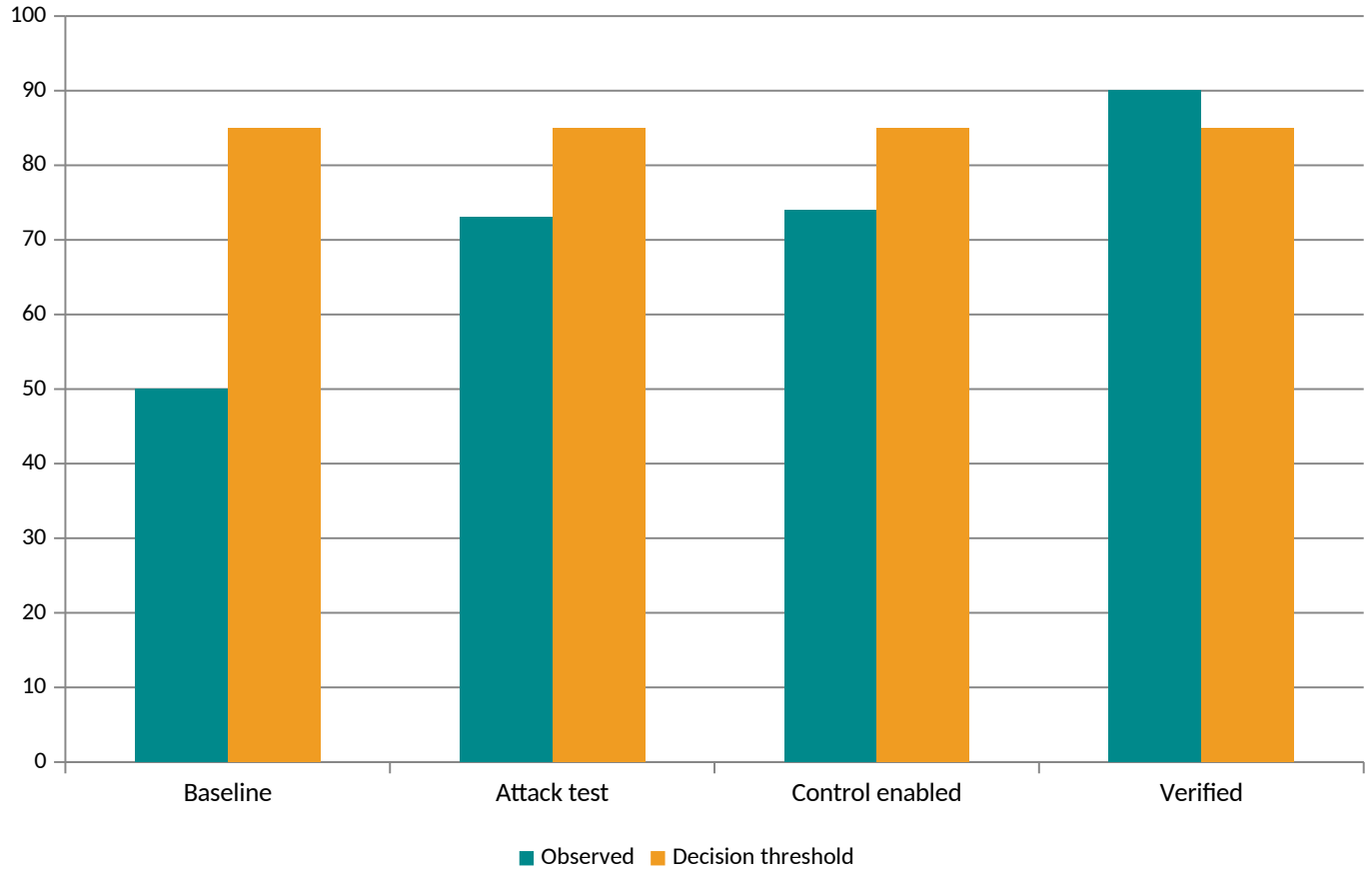
Supplier assurance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	supplier_id	Required in decision and audit evidence
field_2	service	Required in decision and audit evidence
field_3	data_region	Required in decision and audit evidence
field_4	subprocessors[]	Required in decision and audit evidence
field_5	SLA	Required in decision and audit evidence
field_6	exit_plan	Required in decision and audit evidence

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Supplier assurance: evidence and acceptance decision



DECISION RULE

Accept only when Quarterly review closes high-risk findings by due date. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Risk appetite: architecture and control boundary

1

Risk Scenario

State and evidence for risk scenario

2

Inherent Score

State and evidence for inherent score

3

Appetite Threshold

State and evidence for appetite threshold

4

Response Decision

State and evidence for response decision

Control objective: Escalate residual_score>appetite to the risk owner

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Risk appetite: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
risk scenario	trace_id + agent_id + policy_version	Escalate residual_score>appetite to the risk owner	response decision	No above-appetite risk lacks a signed treatment

Failure to detect: Teams accept high residual risk without authority

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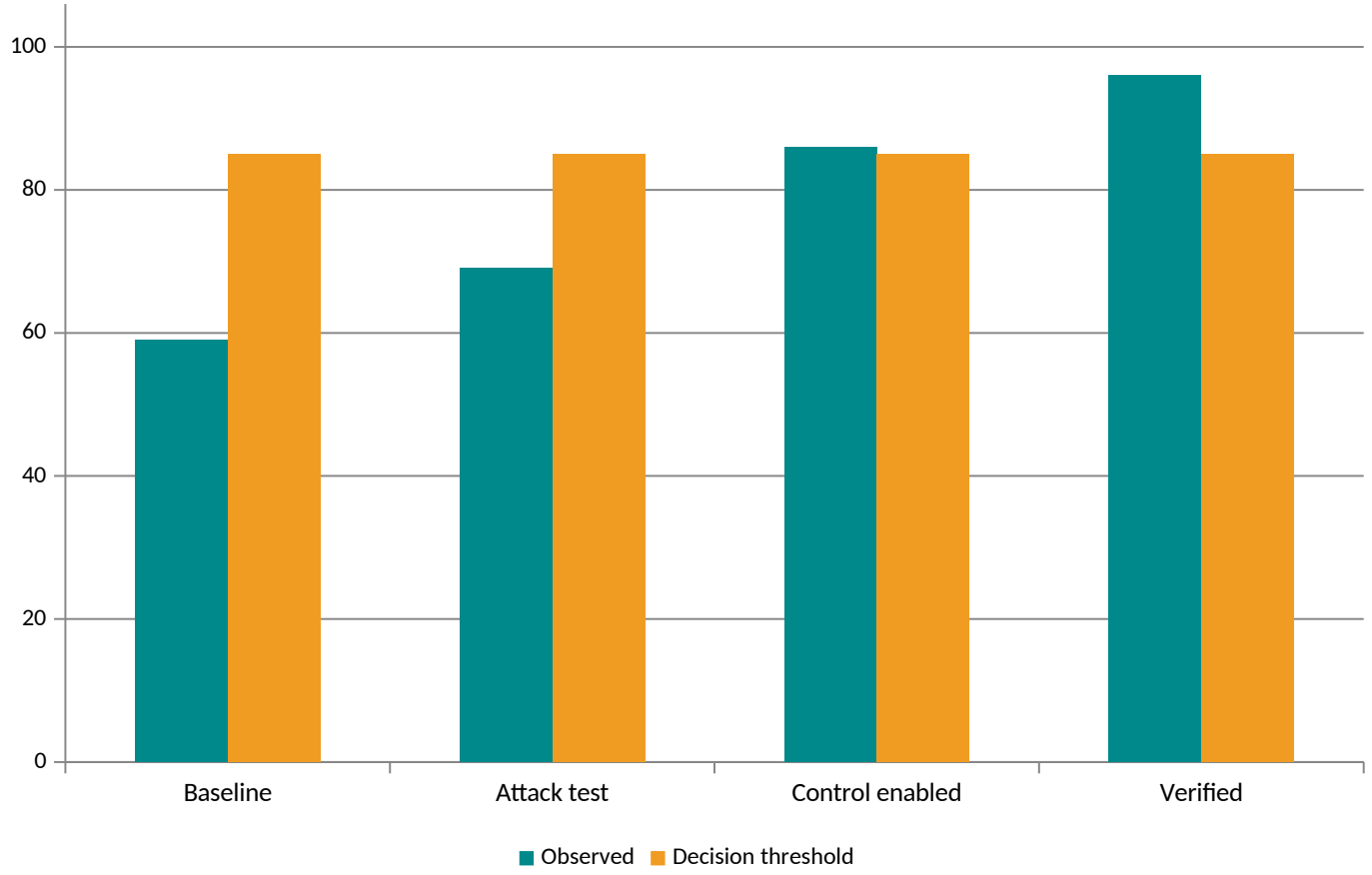
Risk appetite: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	likelihood_1_5	Required in decision and audit evidence
field_2	impact_1_5	Required in decision and audit evidence
field_3	inherent_score	Required in decision and audit evidence
field_4	appetite	Required in decision and audit evidence
field_5	treatment	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Risk appetite: evidence and acceptance decision



DECISION RULE

Accept only when No above-appetite risk lacks a signed treatment. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Confidentiality objective: architecture and control boundary

1

Sensitive Source

State and evidence for sensitive source

2

Authorised Context

State and evidence for authorised context

3

Controlled Output

State and evidence for controlled output

4

Recipient

State and evidence for recipient

Control objective: Apply semantic DLP plus recipient and purpose checks

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Confidentiality objective: runtime decision trace

1 — Observe

sensitive source

2 — Correlate

trace_id + agent_id + policy_version

3 — Decide

Apply semantic DLP plus recipient and purpose checks

4 — Act

recipient

5 — Prove

Leak test score stays below the approved threshold

Failure to detect: Agent paraphrases sensitive content into an allowed channel

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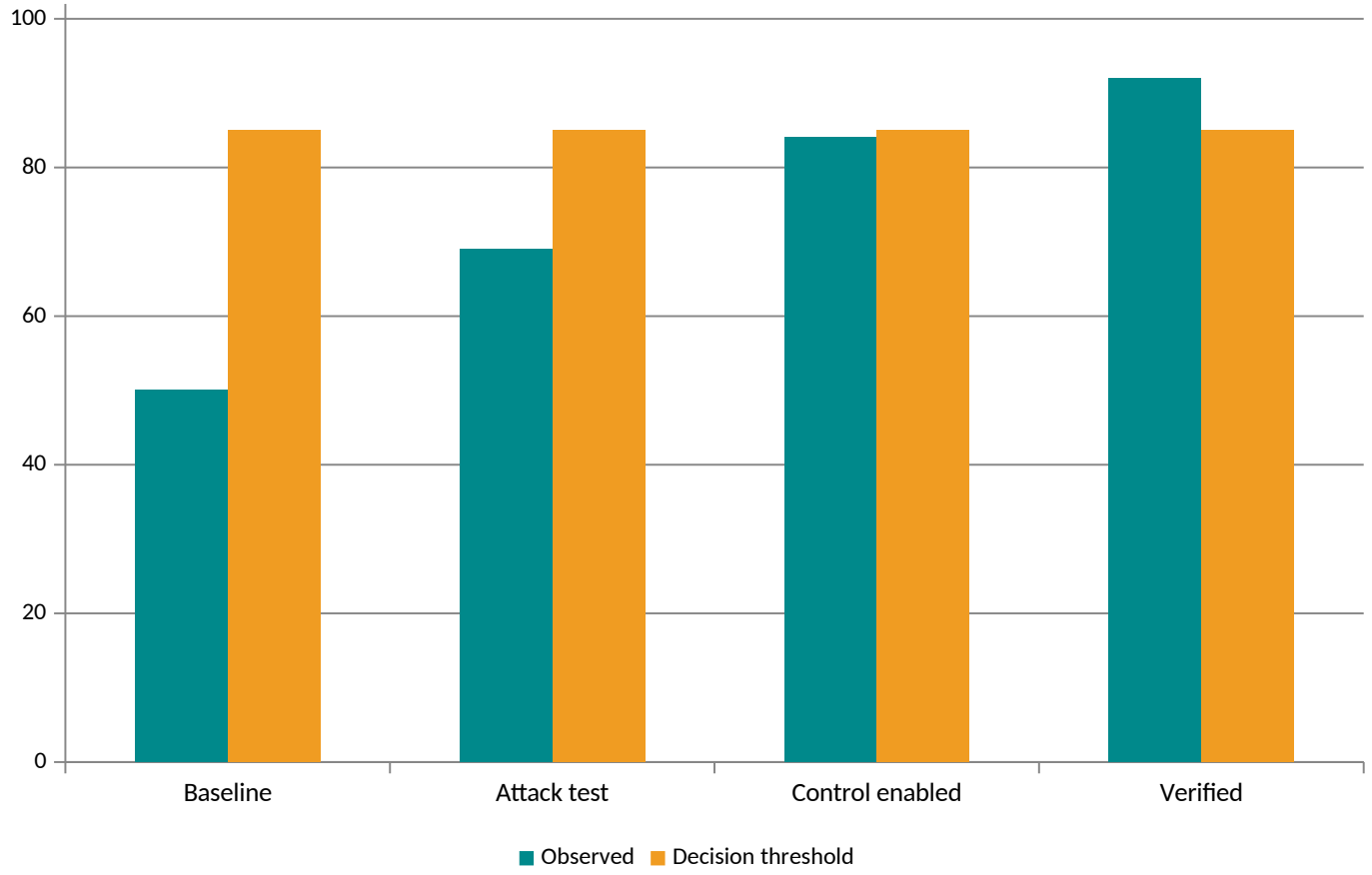
Confidentiality objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	classification	Required in decision and audit evidence
field_2	authorised_purposes[]	Required in decision and audit evidence
field_3	recipient	Required in decision and audit evidence
field_4	redactions[]	Required in decision and audit evidence
field_5	leak_score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Confidentiality objective: evidence and acceptance decision



DECISION RULE

Accept only when Leak test score stays below the approved threshold. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Integrity objective: architecture and control boundary

1

Trusted Instruction

State and evidence for trusted instruction

2

Planner

State and evidence for planner

3

Tool Arguments

State and evidence for tool arguments

4

State Change

State and evidence for state change

5

Verification

State and evidence for verification

Control objective: Bind approved intent to arguments and verify post-condition

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Integrity objective: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
trusted instruction	trace_id + agent_id + policy_version	Bind approved intent to arguments and verify post-condition	verification	Approved and executed argument hashes match

Failure to detect: Prompt injection changes the intended transaction parameters

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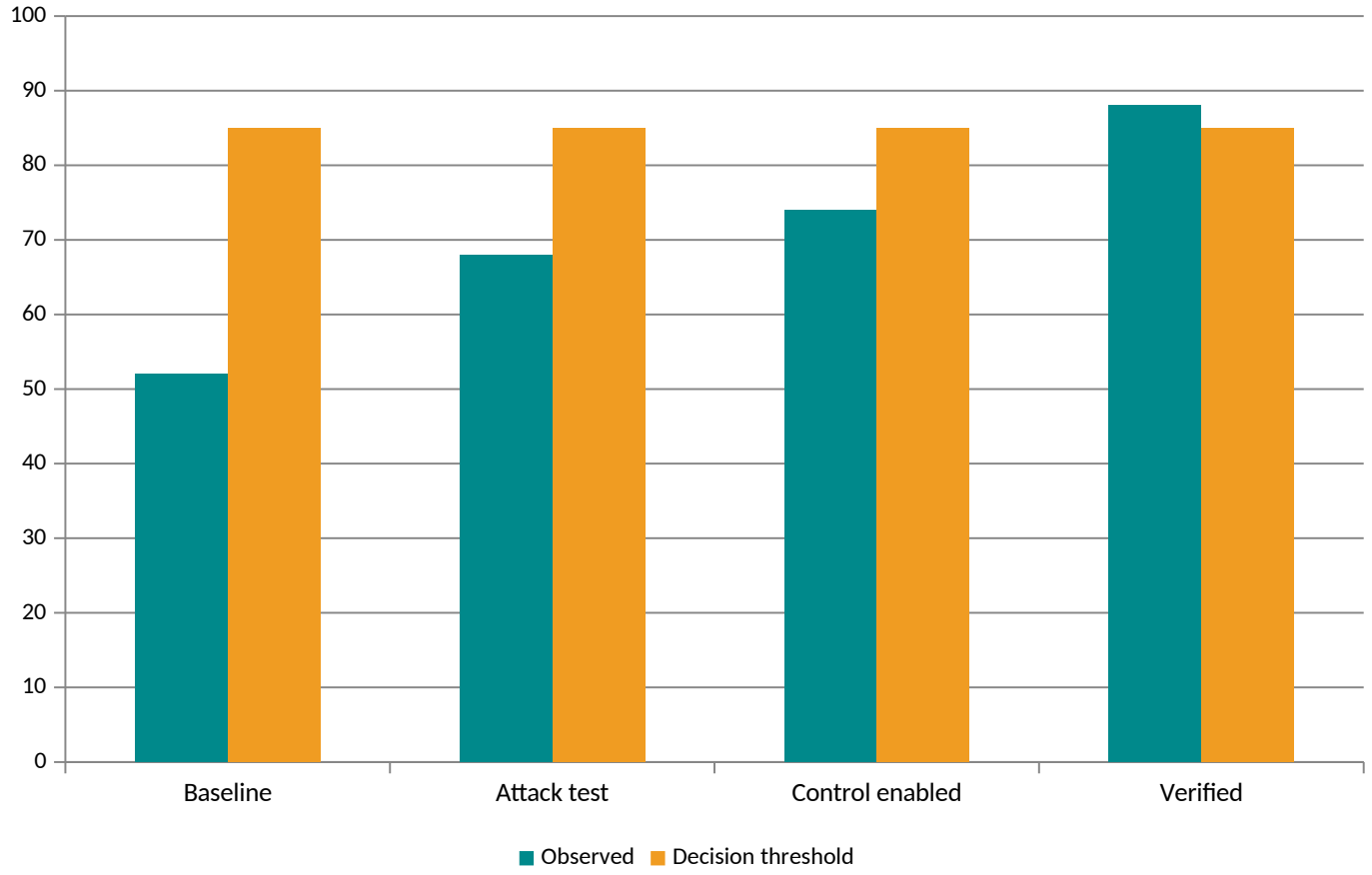
Integrity objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	request_hash	Required in decision and audit evidence
field_2	policy_hash	Required in decision and audit evidence
field_3	argument_hash	Required in decision and audit evidence
field_4	result_hash	Required in decision and audit evidence
field_5	actor	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Integrity objective: evidence and acceptance decision



DECISION RULE

Accept only when Approved and executed argument hashes match. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Authenticity objective: architecture and control boundary

1

Human Or Service

State and evidence for human or service

2

Identity Proof

State and evidence for identity proof

3

Signed Instruction

State and evidence for signed instruction

4

Agent

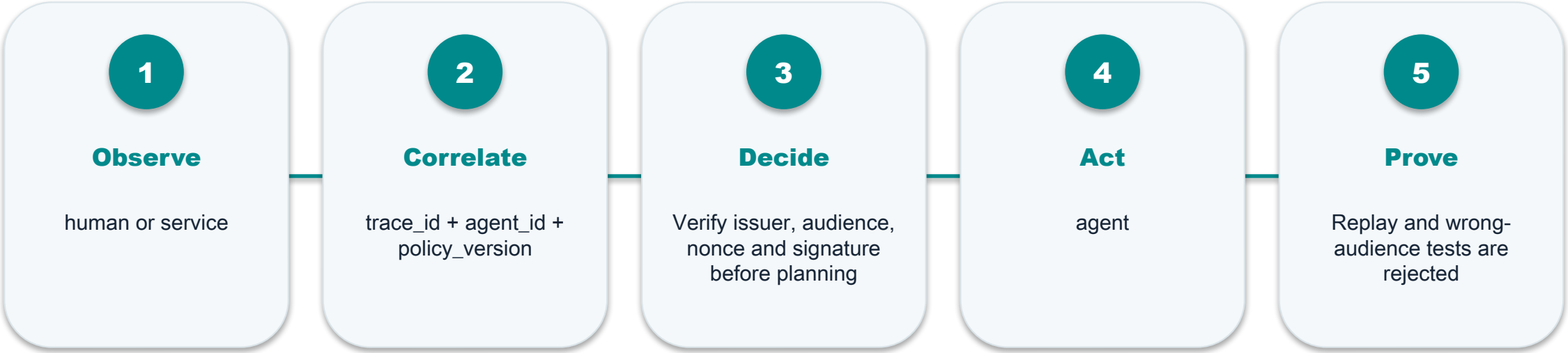
State and evidence for agent

Control objective: Verify issuer, audience, nonce and signature before planning

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Authenticity objective: runtime decision trace



Failure to detect: Spoofed upstream service injects a privileged request

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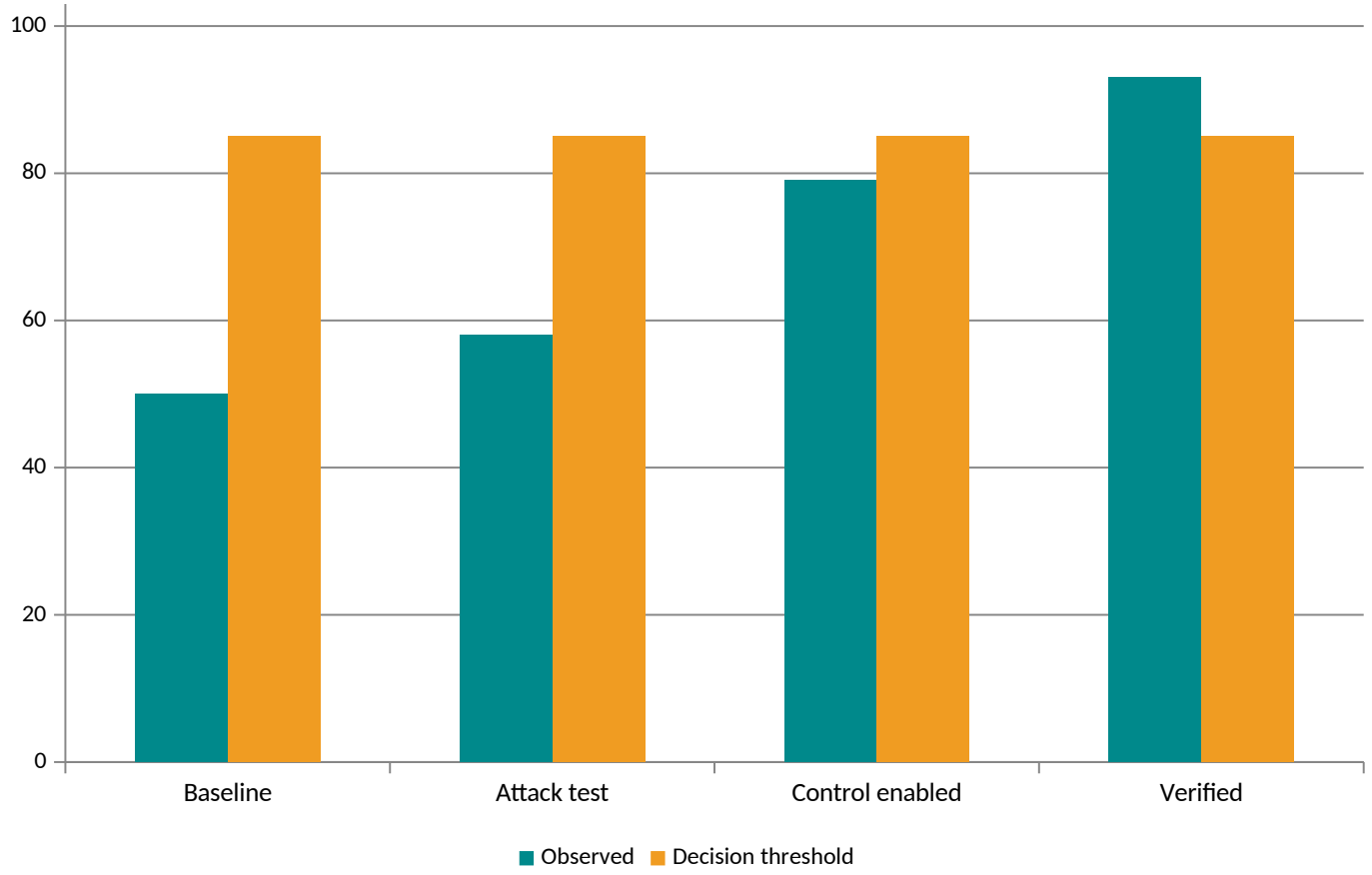
Authenticity objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	issuer	Required in decision and audit evidence
field_2	subject	Required in decision and audit evidence
field_3	audience	Required in decision and audit evidence
field_4	issued_at	Required in decision and audit evidence
field_5	nonce	Required in decision and audit evidence
field_6	signature	Required in decision and audit evidence

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Authenticity objective: evidence and acceptance decision



DECISION RULE

Accept only when Replay and wrong-audience tests are rejected. Otherwise record a gap, owner and due date.

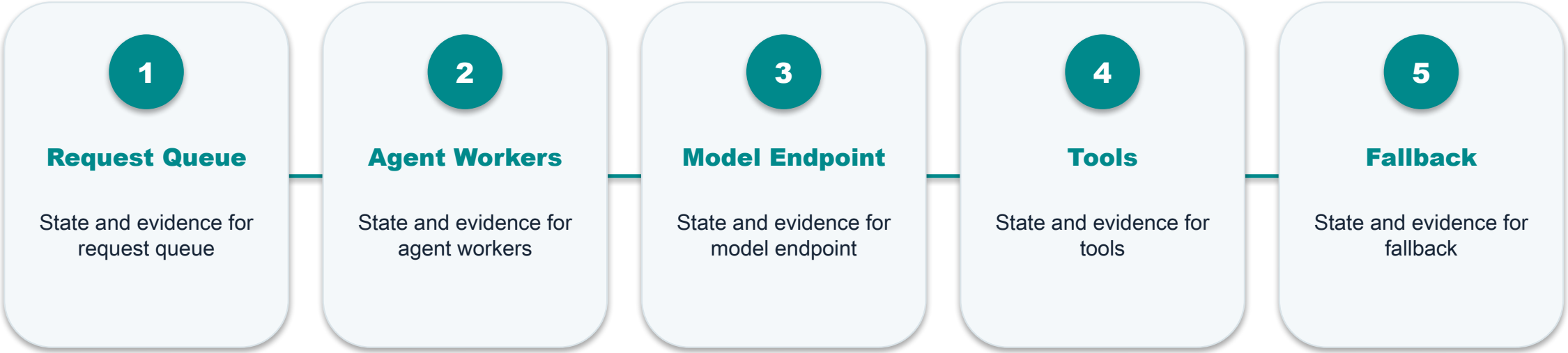
EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Availability objective: architecture and control boundary



Control objective: Use bounded retries, bulkheads and circuit breakers

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Availability objective: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
request queue	trace_id + agent_id + policy_version	Use bounded retries, bulkheads and circuit breakers	fallback	Error budget and queue latency remain inside SLO

Failure to detect: Retry storm exhausts model and tool quotas

Supplied AI-security reference corpus · MITRE ATLAS

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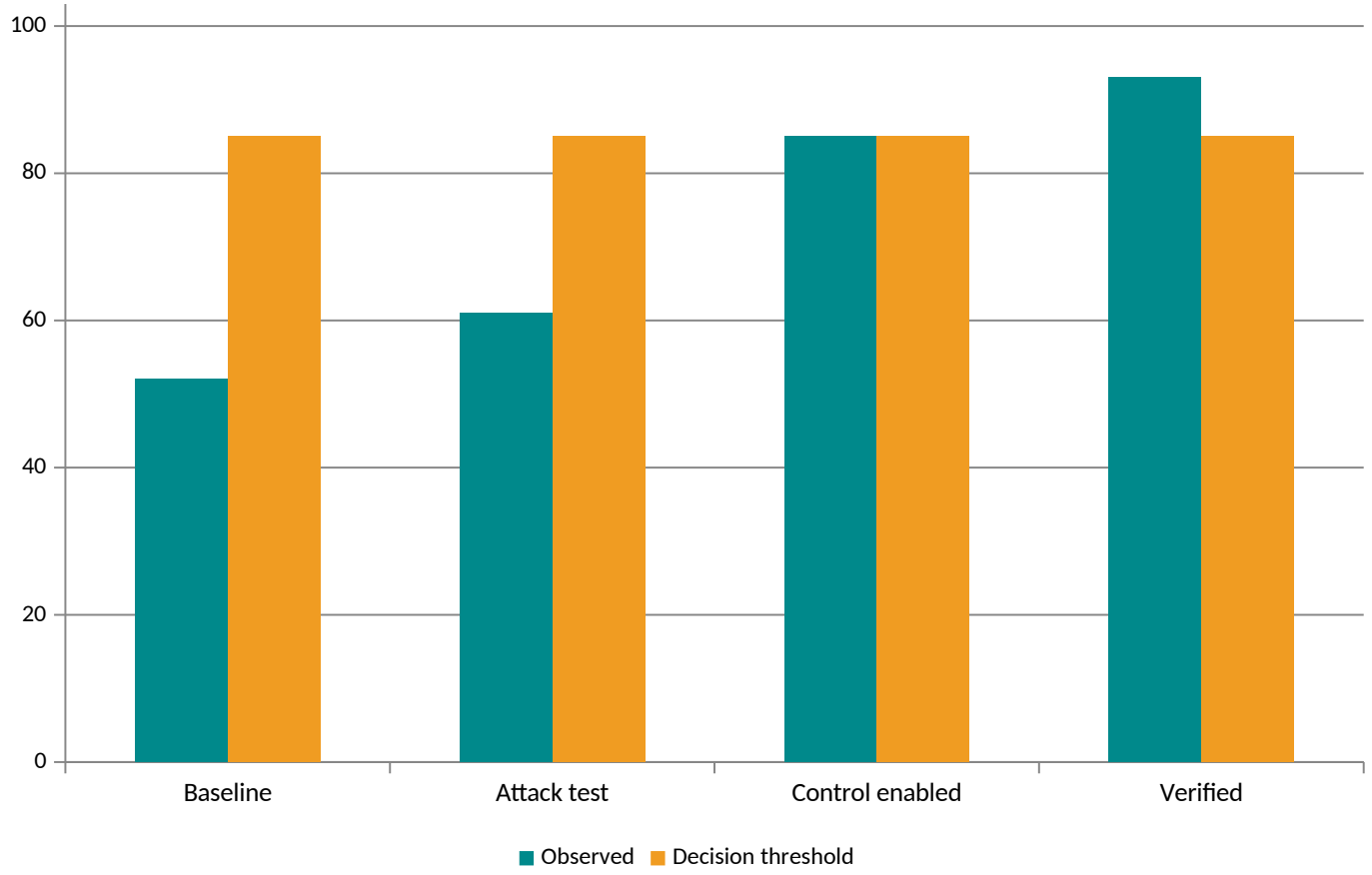
Availability objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	queue_depth	Required in decision and audit evidence
field_2	concurrency	Required in decision and audit evidence
field_3	timeout_ms	Required in decision and audit evidence
field_4	retry_budget	Required in decision and audit evidence
field_5	circuit_state	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Availability objective: evidence and acceptance decision



DECISION RULE

Accept only when Error budget and queue latency remain inside SLO. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Auditability objective: architecture and control boundary



Control objective: Correlate decision and execution events under one trace_id

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Auditability objective: runtime decision trace



Failure to detect: Incident team cannot reconstruct why a tool ran

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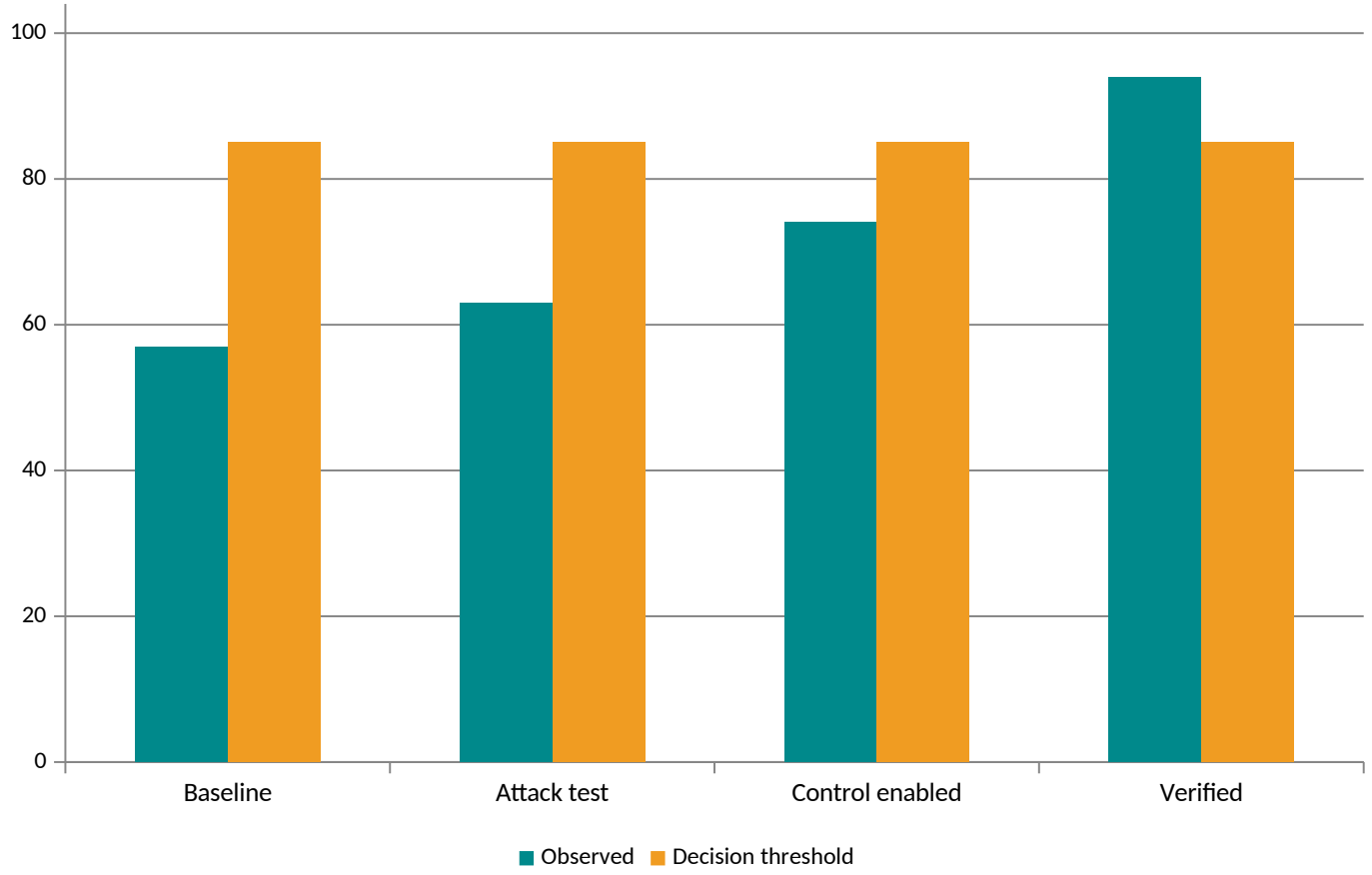
Auditability objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	trace_id	Required in decision and audit evidence
field_2	parent_span	Required in decision and audit evidence
field_3	actor	Required in decision and audit evidence
field_4	action	Required in decision and audit evidence
field_5	resource	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

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Auditability objective: evidence and acceptance decision



DECISION RULE

Accept only when Random sample replays reproduce the ordered action chain. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Kill-switch baseline: architecture and control boundary



Control objective: Implement tested global and per-agent stop paths

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Kill-switch baseline: runtime decision trace

1 — Observe

SOC command

2 — Correlate

trace_id + agent_id + policy_version

3 — Decide

Implement tested global and per-agent stop paths

4 — Act

agent stop

5 — Prove

P95 containment time meets the defined target

Failure to detect: Agent continues acting after a critical alert

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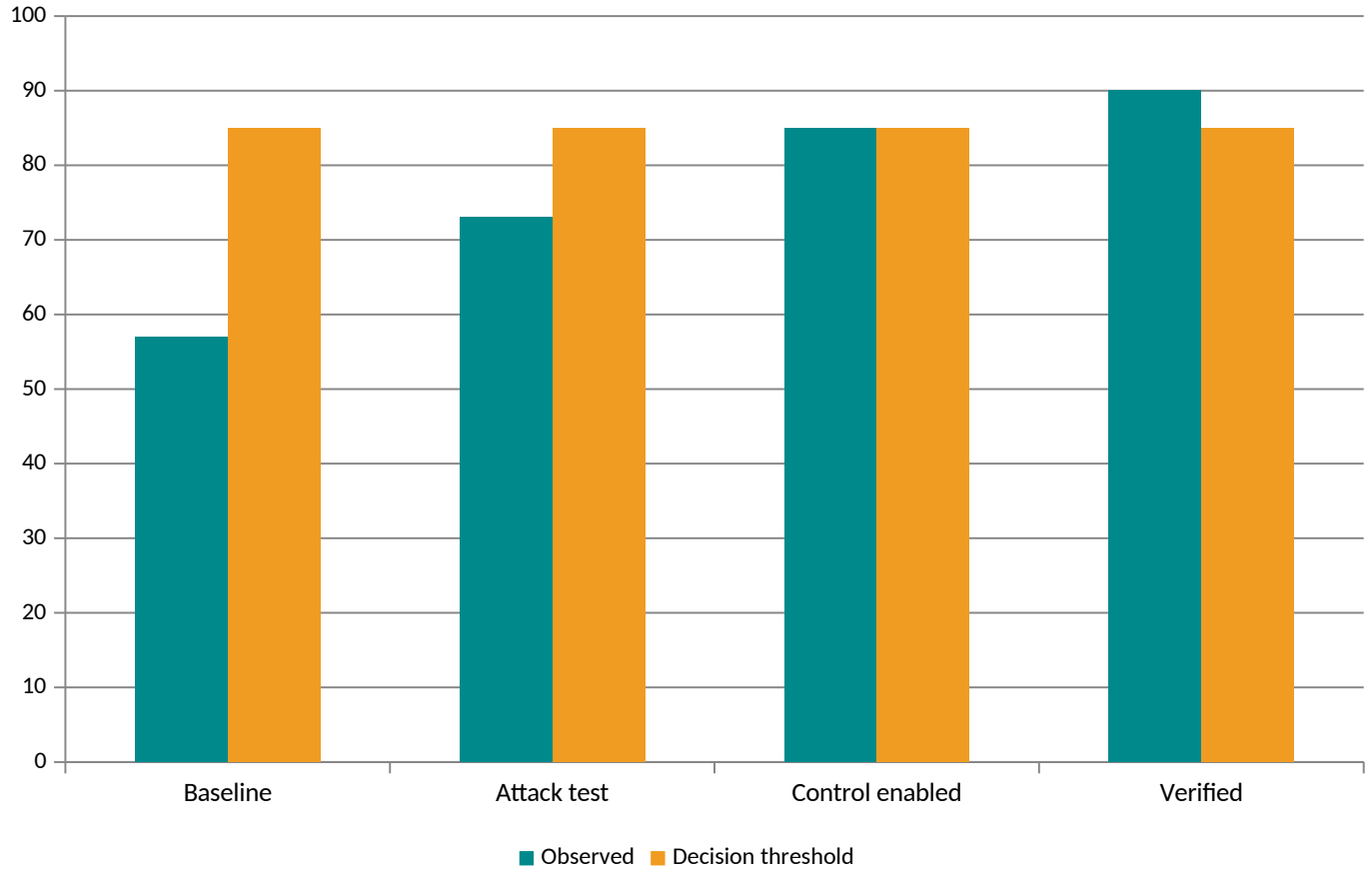
Kill-switch baseline: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	scope	Required in decision and audit evidence
field_3	initiated_by	Required in decision and audit evidence
field_4	reason	Required in decision and audit evidence
field_5	effective_at	Required in decision and audit evidence
field_6	confirmation	Required in decision and audit evidence

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Kill-switch baseline: evidence and acceptance decision



DECISION RULE

Accept only when P95 containment time meets the defined target. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Activity 01 — Build an Agent Inventory and Security Objectives

1 **Scenario**

Use synthetic data in activity-01-agent-inventory to make a security-operations decision.

2 **Input contract**

mock_agents.json

3 **Runnable evidence**

activity.py

4 **Output artifact**

agent_security_baseline.md

5 **Acceptance**

All active agents have an owner, purpose, data class, risk tier and measurable security objective.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 01: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_agents.json	required fields and types	control and decision rules	agent_security_baseline.md	All active agents have an owner, purpose, data class, risk tier and measurable security objective.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 01: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_agents.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	agent_security_baseline.md	Generated locally and reproducible
Criterion	A1	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete agent_security_baseline.md only	Retain script and mock data

Aligned activity files in labs/

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Activity 02 — Threat-Model Agent Trust Boundaries

1 **Scenario**

Use synthetic data in activity-02-trust-boundary-threat-model to make a security-operations decision.

2 **Input contract**

mock_dataflows.json

3 **Runnable evidence**

activity.py

4 **Output artifact**

threat_model.json

5 **Acceptance**

Every cross-zone flow has authentication, authorisation, encryption and logging controls.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 02: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_dataflows.json	required fields and types	control and decision rules	threat_model.json	Every cross-zone flow has authentication, authorisation, encryption and logging controls.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 02: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_dataflows.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	threat_model.json	Generated locally and reproducible
Criterion	A1	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete threat_model.json only	Retain script and mock data

Aligned activity files in labs/

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Activity 03 — Enforce Tool and Data Guardrails

1 **Scenario**

Use synthetic data in activity-03-policy-as-code to make a security-operations decision.

2 **Input contract**

mock_requests.json

3 **Runnable evidence**

activity.py

4 **Output artifact**

policy_decisions.csv

5 **Acceptance**

Allowed requests execute; restricted data, unlisted tools and missing approvals are denied with reasons.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 03: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 03: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_requests.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	policy_decisions.csv	Generated locally and reproducible
Criterion	A2, A3	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete policy_decisions.csv only	Retain script and mock data

Aligned activity files in labs/

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Topic 01 technical recap

1 **Confidentiality objective**

Leak test score stays below the approved threshold

2 **Integrity objective**

Approved and executed argument hashes match

3 **Authenticity objective**

Replay and wrong-audience tests are rejected

4 **Availability objective**

Error budget and queue latency remain inside SLO

5 **Auditability objective**

Random sample replays reproduce the ordered action chain

6 **Kill-switch baseline**

P95 containment time meets the defined target

Topic technical-anchor inventory

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TOPIC 02

Agent Security Policies, Governance and Compliance

Policy-as-code · evidence · permissions · assurance · communication

LO2 · K3, K4, A3, A4



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LO2 and assessment criteria

1 **K3**

Best practices in information security policies

2 **K4**

Best practices and emerging technologies in security control

3 **A3**

Manage compliance with information-security guidelines and classification or permission rules

4 **A4**

Lead communication of security goals and objectives to the organisation

5 **LO2**

LO2: Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.

Approved course proposal criteria adapted to the autonomous-agent context

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Agent Security Policies, Governance and Compliance: operating model

1. **Boundary**
   Components, interfaces and ownership
2. **Runtime**
   States, transitions and decision points
3. **Contract**
   Fields, rules, thresholds and evidence
4. **Failure**
   Attack path, diagnostic and control
5. **Verify**
   Acceptance test and activity artifact

Every mechanism in this topic is taught through all five layers.

Courseware-build mechanism-level coverage contract

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Policy hierarchy: architecture and control boundary

1

Board Principles

State and evidence for board principles

2

Enterprise Standard

State and evidence for enterprise standard

3

Agent Policy

State and evidence for agent policy

4

Runtime Rule

State and evidence for runtime rule

Control objective: Trace every executable rule to an approved parent policy

Supplied AI-security reference corpus · NIST AI RMF · OWASP ASI

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Policy hierarchy: runtime decision trace

1
Observe
board principles

2
Correlate
trace_id + agent_id + policy_version

3
Decide
Trace every executable rule to an approved parent policy

4
Act
runtime rule

5
Prove
No active rule has an orphaned parent_id

Failure to detect: Runtime rules contradict enterprise requirements

Supplied AI-security reference corpus · MITRE ATLAS

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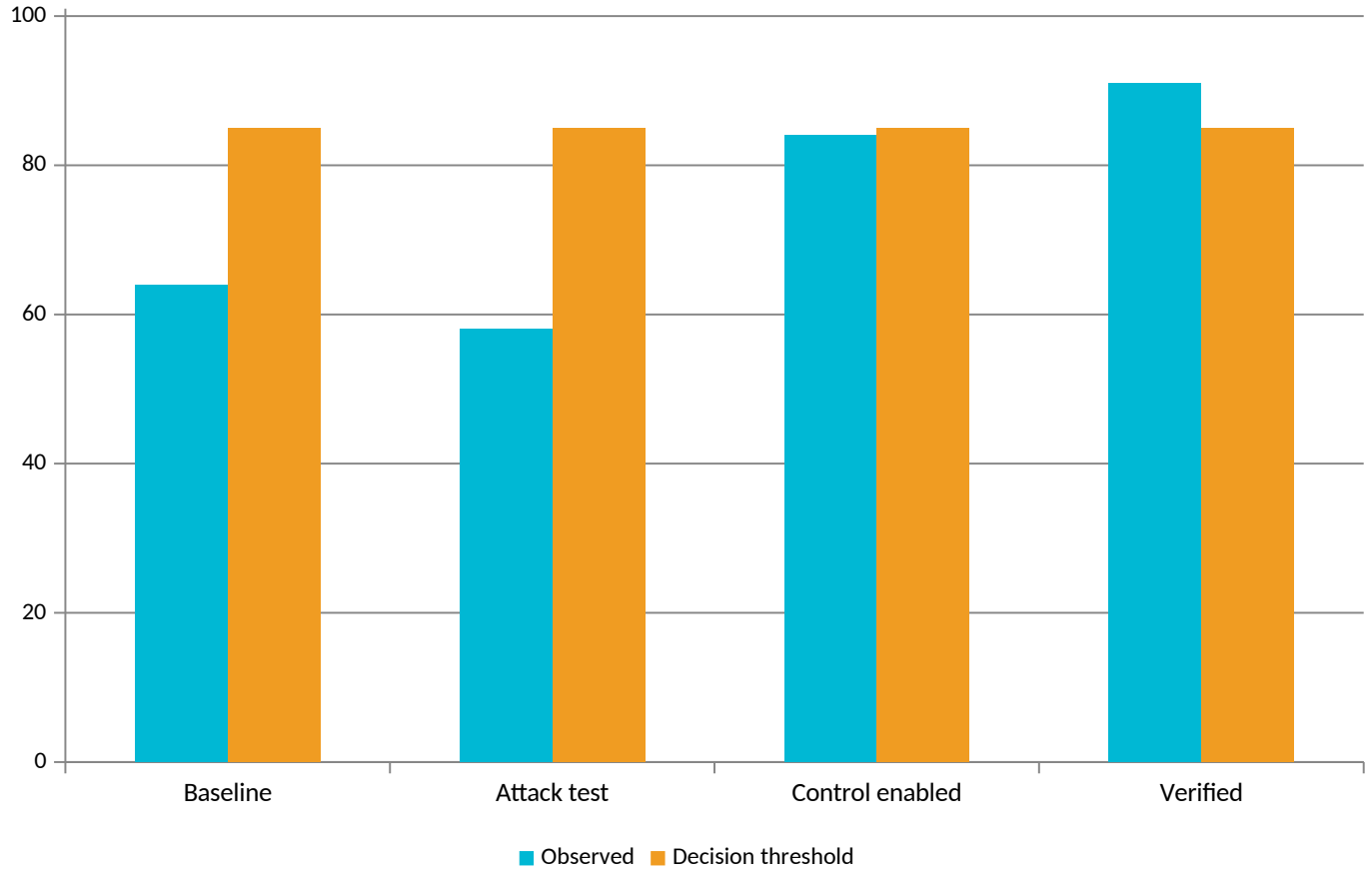
Policy hierarchy: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	policy_id	Required in decision and audit evidence
field_2	parent_id	Required in decision and audit evidence
field_3	scope	Required in decision and audit evidence
field_4	owner	Required in decision and audit evidence
field_5	version	Required in decision and audit evidence
field_6	effective_date	Required in decision and audit evidence

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Policy hierarchy: evidence and acceptance decision



DECISION RULE

Accept only when No active rule has an orphaned parent_id. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Policy-as-code: architecture and control boundary

1. **Repository**
   State and evidence for repository
2. **Review**
   State and evidence for review
3. **Test**
   State and evidence for test
4. **Signed Bundle**
   State and evidence for signed bundle
5. **Policy Decision Point**
   State and evidence for policy decision point

Control objective: Version, test and sign machine-enforced policy bundles

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Policy-as-code: runtime decision trace



Failure to detect: Manual configuration drifts from approved policy

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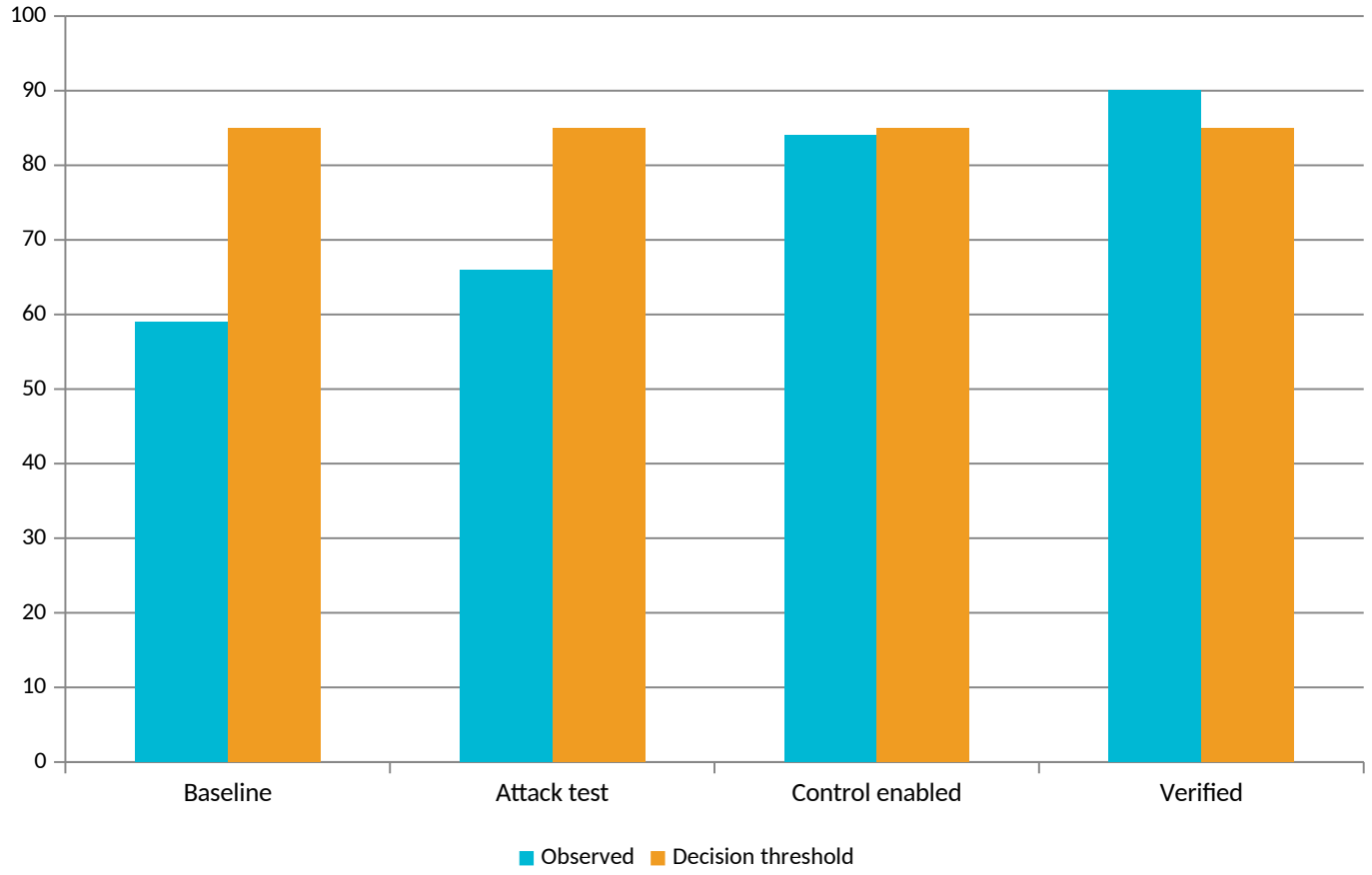
Policy-as-code: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	rule_id	Required in decision and audit evidence
field_2	input_schema	Required in decision and audit evidence
field_3	effect	Required in decision and audit evidence
field_4	conditions[]	Required in decision and audit evidence
field_5	tests[]	Required in decision and audit evidence
field_6	bundle_hash	Required in decision and audit evidence

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Policy-as-code: evidence and acceptance decision



DECISION RULE

Accept only when Policy tests pass and deployed hash matches release. Otherwise record a gap, owner and due date.

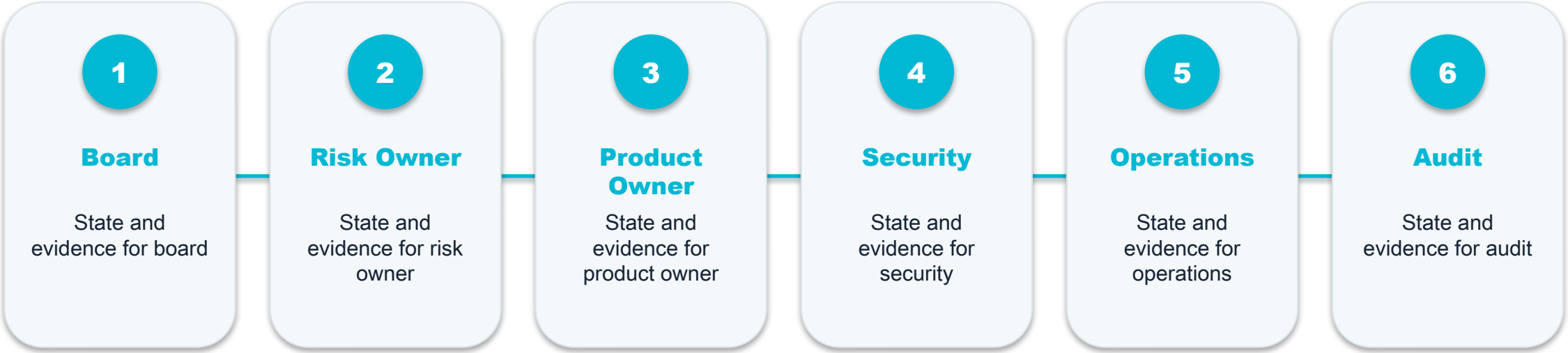
EVIDENCE STATUS

PASS

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Governance RACI: architecture and control boundary



Control objective: Assign one accountable role per lifecycle decision

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Governance RACI: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
board	trace_id + agent_id + policy_version	Assign one accountable role per lifecycle decision	audit	Every critical decision has exactly one accountable owner

Failure to detect: Two teams assume the other owns a risky agent

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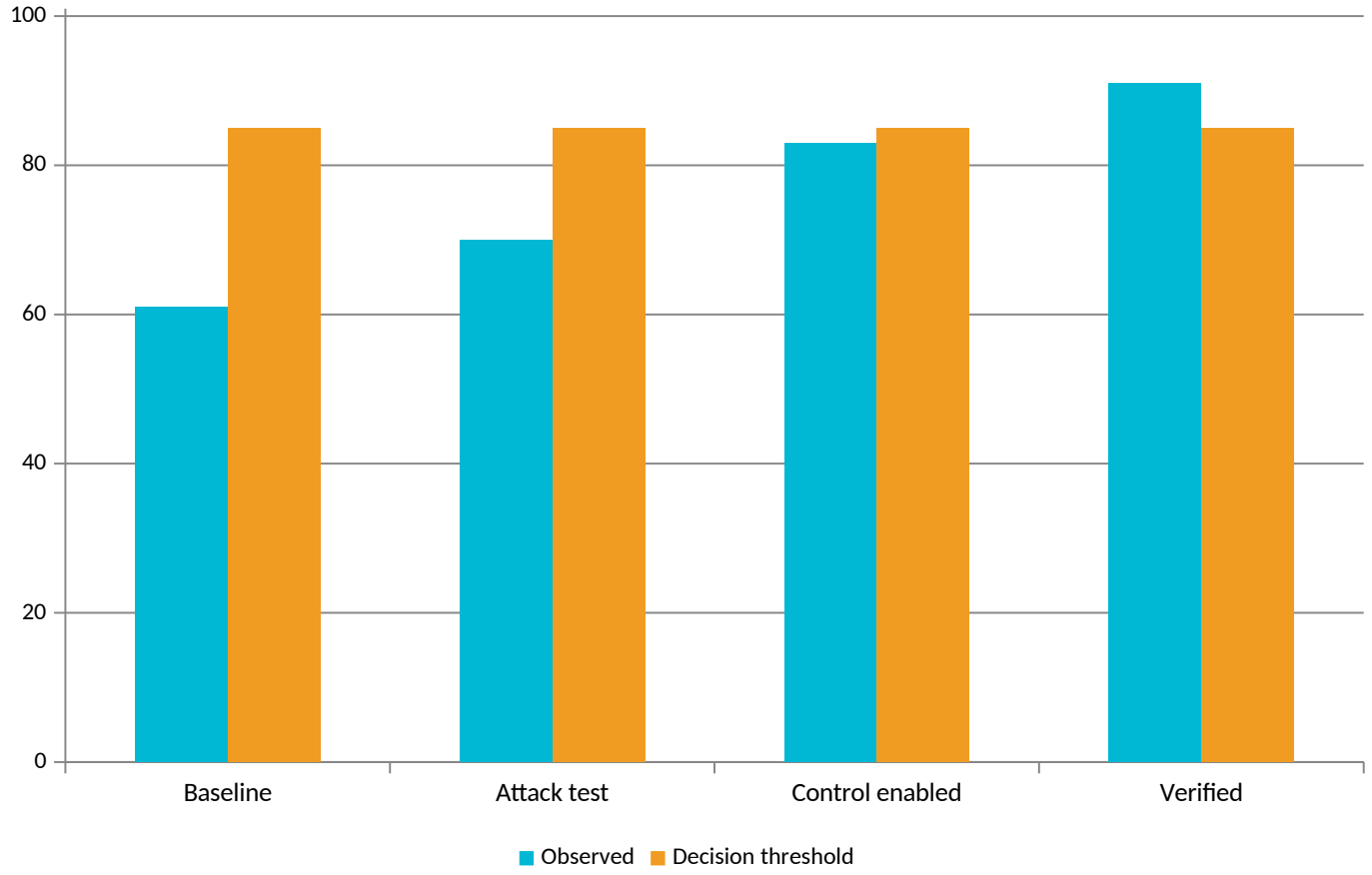
Governance RACI: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	decision	Required in decision and audit evidence
field_2	accountable	Required in decision and audit evidence
field_3	responsible	Required in decision and audit evidence
field_4	consulted[]	Required in decision and audit evidence
field_5	informed[]	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Governance RACI: evidence and acceptance decision



DECISION RULE

Accept only when Every critical decision has exactly one accountable owner. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Information handling rules: architecture and control boundary

1

Classification Label

State and evidence for classification label

2

Permitted Model

State and evidence for permitted model

3

Permitted Tools

State and evidence for permitted tools

4

Output Channel

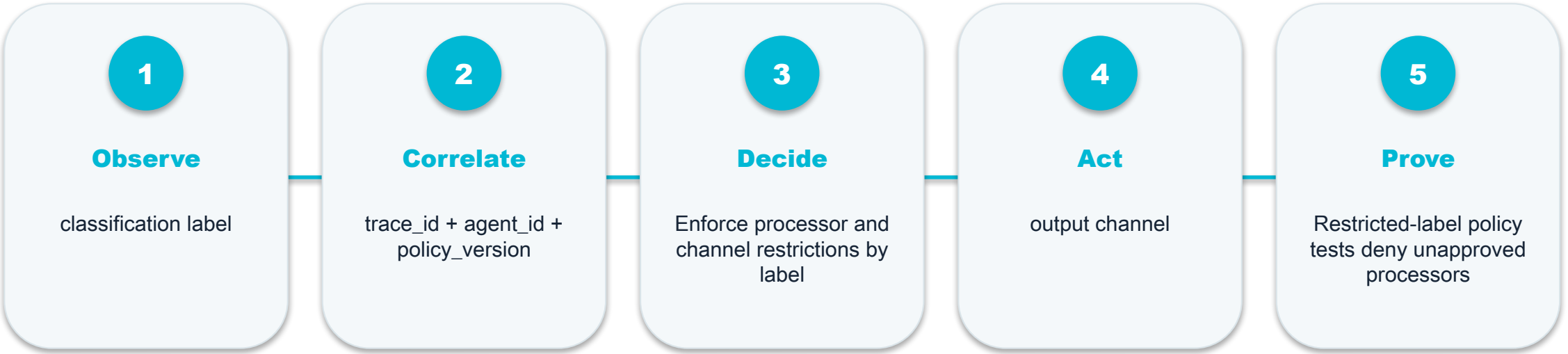
State and evidence for output channel

Control objective: Enforce processor and channel restrictions by label

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Information handling rules: runtime decision trace



Failure to detect: Restricted records flow to a consumer-grade model endpoint

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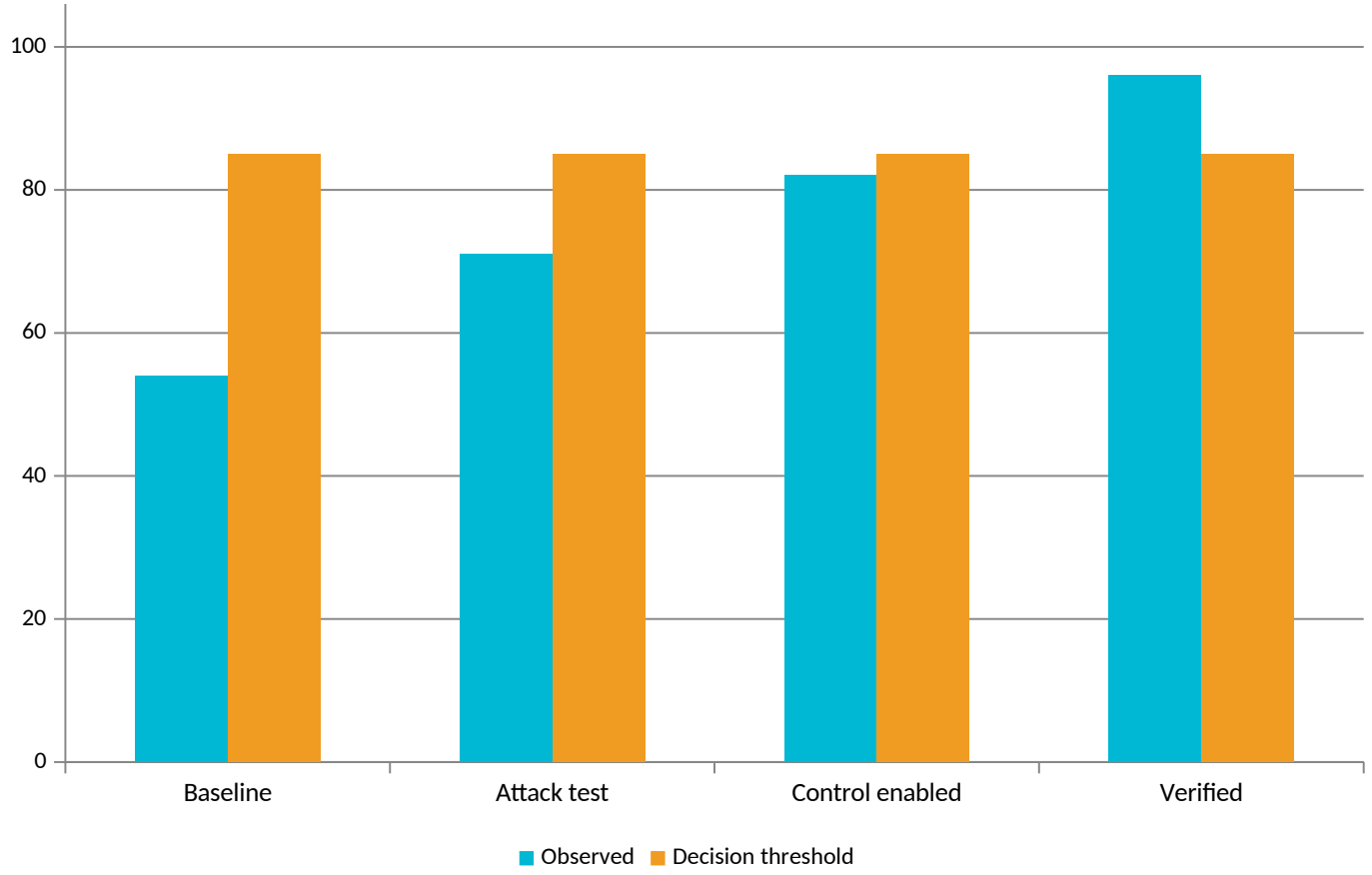
Information handling rules: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	class	Required in decision and audit evidence
field_2	allowed_processors[]	Required in decision and audit evidence
field_3	allowed_actions[]	Required in decision and audit evidence
field_4	retention_days	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Information handling rules: evidence and acceptance decision



DECISION RULE

Accept only when Restricted-label policy tests deny unapproved processors. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Permission recertification: architecture and control boundary

1

Entitlement Inventory

State and evidence for entitlement inventory

2

Owner Review

State and evidence for owner review

3

Revoke/Retain

State and evidence for revoke/retain

4

Evidence

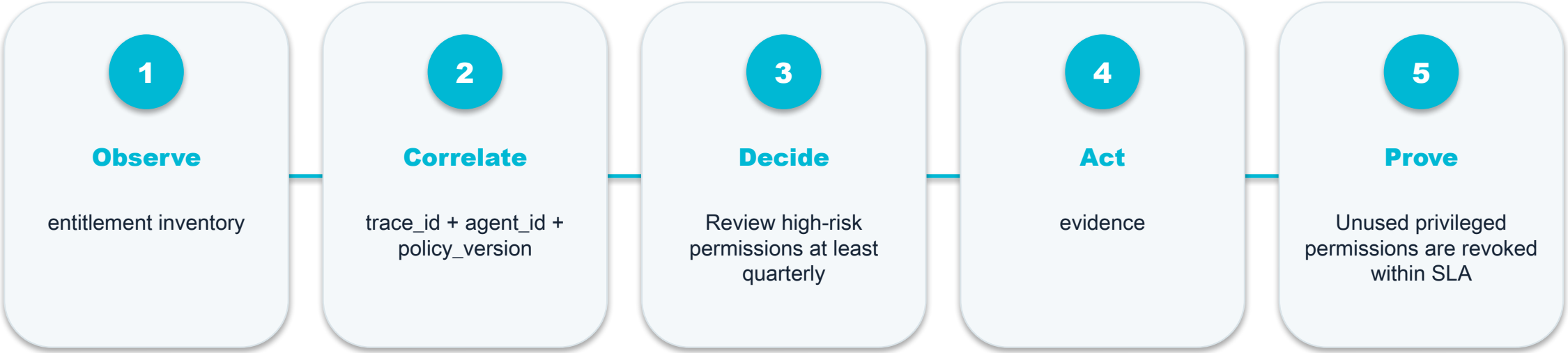
State and evidence for evidence

Control objective: Review high-risk permissions at least quarterly

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Permission recertification: runtime decision trace



Failure to detect: Dormant high-privilege tool scopes remain active

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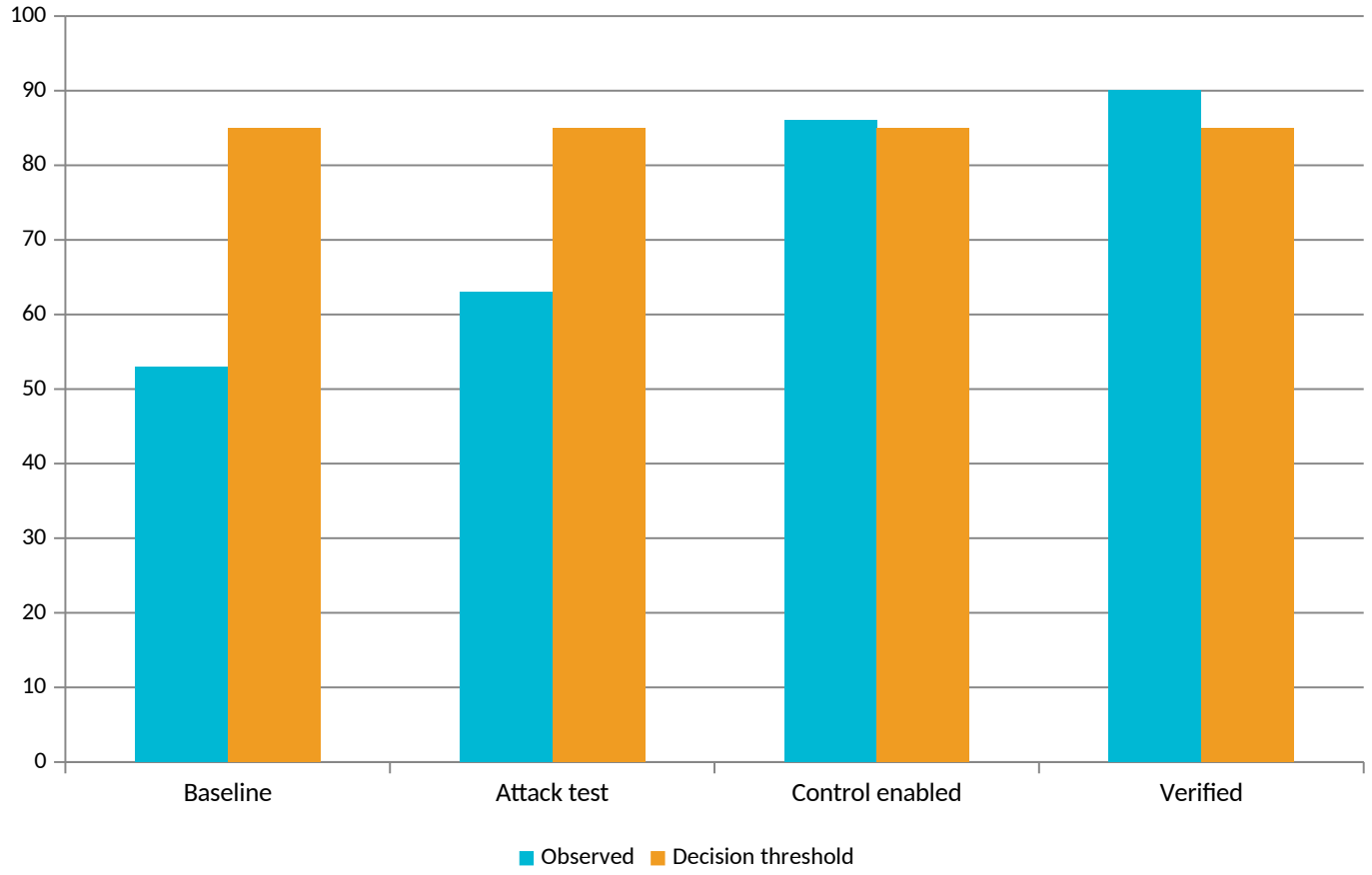
Permission recertification: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	principal	Required in decision and audit evidence
field_2	permission	Required in decision and audit evidence
field_3	resource	Required in decision and audit evidence
field_4	last_used	Required in decision and audit evidence
field_5	reviewer	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

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Permission recertification: evidence and acceptance decision



DECISION RULE

Accept only when Unused privileged permissions are revoked within SLA. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Control ownership: architecture and control boundary

1

Control Objective

State and evidence for control objective

2

Control Operator

State and evidence for control operator

3

Evidence Producer

State and evidence for evidence producer

4

Risk Owner

State and evidence for risk owner

Control objective: Bind control frequency and evidence to named roles

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Control ownership: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
control objective	trace_id + agent_id + policy_version	Bind control frequency and evidence to named roles	risk owner	Missed control runs create owner alerts

Failure to detect: A control exists on paper but nobody operates it

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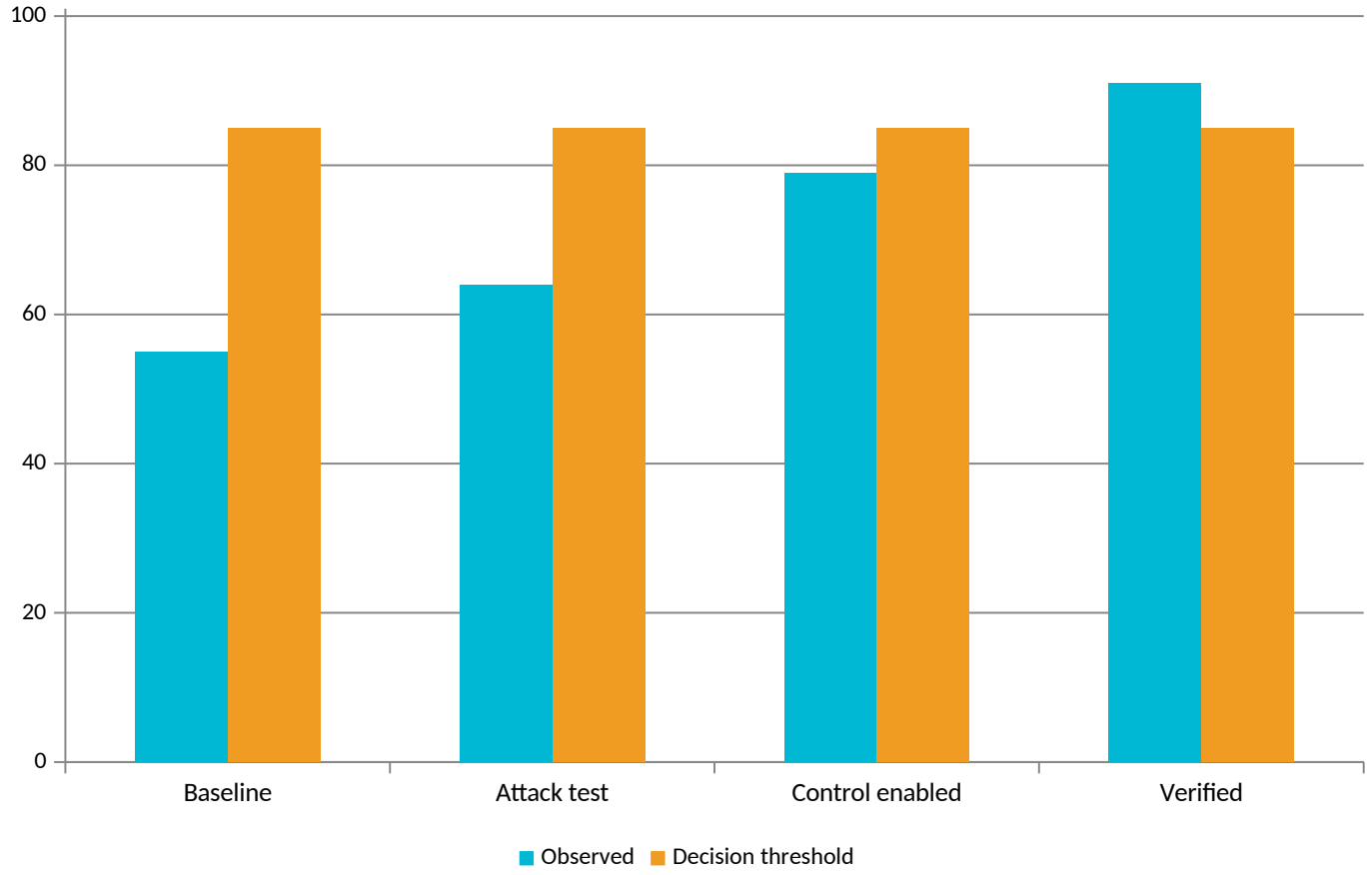
Control ownership: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	control_id	Required in decision and audit evidence
field_2	operator	Required in decision and audit evidence
field_3	frequency	Required in decision and audit evidence
field_4	evidence	Required in decision and audit evidence
field_5	exception_route	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

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Control ownership: evidence and acceptance decision



DECISION RULE

Accept only when Missed control runs create owner alerts. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Evidence retention: architecture and control boundary

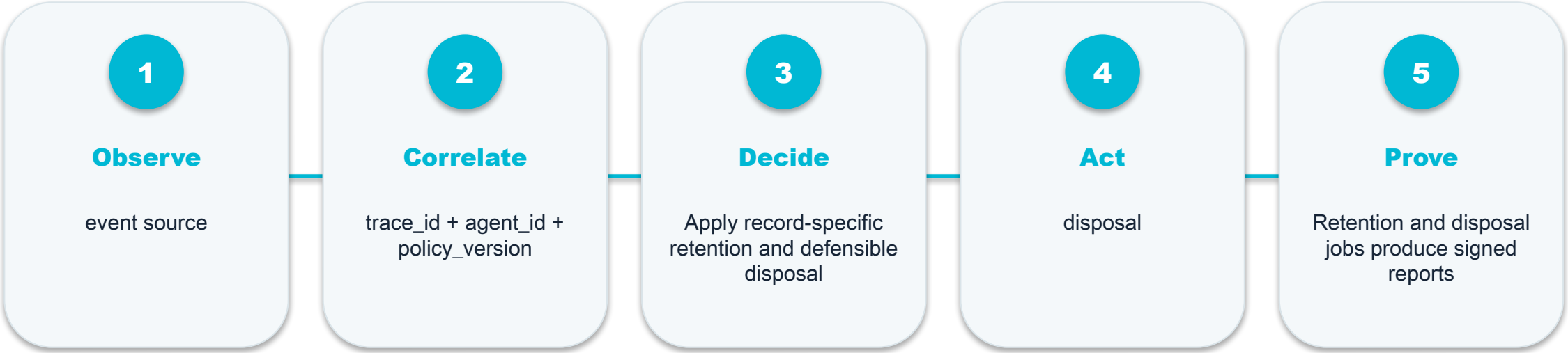


Control objective: Apply record-specific retention and defensible disposal

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Evidence retention: runtime decision trace



Failure to detect: Evidence is deleted before an investigation or kept indefinitely

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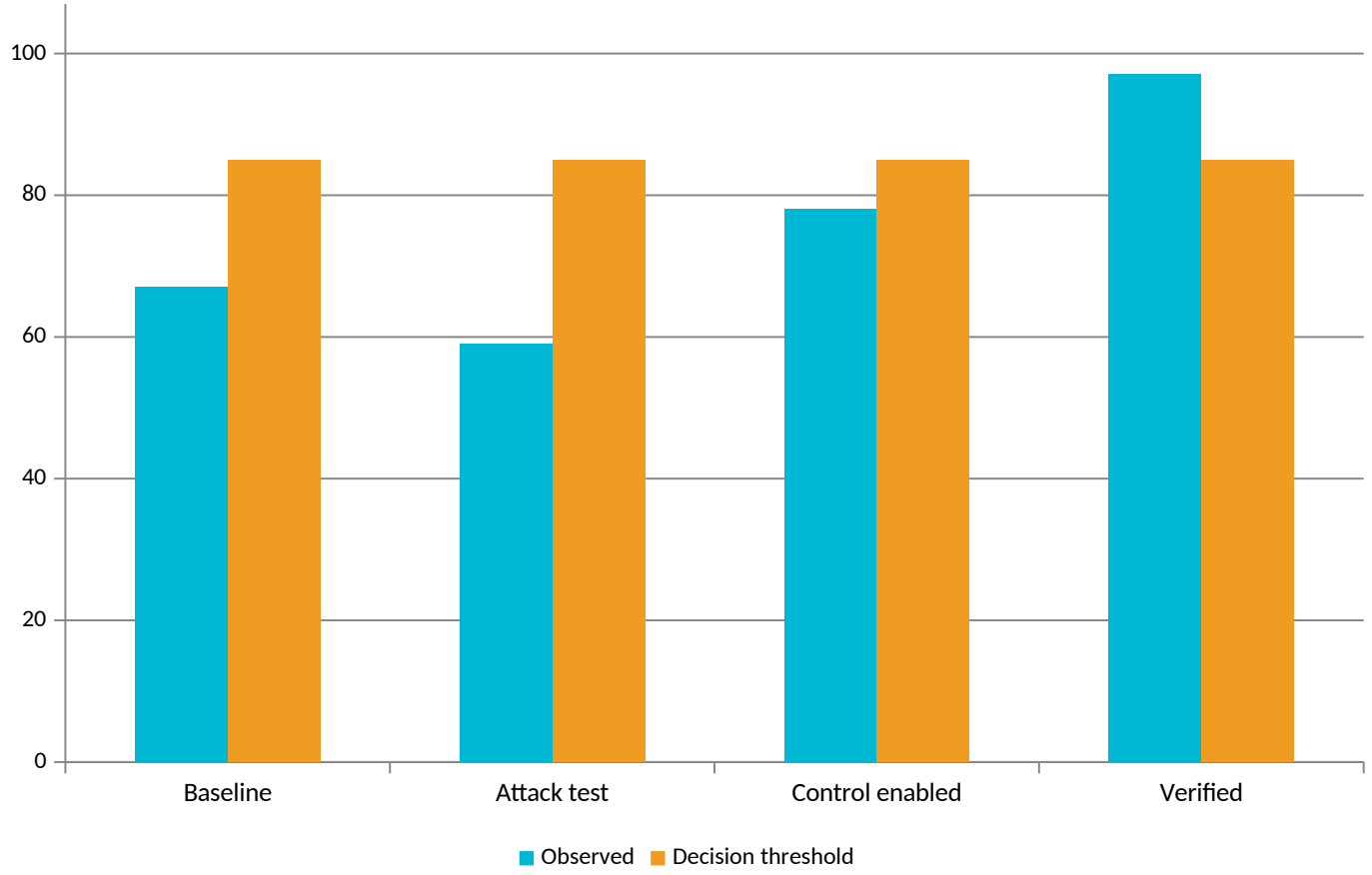
Evidence retention: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	record_type	Required in decision and audit evidence
field_2	retention_days	Required in decision and audit evidence
field_3	legal_hold	Required in decision and audit evidence
field_4	hash	Required in decision and audit evidence
field_5	disposal_date	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Evidence retention: evidence and acceptance decision



DECISION RULE

Accept only when Retention and disposal jobs produce signed reports. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Audit event schema: architecture and control boundary

1

Agent Runtime

State and evidence for agent runtime

2

Gateway

State and evidence for gateway

3

Tools

State and evidence for tools

4

Normaliser

State and evidence for normaliser

5

Evidence Lake

State and evidence for evidence lake

Control objective: Adopt required correlation and identity fields

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Audit event schema: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
agent runtime	trace_id + agent_id + policy_version	Adopt required correlation and identity fields	evidence lake	Schema conformance ≥99.5% for critical events

Failure to detect: Logs cannot be joined across components

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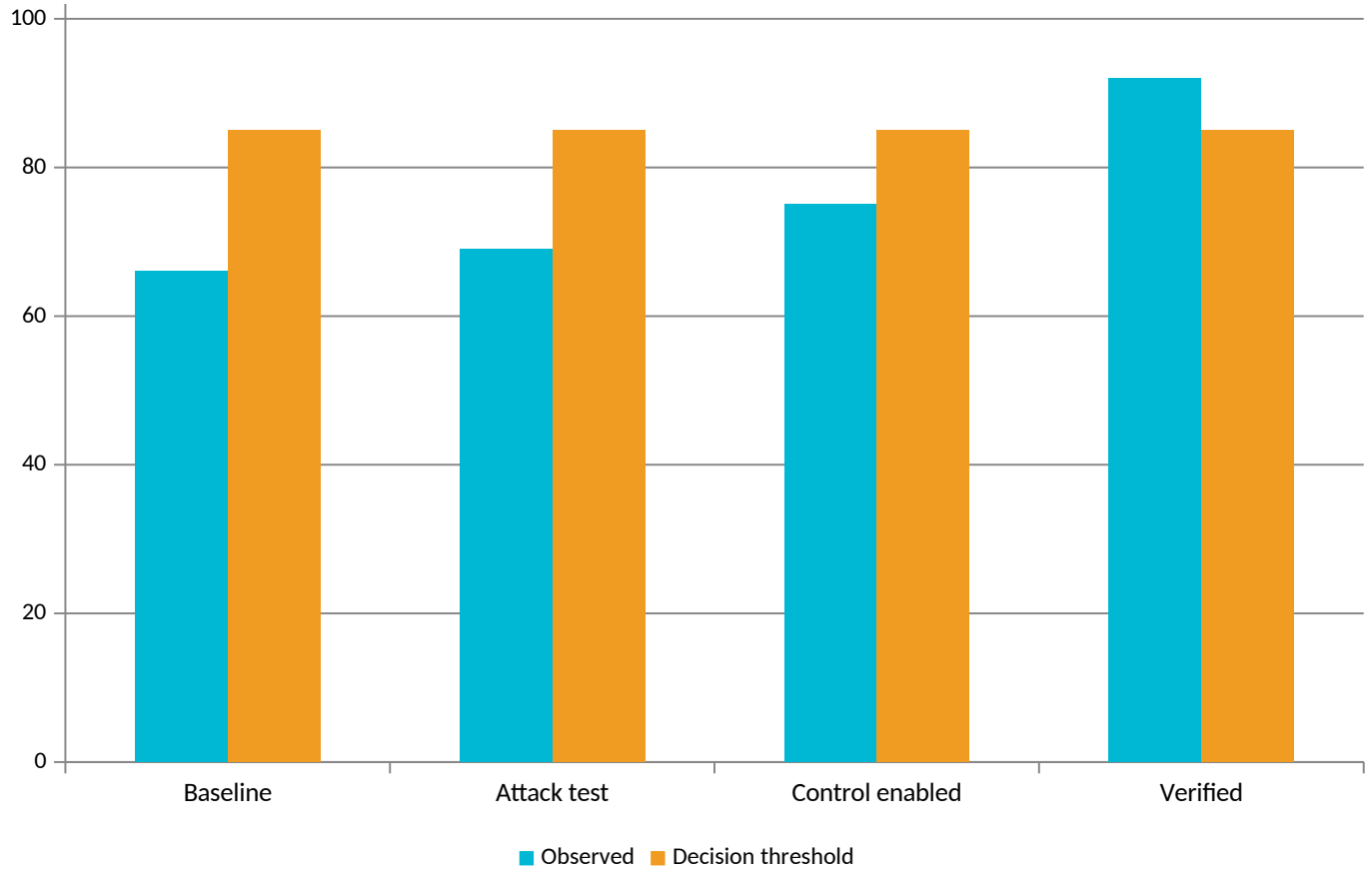
Audit event schema: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	event_time	Required in decision and audit evidence
field_2	trace_id	Required in decision and audit evidence
field_3	agent_id	Required in decision and audit evidence
field_4	tool_id	Required in decision and audit evidence
field_5	action	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

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Audit event schema: evidence and acceptance decision



DECISION RULE

Accept only when Schema conformance ≥99.5% for critical events. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Privacy minimisation: architecture and control boundary

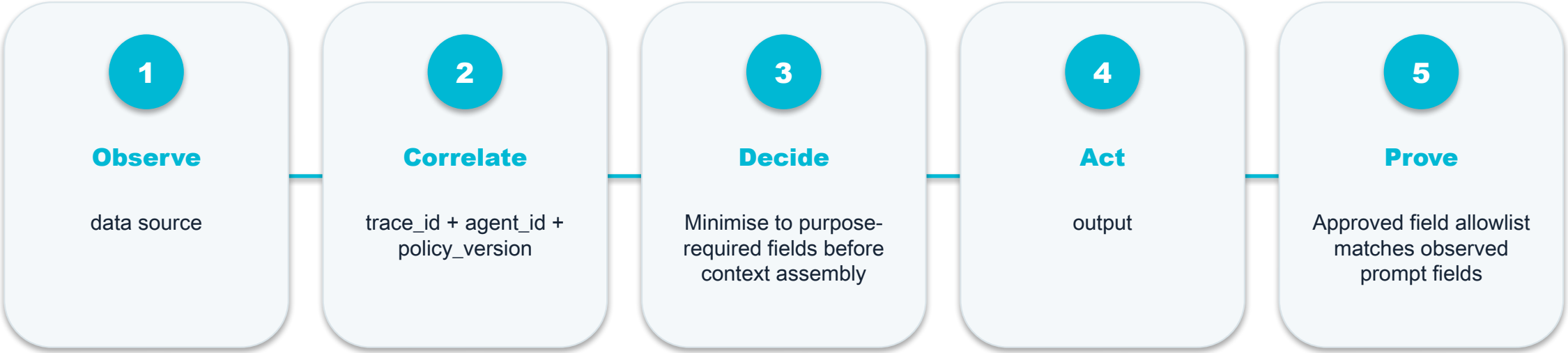


Control objective: Minimise to purpose-required fields before context assembly

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Privacy minimisation: runtime decision trace



Failure to detect: Entire customer record is sent for a narrow task

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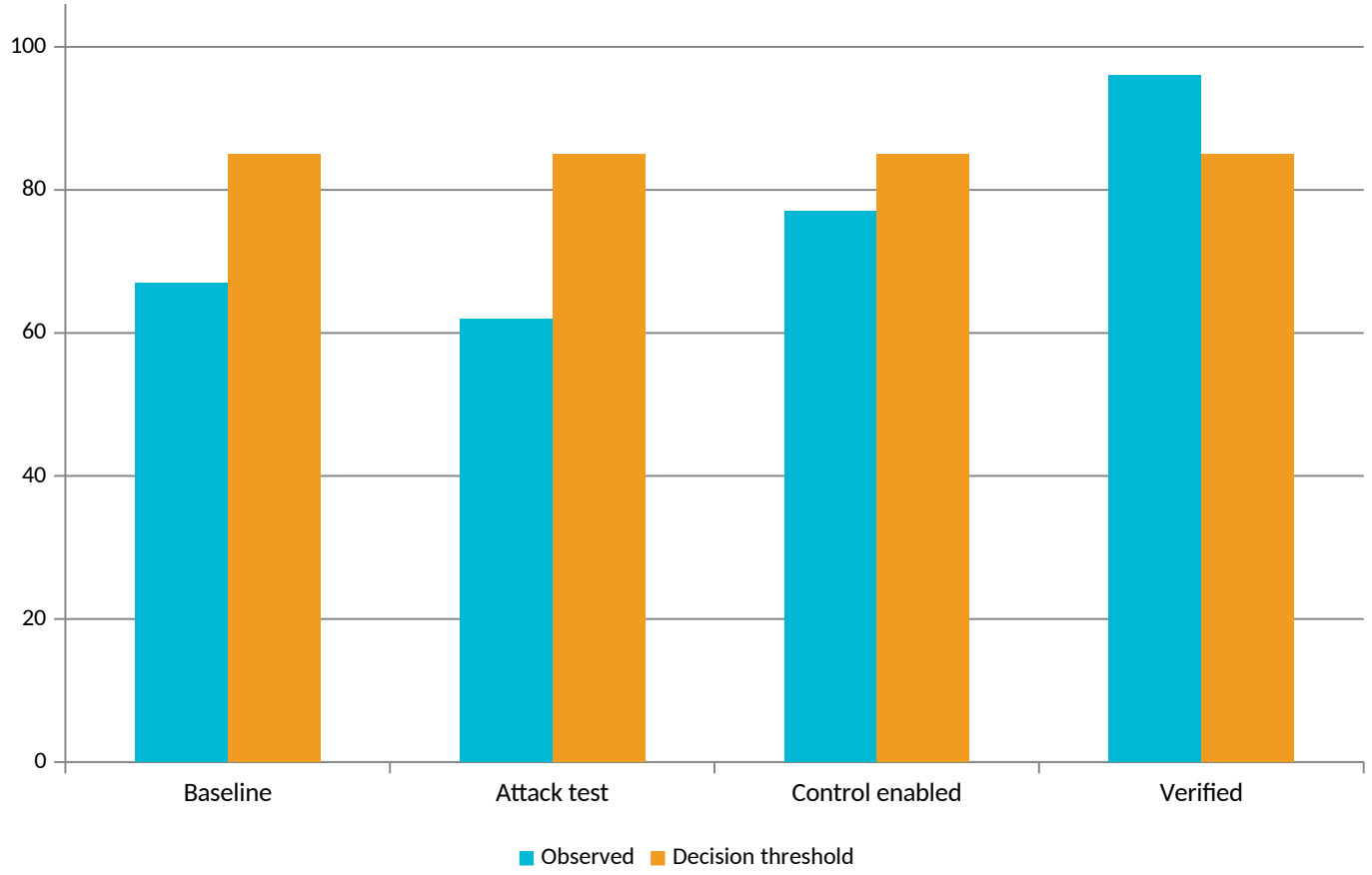
Privacy minimisation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	field	Required in decision and audit evidence
field_2	purpose	Required in decision and audit evidence
field_3	necessity	Required in decision and audit evidence
field_4	transformation	Required in decision and audit evidence
field_5	retention	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Privacy minimisation: evidence and acceptance decision



DECISION RULE

Accept only when Approved field allowlist matches observed prompt fields. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Purpose limitation: architecture and control boundary

1

Request Purpose

State and evidence for request purpose

2

Authorised Use

State and evidence for authorised use

3

Tool Access

State and evidence for tool access

4

Evidence

State and evidence for evidence

Control objective: Bind data and tool permissions to a declared purpose

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Purpose limitation: runtime decision trace

1 — Observe

request purpose

2 — Correlate

trace_id + agent_id + policy_version

3 — Decide

Bind data and tool permissions to a declared purpose

4 — Act

evidence

5 — Prove

Mismatched purpose_id causes a deny decision

Failure to detect: Agent reuses data for an unrelated task

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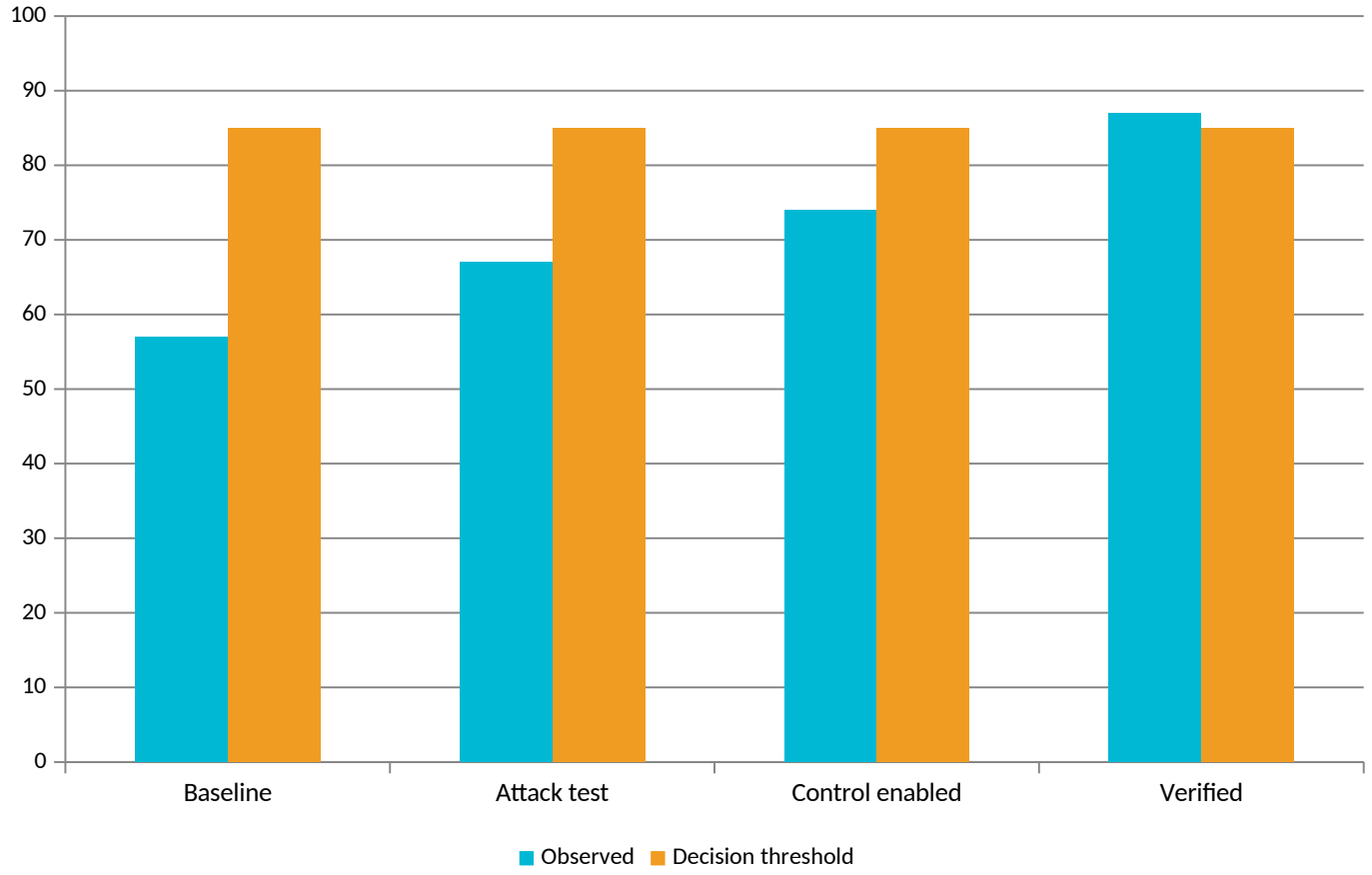
Purpose limitation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	purpose_id	Required in decision and audit evidence
field_2	lawful_basis	Required in decision and audit evidence
field_3	allowed_data[]	Required in decision and audit evidence
field_4	allowed_actions[]	Required in decision and audit evidence
field_5	expiry	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Purpose limitation: evidence and acceptance decision



DECISION RULE

Accept only when Mismatched purpose_id causes a deny decision. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Change management: architecture and control boundary

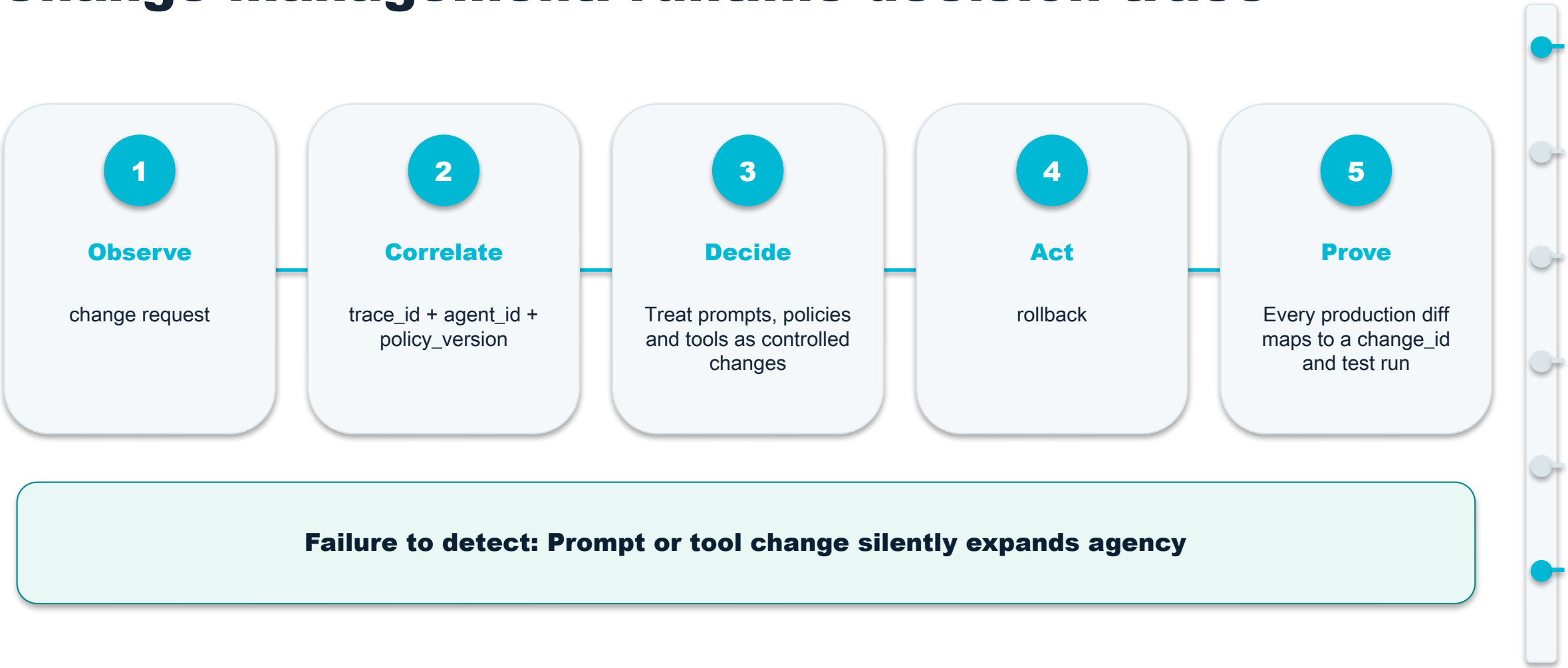
1. **Change Request**
   State and evidence for change request
2. **Impact Analysis**
   State and evidence for impact analysis
3. **Approval**
   State and evidence for approval
4. **Canary**
   State and evidence for canary
5. **Rollback**
   State and evidence for rollback

Control objective: Treat prompts, policies and tools as controlled changes

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Change management: runtime decision trace



Failure to detect: Prompt or tool change silently expands agency

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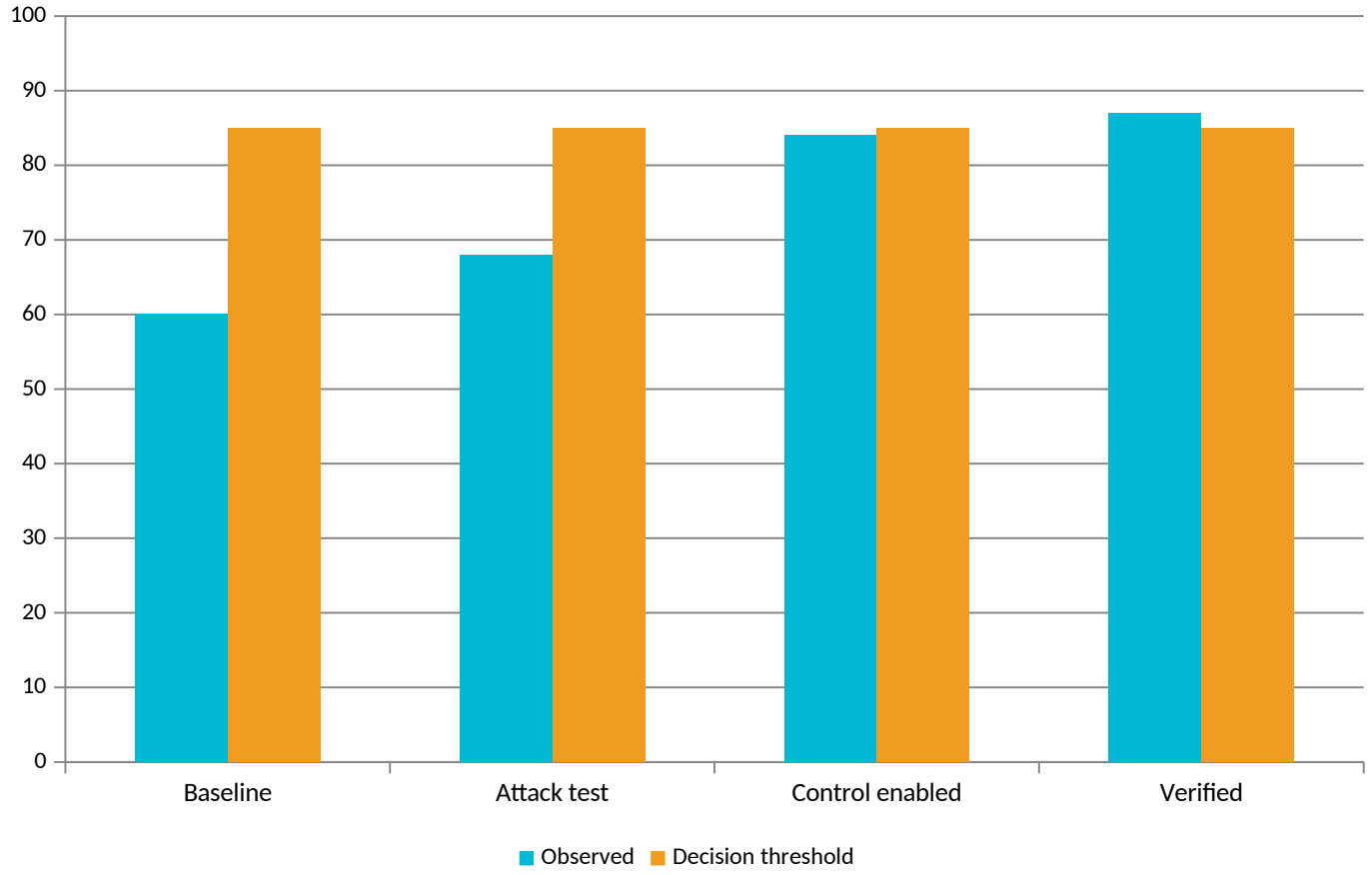
Change management: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	change_id	Required in decision and audit evidence
field_2	components[]	Required in decision and audit evidence
field_3	risk	Required in decision and audit evidence
field_4	tests[]	Required in decision and audit evidence
field_5	approver	Required in decision and audit evidence
field_6	rollback_ref	Required in decision and audit evidence

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Change management: evidence and acceptance decision



DECISION RULE

Accept only when Every production diff maps to a change_id and test run. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Release gate: architecture and control boundary



Control objective: Block on critical failures or missing evidence

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Release gate: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
source commit	trace_id + agent_id + policy_version	Block on critical failures or missing evidence	deploy	Gate records show zero waived critical failures

Failure to detect: Known prompt-injection failure ships to production

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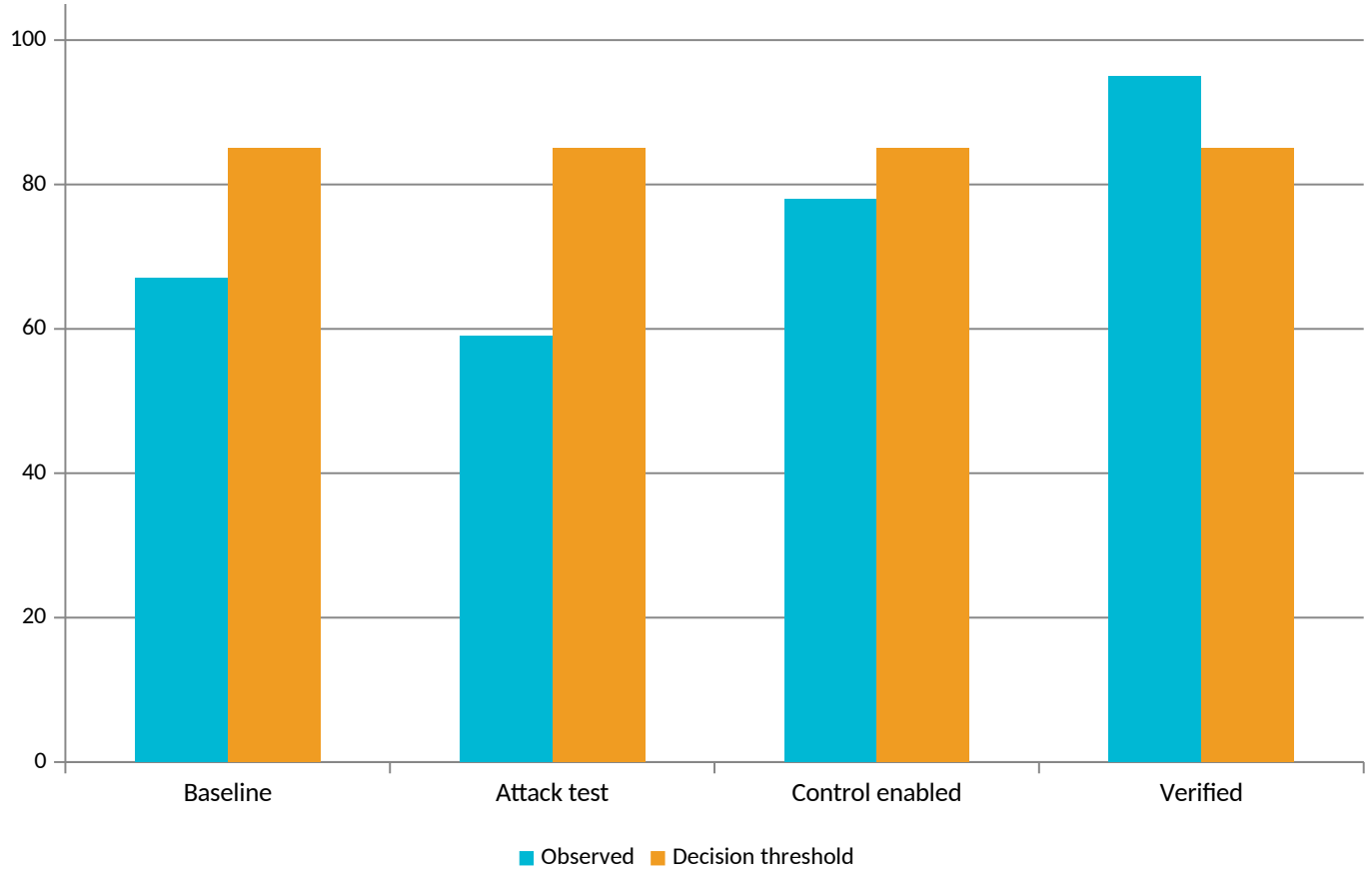
Release gate: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	commit	Required in decision and audit evidence
field_2	test_pass_rate	Required in decision and audit evidence
field_3	critical_failures	Required in decision and audit evidence
field_4	signer	Required in decision and audit evidence
field_5	release_decision	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Release gate: evidence and acceptance decision



DECISION RULE

Accept only when Gate records show zero waived critical failures. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Red-team acceptance: architecture and control boundary

1

Attack Catalogue

State and evidence for attack catalogue

2

Test Harness

State and evidence for test harness

3

Agent Under Test

State and evidence for agent under test

4

Findings

State and evidence for findings

5

Fix

State and evidence for fix

Control objective: Maintain regression tests for direct and indirect injection

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Red-team acceptance: runtime decision trace



Failure to detect: Security claims are not challenged with adversarial inputs

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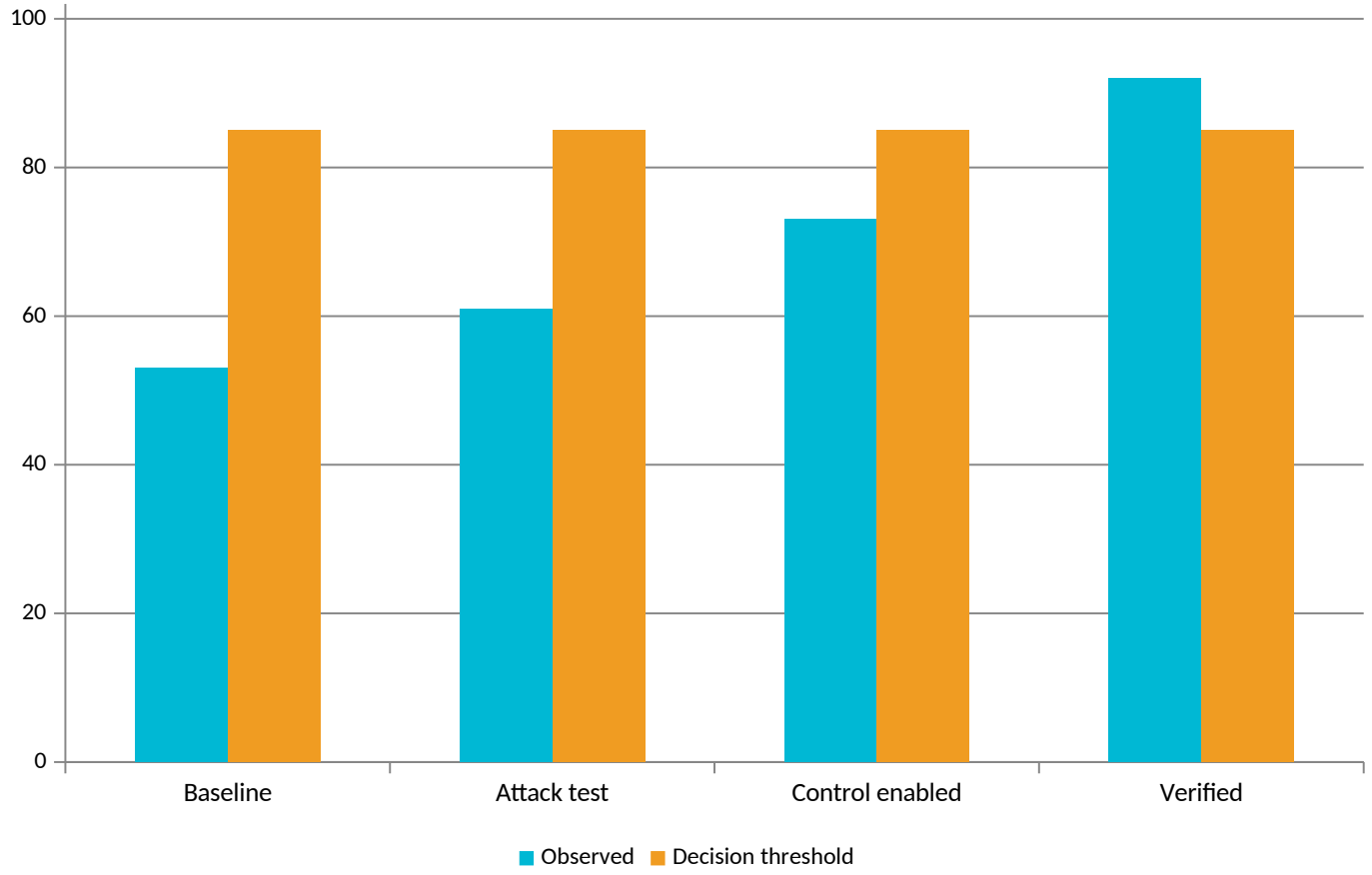
Red-team acceptance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	test_id	Required in decision and audit evidence
field_2	attack_class	Required in decision and audit evidence
field_3	expected_control	Required in decision and audit evidence
field_4	observed_result	Required in decision and audit evidence
field_5	severity	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Red-team acceptance: evidence and acceptance decision



DECISION RULE

Accept only when Critical attack success rate equals zero at release. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Third-party service level: architecture and control boundary

1. **Agent**
   State and evidence for agent
2. **External Service**
   State and evidence for external service
3. **Monitoring**
   State and evidence for monitoring
4. **Escalation**
   State and evidence for escalation
5. **Exit**
   State and evidence for exit

Control objective: Contract measurable security and exit obligations

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Third-party service level: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
agent	trace_id + agent_id + policy_version	Contract measurable security and exit obligations	exit	Provider evidence meets each contracted SLA

Failure to detect: Provider outage or breach cannot be investigated

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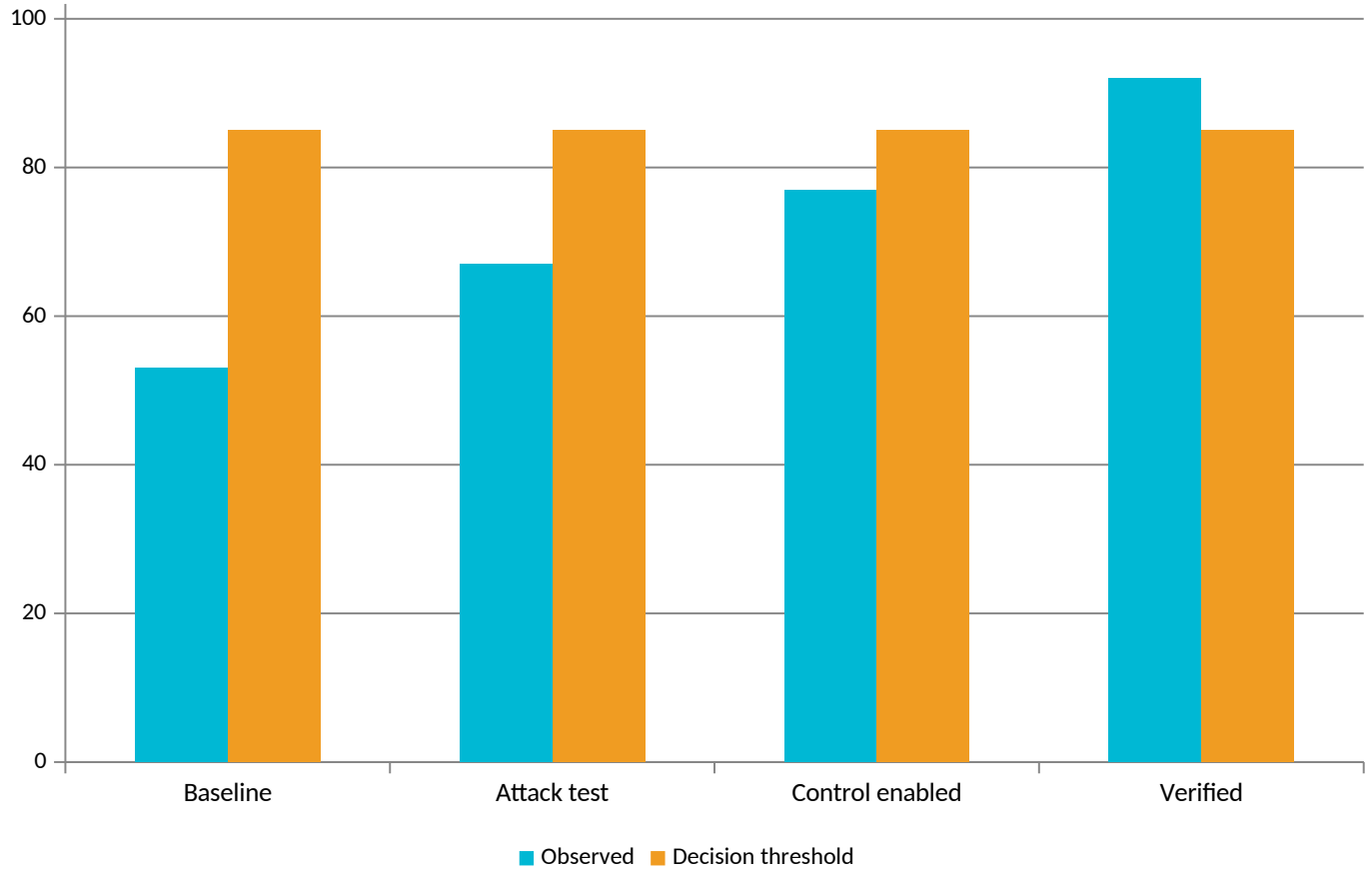
Third-party service level: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	availability_slo	Required in decision and audit evidence
field_2	log_access_sla	Required in decision and audit evidence
field_3	breach_notice_hours	Required in decision and audit evidence
field_4	RTO	Required in decision and audit evidence
field_5	export_format	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Third-party service level: evidence and acceptance decision



DECISION RULE

Accept only when Provider evidence meets each contracted SLA. Otherwise record a gap, owner and due date.

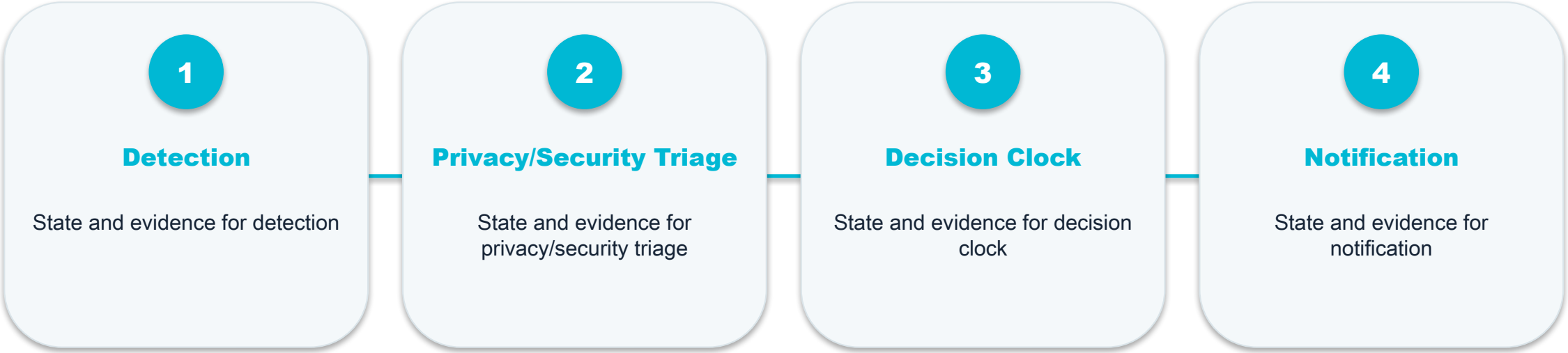
EVIDENCE STATUS

PASS

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Incident notification: architecture and control boundary



Control objective: Start jurisdiction-specific clocks at verified discovery

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Incident notification: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
detection	trace_id + agent_id + policy_version	Start jurisdiction-specific clocks at verified discovery	notification	Decision and notification timestamps are auditable

Failure to detect: Notification deadline is missed because ownership is unclear

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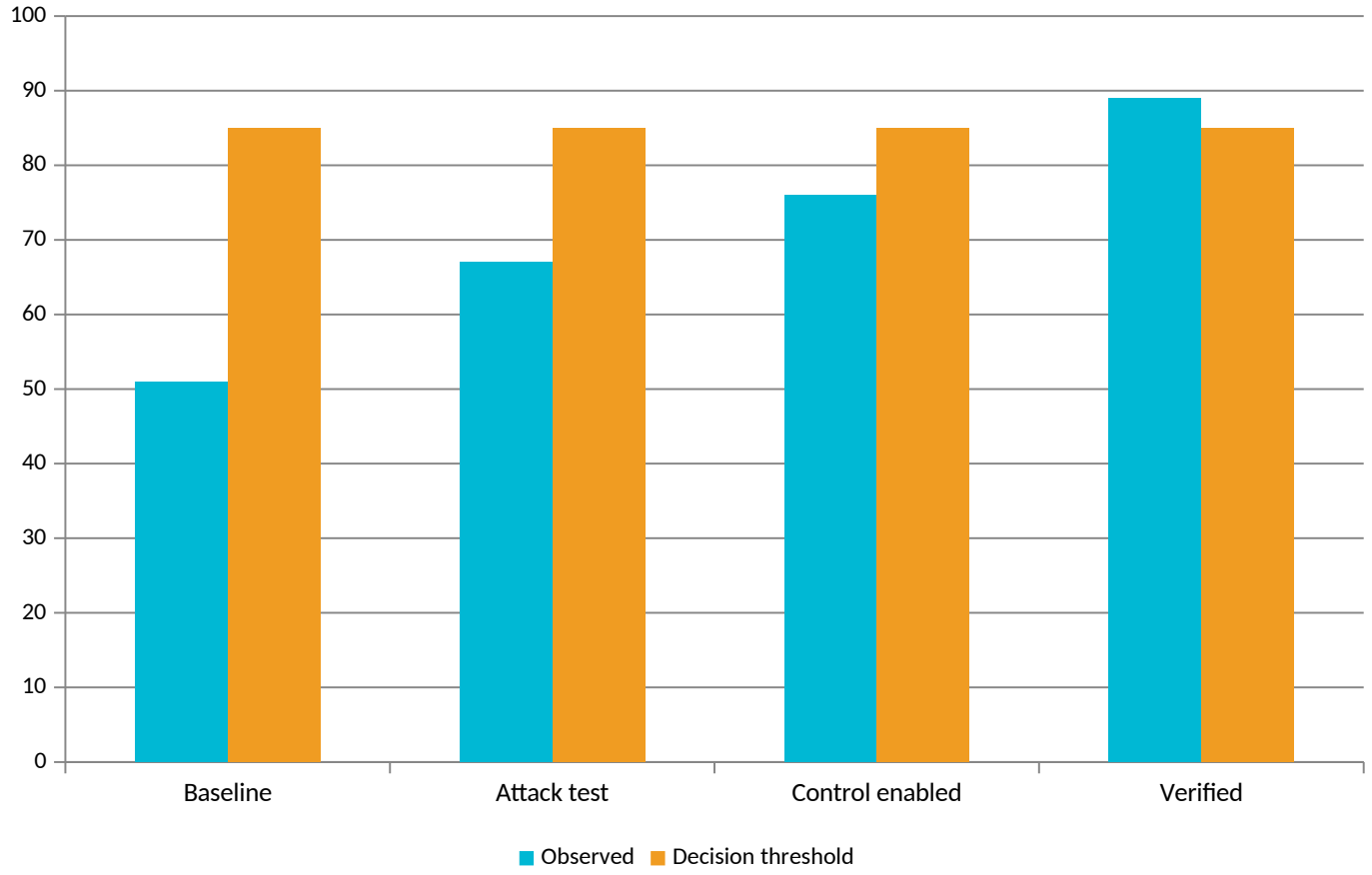
Incident notification: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	incident_id	Required in decision and audit evidence
field_2	discovered_at	Required in decision and audit evidence
field_3	jurisdictions[]	Required in decision and audit evidence
field_4	impact	Required in decision and audit evidence
field_5	deadline	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

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Incident notification: evidence and acceptance decision



DECISION RULE

Accept only when Decision and notification timestamps are auditable. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Exception workflow: architecture and control boundary

1

Control Failure

State and evidence for control failure

2

Exception Request

State and evidence for exception request

3

Compensating Control

State and evidence for compensating control

4

Expiry

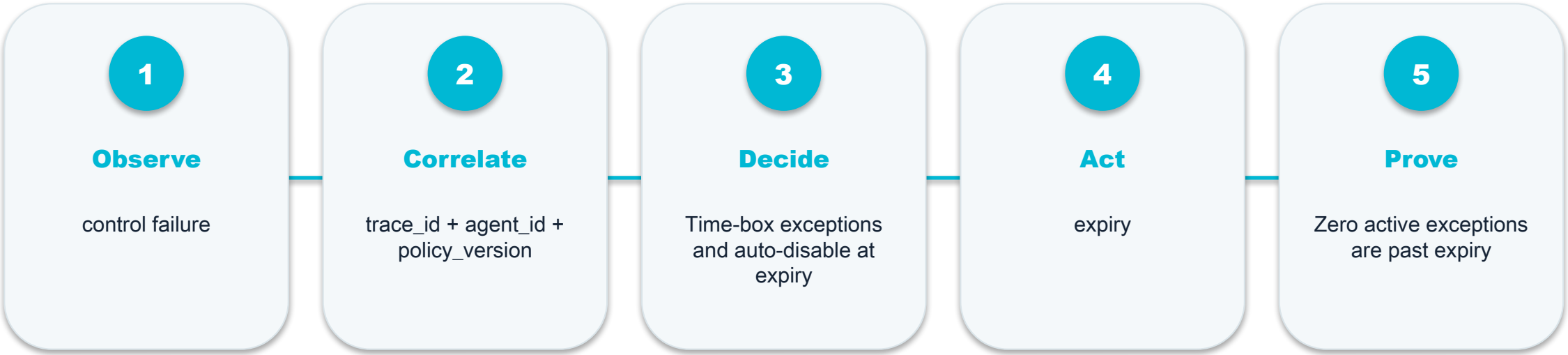
State and evidence for expiry

Control objective: Time-box exceptions and auto-disable at expiry

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Exception workflow: runtime decision trace



Failure to detect: Temporary bypass becomes permanent shadow policy

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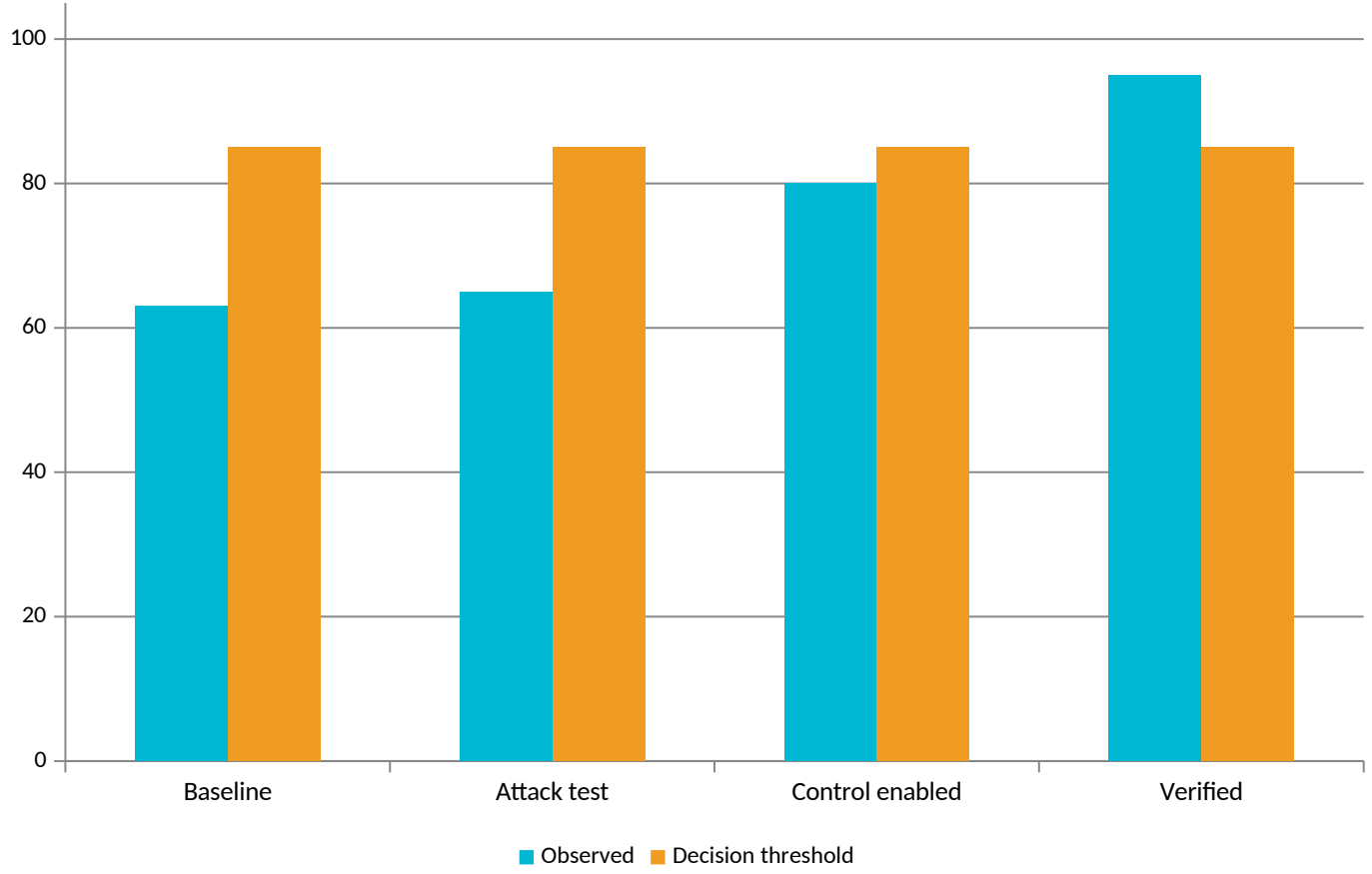
Exception workflow: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	exception_id	Required in decision and audit evidence
field_2	control_id	Required in decision and audit evidence
field_3	rationale	Required in decision and audit evidence
field_4	risk_owner	Required in decision and audit evidence
field_5	expiry	Required in decision and audit evidence
field_6	compensation	Required in decision and audit evidence

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Exception workflow: evidence and acceptance decision



DECISION RULE

Accept only when Zero active exceptions are past expiry. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Framework crosswalk: architecture and control boundary

1

Requirement

State and evidence for requirement

2

Control Objective

State and evidence for control objective

3

Implementation

State and evidence for implementation

4

Evidence

State and evidence for evidence

Control objective: Maintain one control-to-framework crosswalk

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Framework crosswalk: runtime decision trace



Failure to detect: Multiple audits ask for duplicate evidence with inconsistent answers

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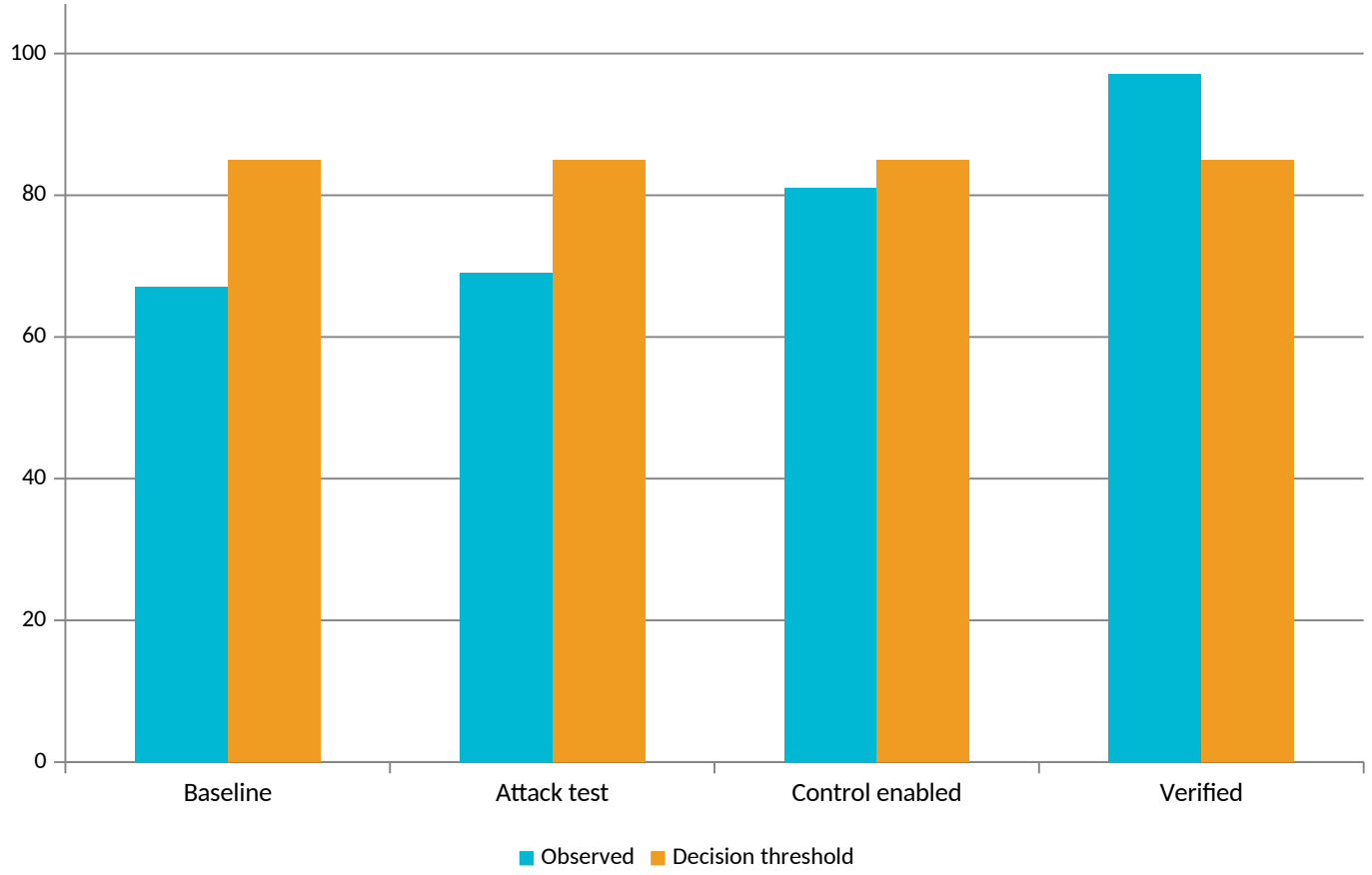
Framework crosswalk: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	framework	Required in decision and audit evidence
field_2	clause	Required in decision and audit evidence
field_3	control_id	Required in decision and audit evidence
field_4	evidence_id	Required in decision and audit evidence
field_5	coverage_status	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Framework crosswalk: evidence and acceptance decision



DECISION RULE

Accept only when Each applicable clause has current evidence or a gap. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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NIST AI RMF operations: architecture and control boundary

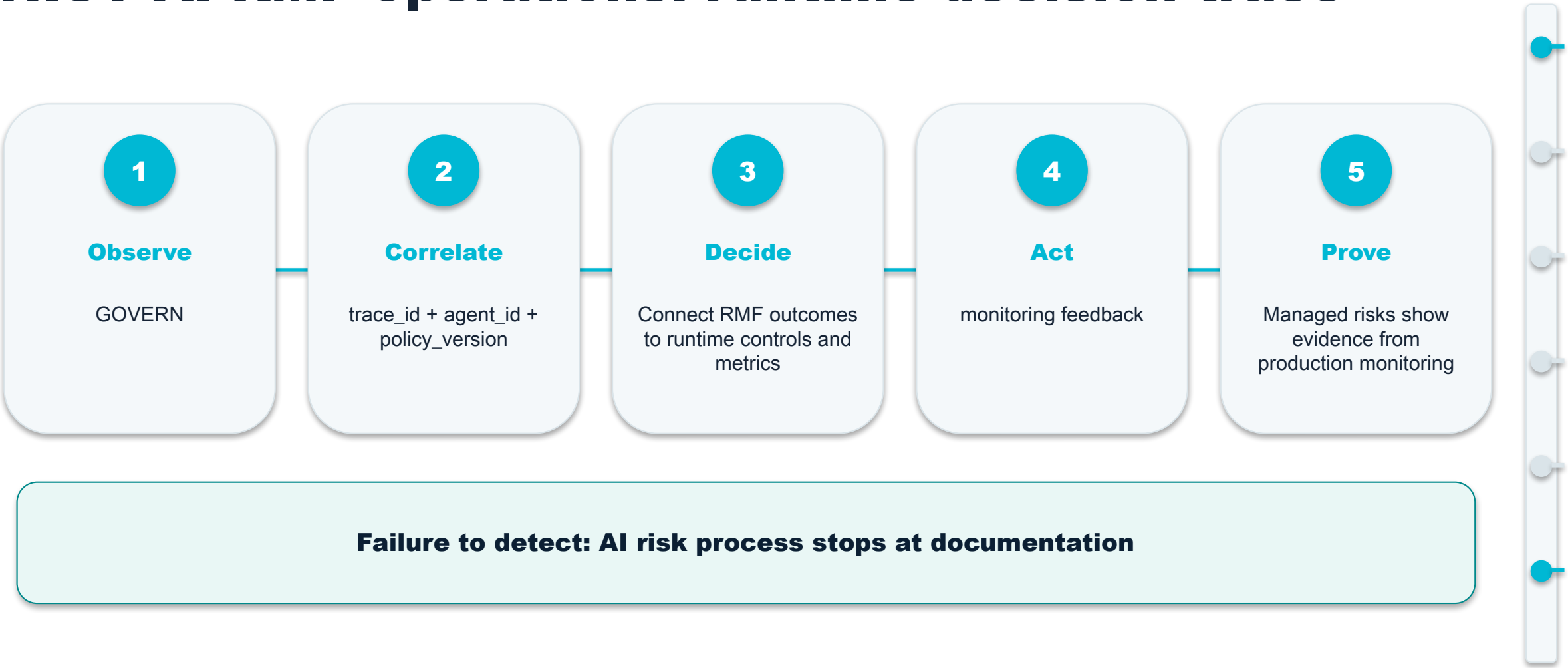


Control objective: Connect RMF outcomes to runtime controls and metrics

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NIST AI RMF operations: runtime decision trace



Failure to detect: AI risk process stops at documentation

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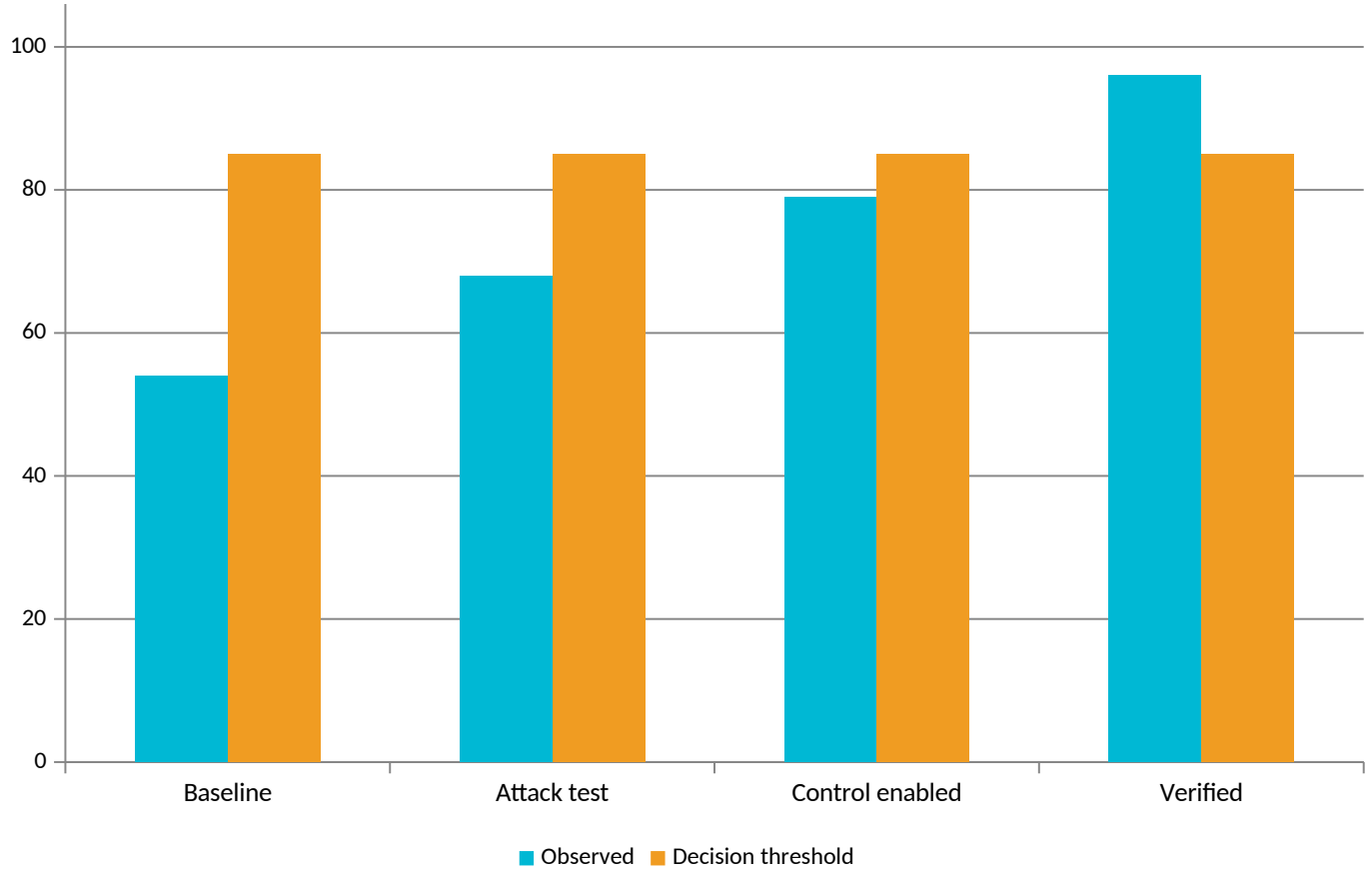
NIST AI RMF operations: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	function	Required in decision and audit evidence
field_2	category	Required in decision and audit evidence
field_3	risk_scenario	Required in decision and audit evidence
field_4	metric	Required in decision and audit evidence
field_5	response	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

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NIST AI RMF operations: evidence and acceptance decision



DECISION RULE

Accept only when Managed risks show evidence from production monitoring. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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ISO 27001 integration: architecture and control boundary

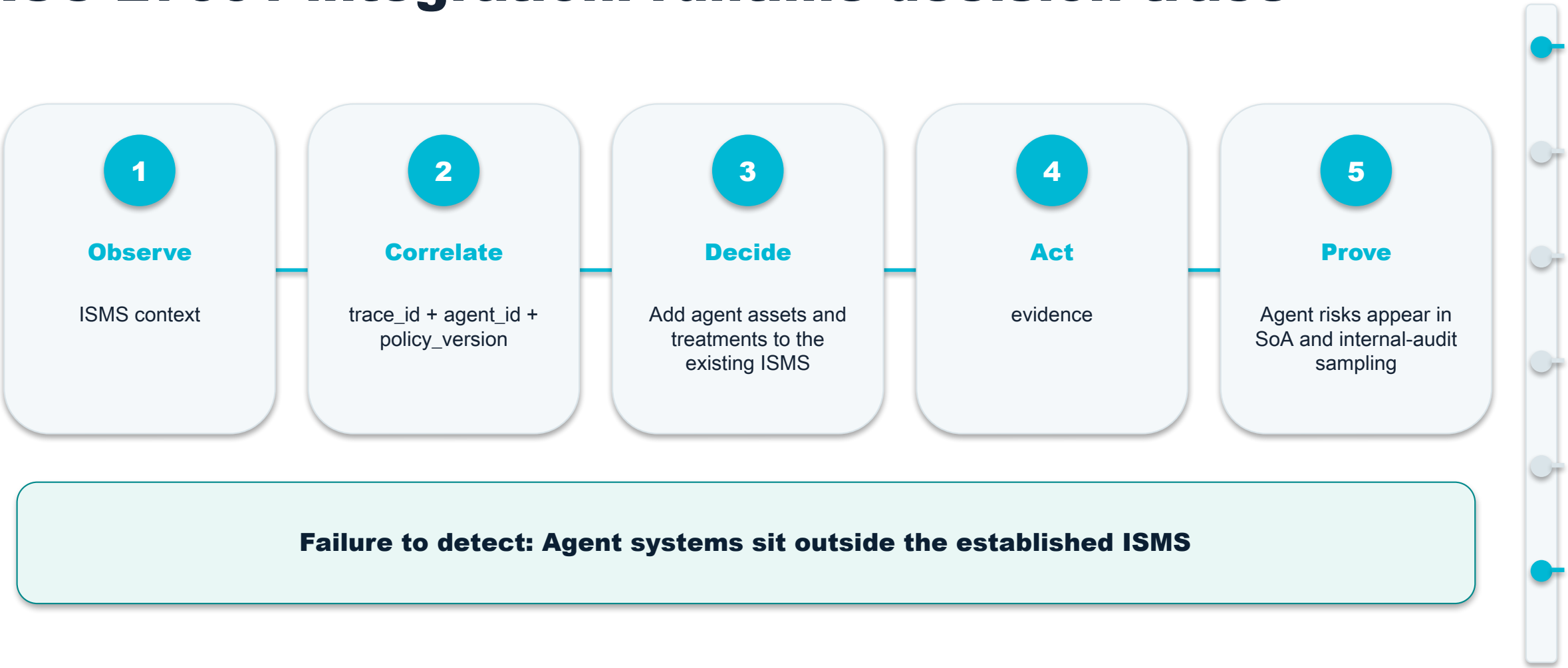


Control objective: Add agent assets and treatments to the existing ISMS

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ISO 27001 integration: runtime decision trace



Failure to detect: Agent systems sit outside the established ISMS

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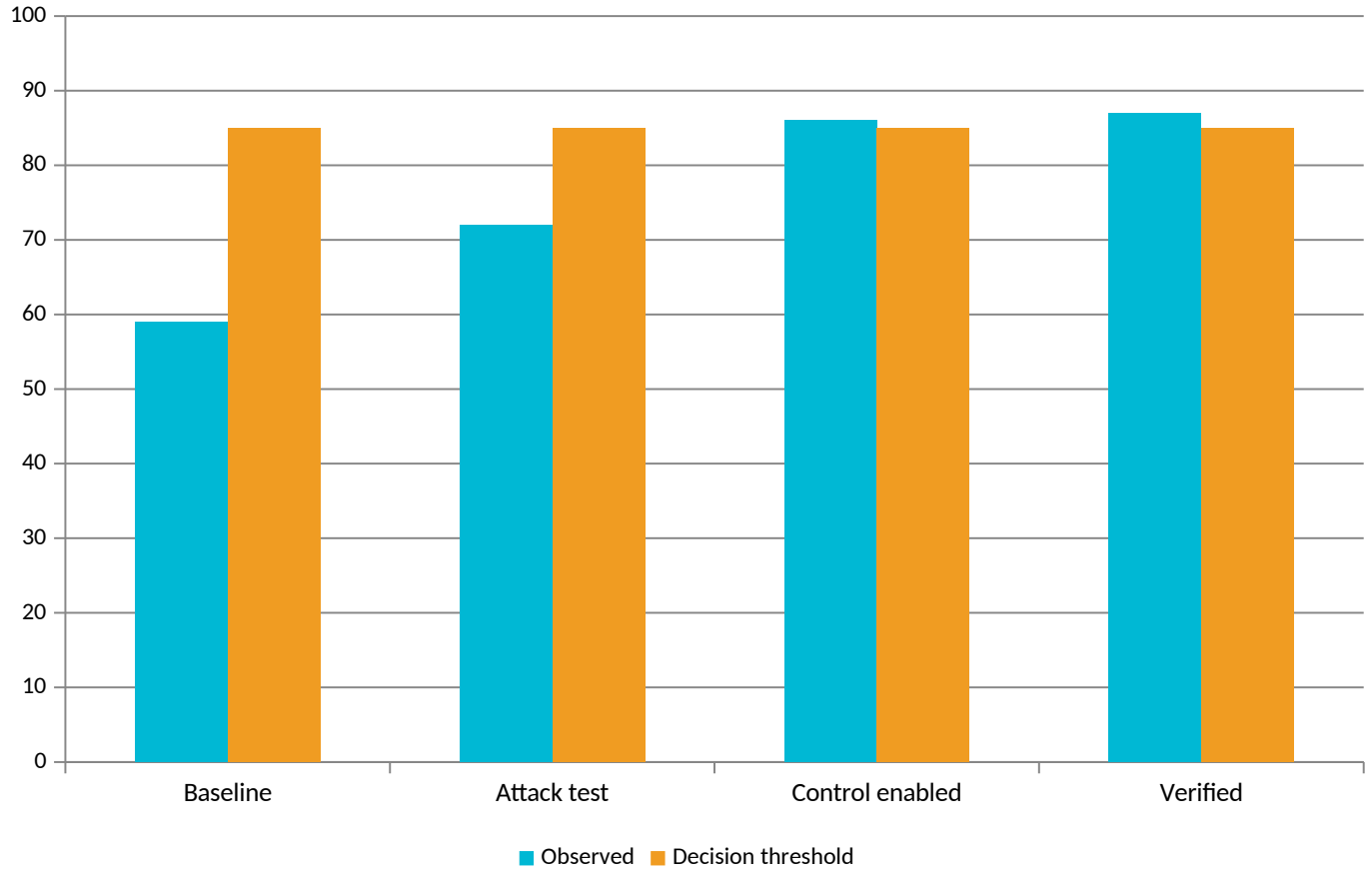
ISO 27001 integration: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	asset	Required in decision and audit evidence
field_2	risk_id	Required in decision and audit evidence
field_3	control_ref	Required in decision and audit evidence
field_4	statement_of_applicability	Required in decision and audit evidence
field_5	evidence	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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ISO 27001 integration: evidence and acceptance decision



DECISION RULE

Accept only when Agent risks appear in SoA and internal-audit sampling. Otherwise record a gap, owner and due date.

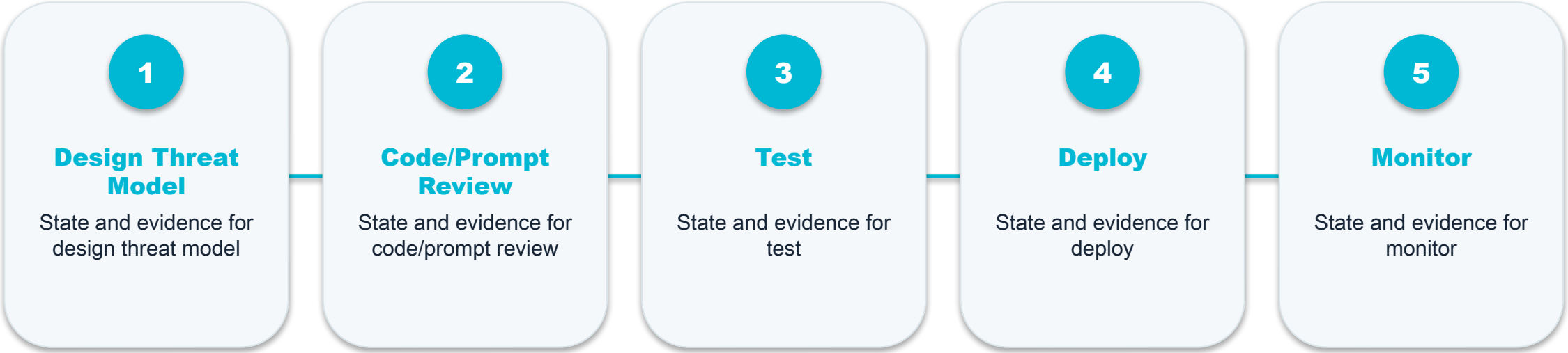
EVIDENCE STATUS

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Secure development lifecycle: architecture and control boundary

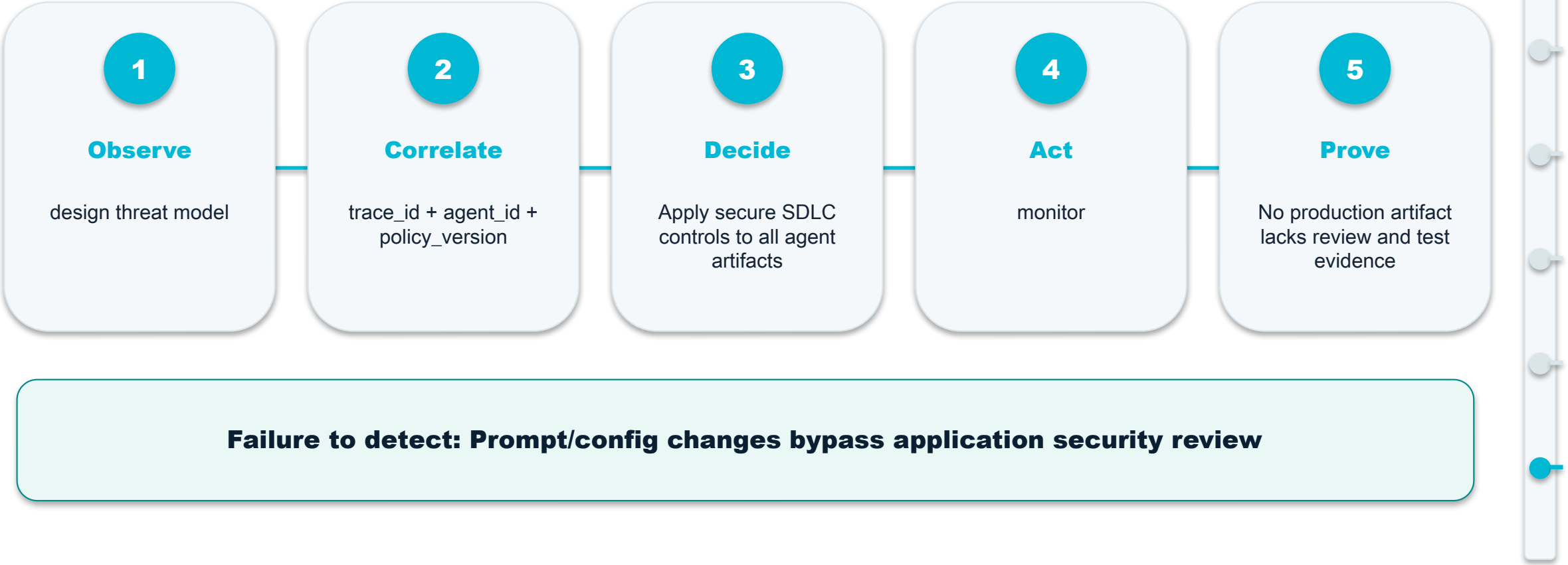


Control objective: Apply secure SDLC controls to all agent artifacts

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Secure development lifecycle: runtime decision trace



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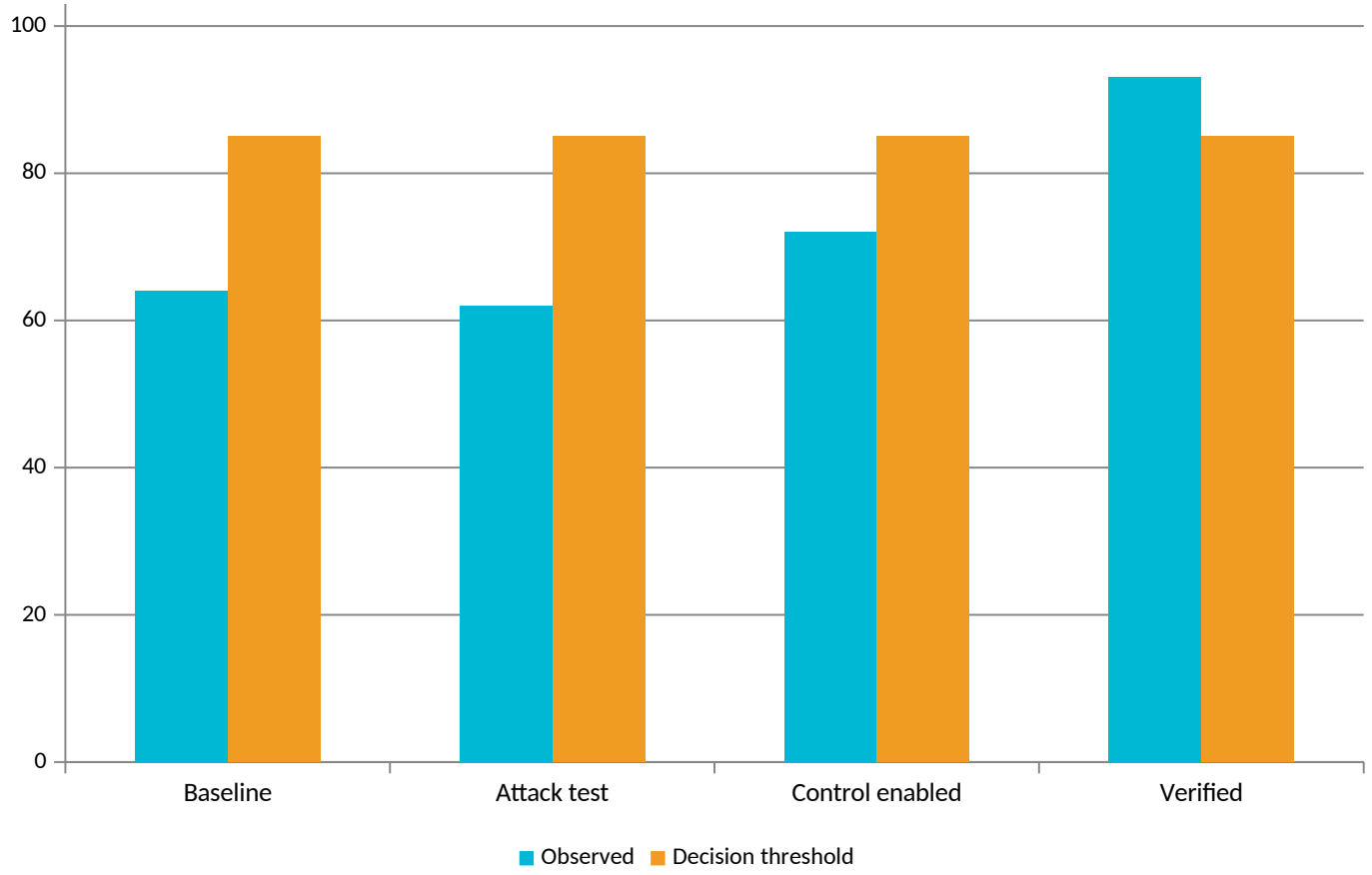
Secure development lifecycle: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	artifact_type	Required in decision and audit evidence
field_2	reviewer	Required in decision and audit evidence
field_3	tests	Required in decision and audit evidence
field_4	vulnerabilities	Required in decision and audit evidence
field_5	approval	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

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Secure development lifecycle: evidence and acceptance decision



DECISION RULE

Accept only when No production artifact lacks review and test evidence. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Model/version governance: architecture and control boundary

1

Approved Model Catalogue

State and evidence for approved model catalogue

2

Evaluation

State and evidence for evaluation

3

Pinning

State and evidence for pinning

4

Monitoring

State and evidence for monitoring

Control objective: Pin versions where possible and re-evaluate changes

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Model/version governance: runtime decision trace

1 — Observe

approved model catalogue

2 — Correlate

trace_id + agent_id + policy_version

3 — Decide

Pin versions where possible and re-evaluate changes

4 — Act

monitoring

5 — Prove

Behavioural regression stays within tolerance

Failure to detect: Provider model update changes behaviour unexpectedly

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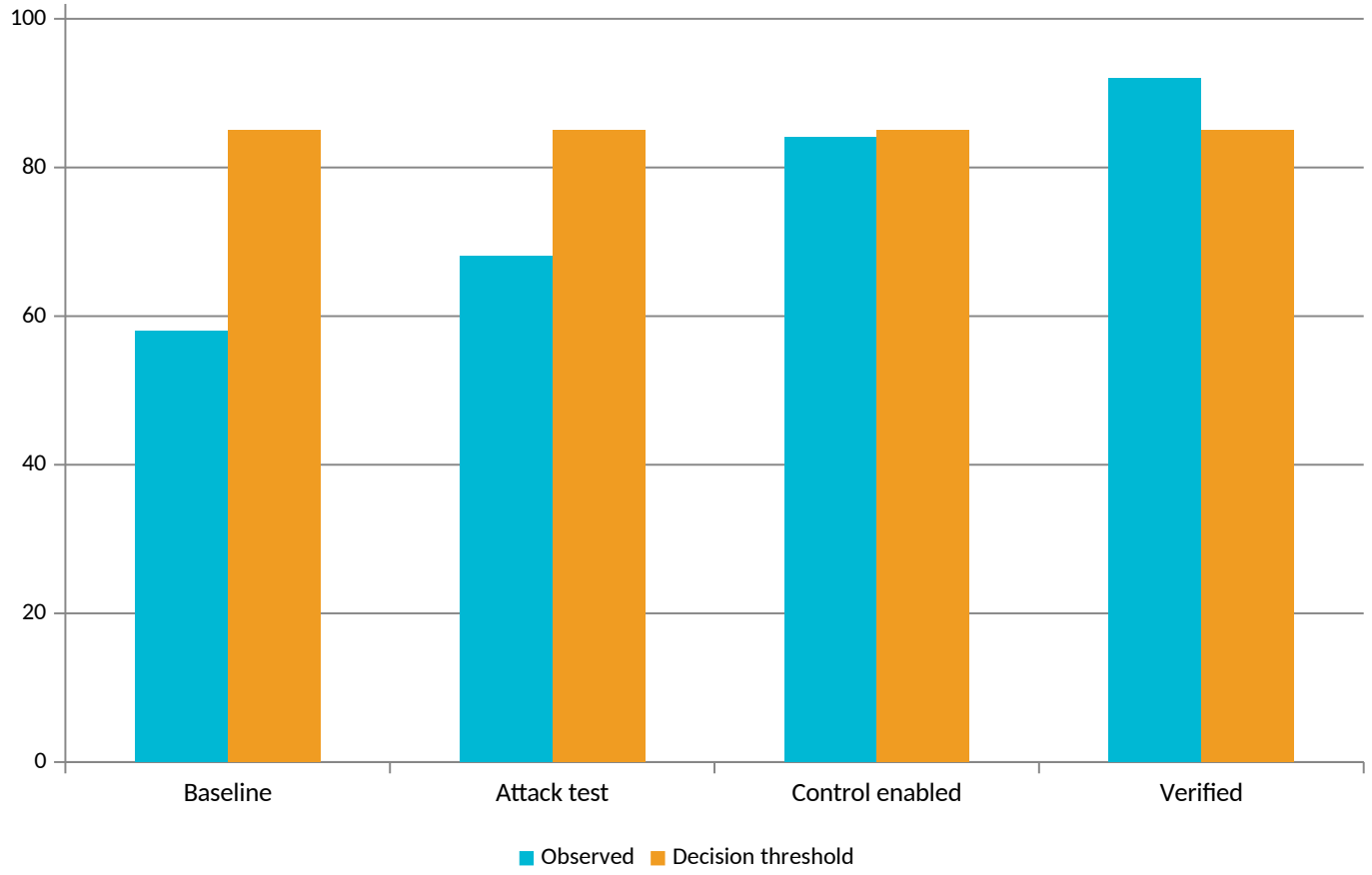
Model/version governance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	provider	Required in decision and audit evidence
field_2	model_id	Required in decision and audit evidence
field_3	version	Required in decision and audit evidence
field_4	eval_set	Required in decision and audit evidence
field_5	score	Required in decision and audit evidence
field_6	retirement_date	Required in decision and audit evidence

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Model/version governance: evidence and acceptance decision



DECISION RULE

Accept only when Behavioural regression stays within tolerance. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Security communication cadence: architecture and control boundary

1

Control Owners

State and evidence for control owners

2

Operational Review

State and evidence for operational review

3

Risk Committee

State and evidence for risk committee

4

Board

State and evidence for board

Control objective: Tailor evidence and decisions to each governance layer

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Security communication cadence: runtime decision trace

1

Observe

control owners

2

Correlate

trace_id + agent_id + policy_version

3

Decide

Tailor evidence and decisions to each governance layer

4

Act

board

5

Prove

Actions have owners and due dates after every review

Failure to detect: Technical alerts never become business decisions

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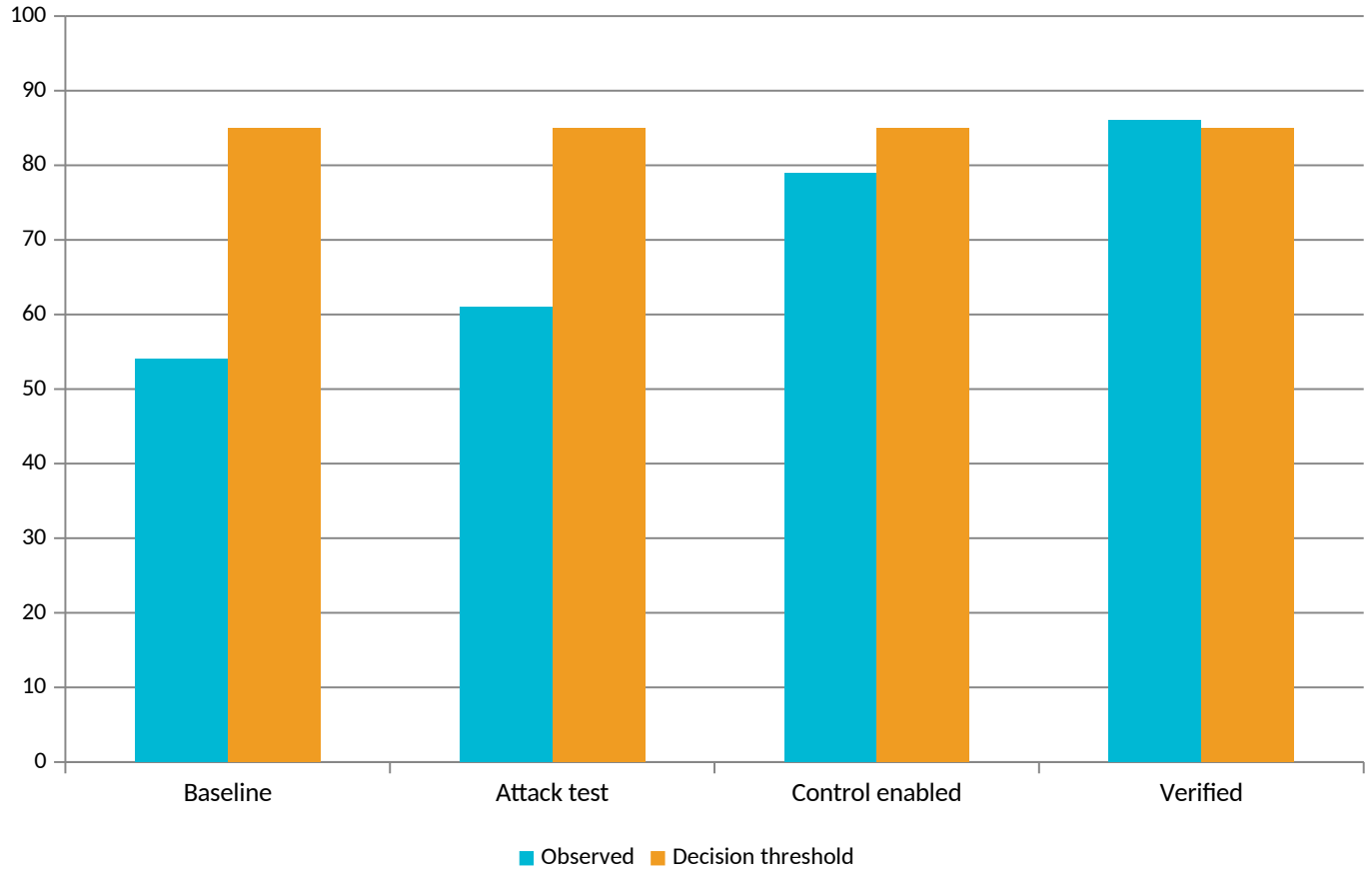
Security communication cadence: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	audience	Required in decision and audit evidence
field_2	frequency	Required in decision and audit evidence
field_3	metrics[]	Required in decision and audit evidence
field_4	decisions[]	Required in decision and audit evidence
field_5	action_owner	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Security communication cadence: evidence and acceptance decision



DECISION RULE

Accept only when Actions have owners and due dates after every review. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Assurance dashboard: architecture and control boundary

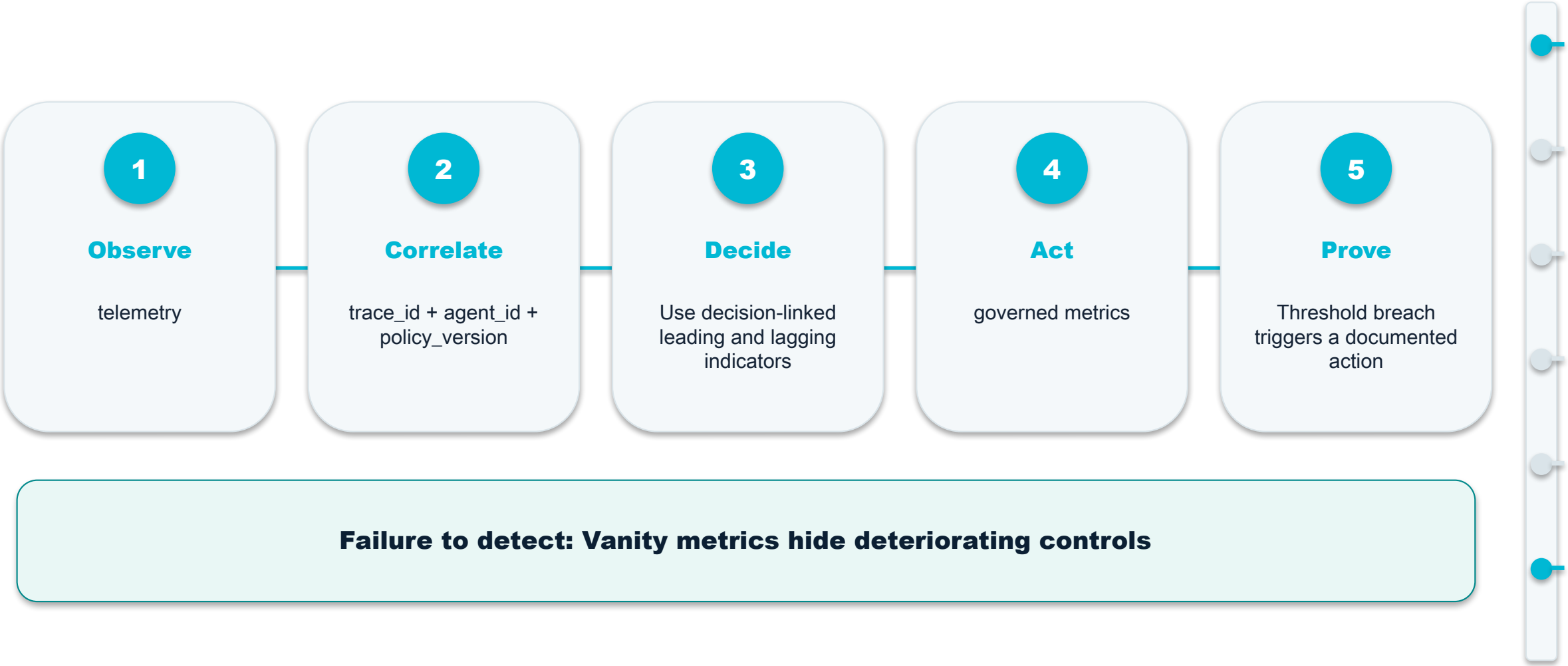


Control objective: Use decision-linked leading and lagging indicators

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Assurance dashboard: runtime decision trace



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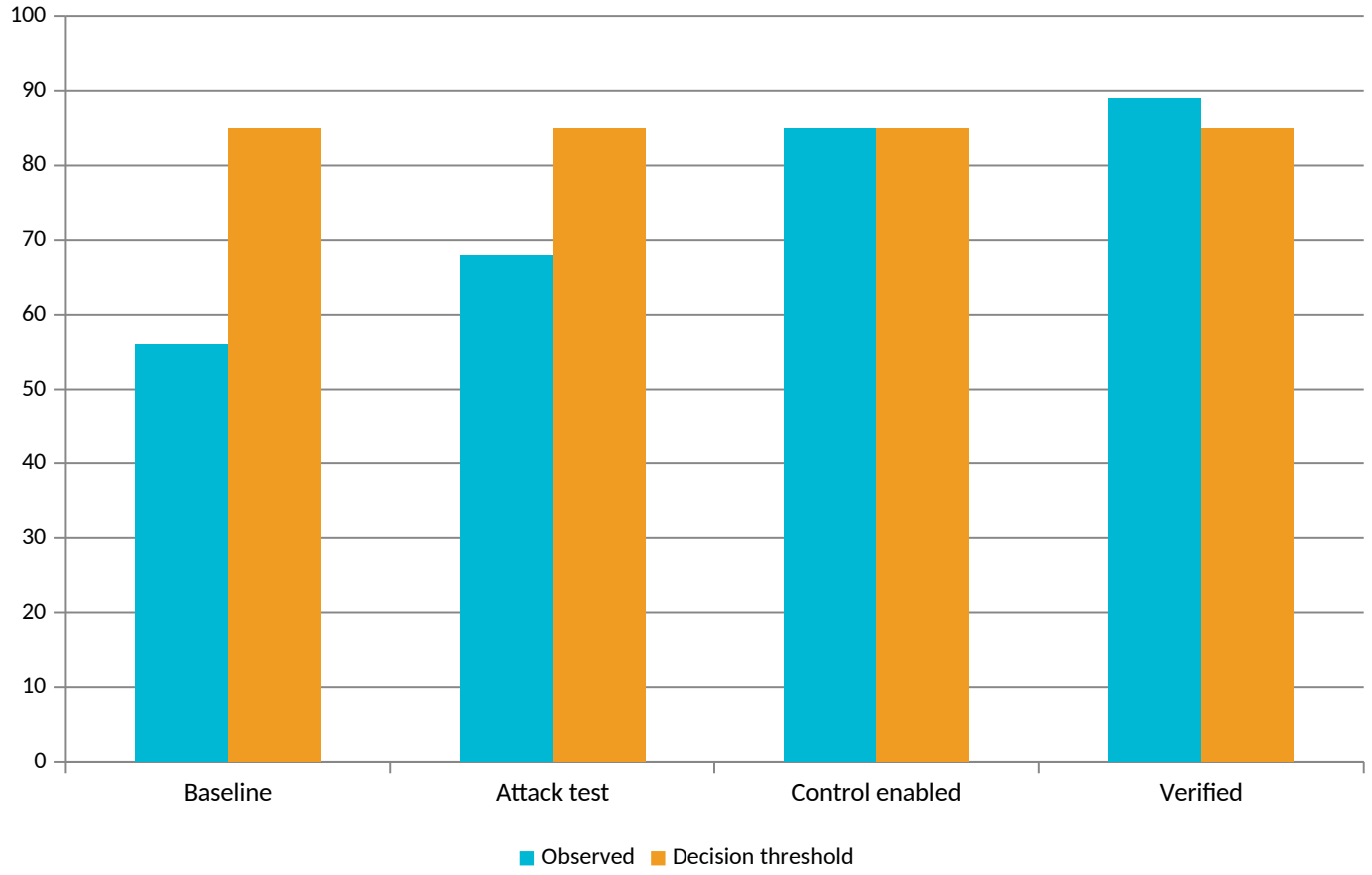
Assurance dashboard: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	metric_id	Required in decision and audit evidence
field_2	numerator	Required in decision and audit evidence
field_3	denominator	Required in decision and audit evidence
field_4	threshold	Required in decision and audit evidence
field_5	trend	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

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Assurance dashboard: evidence and acceptance decision



DECISION RULE

Accept only when Threshold breach triggers a documented action. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Training and tabletop: architecture and control boundary

1. **Role Profile**
   State and evidence for role profile
2. **Scenario**
   State and evidence for scenario
3. **Injects**
   State and evidence for injects
4. **Decisions**
   State and evidence for decisions
5. **After-Action Plan**
   State and evidence for after-action plan

Control objective: Rehearse prompt injection and tool abuse quarterly

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Training and tabletop: runtime decision trace

1. **Observe**
   role profile
2. **Correlate**
   trace_id + agent_id + policy_version
3. **Decide**
   Rehearse prompt injection and tool abuse quarterly
4. **Act**
   after-action plan
5. **Prove**
   Closure rate and containment time improve across exercises

Failure to detect: Teams know policy but cannot execute containment

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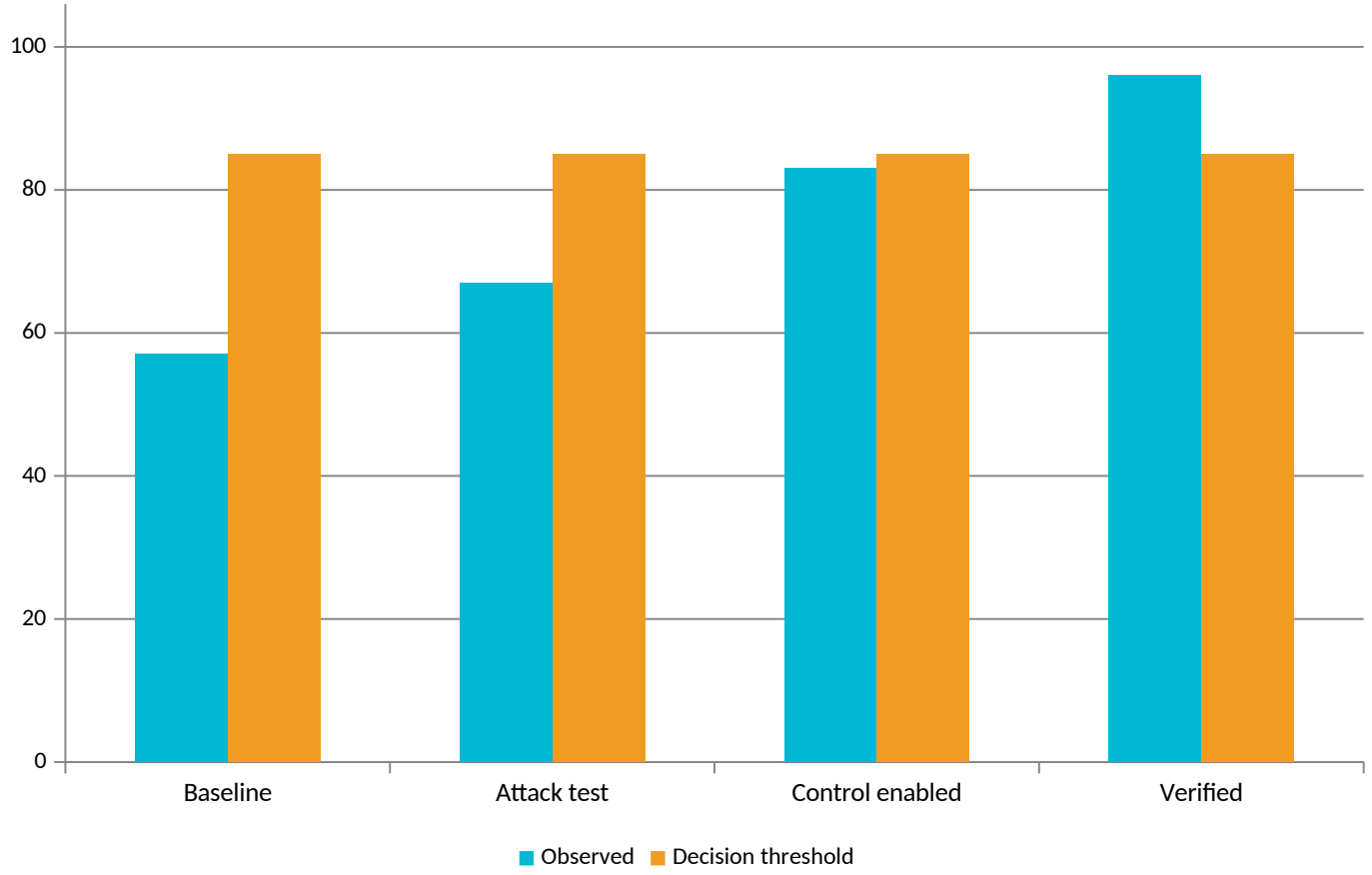
Training and tabletop: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	exercise_id	Required in decision and audit evidence
field_2	roles[]	Required in decision and audit evidence
field_3	inject_time	Required in decision and audit evidence
field_4	decision	Required in decision and audit evidence
field_5	gap	Required in decision and audit evidence
field_6	due_date	Required in decision and audit evidence

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Training and tabletop: evidence and acceptance decision



DECISION RULE

Accept only when Closure rate and containment time improve across exercises. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Activity 04 — Assemble a Compliance Evidence Pack

1 **Scenario**

Use synthetic data in activity-04-compliance-evidence to make a security-operations decision.

2 **Input contract**

mock_evidence.json

3 **Runnable evidence**

activity.py

4 **Output artifact**

compliance_report.md

5 **Acceptance**

Each required control maps to current evidence, an owner and a status; gaps are explicit.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 04: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_evidence.json	required fields and types	control and decision rules	compliance_report.md	Each required control maps to current evidence, an owner and a status; gaps are explicit.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 04: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_evidence.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	compliance_report.md	Generated locally and reproducible
Criterion	A3	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete compliance_report.md only	Retain script and mock data

Aligned activity files in labs/

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Activity 05 — Communicate Agent Security Posture

1 **Scenario**

Use synthetic data in activity-05-security-communication to make a security-operations decision.

2 **Input contract**

mock_metrics.csv

3 **Runnable evidence**

activity.py

4 **Output artifact**

security_brief.html

5 **Acceptance**

The generated brief shows thresholds, current values, decisions, owners and due dates for two audiences.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 05: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_metrics.csv	required fields and types	control and decision rules	security_brief.html	The generated brief shows thresholds, current values, decisions, owners and due dates for two audiences.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 05: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_metrics.csv	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	security_brief.html	Generated locally and reproducible
Criterion	A4	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete security_brief.html only	Retain script and mock data

Aligned activity files in labs/

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Topic 02 technical recap

1 **ISO 27001 integration**

Agent risks appear in SoA and internal-audit sampling

2 **Secure development lifecycle**

No production artifact lacks review and test evidence

3 **Model/version governance**

Behavioural regression stays within tolerance

4 **Security communication cadence**

Actions have owners and due dates after every review

5 **Assurance dashboard**

Threshold breach triggers a documented action

6 **Training and tabletop**

Closure rate and containment time improve across exercises

Topic technical-anchor inventory

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TOPIC 03

AI Agent Risk Assessment and Security Control Evaluation

Threat paths · risk quantification · control effectiveness · remediation

LO3 · K5, K6, A5, A6



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LO3 and assessment criteria

1 **K5**

Knowledge of security management benchmarks

2 **K6**

Gap analysis in organisation security

3 **A5**

Review existing security controls against current and future business costs and risks

4 **A6**

Develop strategies and plans to resolve security gaps

5 **LO3**

LO3: Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.

Approved course proposal criteria adapted to the autonomous-agent context

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AI Agent Risk Assessment and Security Control Evaluation: operating model

1. **Boundary** — Components, interfaces and ownership
2. **Runtime** — States, transitions and decision points
3. **Contract** — Fields, rules, thresholds and evidence
4. **Failure** — Attack path, diagnostic and control
5. **Verify** — Acceptance test and activity artifact

Every mechanism in this topic is taught through all five layers.

Courseware-build mechanism-level coverage contract

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Agent threat model: architecture and control boundary

1. **Assets**
   State and evidence for assets
2. **Trust Boundaries**
   State and evidence for trust boundaries
3. **Attacker Goals**
   State and evidence for attacker goals
4. **Abuse Paths**
   State and evidence for abuse paths
5. **Controls**
   State and evidence for controls

Control objective: Model runtime, tools, memory, identity and suppliers

Supplied AI-security reference corpus · NIST AI RMF · OWASP ASI

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Agent threat model: runtime decision trace



Failure to detect: Threat model treats the LLM as the only attack surface

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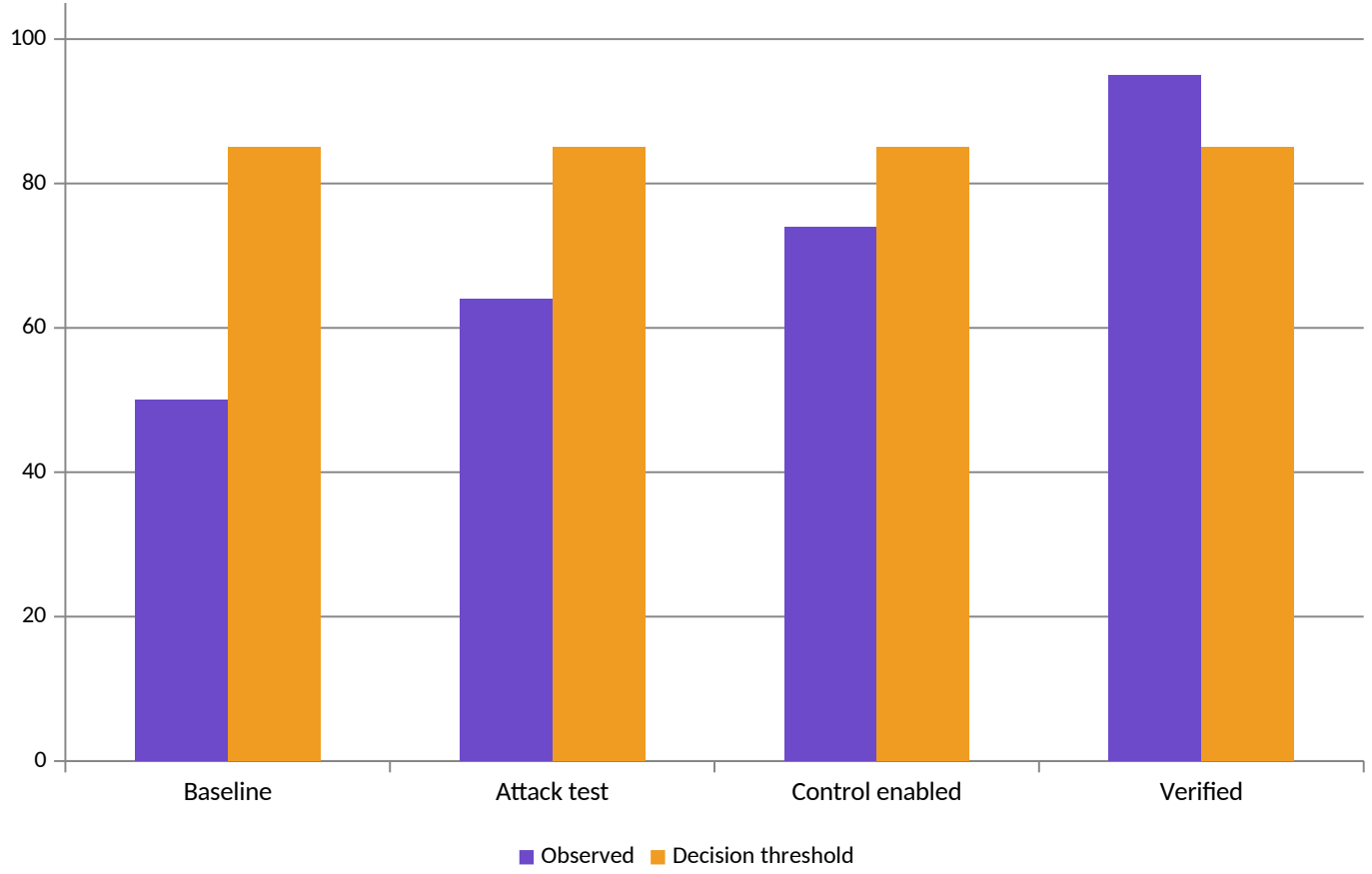
Agent threat model: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	asset	Required in decision and audit evidence
field_2	entry_point	Required in decision and audit evidence
field_3	precondition	Required in decision and audit evidence
field_4	technique	Required in decision and audit evidence
field_5	impact	Required in decision and audit evidence
field_6	mitigation	Required in decision and audit evidence

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Agent threat model: evidence and acceptance decision



DECISION RULE

Accept only when Every high-value asset has at least one tested abuse path. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Direct prompt injection: architecture and control boundary

1

User Input

State and evidence for user input

2

Context

State and evidence for context

3

Planner

State and evidence for planner

4

Prohibited Tool Proposal

State and evidence for prohibited tool proposal

Control objective: Separate policy from content and enforce action authorisation

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Direct prompt injection: runtime decision trace



Failure to detect: User instruction overrides policy and triggers a sensitive action

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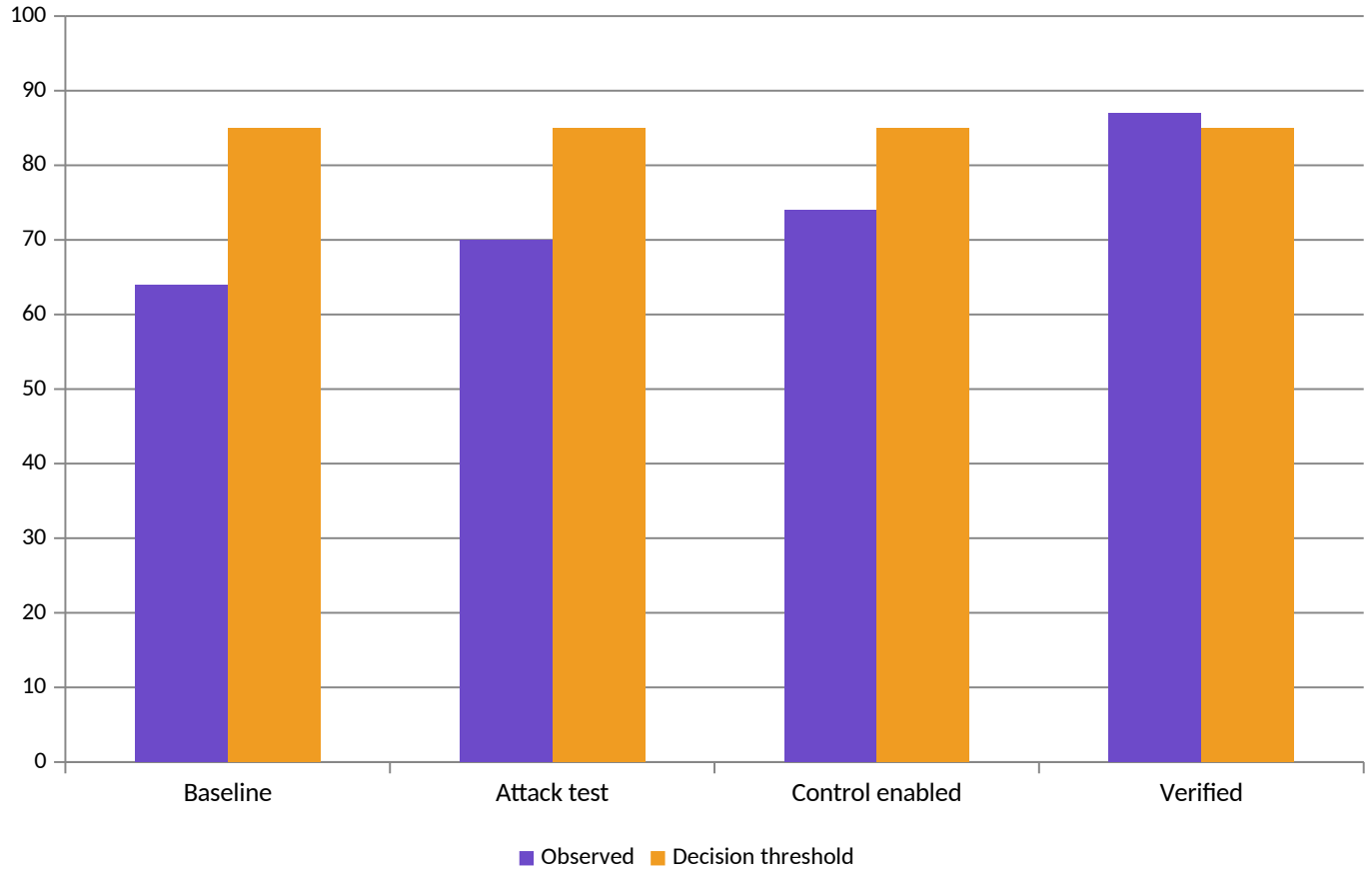
Direct prompt injection: concrete control contract

Field / key	Expected value or type	Enforcement
input_source	user	Required in decision and audit evidence
field_2	injection_score	Required in decision and audit evidence
field_3	target_instruction	Required in decision and audit evidence
field_4	decision	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Direct prompt injection: evidence and acceptance decision



DECISION RULE

Accept only when Injection suite cannot produce an unauthorised tool call. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Indirect prompt injection: architecture and control boundary

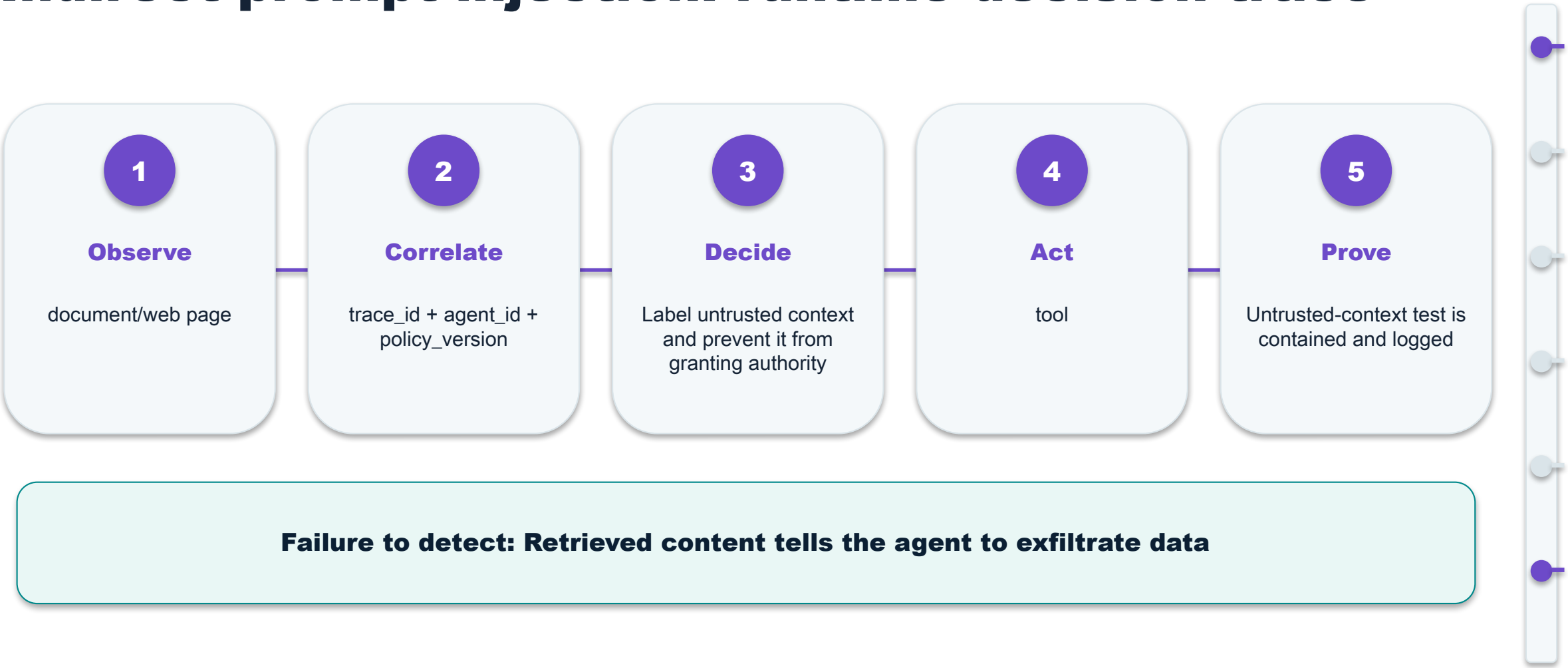


Control objective: Label untrusted context and prevent it from granting authority

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Indirect prompt injection: runtime decision trace



Failure to detect: Retrieved content tells the agent to exfiltrate data

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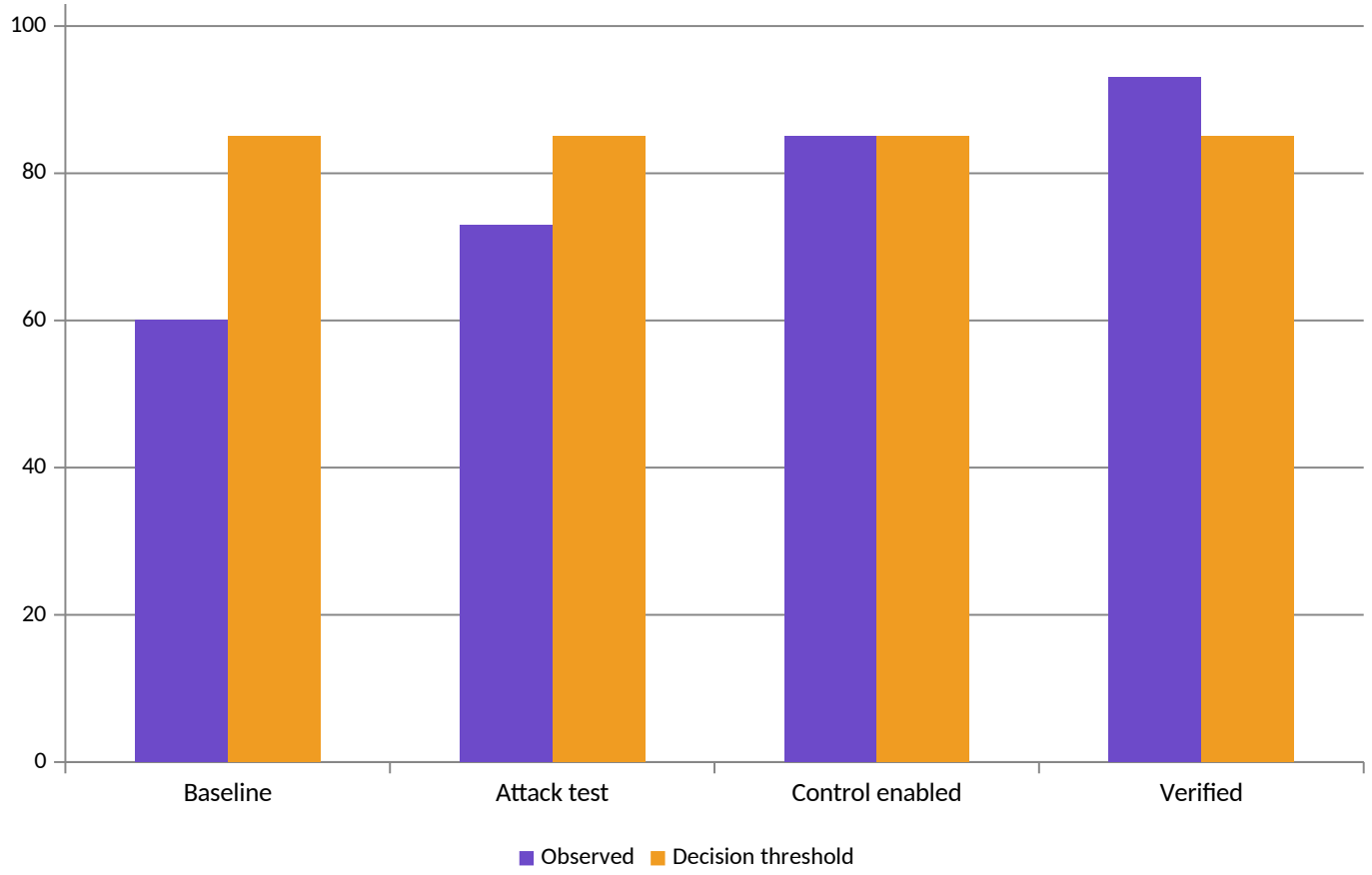
Indirect prompt injection: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	source_uri	Required in decision and audit evidence
field_2	trust_label	Required in decision and audit evidence
field_3	content_hash	Required in decision and audit evidence
field_4	instruction_markers	Required in decision and audit evidence
field_5	action	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Indirect prompt injection: evidence and acceptance decision



DECISION RULE

Accept only when Untrusted-context test is contained and logged. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Tool poisoning: architecture and control boundary

1

Tool Metadata

State and evidence for tool metadata

2

Planner Selection

State and evidence for planner selection

3

Malicious Output

State and evidence for malicious output

4

Next Action

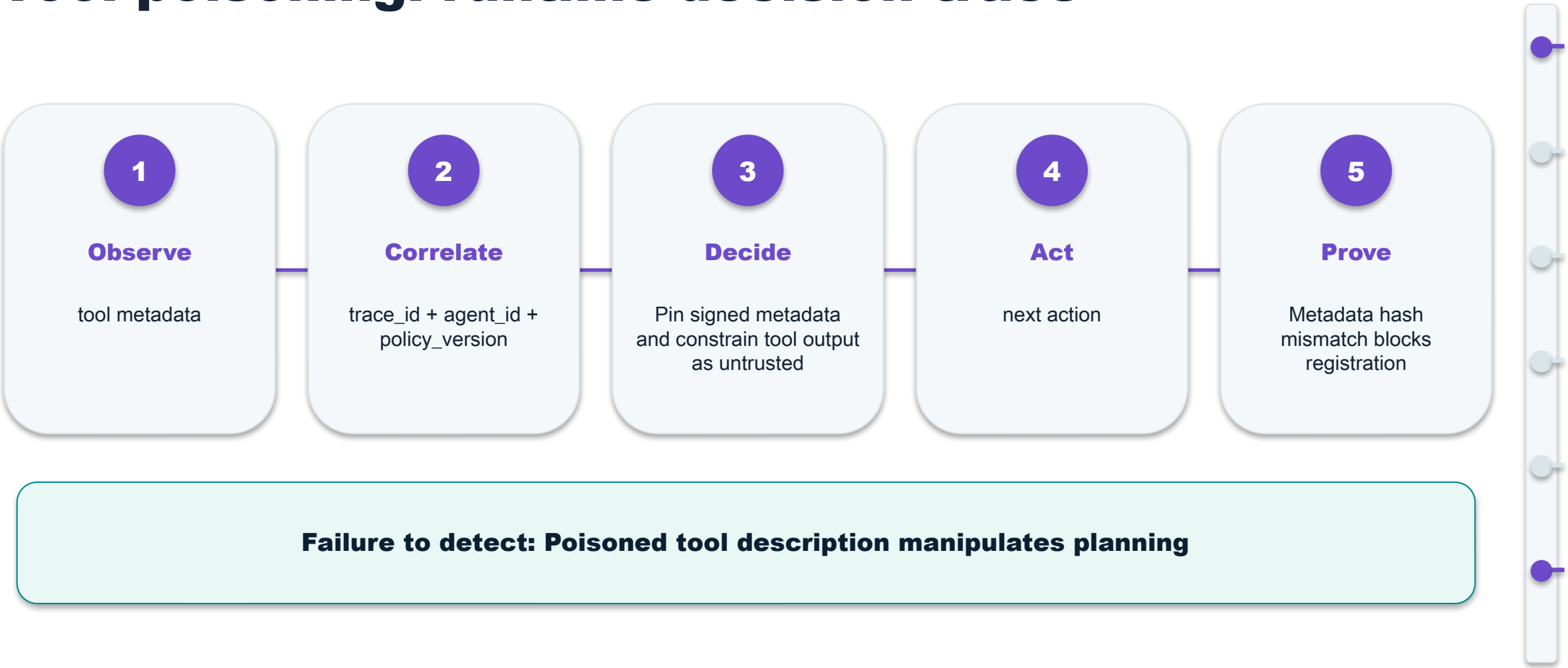
State and evidence for next action

Control objective: Pin signed metadata and constrain tool output as untrusted

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Tool poisoning: runtime decision trace



Failure to detect: Poisoned tool description manipulates planning

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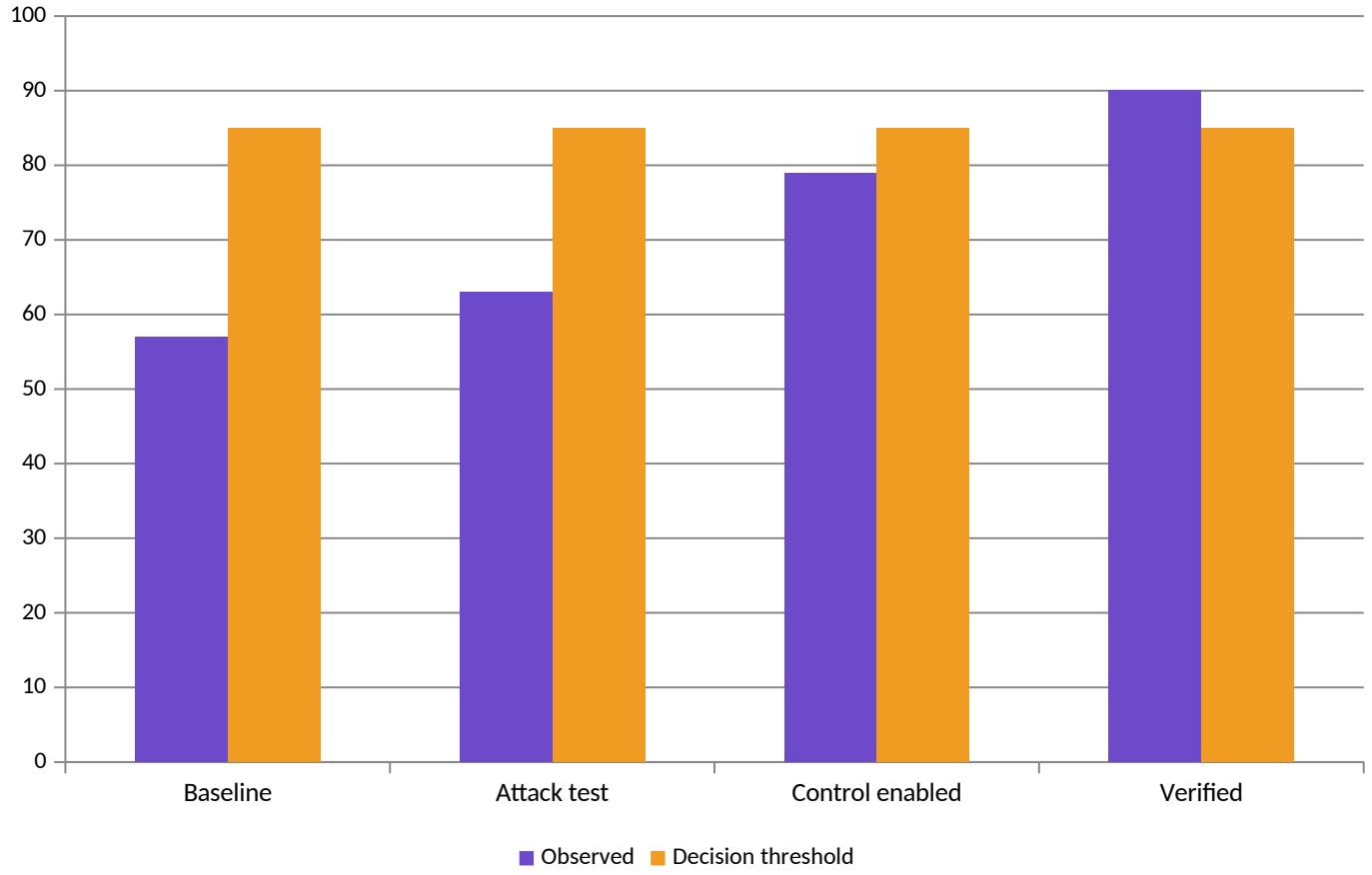
Tool poisoning: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tool_id	Required in decision and audit evidence
field_2	description_hash	Required in decision and audit evidence
field_3	publisher	Required in decision and audit evidence
field_4	output_trust	Required in decision and audit evidence
field_5	version	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Tool poisoning: evidence and acceptance decision



DECISION RULE

Accept only when Metadata hash mismatch blocks registration. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Context poisoning: architecture and control boundary

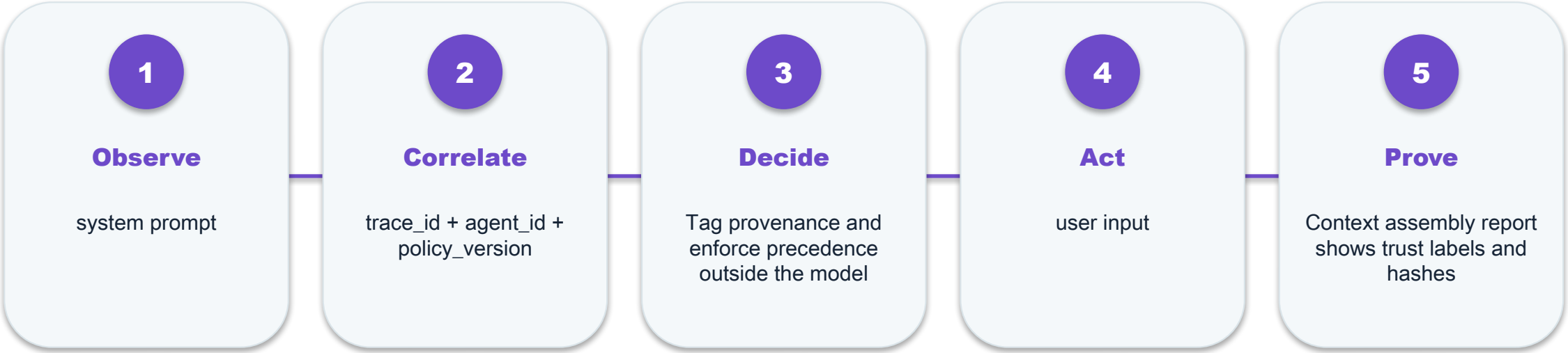


Control objective: Tag provenance and enforce precedence outside the model

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Context poisoning: runtime decision trace



Failure to detect: Low-trust context persists as high-trust instruction

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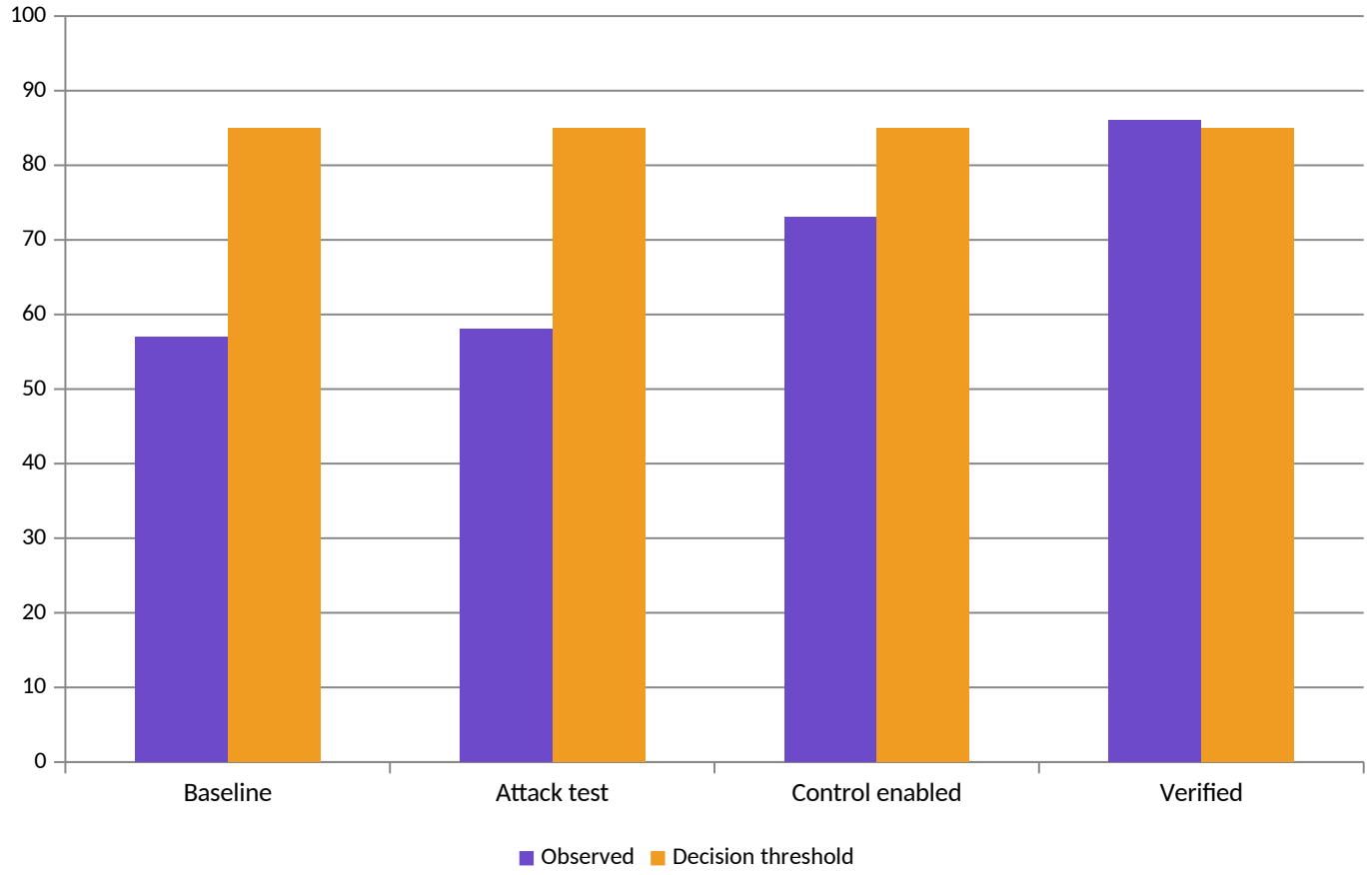
Context poisoning: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	segment_type	Required in decision and audit evidence
field_2	source	Required in decision and audit evidence
field_3	precedence	Required in decision and audit evidence
field_4	integrity_hash	Required in decision and audit evidence
field_5	trust_level	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Context poisoning: evidence and acceptance decision



DECISION RULE

Accept only when Context assembly report shows trust labels and hashes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Memory poisoning: architecture and control boundary

1

Conversation

State and evidence for conversation

2

Memory Write Policy

State and evidence for memory write policy

3

Vector Store

State and evidence for vector store

4

Later Retrieval

State and evidence for later retrieval

Control objective: Validate writes, isolate tenants and expire low-confidence memory

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Memory poisoning: runtime decision trace



Failure to detect: Adversarial fact persists and steers future sessions

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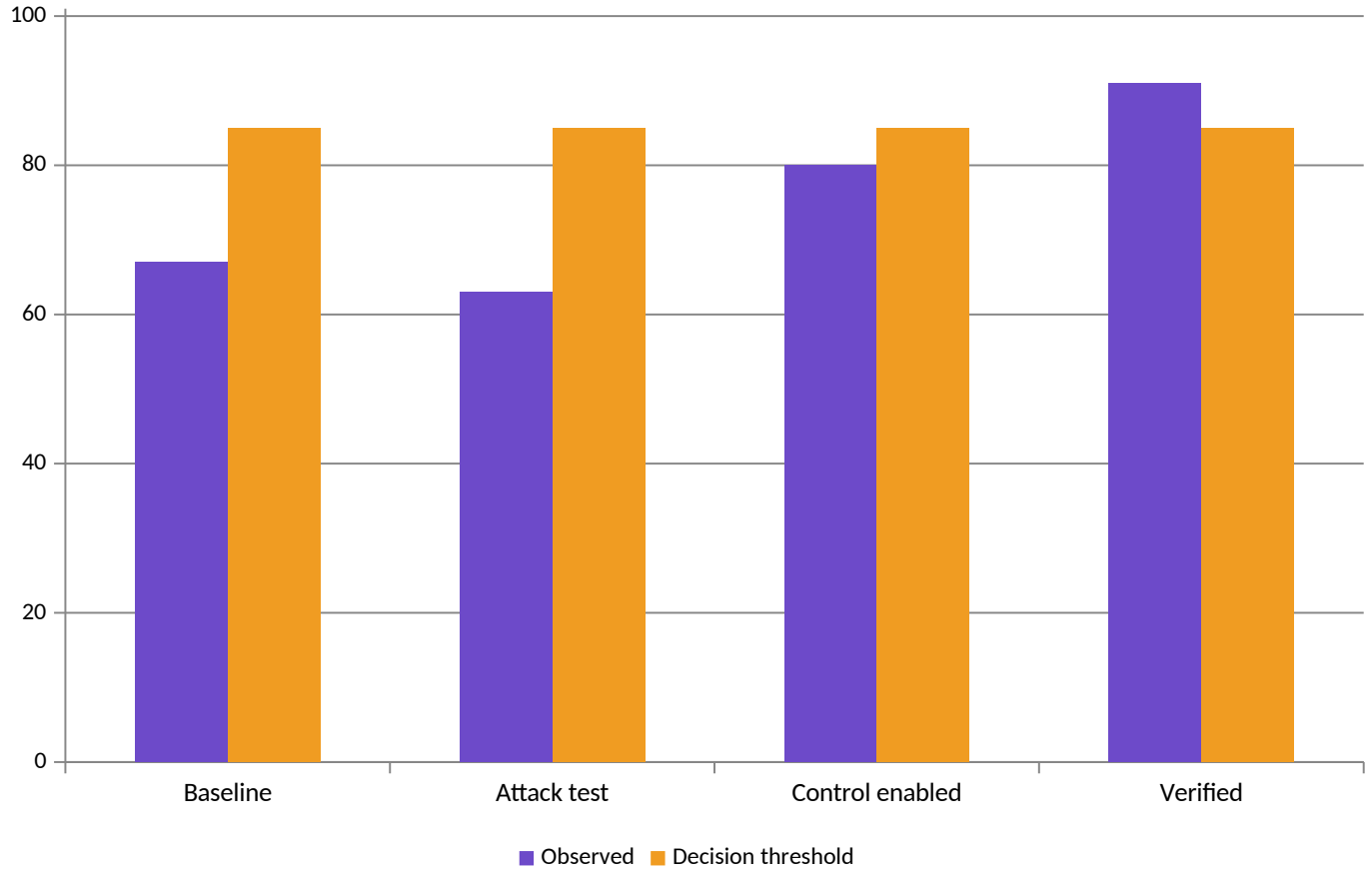
Memory poisoning: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	memory_id	Required in decision and audit evidence
field_2	author	Required in decision and audit evidence
field_3	tenant	Required in decision and audit evidence
field_4	confidence	Required in decision and audit evidence
field_5	ttl	Required in decision and audit evidence
field_6	approved_for_reuse	Required in decision and audit evidence

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Memory poisoning: evidence and acceptance decision



DECISION RULE

Accept only when Poison seed is absent from unrelated session retrieval. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Data exfiltration: architecture and control boundary

1

Sensitive Store

State and evidence for sensitive store

2

Agent Context

State and evidence for agent context

3

Transformation

State and evidence for transformation

4

Outbound Tool

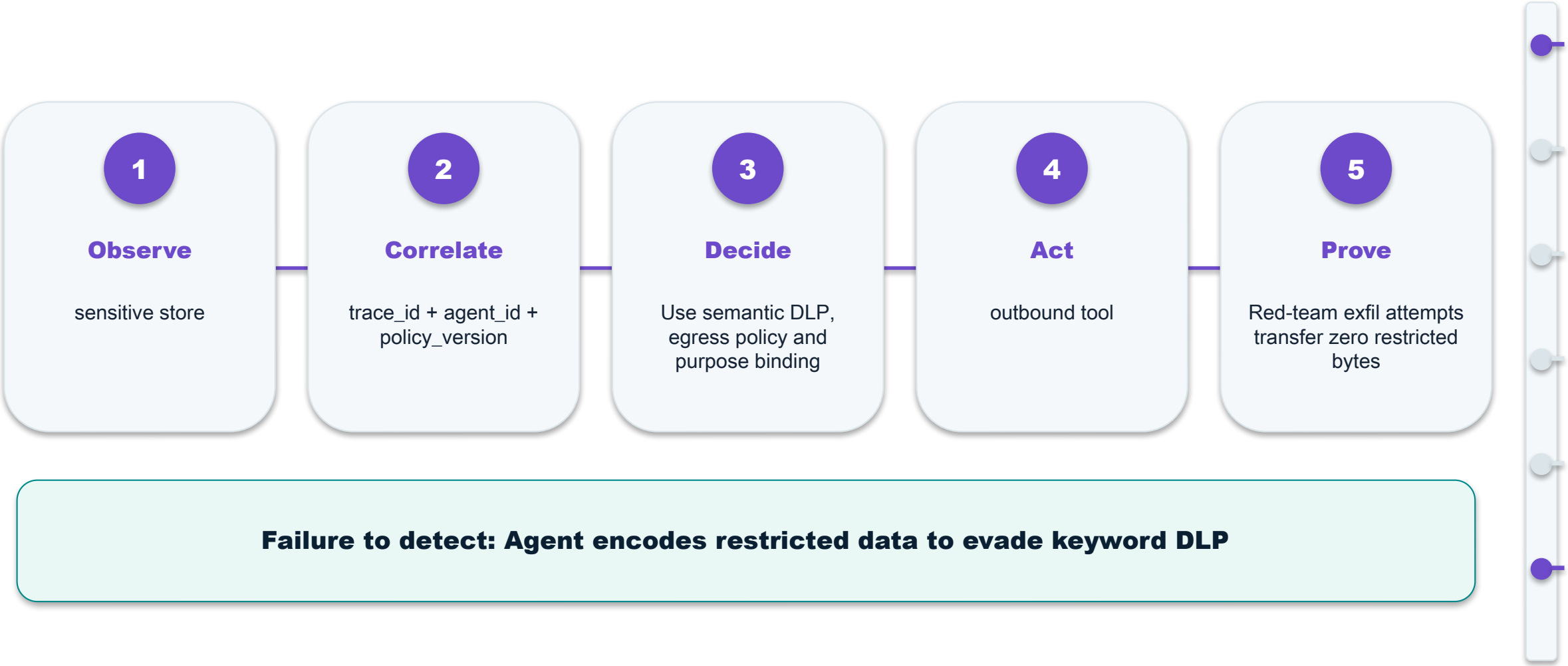
State and evidence for outbound tool

Control objective: Use semantic DLP, egress policy and purpose binding

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Data exfiltration: runtime decision trace



Supplied AI-security reference corpus · MITRE ATLAS

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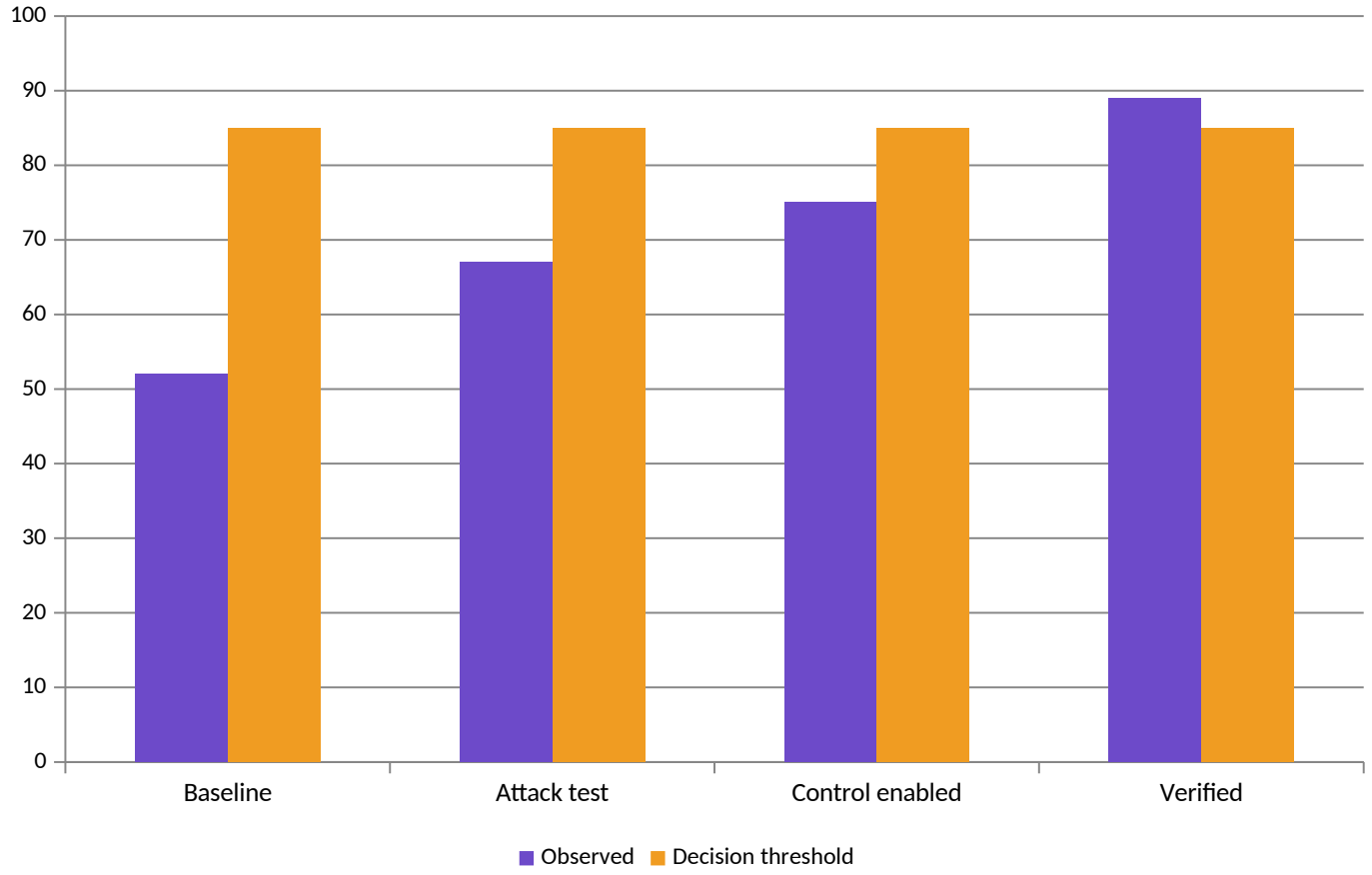
Data exfiltration: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	data_class	Required in decision and audit evidence
field_2	bytes_read	Required in decision and audit evidence
field_3	destination	Required in decision and audit evidence
field_4	bytes_out	Required in decision and audit evidence
field_5	leak_score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

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Data exfiltration: evidence and acceptance decision



DECISION RULE

Accept only when Red-team exfil attempts transfer zero restricted bytes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Credential harvesting: architecture and control boundary

1

Prompt/Tool Output

State and evidence for prompt/tool output

2

Context

State and evidence for context

3

Secret Lookup

State and evidence for secret lookup

4

Outbound Channel

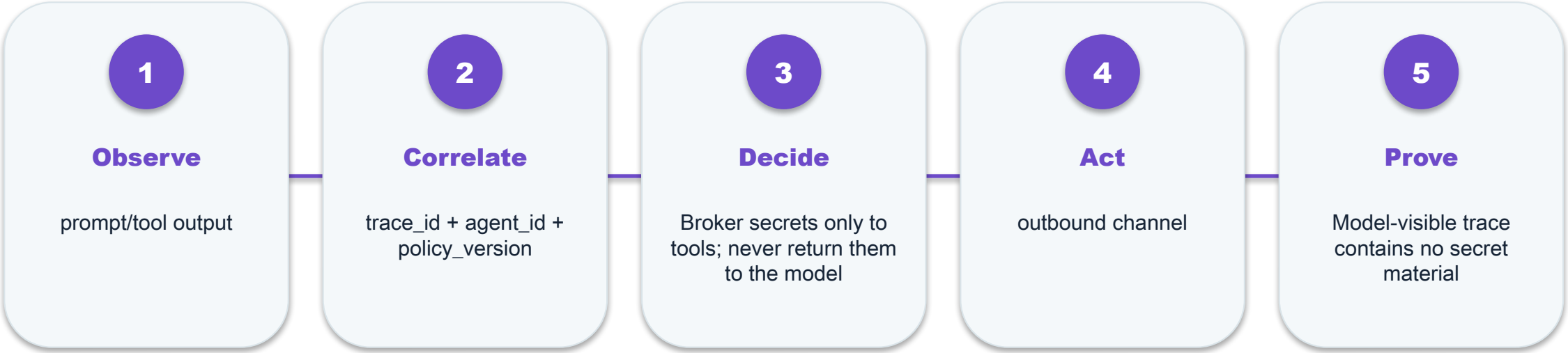
State and evidence for outbound channel

Control objective: Broker secrets only to tools; never return them to the model

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Credential harvesting: runtime decision trace



Failure to detect: Agent is tricked into retrieving and revealing a token

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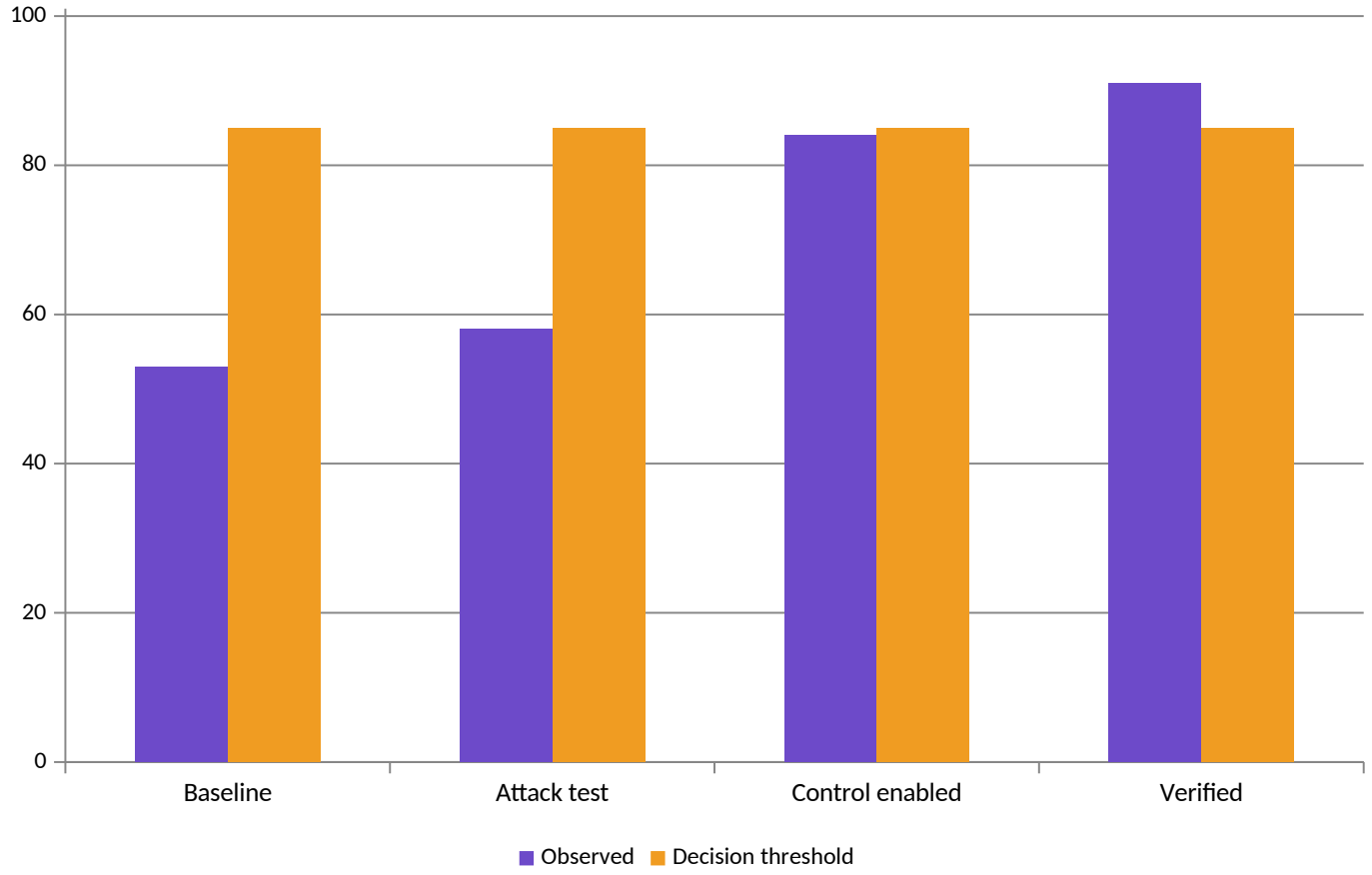
Credential harvesting: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	secret_ref	Required in decision and audit evidence
field_2	access_reason	Required in decision and audit evidence
field_3	requester	Required in decision and audit evidence
field_4	destination	Required in decision and audit evidence
field_5	decision	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Credential harvesting: evidence and acceptance decision



DECISION RULE

Accept only when Model-visible trace contains no secret material. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Excessive agency: architecture and control boundary

1

Broad Goal

State and evidence for broad goal

2

Planner

State and evidence for planner

3

Long Action Chain

State and evidence for long action chain

4

Irreversible Side Effect

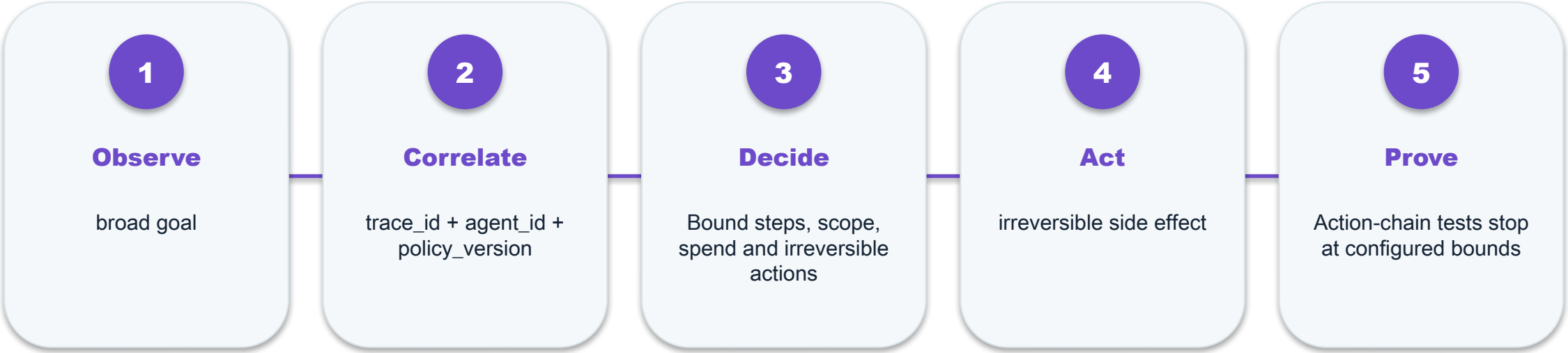
State and evidence for irreversible side effect

Control objective: Bound steps, scope, spend and irreversible actions

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Excessive agency: runtime decision trace



Failure to detect: Agent performs destructive actions beyond user intent

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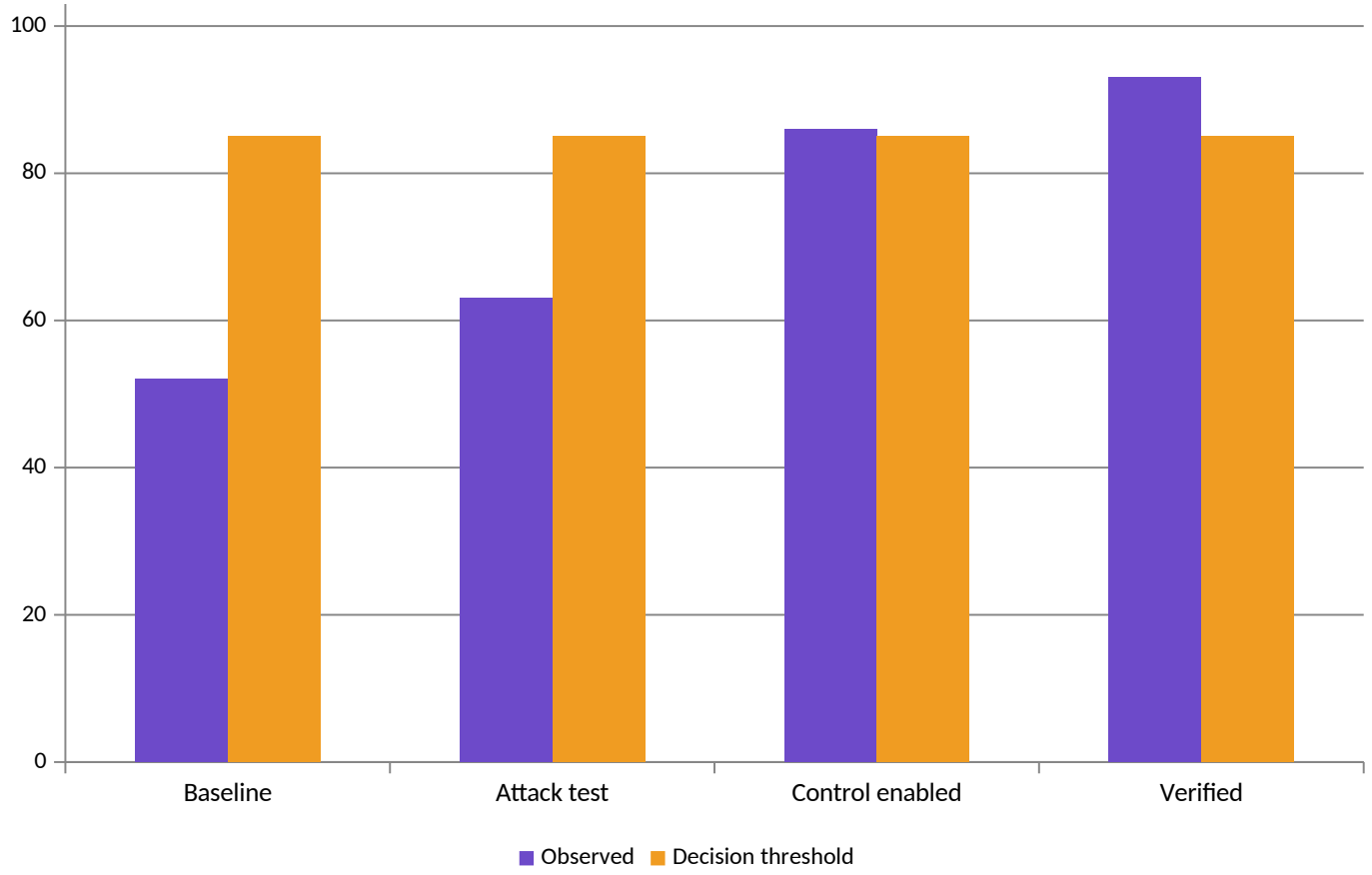
Excessive agency: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	max_steps	Required in decision and audit evidence
field_2	allowed_tools	Required in decision and audit evidence
field_3	spend_limit	Required in decision and audit evidence
field_4	approval_tier	Required in decision and audit evidence
field_5	stop_reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Excessive agency: evidence and acceptance decision



DECISION RULE

Accept only when Action-chain tests stop at configured bounds. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Privilege escalation: architecture and control boundary

1

Low-Privilege Agent

State and evidence for low-privilege agent

2

Mis-Scoped Role

State and evidence for mis-scoped role

3

Admin Api

State and evidence for admin API

4

State Change

State and evidence for state change

Control objective: Eliminate wildcard scopes and validate resource ownership

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Privilege escalation: runtime decision trace



Failure to detect: Wildcard permission makes admin action reachable

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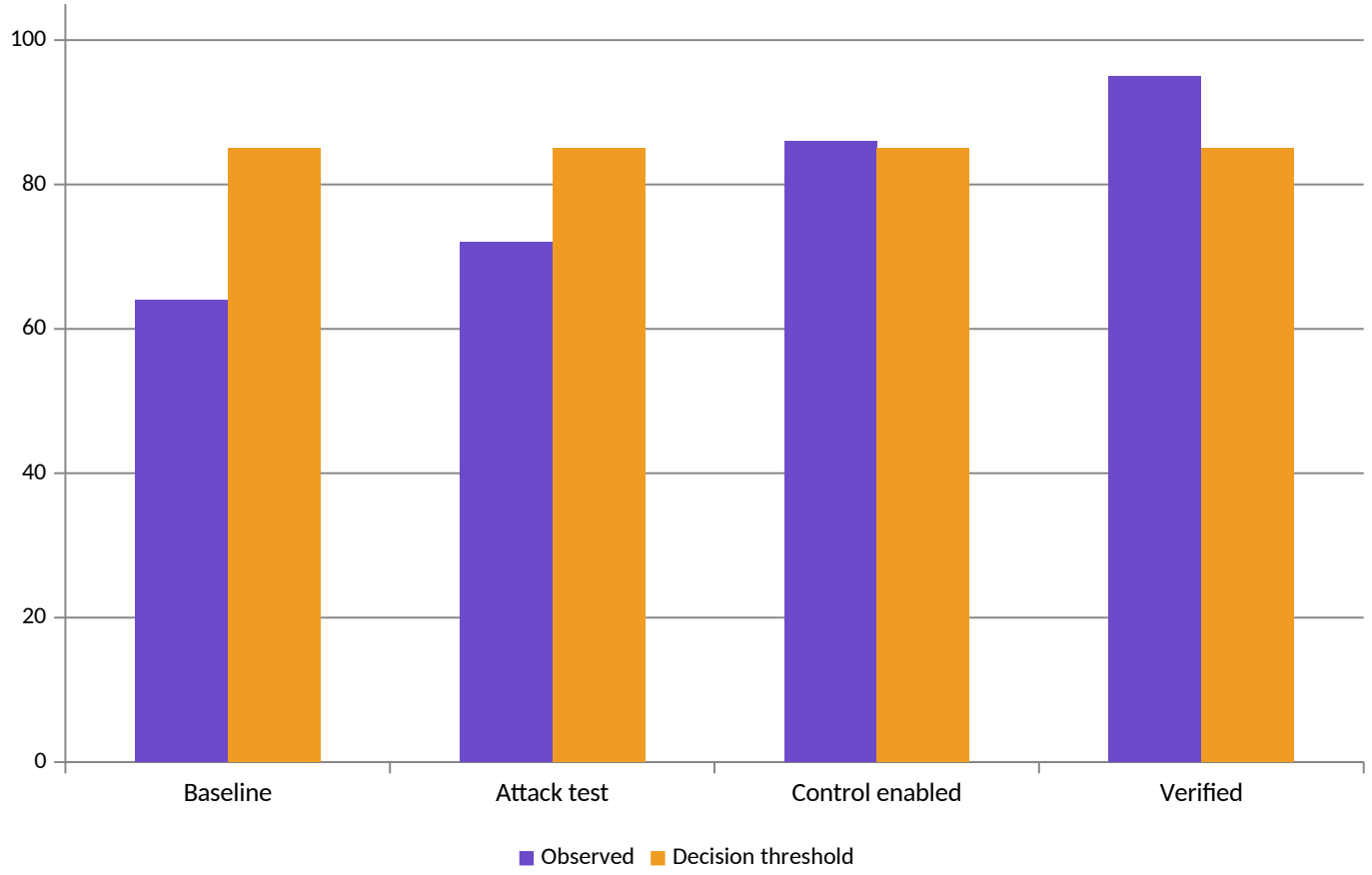
Privilege escalation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	principal	Required in decision and audit evidence
field_2	role	Required in decision and audit evidence
field_3	scope	Required in decision and audit evidence
field_4	resource	Required in decision and audit evidence
field_5	requested_action	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

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Privilege escalation: evidence and acceptance decision



DECISION RULE

Accept only when Privilege-path graph has no unintended route. Otherwise record a gap, owner and due date.

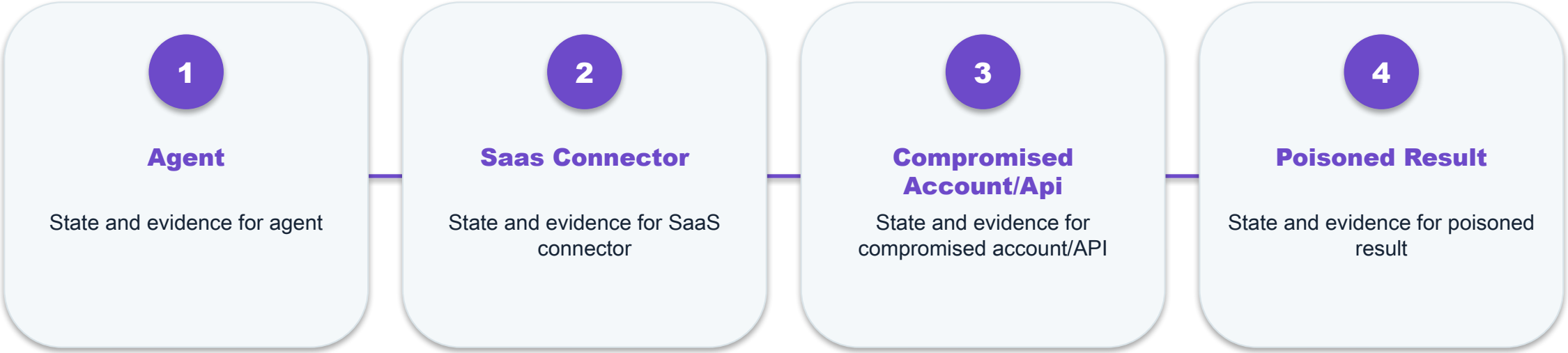
EVIDENCE STATUS

PASS

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Compromised integration: architecture and control boundary



Control objective: Treat external output as untrusted and monitor connector identity

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Compromised integration: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
agent	trace_id + agent_id + policy_version	Treat external output as untrusted and monitor connector identity	poisoned result	Connector anomaly triggers quarantine

Failure to detect: Trusted integration returns malicious instructions or data

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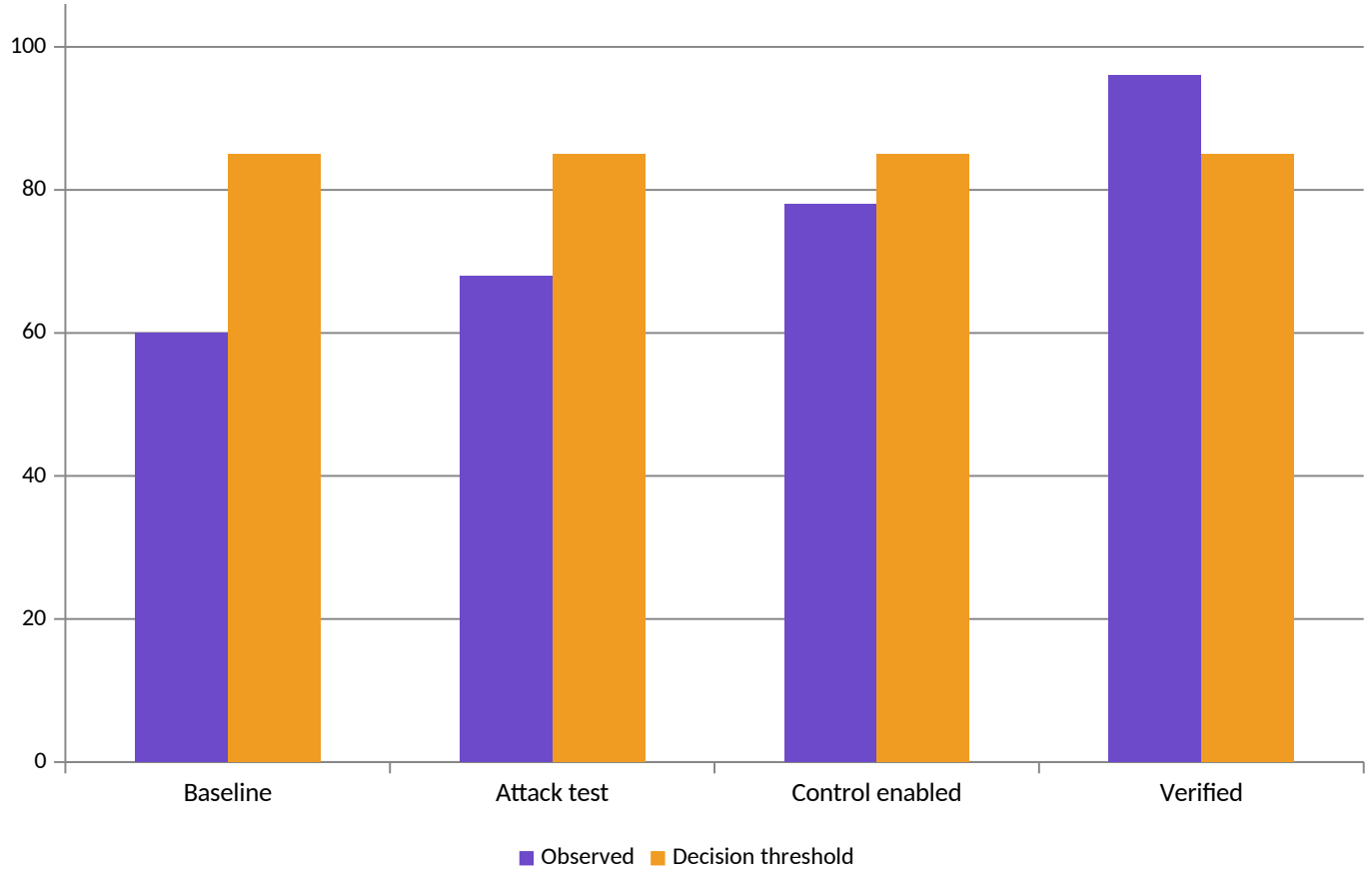
Compromised integration: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	connector_id	Required in decision and audit evidence
field_2	auth_subject	Required in decision and audit evidence
field_3	endpoint	Required in decision and audit evidence
field_4	tls_peer	Required in decision and audit evidence
field_5	response_hash	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Compromised integration: evidence and acceptance decision



DECISION RULE

Accept only when Connector anomaly triggers quarantine. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Supply-chain compromise: architecture and control boundary

1

Library/Model/Tool Image

State and evidence for library/model/tool image

2

Build

State and evidence for build

3

Registry

State and evidence for registry

4

Runtime

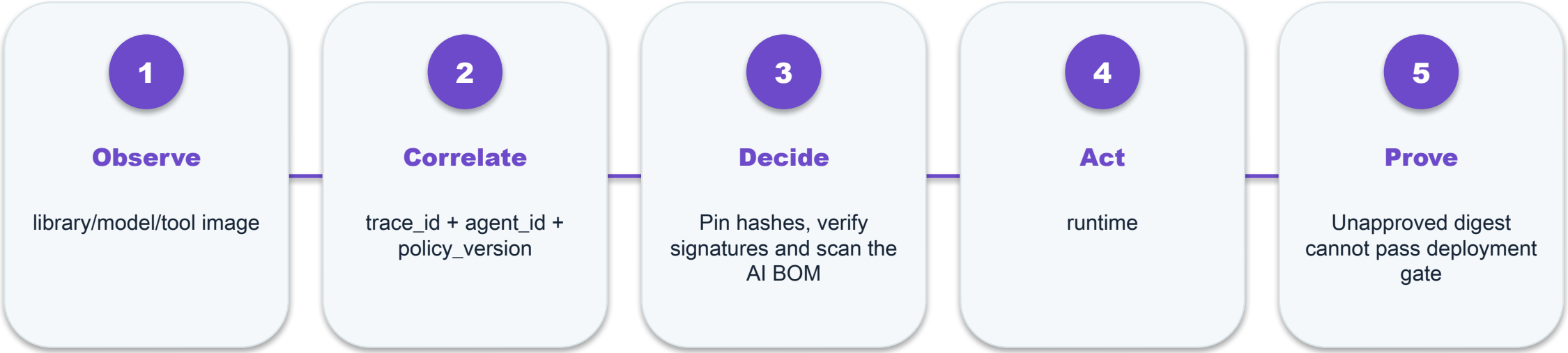
State and evidence for runtime

Control objective: Pin hashes, verify signatures and scan the AI BOM

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Supply-chain compromise: runtime decision trace



Failure to detect: Typosquatted dependency executes inside the agent runtime

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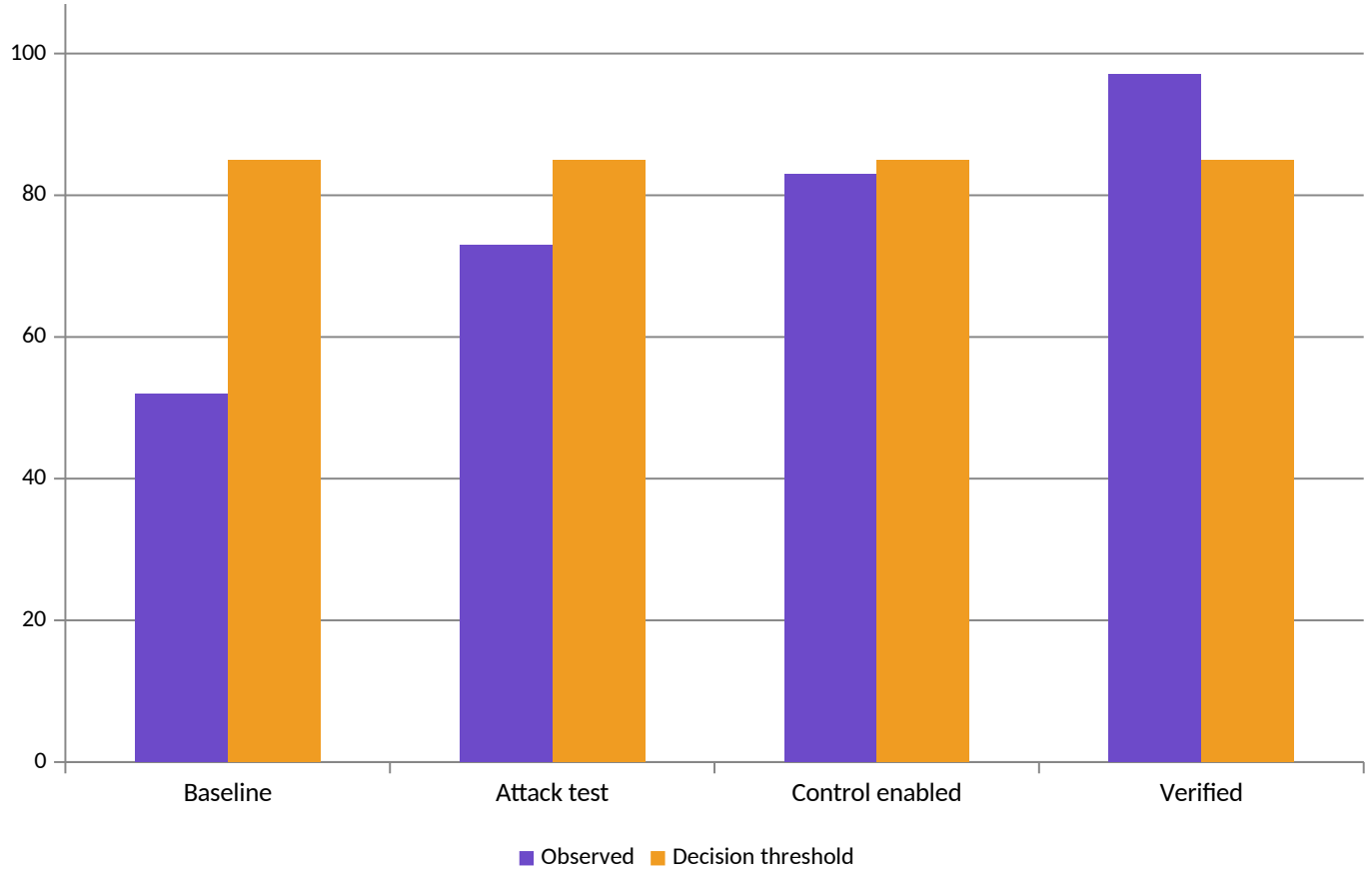
Supply-chain compromise: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	package	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	digest	Required in decision and audit evidence
field_4	signer	Required in decision and audit evidence
field_5	SBOM_ref	Required in decision and audit evidence
field_6	vuln_status	Required in decision and audit evidence

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Supply-chain compromise: evidence and acceptance decision



DECISION RULE

Accept only when Unapproved digest cannot pass deployment gate. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Unbounded consumption: architecture and control boundary

1

Request

State and evidence for request

2

Recursive Planning

State and evidence for recursive planning

3

Model/Tool Calls

State and evidence for model/tool calls

4

Billing

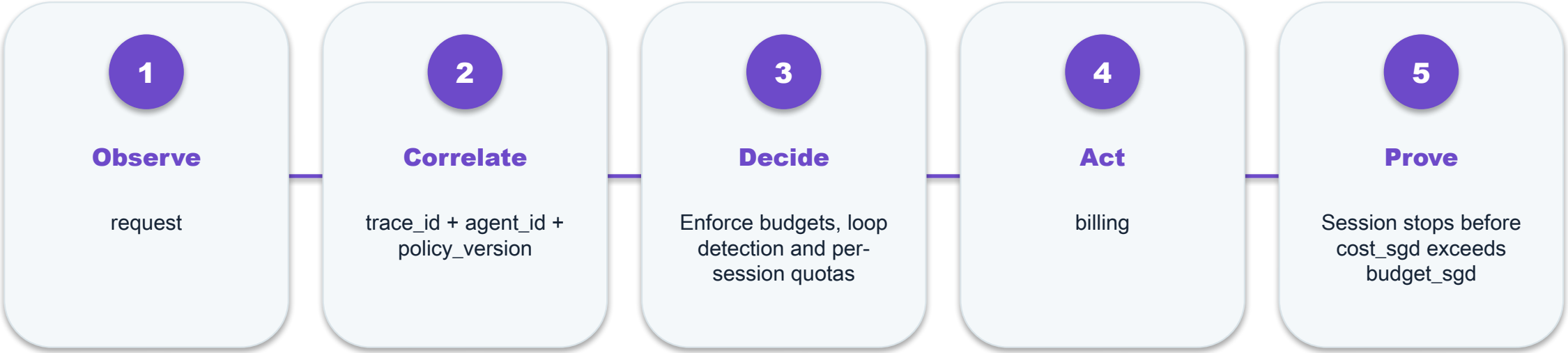
State and evidence for billing

Control objective: Enforce budgets, loop detection and per-session quotas

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Unbounded consumption: runtime decision trace



Failure to detect: Loop consumes quota and creates unexpected cost

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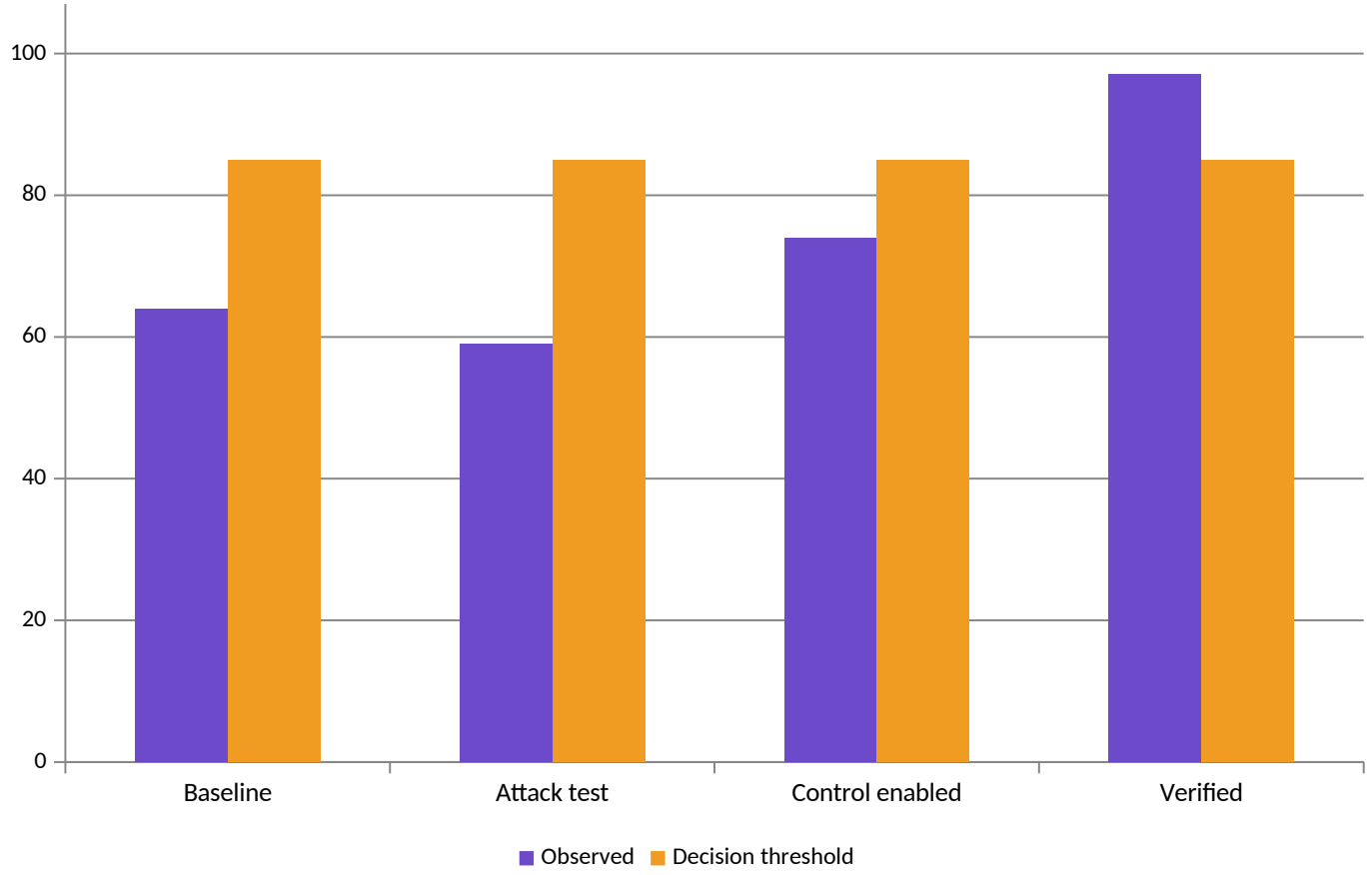
Unbounded consumption: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	call_count	Required in decision and audit evidence
field_2	token_count	Required in decision and audit evidence
field_3	elapsed_ms	Required in decision and audit evidence
field_4	cost_sgd	Required in decision and audit evidence
field_5	budget_sgd	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Unbounded consumption: evidence and acceptance decision



DECISION RULE

Accept only when Session stops before cost_sgd exceeds budget_sgd. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Service denial: architecture and control boundary

1

Traffic Source

State and evidence for traffic source

2

Queue

State and evidence for queue

3

Agent Worker

State and evidence for agent worker

4

Model/Tool Dependencies

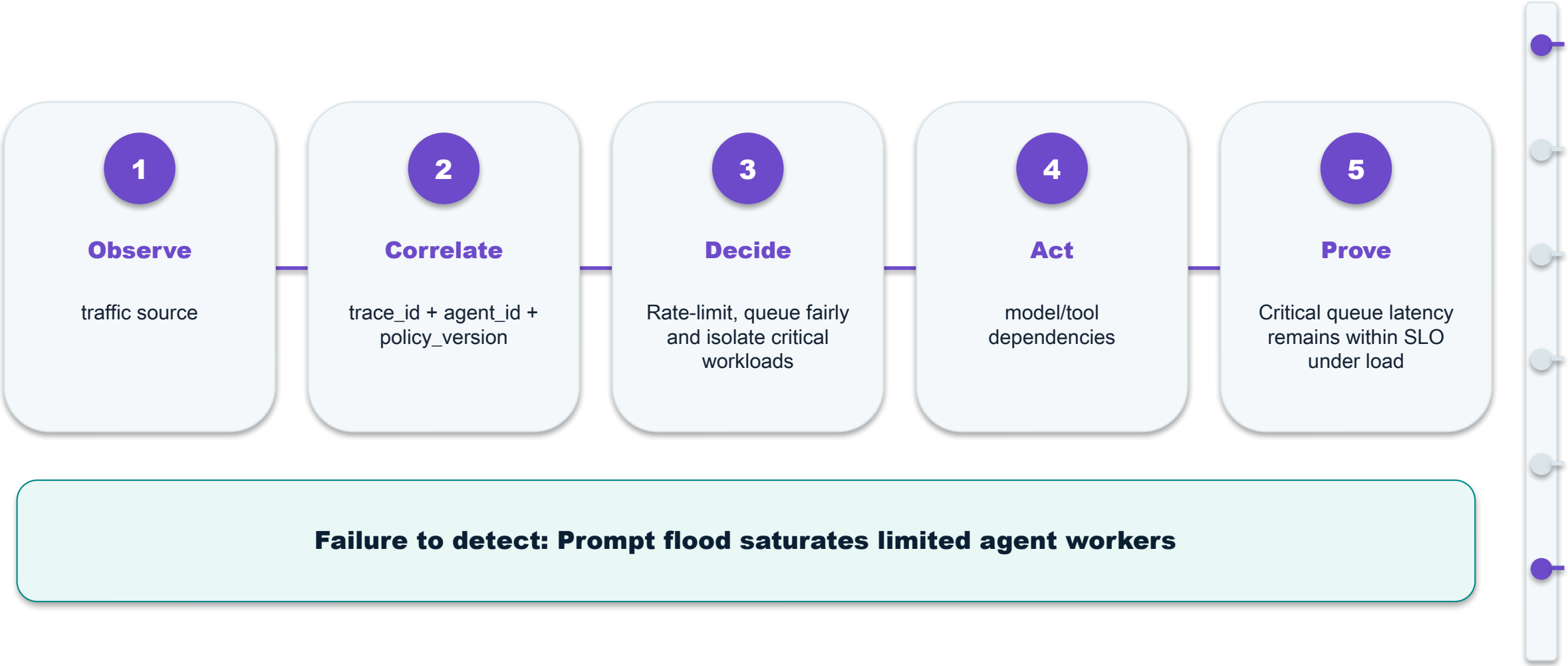
State and evidence for model/tool dependencies

Control objective: Rate-limit, queue fairly and isolate critical workloads

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Service denial: runtime decision trace



Failure to detect: Prompt flood saturates limited agent workers

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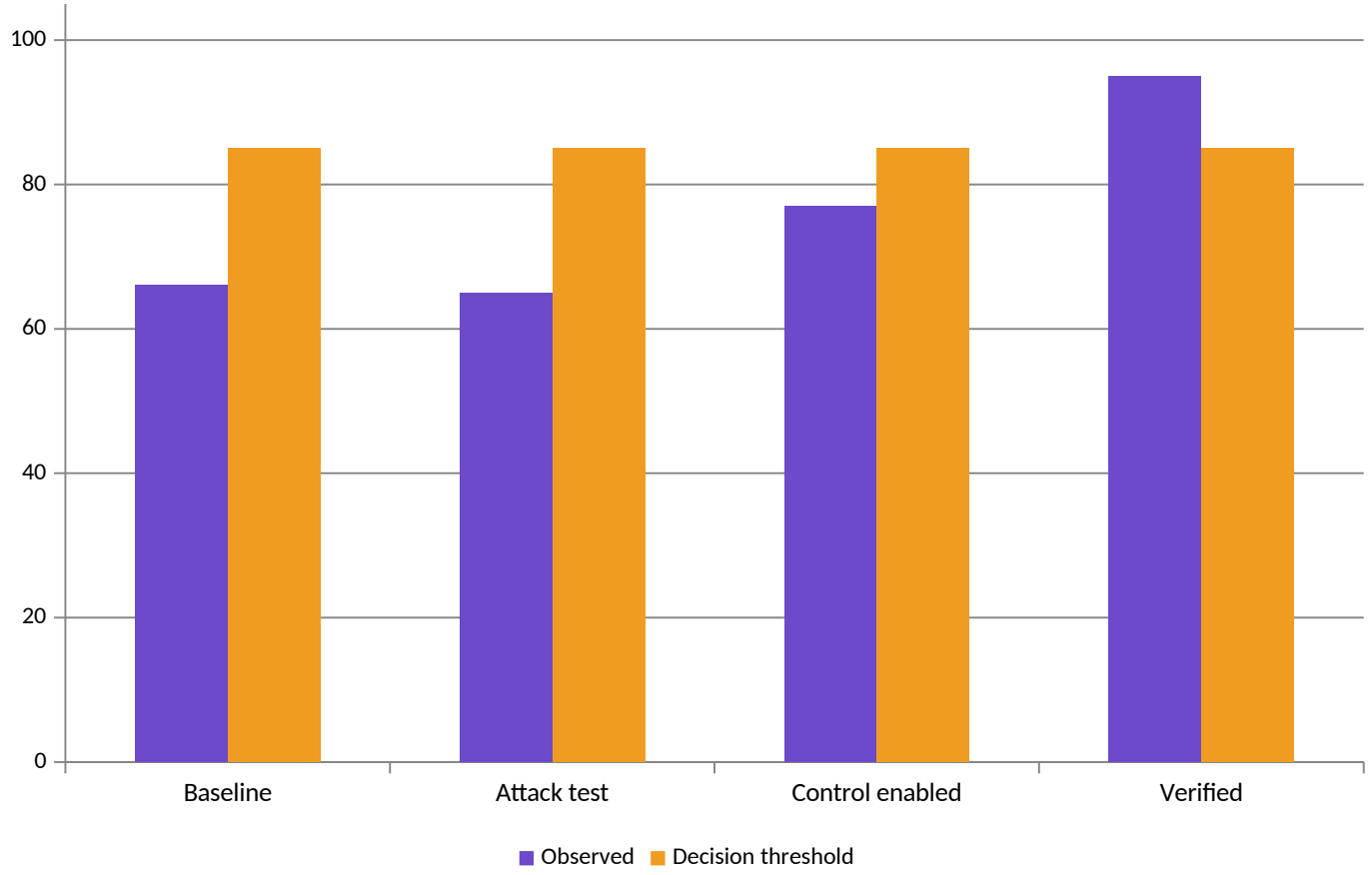
Service denial: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	request_rate	Required in decision and audit evidence
field_2	queue_depth	Required in decision and audit evidence
field_3	timeout_rate	Required in decision and audit evidence
field_4	dependency_health	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Service denial: evidence and acceptance decision



DECISION RULE

Accept only when Critical queue latency remains within SLO under load. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Hallucinated action: architecture and control boundary

1

Ambiguous Request

State and evidence for ambiguous request

2

Fabricated Resource

State and evidence for fabricated resource

3

Tool Call

State and evidence for tool call

4

Failure/Impact

State and evidence for failure/impact

Control objective: Resolve and validate identifiers against authoritative data

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Hallucinated action: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
ambiguous request	trace_id + agent_id + policy_version	Resolve and validate identifiers against authoritative data	failure/impact	Unknown resource_id is denied before execution

Failure to detect: Agent invents an account and changes the wrong record

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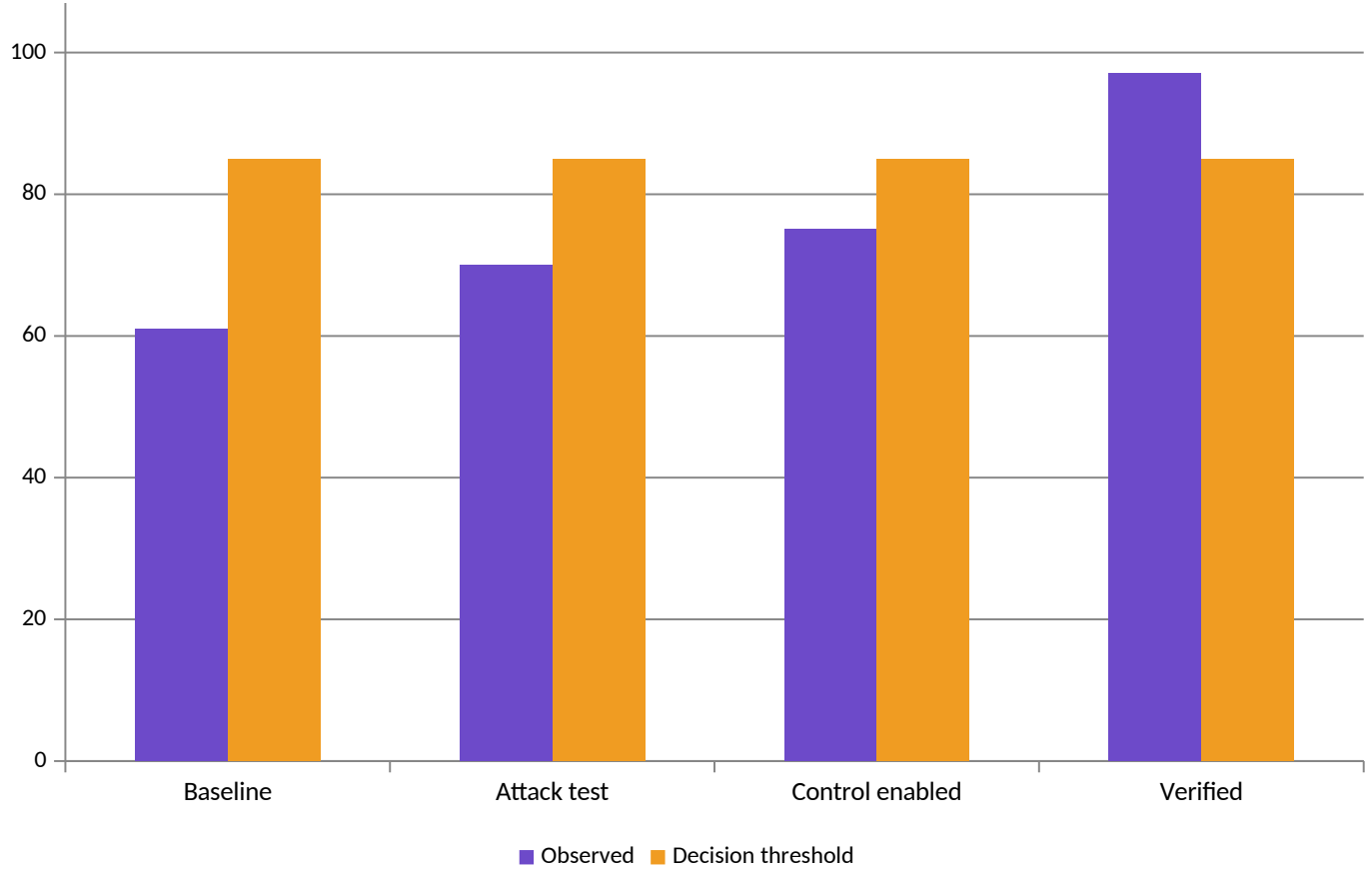
Hallucinated action: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	resource_id	Required in decision and audit evidence
field_2	source_of_truth	Required in decision and audit evidence
field_3	confidence	Required in decision and audit evidence
field_4	validation_result	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Hallucinated action: evidence and acceptance decision



DECISION RULE

Accept only when Unknown resource_id is denied before execution. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Insecure output handling: architecture and control boundary

1

Model Output

State and evidence for model output

2

Parser

State and evidence for parser

3

Downstream Interpreter/Browser/Sql

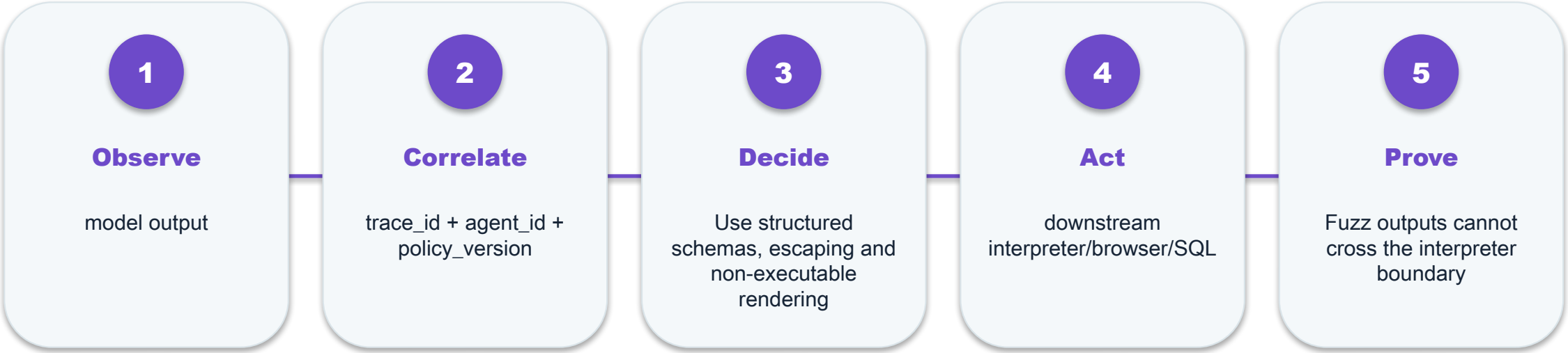
State and evidence for downstream interpreter/browser/SQL

Control objective: Use structured schemas, escaping and non-executable rendering

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Insecure output handling: runtime decision trace



Failure to detect: Generated command or markup executes as code

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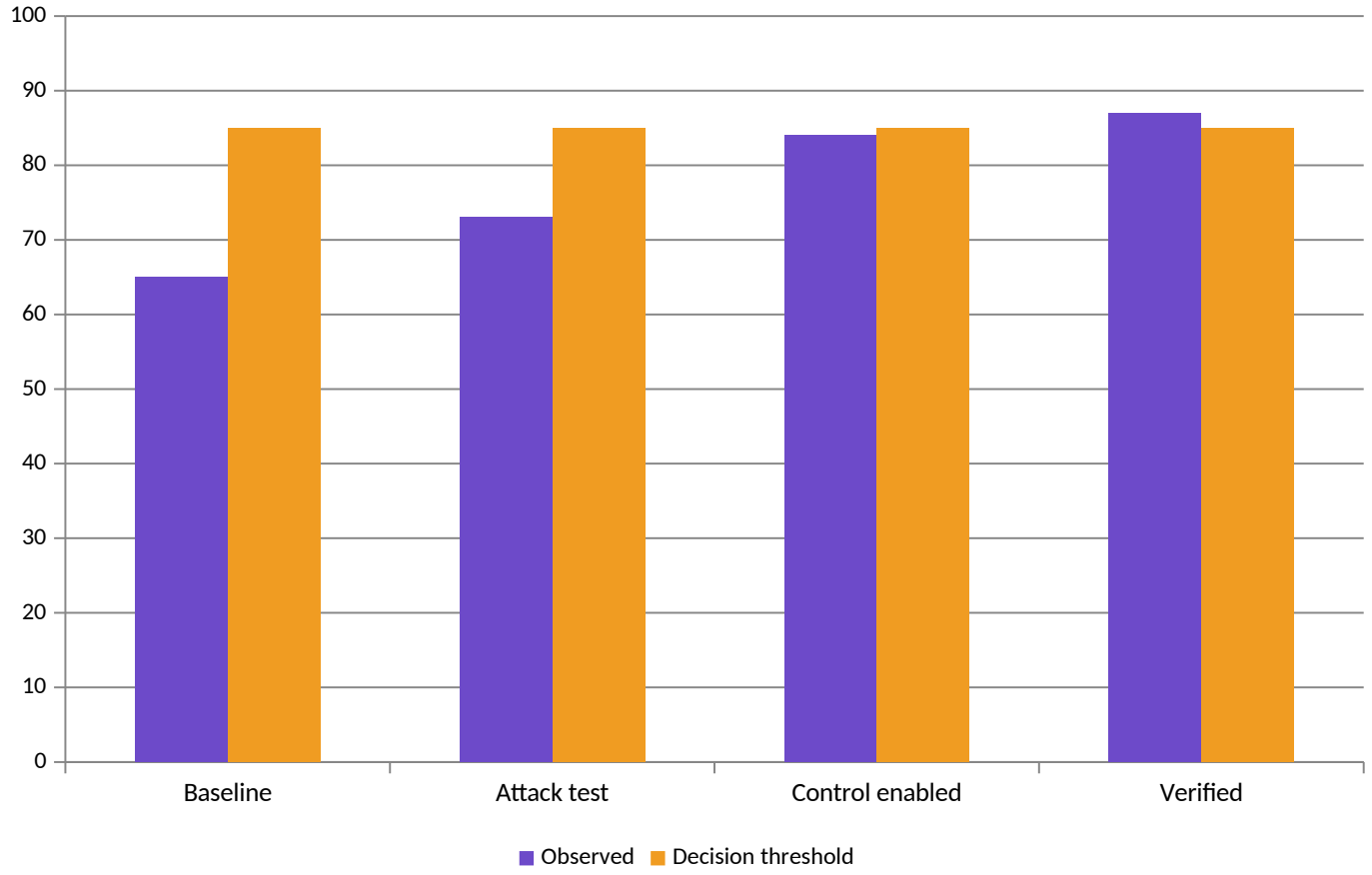
Insecure output handling: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	output_type	Required in decision and audit evidence
field_2	schema_valid	Required in decision and audit evidence
field_3	encoding	Required in decision and audit evidence
field_4	destination	Required in decision and audit evidence
field_5	sanitised	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Insecure output handling: evidence and acceptance decision



DECISION RULE

Accept only when Fuzz outputs cannot cross the interpreter boundary. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Sandbox escape: architecture and control boundary

1

Generated Code

State and evidence for generated code

2

Container Syscall

State and evidence for container syscall

3

Host Kernel

State and evidence for host kernel

4

Adjacent Workload

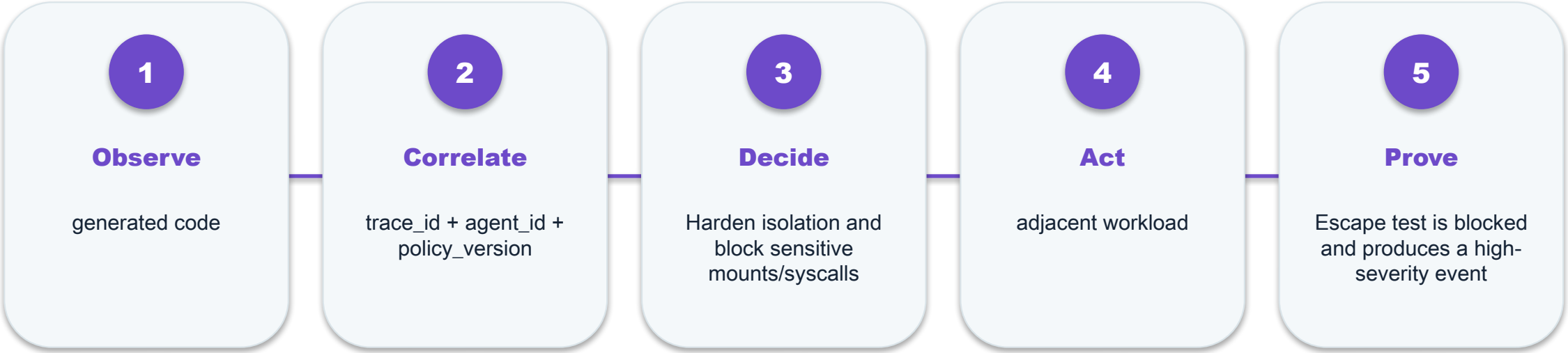
State and evidence for adjacent workload

Control objective: Harden isolation and block sensitive mounts/syscalls

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Sandbox escape: runtime decision trace



Failure to detect: Agent workload accesses host socket or filesystem

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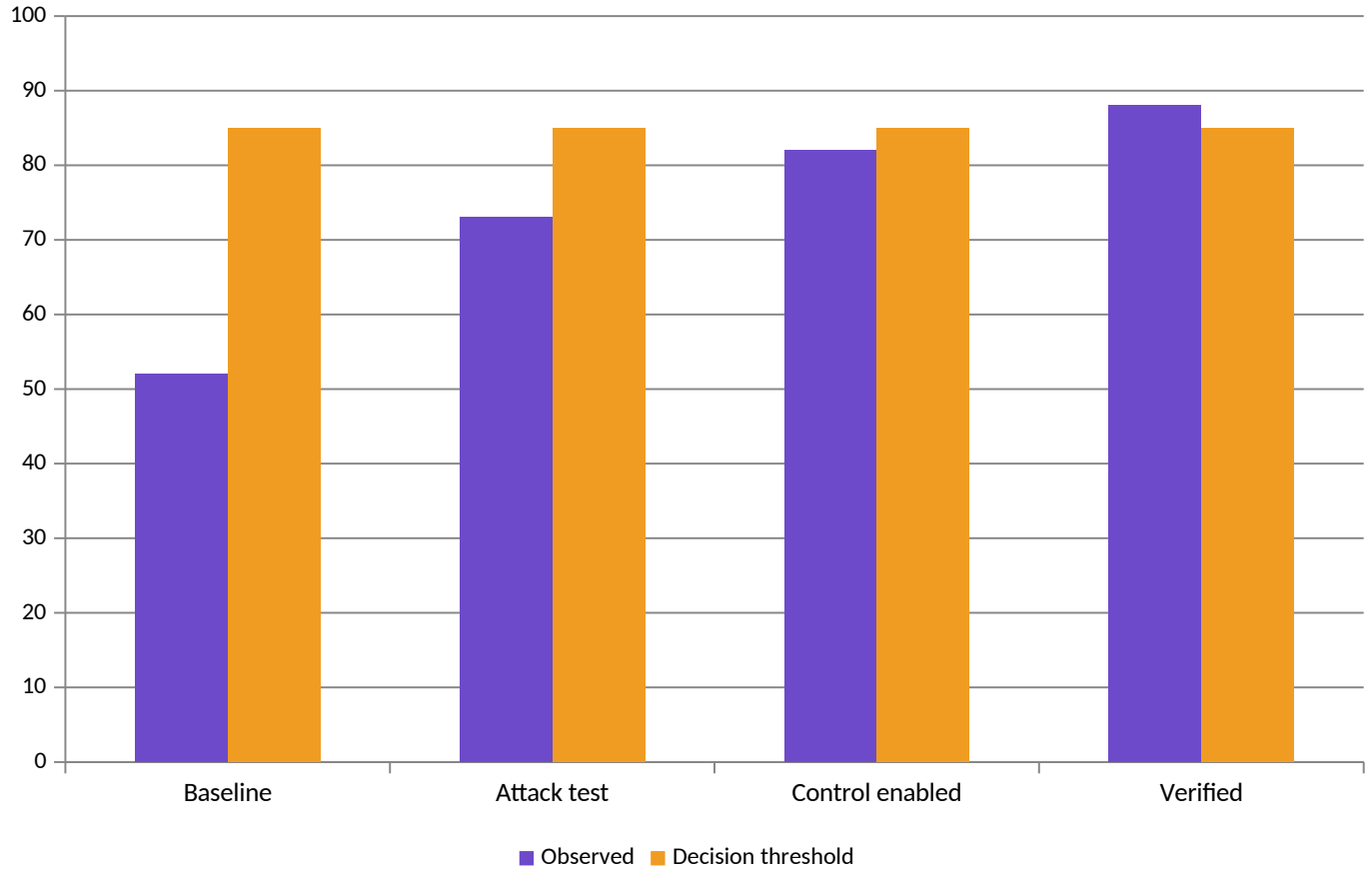
Sandbox escape: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	syscall	Required in decision and audit evidence
field_2	seccomp_action	Required in decision and audit evidence
field_3	uid	Required in decision and audit evidence
field_4	mount	Required in decision and audit evidence
field_5	network_namespace	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Sandbox escape: evidence and acceptance decision



DECISION RULE

Accept only when Escape test is blocked and produces a high-severity event. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Human override failure: architecture and control boundary

1. **Agent Proposal**
   State and evidence for agent proposal
2. **Reviewer Ui**
   State and evidence for reviewer UI
3. **Context**
   State and evidence for context
4. **Decision**
   State and evidence for decision
5. **Execution**
   State and evidence for execution

Control objective: Show exact diff, impact and evidence; support reject and edit

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Human override failure: runtime decision trace



Failure to detect: Reviewer rubber-stamps an opaque or misleading proposal

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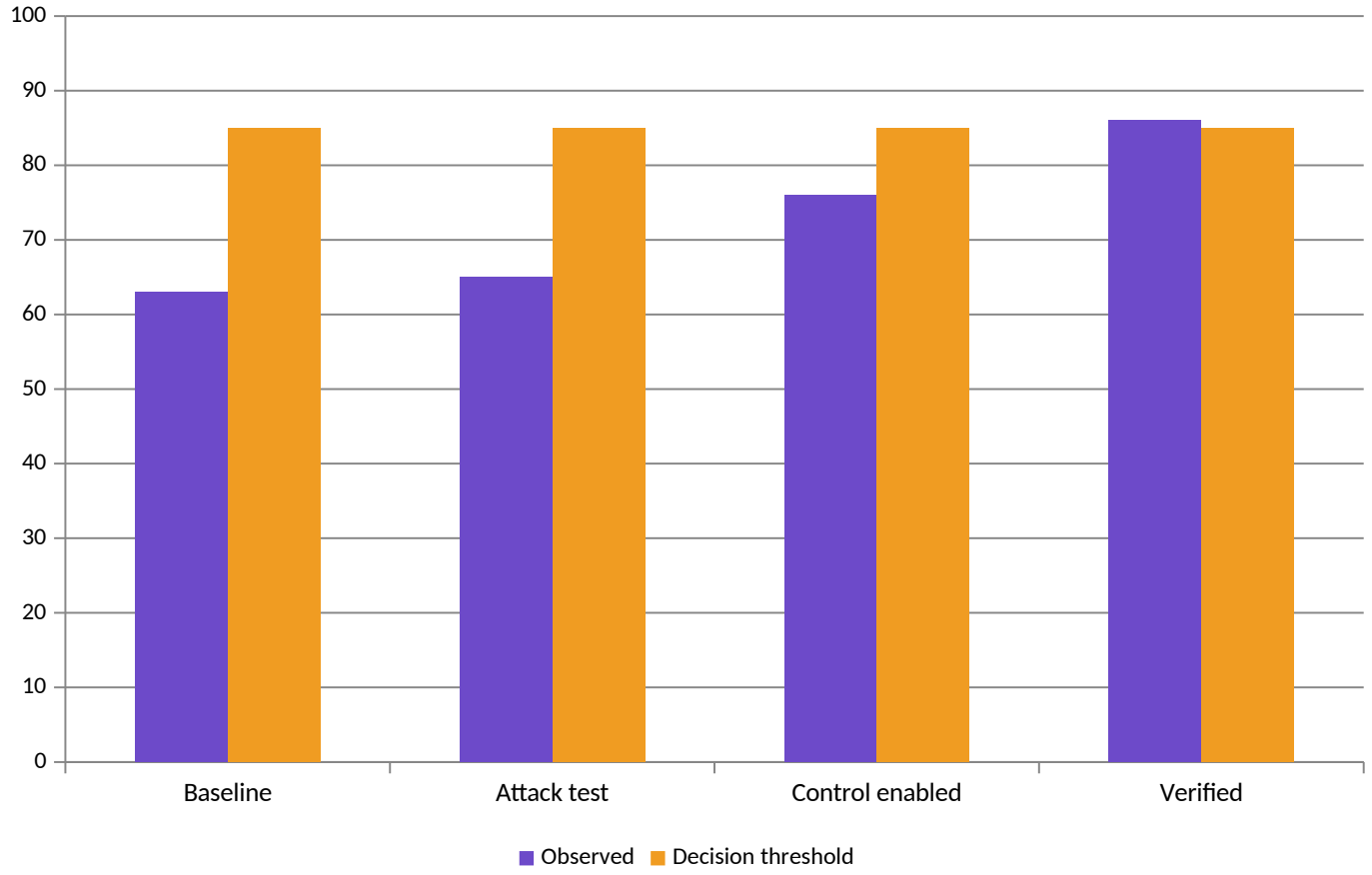
Human override failure: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	approval_id	Required in decision and audit evidence
field_2	action_summary	Required in decision and audit evidence
field_3	risk	Required in decision and audit evidence
field_4	evidence	Required in decision and audit evidence
field_5	reviewer	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

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Human override failure: evidence and acceptance decision



DECISION RULE

Accept only when Review sampling finds complete decision context. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Likelihood-impact scoring: architecture and control boundary

1

Scenario Evidence

State and evidence for scenario evidence

2

Likelihood

State and evidence for likelihood

3

Impact Dimensions

State and evidence for impact dimensions

4

Inherent Score

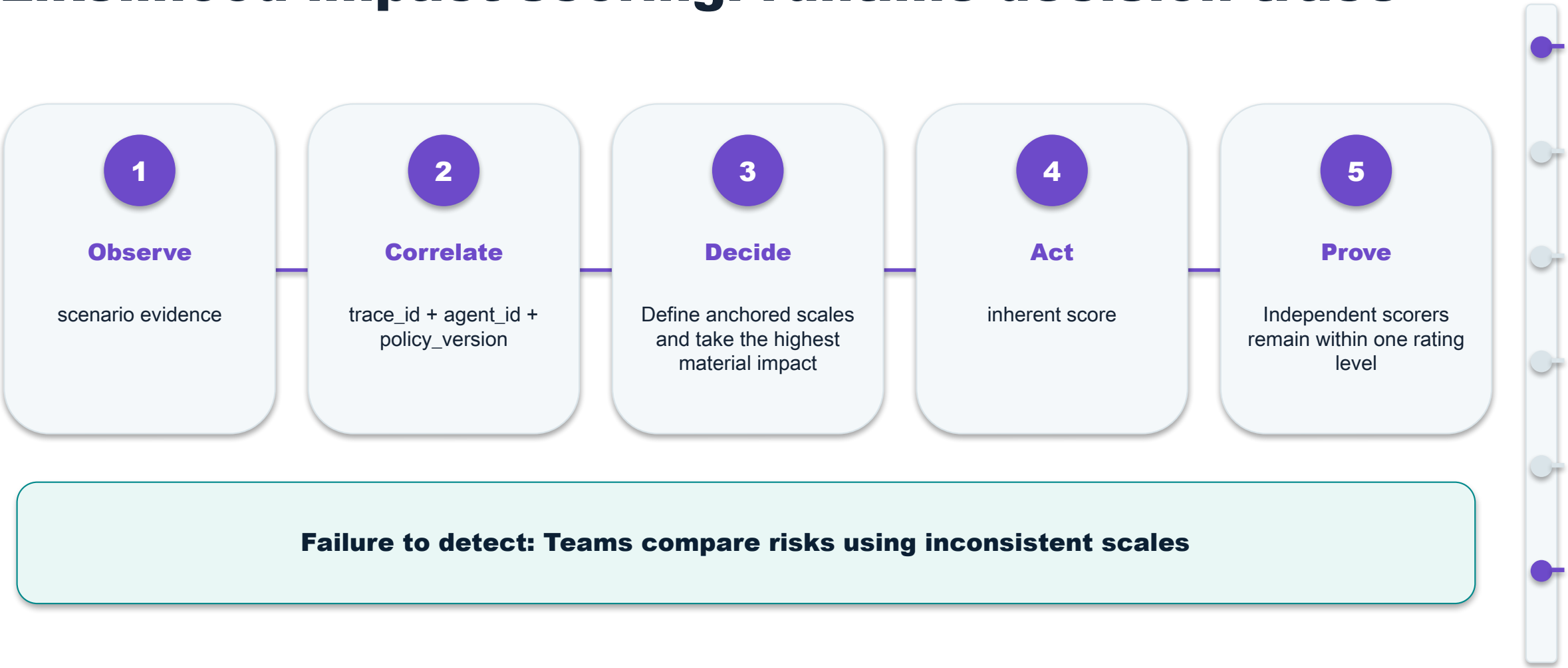
State and evidence for inherent score

Control objective: Define anchored scales and take the highest material impact

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Likelihood-impact scoring: runtime decision trace



Failure to detect: Teams compare risks using inconsistent scales

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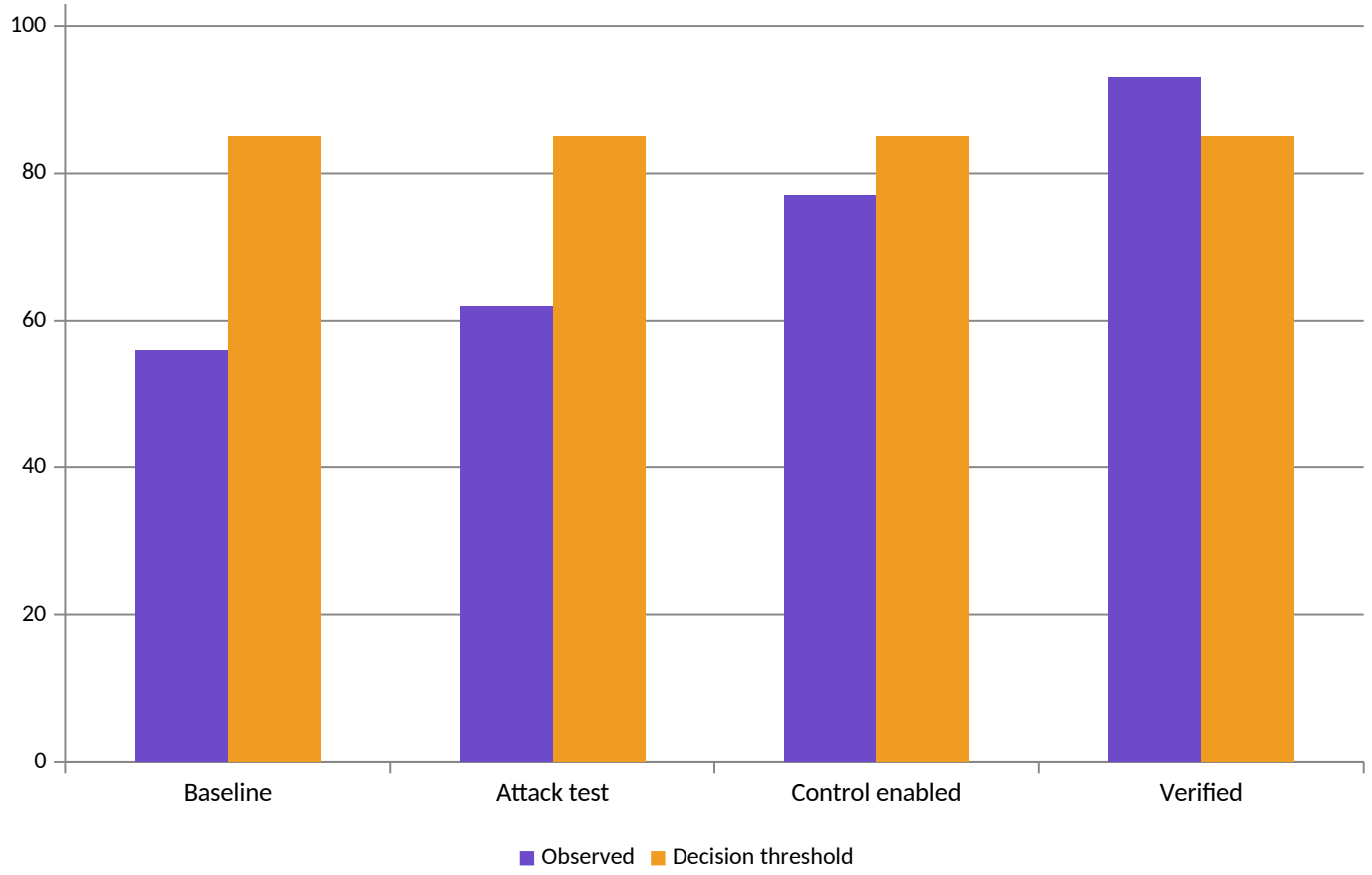
Likelihood-impact scoring: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	likelihood_1_5	Required in decision and audit evidence
field_2	financial	Required in decision and audit evidence
field_3	operational	Required in decision and audit evidence
field_4	legal	Required in decision and audit evidence
field_5	safety	Required in decision and audit evidence
field_6	score	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

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Likelihood-impact scoring: evidence and acceptance decision



DECISION RULE

Accept only when Independent scorers remain within one rating level. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Expected-loss estimate: architecture and control boundary

1

Event Frequency × Loss Magnitude

State and evidence for event frequency × loss magnitude

2

Annualised Loss

State and evidence for annualised loss

Control objective: Use transparent ranges and sensitivity analysis

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Expected-loss estimate: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
event frequency × loss magnitude	trace_id + agent_id + policy_version	Use transparent ranges and sensitivity analysis	annualised loss	expected_loss = frequency × loss; assumptions are documented

Failure to detect: Control spend is justified only by qualitative fear

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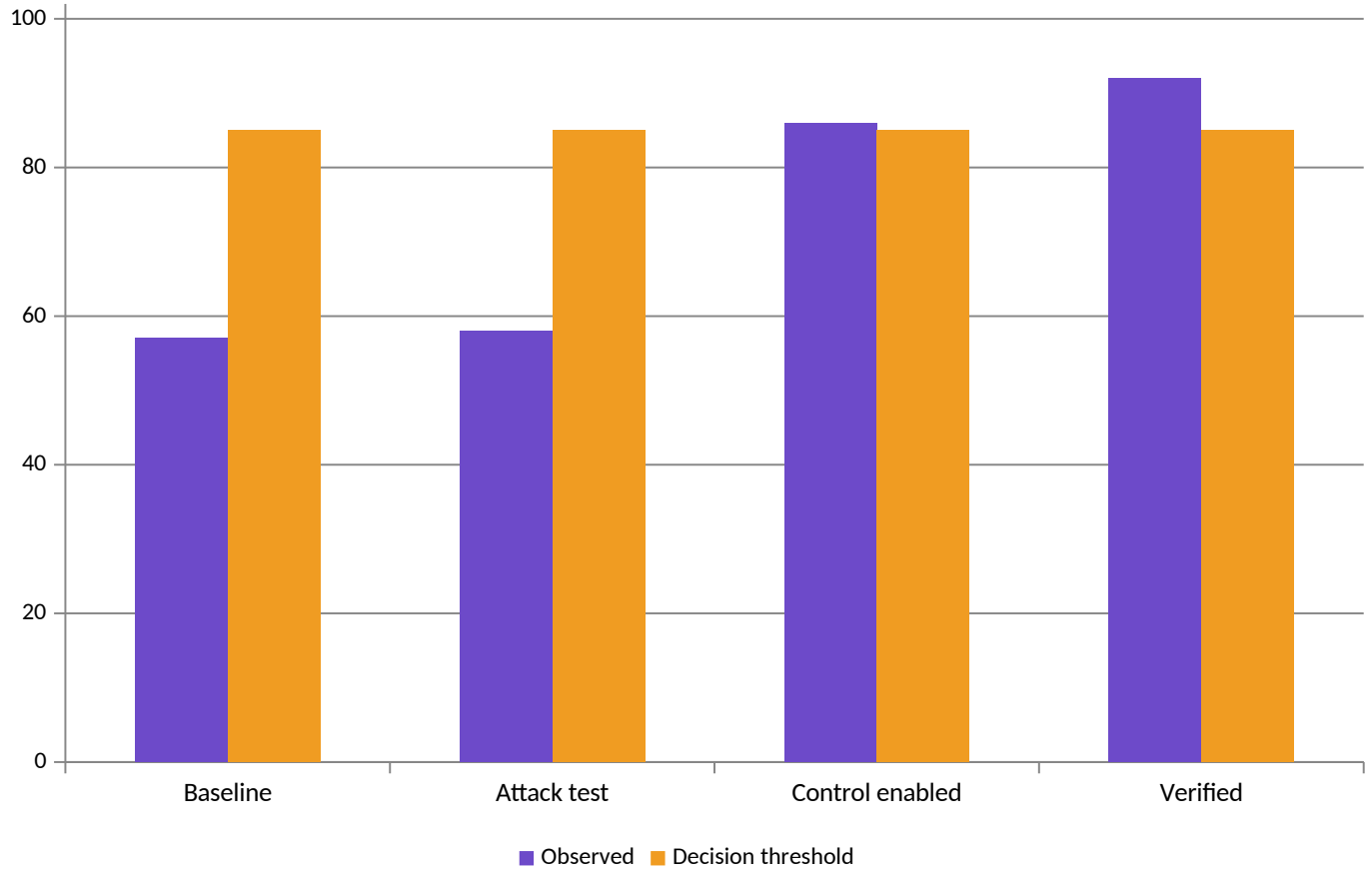
Expected-loss estimate: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	frequency_per_year	Required in decision and audit evidence
field_2	loss_sgd	Required in decision and audit evidence
field_3	expected_loss_sgd	Required in decision and audit evidence
field_4	confidence	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Expected-loss estimate: evidence and acceptance decision



DECISION RULE

Accept only when expected_loss = frequency × loss; assumptions are documented. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Control effectiveness: architecture and control boundary

1

Designed Control

State and evidence for designed control

2

Operating Test

State and evidence for operating test

3

Coverage

State and evidence for coverage

4

Effectiveness

State and evidence for effectiveness

Control objective: Test design, operation and coverage separately

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Control effectiveness: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
designed control	trace_id + agent_id + policy_version	Test design, operation and coverage separately	effectiveness	effective coverage meets the risk-treatment target

Failure to detect: Control exists but operates on only part of the fleet

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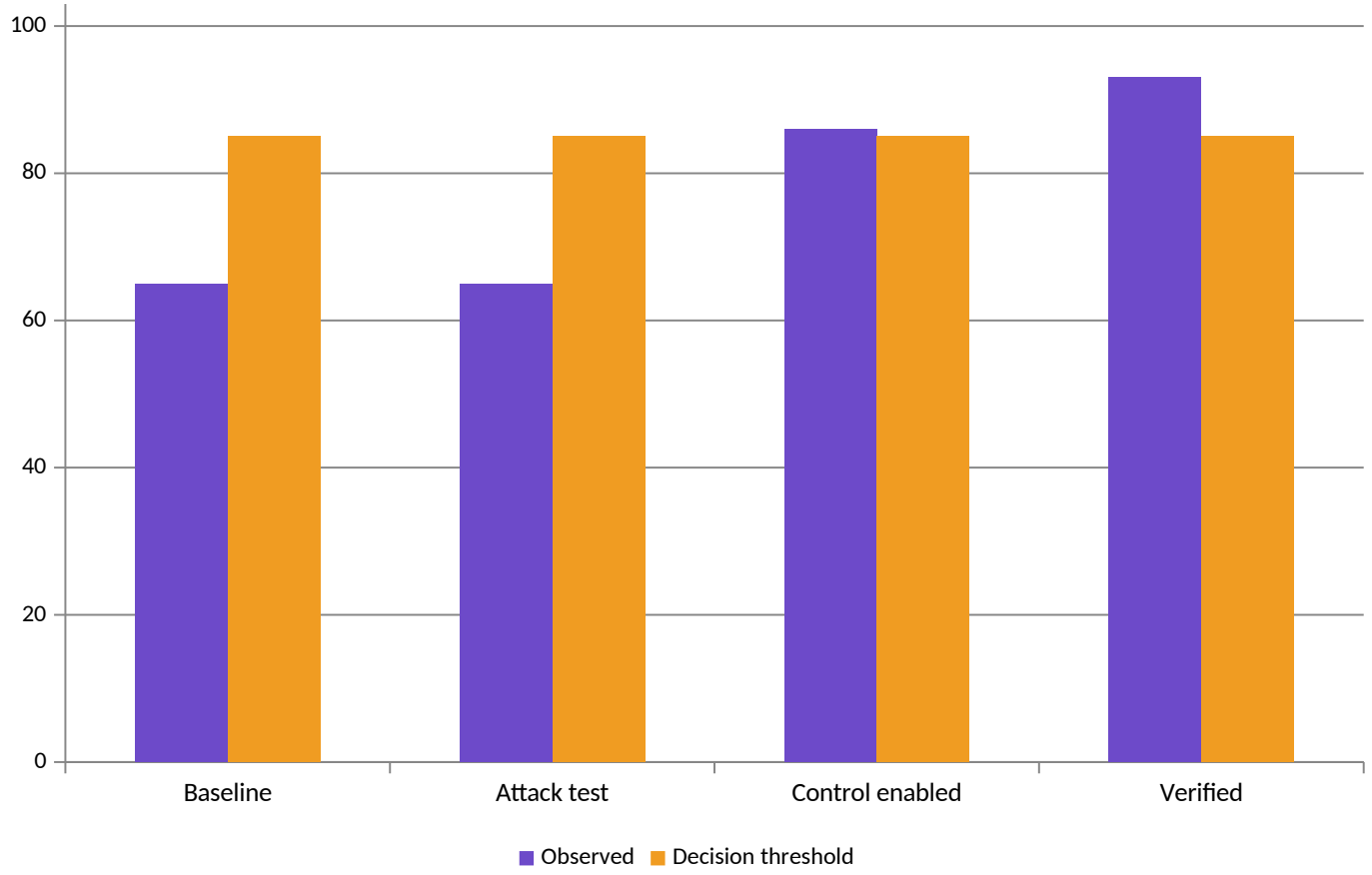
Control effectiveness: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	design_score	Required in decision and audit evidence
field_2	operation_score	Required in decision and audit evidence
field_3	coverage_pct	Required in decision and audit evidence
field_4	effectiveness_pct	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Control effectiveness: evidence and acceptance decision



DECISION RULE

Accept only when effective coverage meets the risk-treatment target. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Residual risk: architecture and control boundary

1

Inherent Risk × (1 − Control Effectiveness)

State and evidence for inherent risk × (1 − control effectiveness)

2

Residual Decision

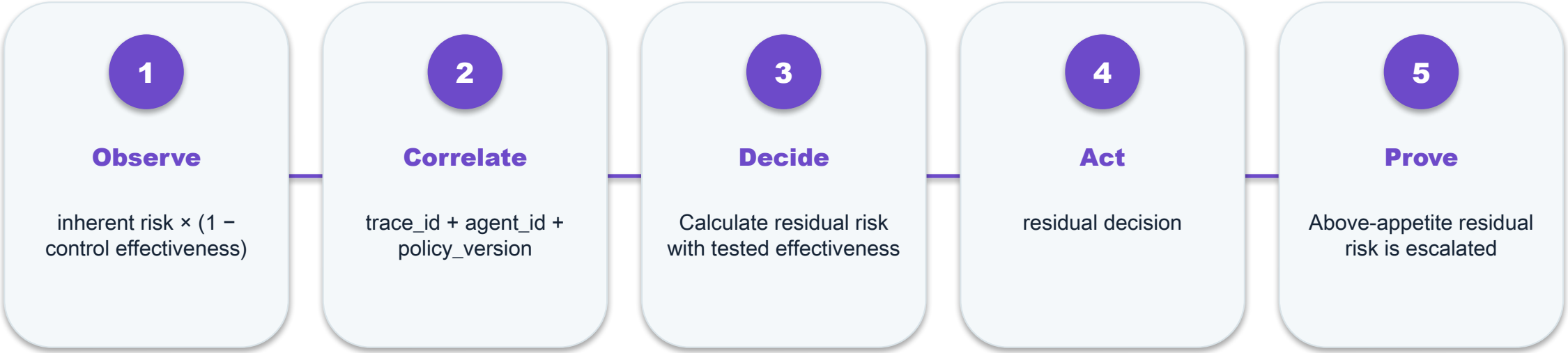
State and evidence for residual decision

Control objective: Calculate residual risk with tested effectiveness

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Residual risk: runtime decision trace



Failure to detect: Reported risk ignores control failure or coverage gaps

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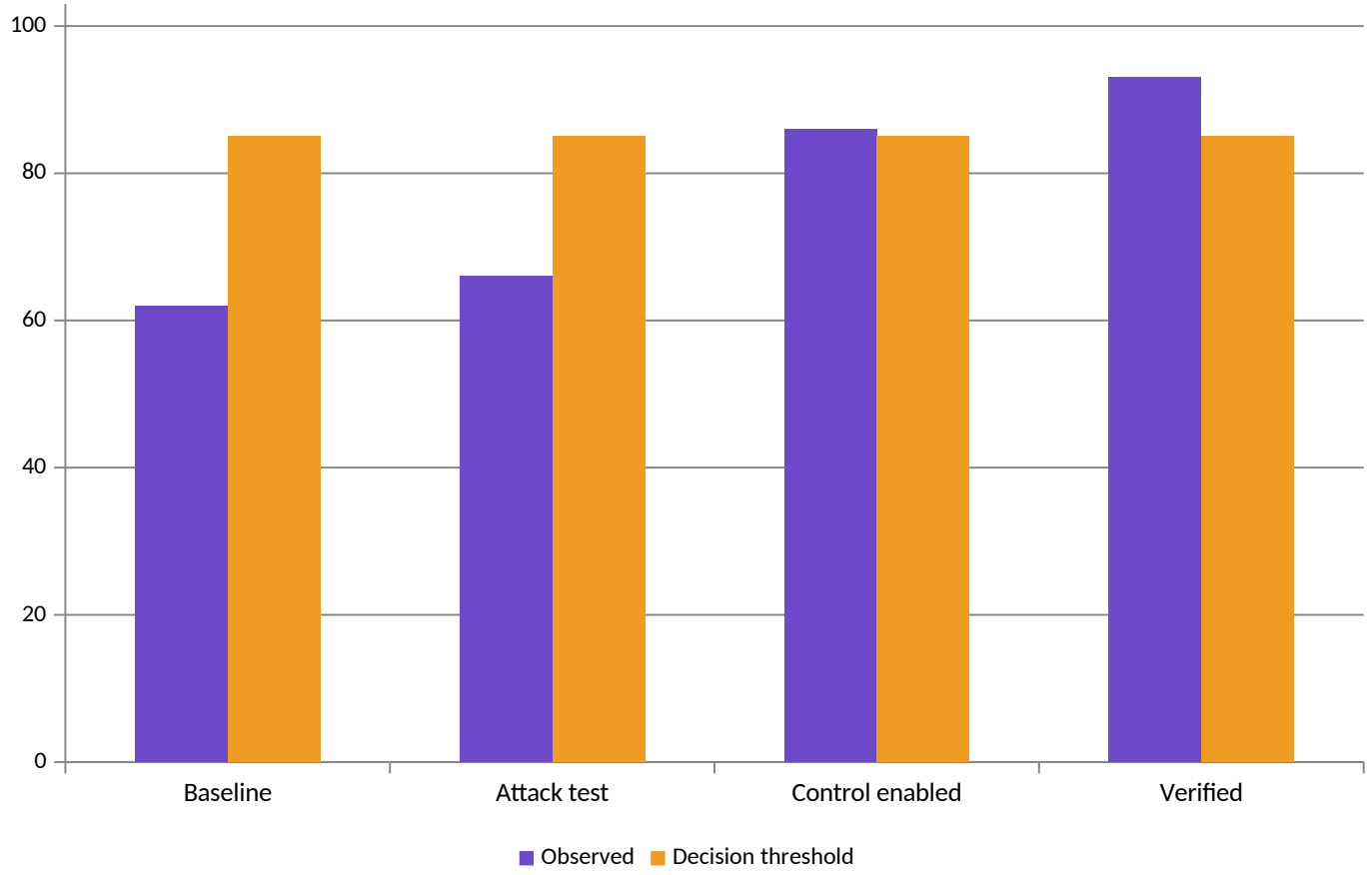
Residual risk: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	inherent_score	Required in decision and audit evidence
field_2	effectiveness_pct	Required in decision and audit evidence
field_3	residual_score	Required in decision and audit evidence
field_4	appetite	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Residual risk: evidence and acceptance decision



DECISION RULE

Accept only when Above-appetite residual risk is escalated. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Gap register: architecture and control boundary

1. **Benchmark/ Control Objective**
   State and evidence for benchmark/control objective
2. **Current State**
   State and evidence for current state
3. **Gap**
   State and evidence for gap
4. **Owner**
   State and evidence for owner
5. **Due Date**
   State and evidence for due date

Control objective: Record evidence, severity, owner, dependency and acceptance test

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Gap register: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
benchmark/control objective	trace_id + agent_id + policy_version	Record evidence, severity, owner, dependency and acceptance test	due date	No high gap is ownerless or overdue without escalation

Failure to detect: Findings lack an actionable path to closure

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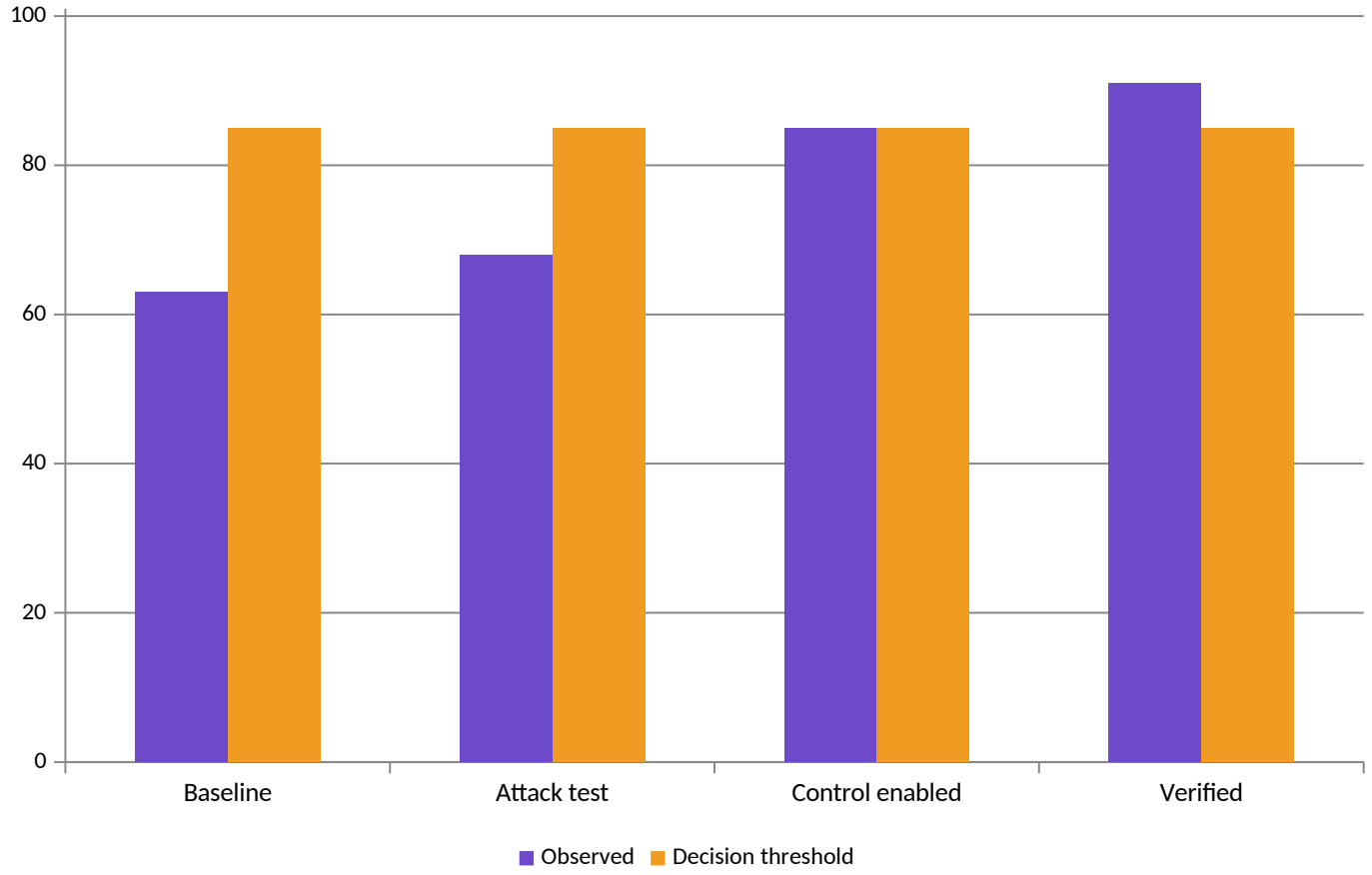
Gap register: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	gap_id	Required in decision and audit evidence
field_2	requirement	Required in decision and audit evidence
field_3	evidence	Required in decision and audit evidence
field_4	severity	Required in decision and audit evidence
field_5	owner	Required in decision and audit evidence
field_6	target_date	Required in decision and audit evidence

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Gap register: evidence and acceptance decision



DECISION RULE

Accept only when No high gap is ownerless or overdue without escalation. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Remediation portfolio: architecture and control boundary

1. **Gaps**
   State and evidence for gaps
2. **Options**
   State and evidence for options
3. **Cost/Risk Reduction**
   State and evidence for cost/risk reduction
4. **Priority**
   State and evidence for priority
5. **Roadmap**
   State and evidence for roadmap

Control objective: Prioritise by risk reduction, feasibility and dependencies

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Remediation portfolio: runtime decision trace

1. **Observe**
   gaps
2. **Correlate**
   trace_id + agent_id + policy_version
3. **Decide**
   Prioritise by risk reduction, feasibility and dependencies
4. **Act**
   roadmap
5. **Prove**
   Selected portfolio fits budget and maximises risk reduction

Failure to detect: Teams fund visible controls instead of highest value

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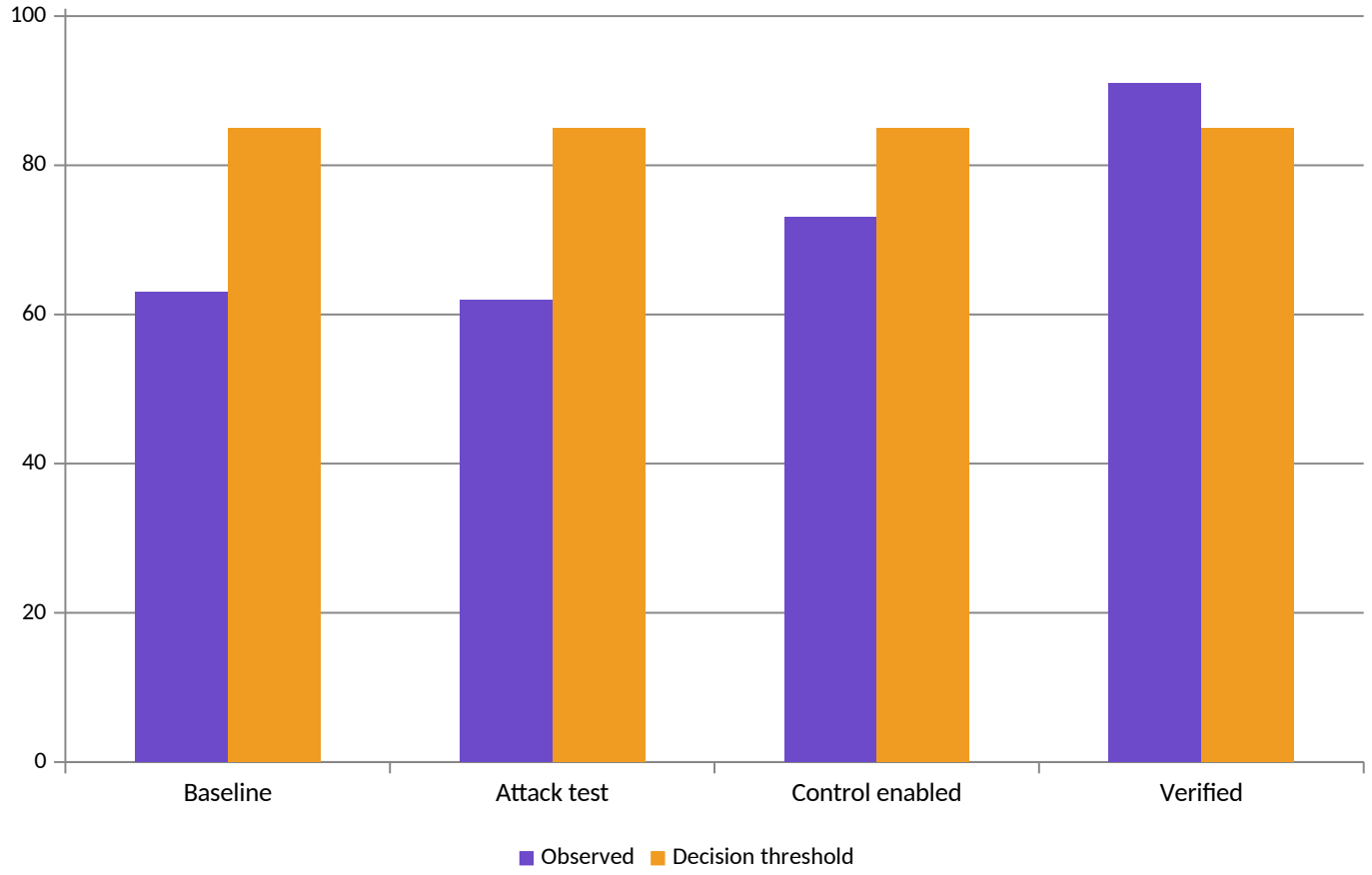
Remediation portfolio: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	option	Required in decision and audit evidence
field_2	cost_sgd	Required in decision and audit evidence
field_3	effort_days	Required in decision and audit evidence
field_4	risk_reduction	Required in decision and audit evidence
field_5	dependency	Required in decision and audit evidence
field_6	priority	Required in decision and audit evidence

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Remediation portfolio: evidence and acceptance decision



DECISION RULE

Accept only when Selected portfolio fits budget and maximises risk reduction. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Activity 06 — Quantify Agent Risks and Control Value

1 **Scenario**

Use synthetic data in activity-06-risk-control-evaluation to make a security-operations decision.

2 **Input contract**

mock_risks.csv

3 **Runnable evidence**

activity.py

4 **Output artifact**

risk_evaluation.csv

5 **Acceptance**

Residual risk and expected annual loss calculations reconcile with the input assumptions.

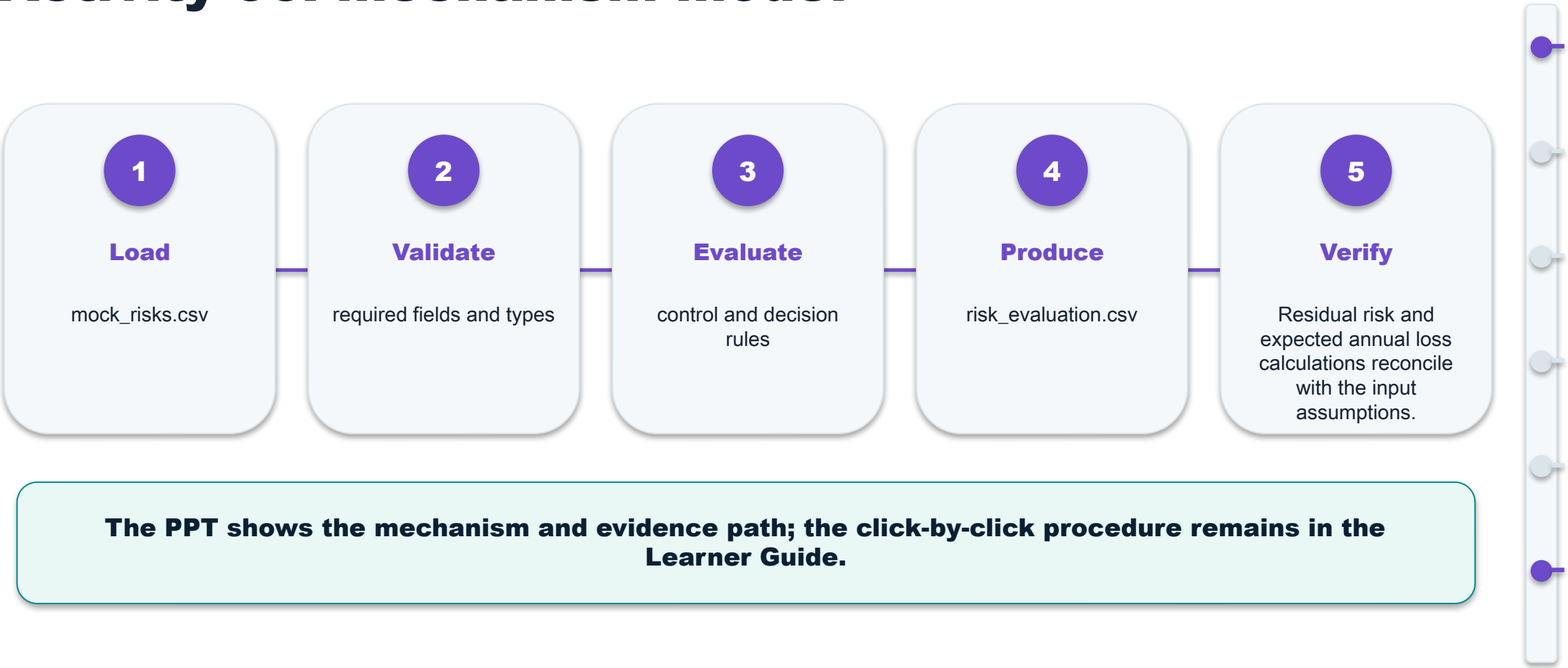
6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 06: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 06: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_risks.csv	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	risk_evaluation.csv	Generated locally and reproducible
Criterion	A5	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete risk_evaluation.csv only	Retain script and mock data

Aligned activity files in labs/

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Activity 07 — Prioritise a Security Gap Remediation Roadmap

1 **Scenario**

Use synthetic data in activity-07-gap-remediation to make a security-operations decision.

2 **Input contract**

mock_gaps.csv

3 **Runnable evidence**

activity.py

4 **Output artifact**

remediation_roadmap.md

5 **Acceptance**

The roadmap stays within the mock budget and prioritises overdue and above-appetite gaps.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 07: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_gaps.csv	required fields and types	control and decision rules	remediation_roadmap.md	The roadmap stays within the mock budget and prioritises overdue and above-appetite gaps.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 07: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_gaps.csv	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	remediation_roadmap.md	Generated locally and reproducible
Criterion	A6	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete remediation_roadmap.md only	Retain script and mock data

Aligned activity files in labs/

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Topic 03 technical recap

1 **Likelihood-impact scoring**

Independent scorers remain within one rating level

2 **Expected-loss estimate**

expected_loss = frequency × loss; assumptions are documented

3 **Control effectiveness**

effective coverage meets the risk-treatment target

4 **Residual risk**

Above-appetite residual risk is escalated

5 **Gap register**

No high gap is ownerless or overdue without escalation

6 **Remediation portfolio**

Selected portfolio fits budget and maximises risk reduction

Topic technical-anchor inventory

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TOPIC 04

Implementing and Monitoring Organisation-Wide Agent Security Operations

Telemetry · detection · incident response · containment · recovery

LO4 · K7, A7



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LO4 and assessment criteria

1 **K7**

Implications and impact of security gaps

2 **A7**

Drive organisation-wide security initiatives in line with internal and external standards

3 **LO4**

LO4: Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.

Approved course proposal criteria adapted to the autonomous-agent context

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Implementing and Monitoring Organisation-Wide Agent Security Operations: operating model

1 — Boundary

Components, interfaces and ownership

2 — Runtime

States, transitions and decision points

3 — Contract

Fields, rules, thresholds and evidence

4 — Failure

Attack path, diagnostic and control

5 — Verify

Acceptance test and activity artifact

Every mechanism in this topic is taught through all five layers.

Courseware-build mechanism-level coverage contract

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Telemetry contract: architecture and control boundary

1

Runtime

State and evidence for runtime

2

Policy Gateway

State and evidence for policy gateway

3

Tools

State and evidence for tools

4

Collector

State and evidence for collector

5

Siem

State and evidence for SIEM

Control objective: Require schema validation at ingestion

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Telemetry contract: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
runtime	trace_id + agent_id + policy_version	Require schema validation at ingestion	SIEM	Critical-event schema conformance ≥99.5%

Failure to detect: Missing fields break investigations and detections

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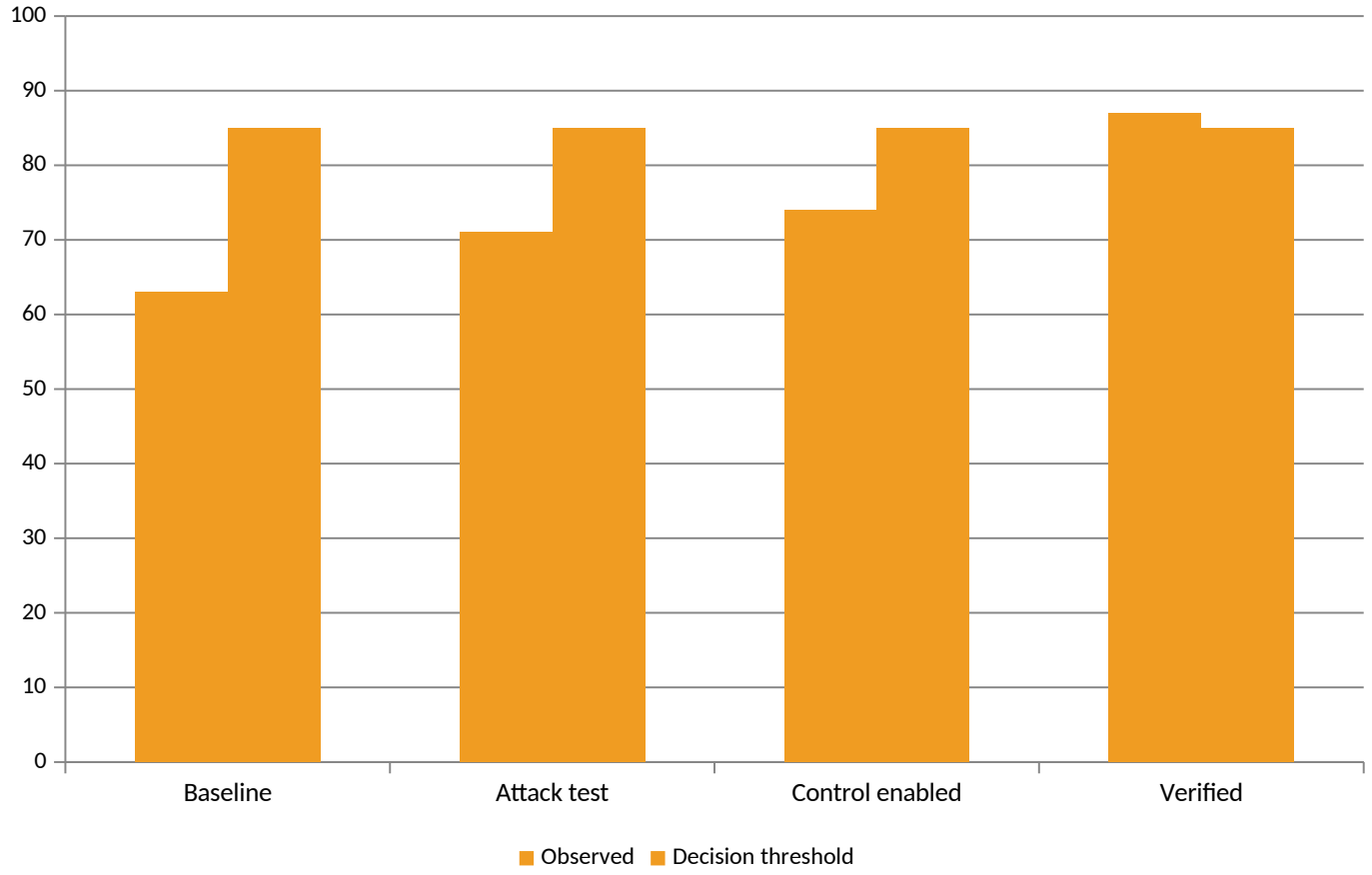
Telemetry contract: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	event_time	Required in decision and audit evidence
field_2	trace_id	Required in decision and audit evidence
field_3	span_id	Required in decision and audit evidence
field_4	agent_id	Required in decision and audit evidence
field_5	session_id	Required in decision and audit evidence
field_6	event_type	Required in decision and audit evidence

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Telemetry contract: evidence and acceptance decision



DECISION RULE

Accept only when Critical-event schema conformance ≥99.5%. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Trace correlation: architecture and control boundary

1. **User Request**
   State and evidence for user request
2. **Plan Spans**
   State and evidence for plan spans
3. **Tool Spans**
   State and evidence for tool spans
4. **Approval Span**
   State and evidence for approval span
5. **Outcome**
   State and evidence for outcome

Control objective: Propagate trace context across every component

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Trace correlation: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
user request	trace_id + agent_id + policy_version	Propagate trace context across every component	outcome	A sampled session reconstructs without orphan spans

Failure to detect: Events exist but cannot be ordered into one incident

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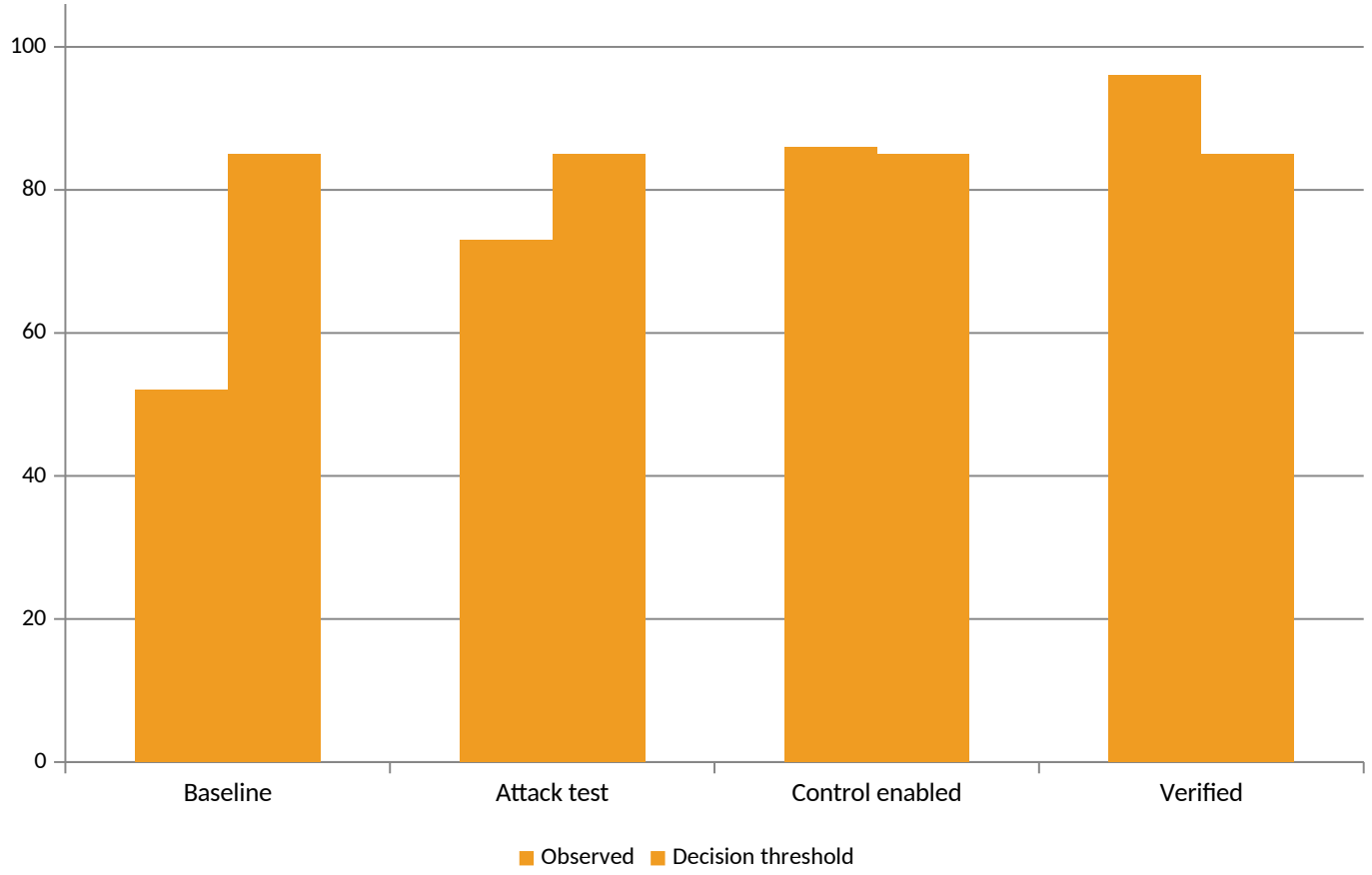
Trace correlation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	trace_id	Required in decision and audit evidence
field_2	parent_span_id	Required in decision and audit evidence
field_3	actor	Required in decision and audit evidence
field_4	action	Required in decision and audit evidence
field_5	resource	Required in decision and audit evidence
field_6	status	Required in decision and audit evidence

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Trace correlation: evidence and acceptance decision



DECISION RULE

Accept only when A sampled session reconstructs without orphan spans. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Immutable audit log: architecture and control boundary

1

Event Producer

State and evidence for event producer

2

Signed Envelope

State and evidence for signed envelope

3

Append-Only Store

State and evidence for append-only store

4

Verifier

State and evidence for verifier

Control objective: Use append-only storage and hash chaining

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Immutable audit log: runtime decision trace



Failure to detect: Attacker deletes or edits incriminating tool events

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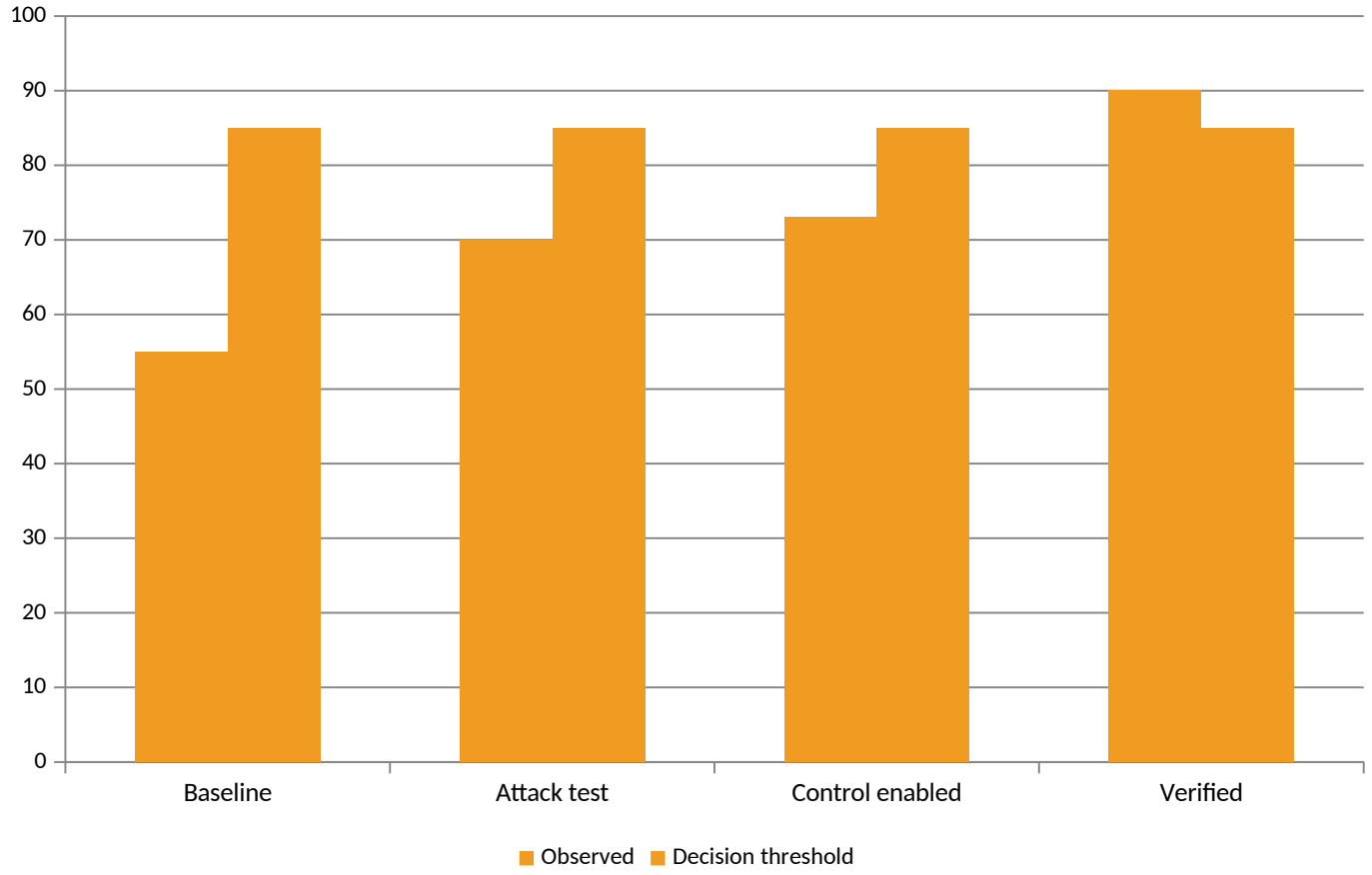
Immutable audit log: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	sequence	Required in decision and audit evidence
field_2	previous_hash	Required in decision and audit evidence
field_3	event_hash	Required in decision and audit evidence
field_4	signer	Required in decision and audit evidence
field_5	timestamp	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Immutable audit log: evidence and acceptance decision



DECISION RULE

Accept only when Verification detects any missing or altered record. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Detection rule lifecycle: architecture and control boundary

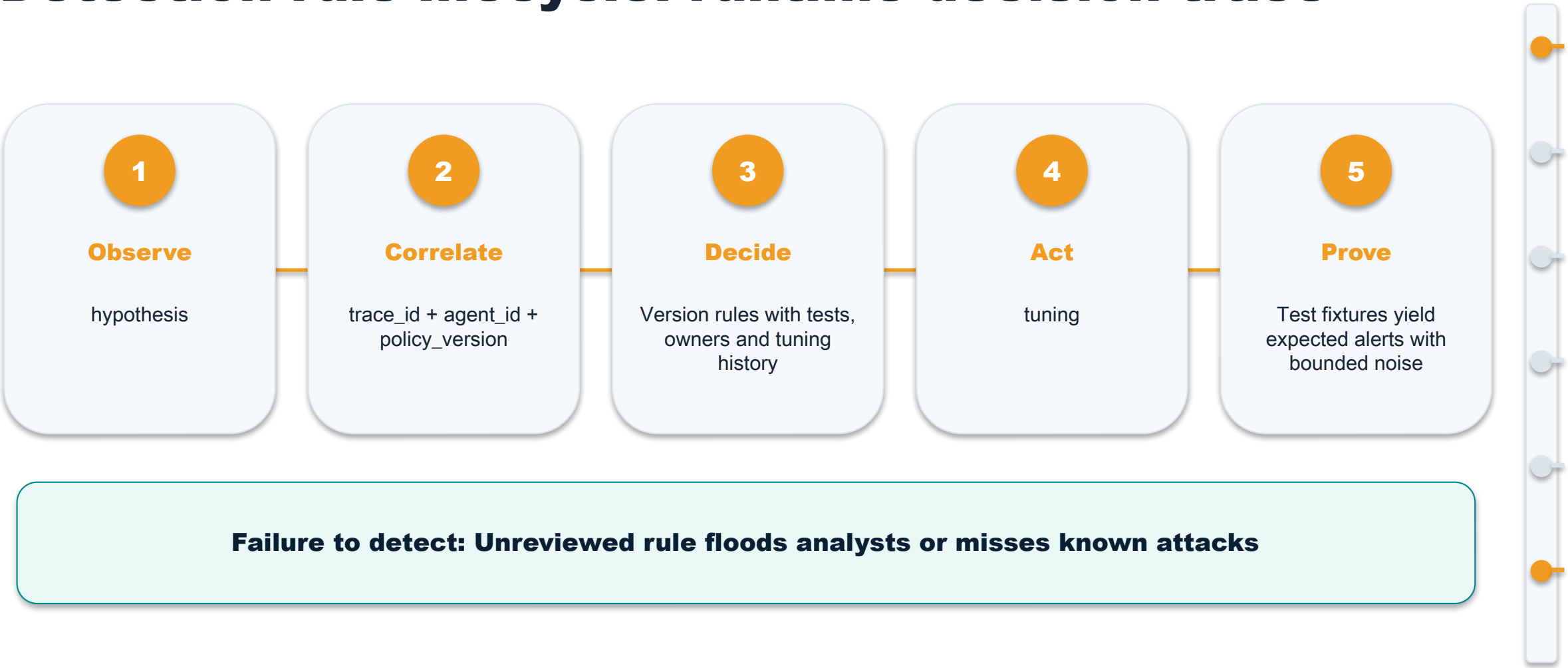
1. **Hypothesis**
   State and evidence for hypothesis
2. **Query**
   State and evidence for query
3. **Test Data**
   State and evidence for test data
4. **Deployment**
   State and evidence for deployment
5. **Tuning**
   State and evidence for tuning

Control objective: Version rules with tests, owners and tuning history

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Detection rule lifecycle: runtime decision trace



Failure to detect: Unreviewed rule floods analysts or misses known attacks

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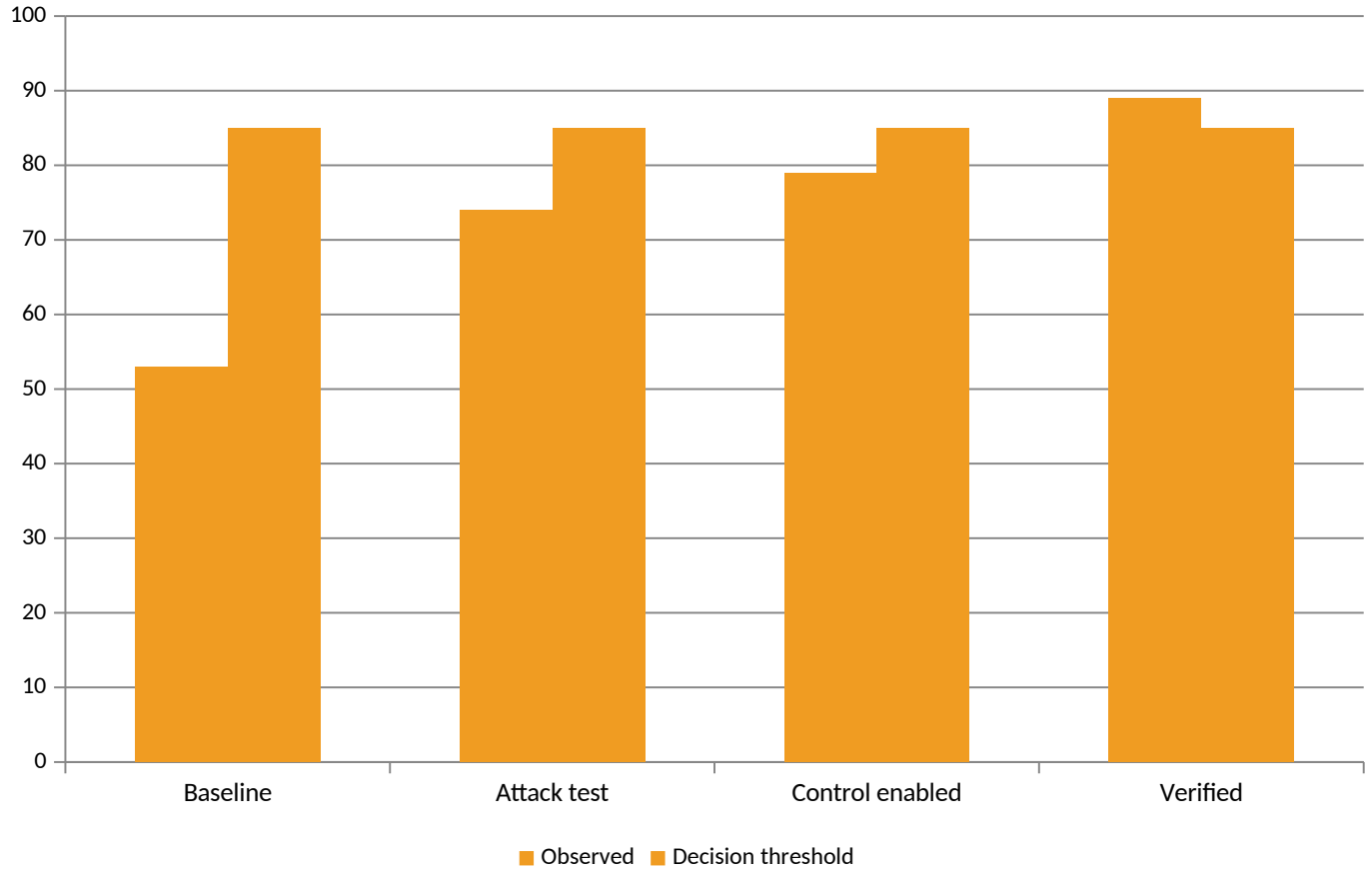
Detection rule lifecycle: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	rule_id	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	query	Required in decision and audit evidence
field_4	severity	Required in decision and audit evidence
field_5	threshold	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

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Detection rule lifecycle: evidence and acceptance decision



DECISION RULE

Accept only when Test fixtures yield expected alerts with bounded noise. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Behaviour anomaly score: architecture and control boundary

1. **Feature Window** — State and evidence for feature window
2. **Baseline** — State and evidence for baseline
3. **Deviation** — State and evidence for deviation
4. **Score** — State and evidence for score
5. **Alert** — State and evidence for alert

Control objective: Combine identity-specific behavioural features

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Behaviour anomaly score: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
feature window	trace_id + agent_id + policy_version	Combine identity-specific behavioural features	alert	score≥0.80 opens an investigation in mock policy

Failure to detect: Compromised agent stays within simple static limits

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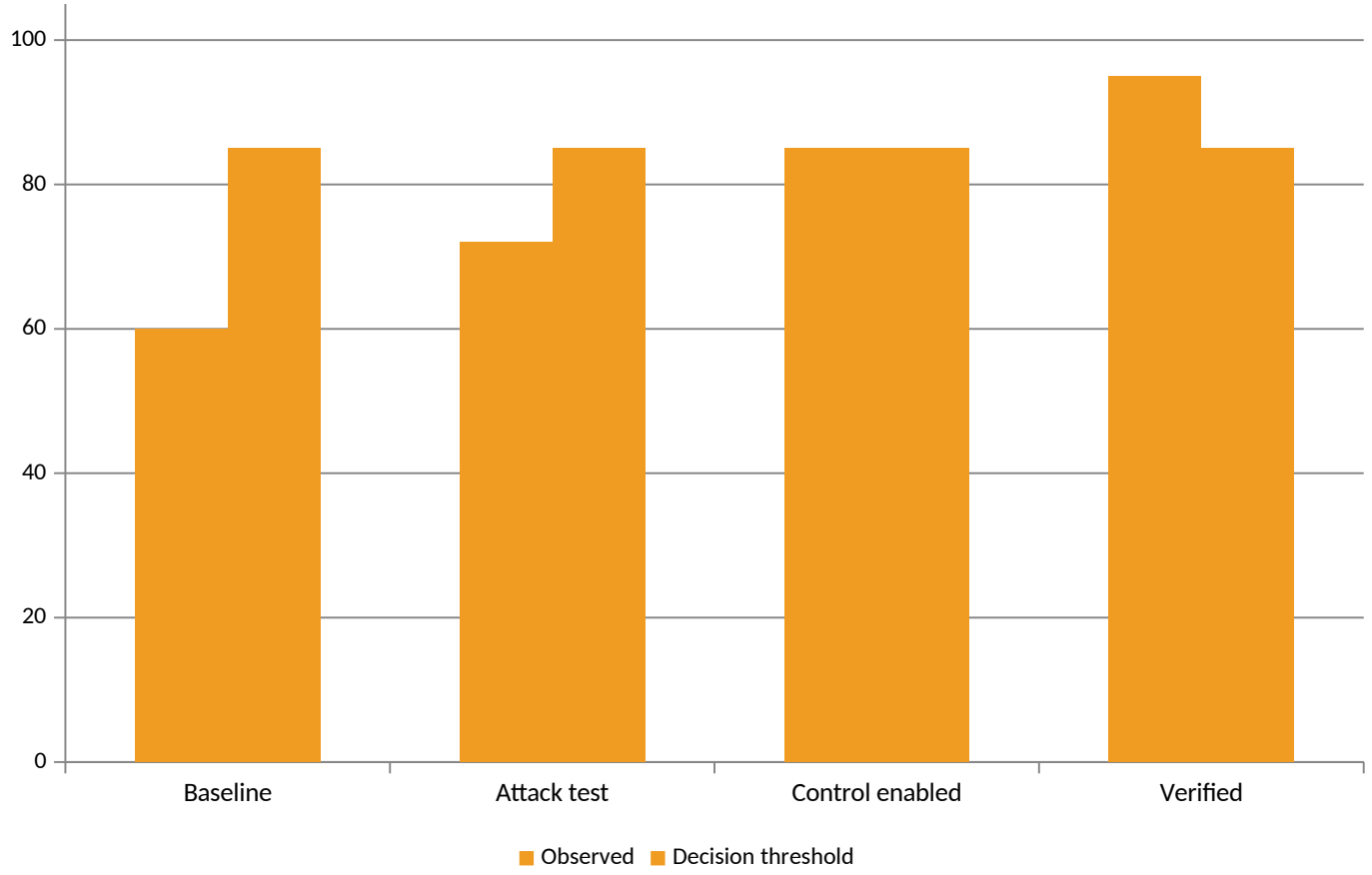
Behaviour anomaly score: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tool_rate_z	Required in decision and audit evidence
field_2	new_destination	Required in decision and audit evidence
field_3	denied_ratio	Required in decision and audit evidence
field_4	step_count	Required in decision and audit evidence
field_5	score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Behaviour anomaly score: evidence and acceptance decision



DECISION RULE

Accept only when score≥0.80 opens an investigation in mock policy. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Prompt-injection signals: architecture and control boundary

1

Input/Context

State and evidence for input/context

2

Detector Ensemble

State and evidence for detector ensemble

3

Policy Gate

State and evidence for policy gate

4

Trace

State and evidence for trace

Control objective: Layer provenance, classifiers and action authorisation

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Prompt-injection signals: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
input/context	trace_id + agent_id + policy_version	Layer provenance, classifiers and action authorisation	trace	High-risk signal plus sensitive action is blocked

Failure to detect: Obfuscated injection bypasses one keyword filter

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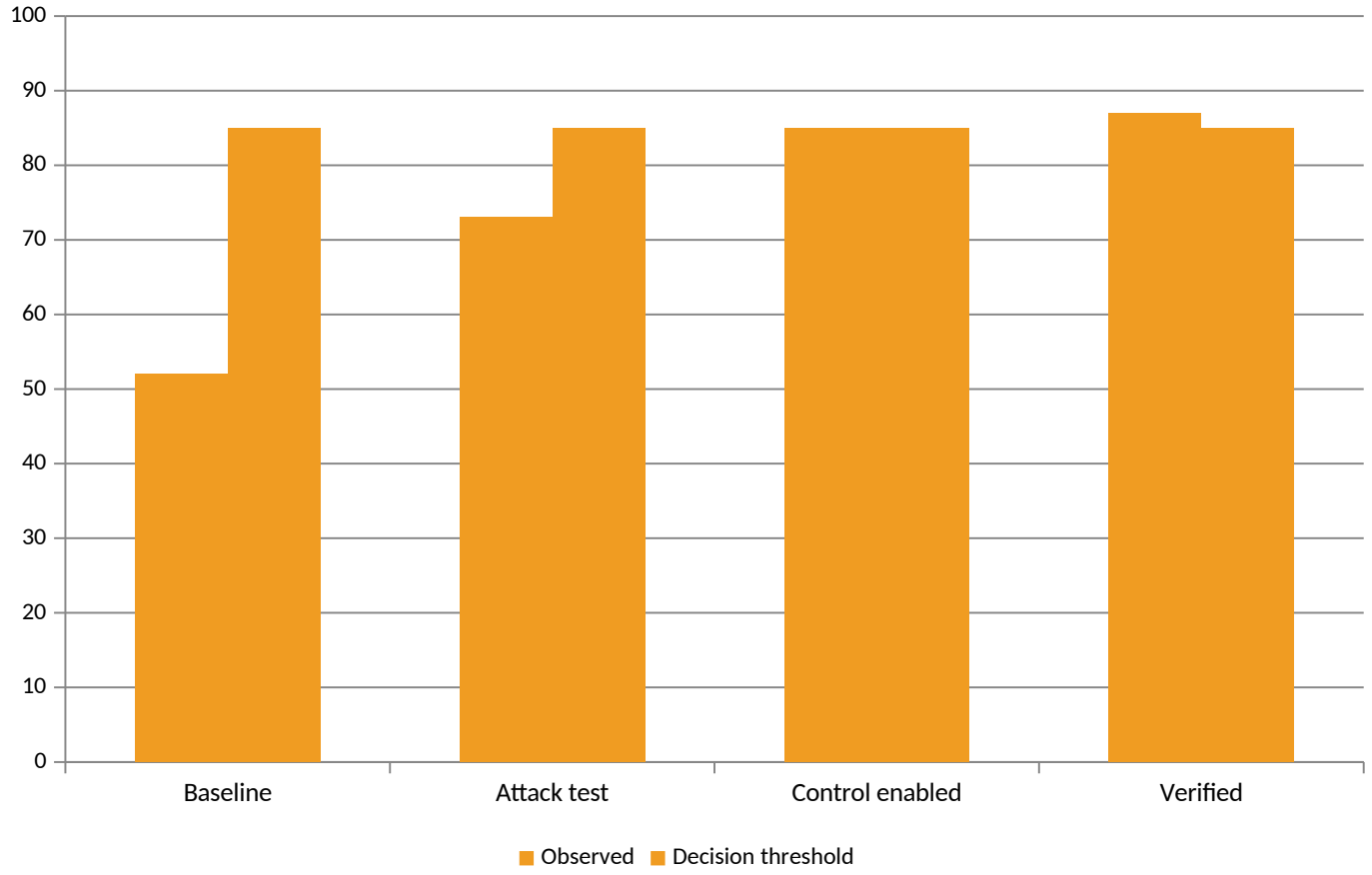
Prompt-injection signals: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	instruction_density	Required in decision and audit evidence
field_2	trust_mismatch	Required in decision and audit evidence
field_3	secret_request	Required in decision and audit evidence
field_4	override_phrase	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Prompt-injection signals: evidence and acceptance decision



DECISION RULE

Accept only when High-risk signal plus sensitive action is blocked. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Tool-sequence graph: architecture and control boundary

1

Tool Events

State and evidence for tool events

2

Directed Graph

State and evidence for directed graph

3

Rare Edge Model

State and evidence for rare edge model

4

Alert

State and evidence for alert

Control objective: Detect unusual action chains and privileged transitions

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Tool-sequence graph: runtime decision trace



Failure to detect: Each call is allowed but the sequence is malicious

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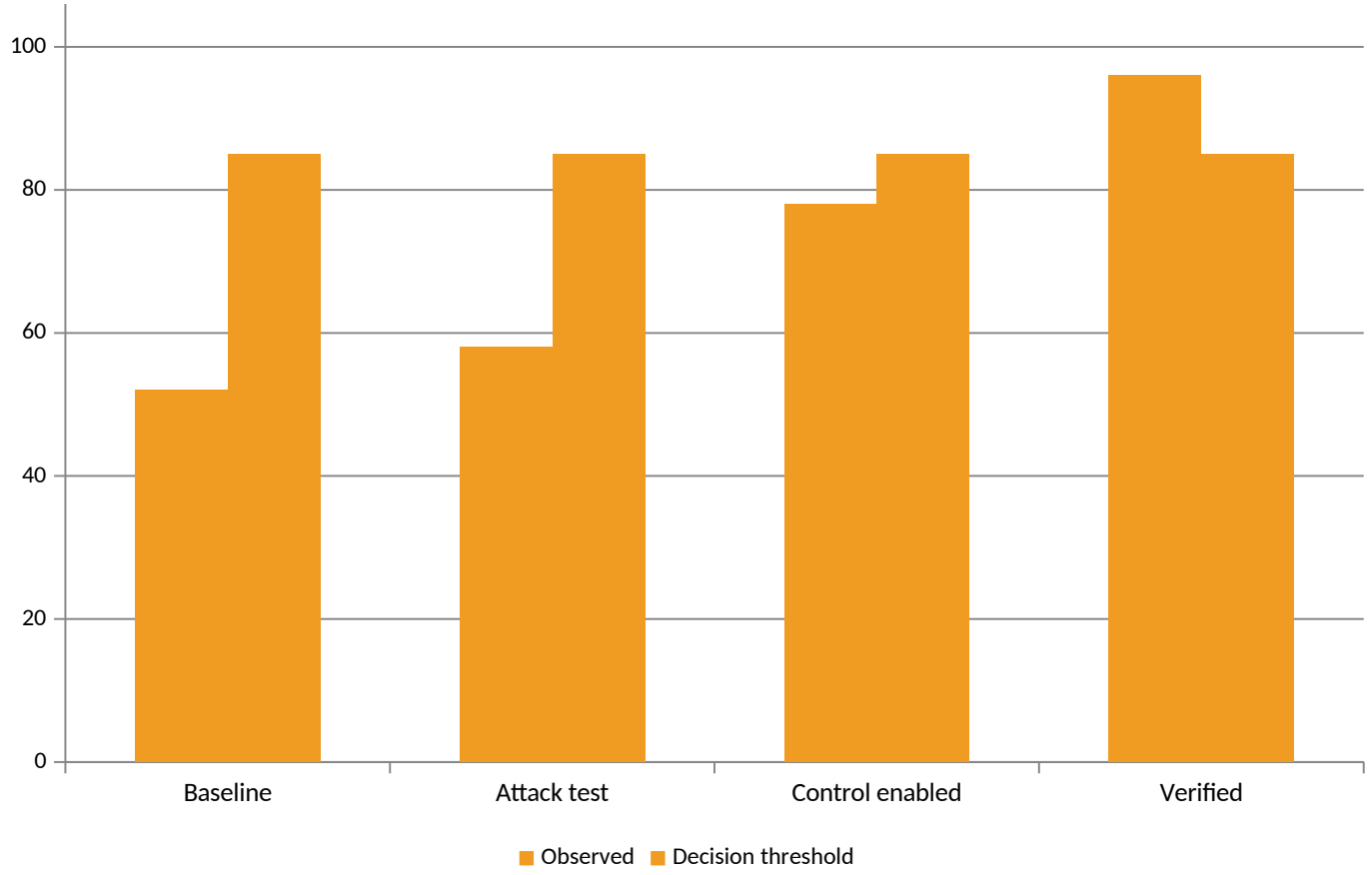
Tool-sequence graph: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	from_tool	Required in decision and audit evidence
field_2	to_tool	Required in decision and audit evidence
field_3	edge_count	Required in decision and audit evidence
field_4	baseline_prob	Required in decision and audit evidence
field_5	risk_weight	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Tool-sequence graph: evidence and acceptance decision



DECISION RULE

Accept only when Rare high-risk edge produces a correlated alert. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Data-egress detection: architecture and control boundary

1

Read Events

State and evidence for read events

2

Classification

State and evidence for classification

3

Outbound Events

State and evidence for outbound events

4

Correlation

State and evidence for correlation

Control objective: Correlate sensitive reads with outbound channels

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Data-egress detection: runtime decision trace



Failure to detect: Agent reads restricted data then sends a small encoded payload

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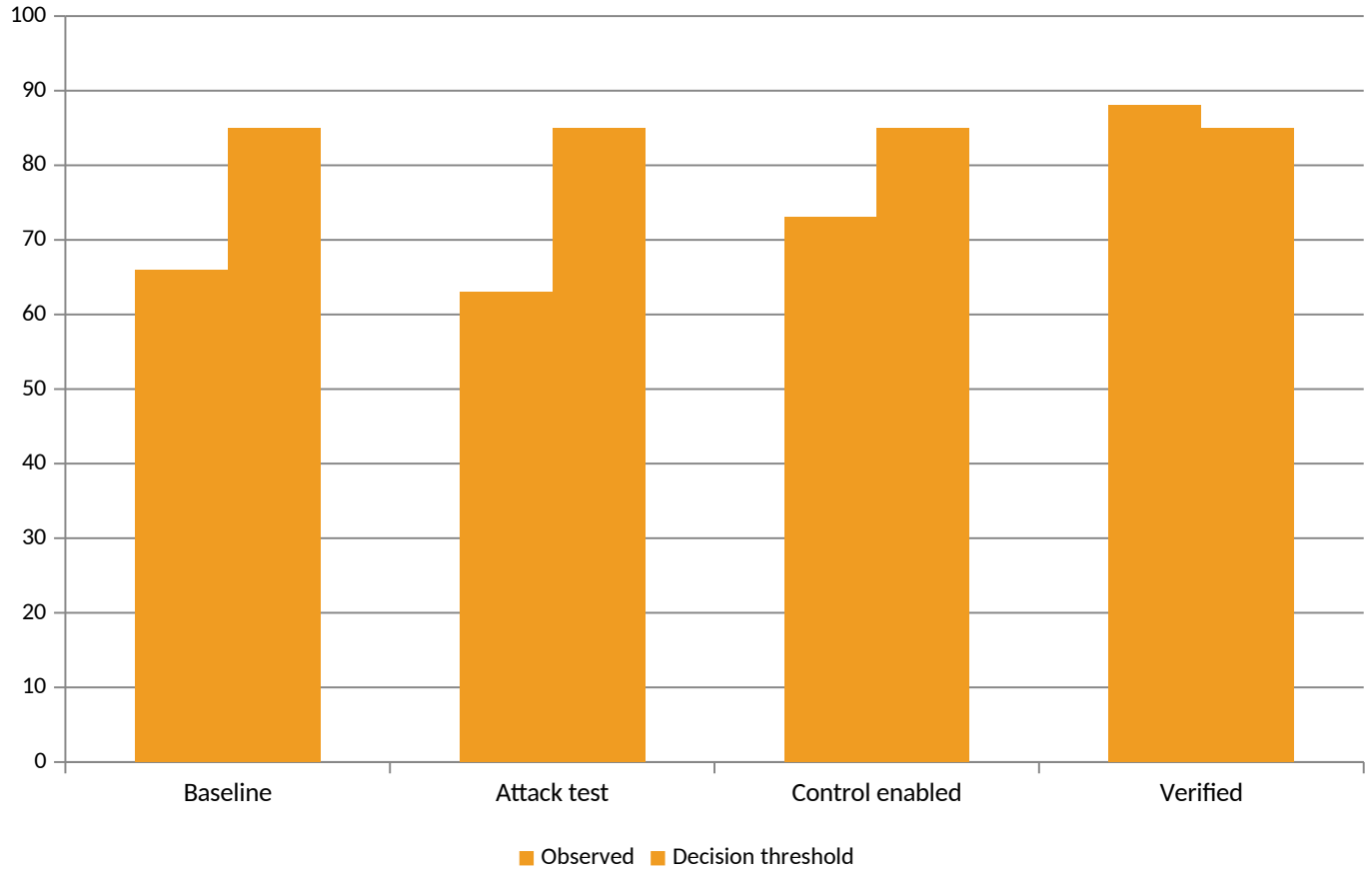
Data-egress detection: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	bytes_read	Required in decision and audit evidence
field_2	data_class	Required in decision and audit evidence
field_3	destination	Required in decision and audit evidence
field_4	bytes_out	Required in decision and audit evidence
field_5	time_delta	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Data-egress detection: evidence and acceptance decision



DECISION RULE

Accept only when Restricted-read + unknown-egress within 5 min alerts. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Secret-access anomaly: architecture and control boundary

1

Vault Access

State and evidence for vault access

2

Purpose

State and evidence for purpose

3

Tool Call

State and evidence for tool call

4

Destination

State and evidence for destination

Control objective: Baseline secret-to-tool relationships and deny mismatches

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Secret-access anomaly: runtime decision trace



Failure to detect: Agent requests a secret unrelated to its normal tools

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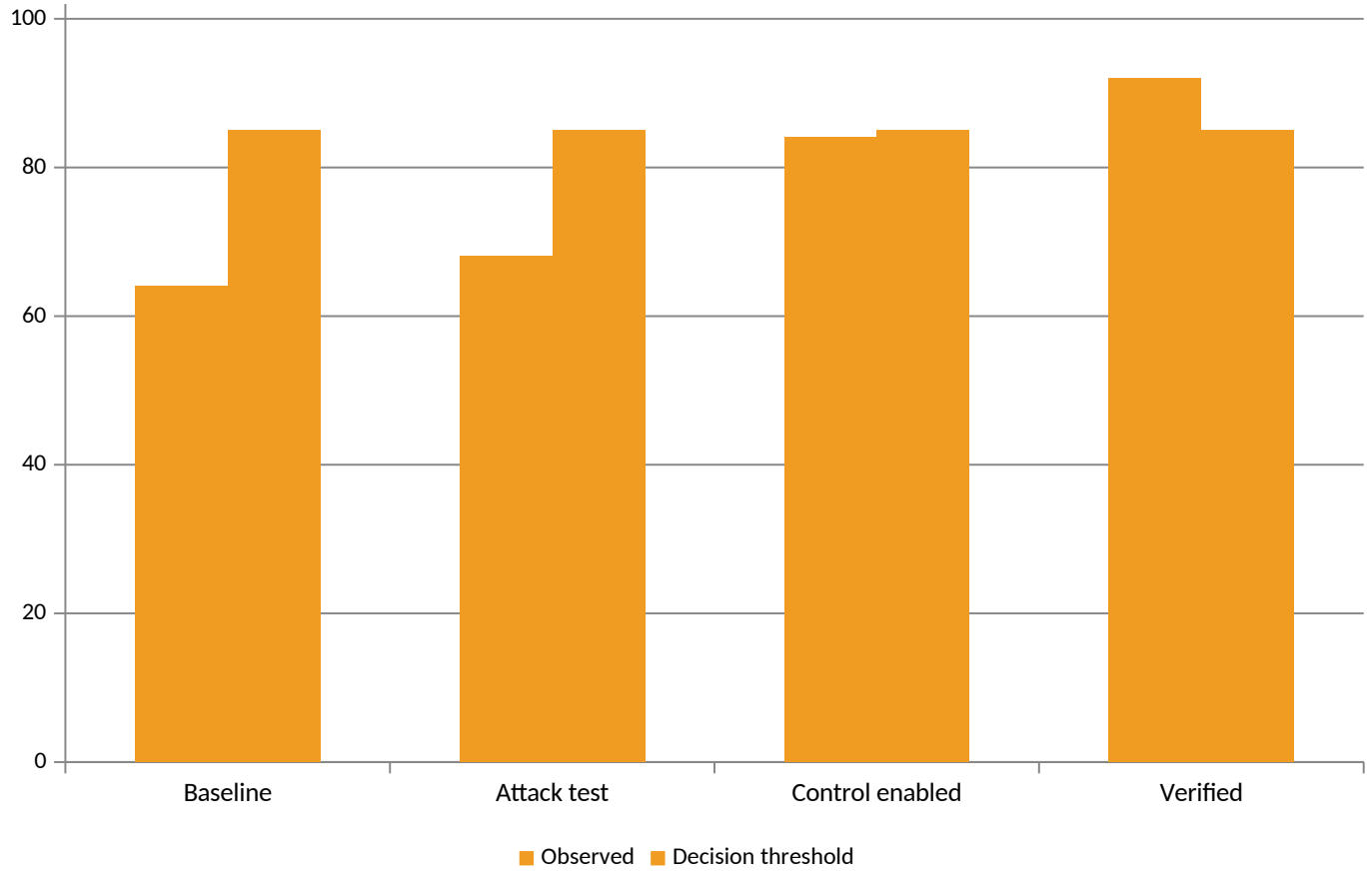
Secret-access anomaly: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	secret_ref	Required in decision and audit evidence
field_2	agent_id	Required in decision and audit evidence
field_3	tool_id	Required in decision and audit evidence
field_4	reason	Required in decision and audit evidence
field_5	usual_for_agent	Required in decision and audit evidence
field_6	result	Required in decision and audit evidence

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Secret-access anomaly: evidence and acceptance decision



DECISION RULE

Accept only when Unexpected secret_ref/tool_id pair is denied. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Session drift: architecture and control boundary

1

Session Features Over Time

State and evidence for session features over time

2

Baseline Comparison

State and evidence for baseline comparison

3

Drift Signal

State and evidence for drift signal

Control objective: Re-authorise when goal or capability footprint changes

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Session drift: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
session features over time	trace_id + agent_id + policy_version	Re-authorise when goal or capability footprint changes	drift signal	drift_score≥0.70 pauses before the next action

Failure to detect: Long session gradually moves away from approved goal

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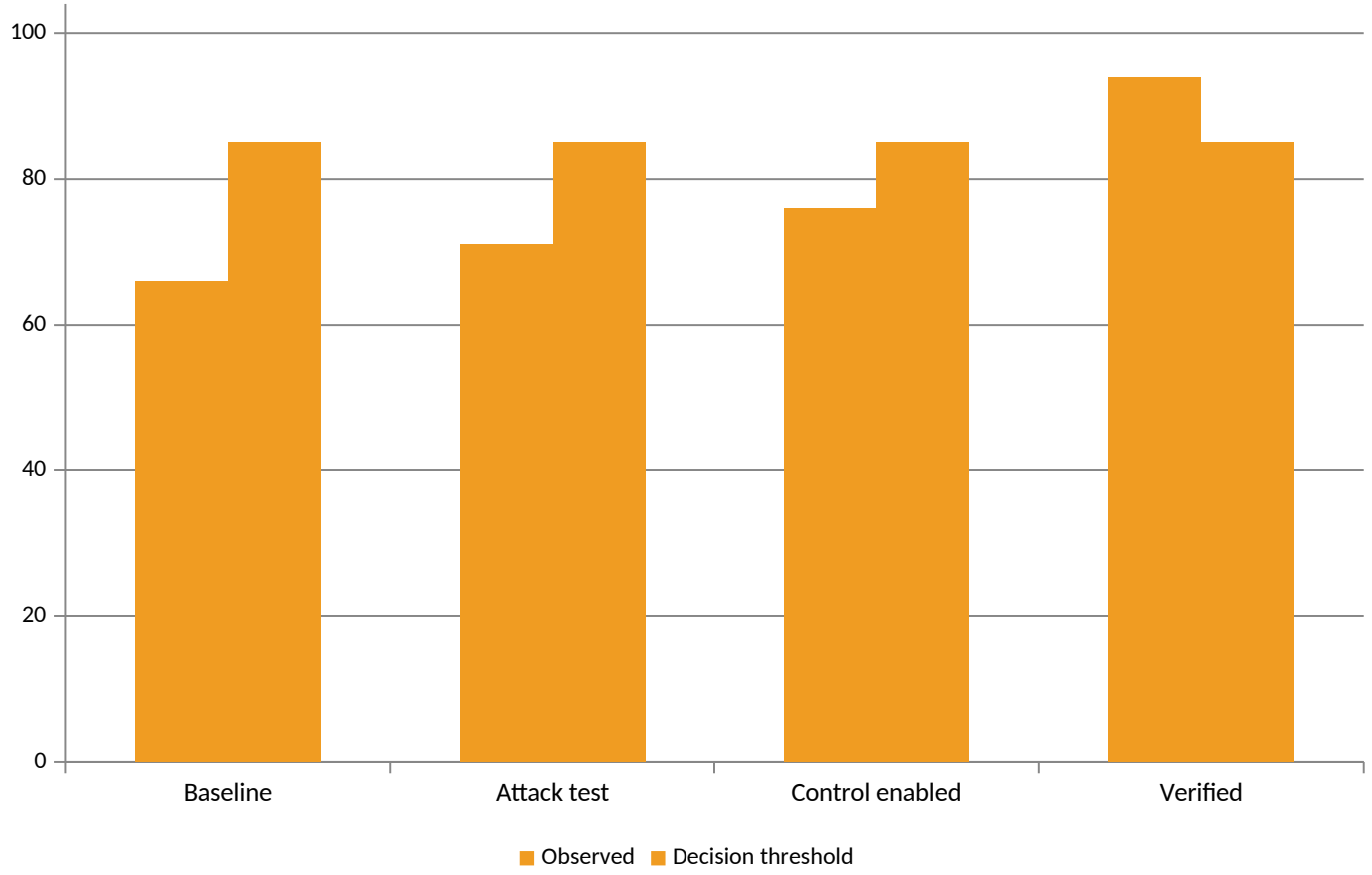
Session drift: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	goal_hash	Required in decision and audit evidence
field_2	active_tools[]	Required in decision and audit evidence
field_3	destinations[]	Required in decision and audit evidence
field_4	step_count	Required in decision and audit evidence
field_5	drift_score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Session drift: evidence and acceptance decision



DECISION RULE

Accept only when drift_score≥0.70 pauses before the next action. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Rate-limit control: architecture and control boundary

1

Identity Bucket

State and evidence for identity bucket

2

Tool Bucket

State and evidence for tool bucket

3

Destination Bucket

State and evidence for destination bucket

4

Decision

State and evidence for decision

Control objective: Limit by agent, tenant, tool and destination

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Rate-limit control: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
identity bucket	trace_id + agent_id + policy_version	Limit by agent, tenant, tool and destination	decision	Load test produces 429/deny at configured thresholds

Failure to detect: Distributed calls evade a single global counter

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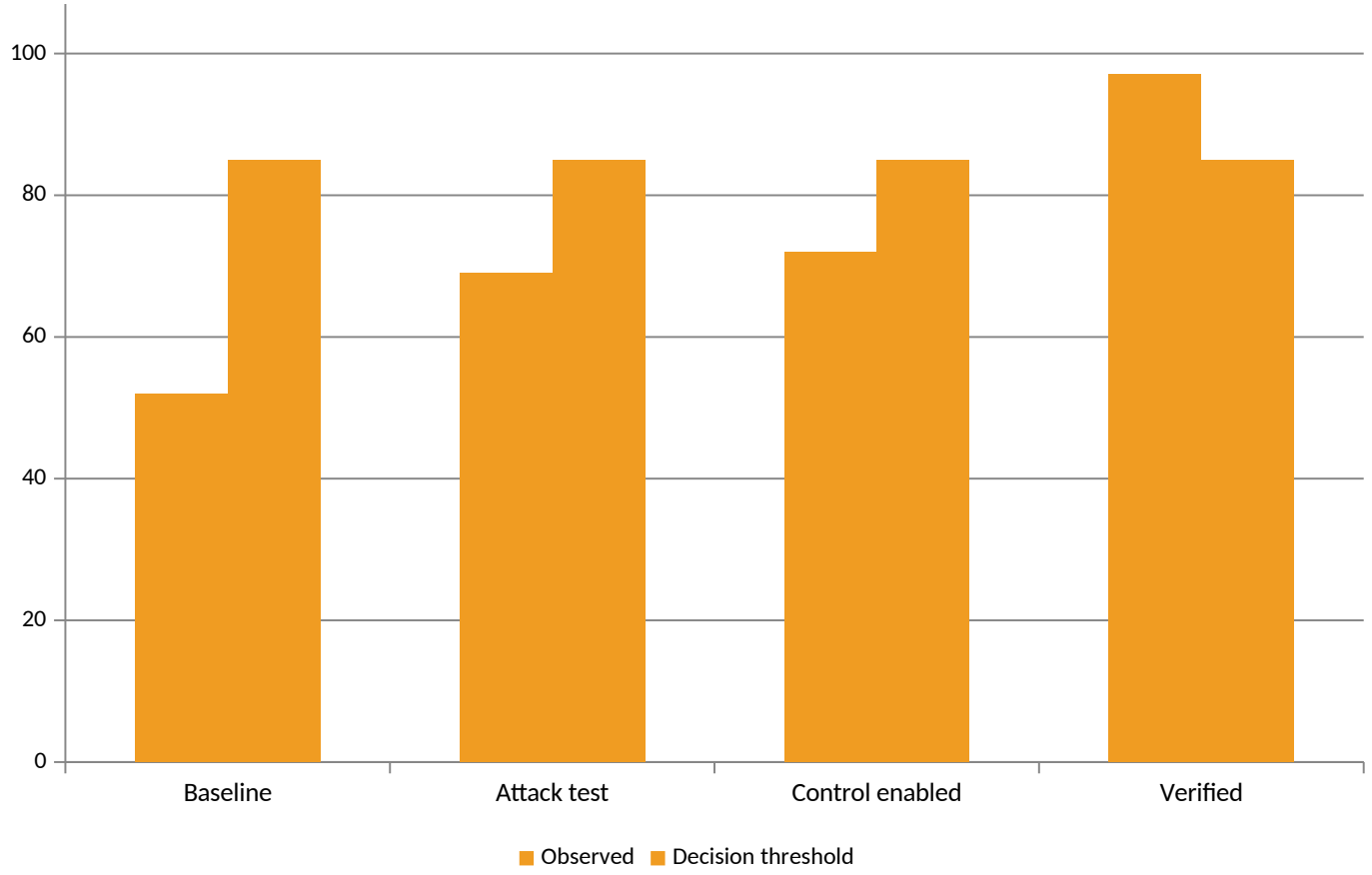
Rate-limit control: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	window_seconds	Required in decision and audit evidence
field_2	max_requests	Required in decision and audit evidence
field_3	current_count	Required in decision and audit evidence
field_4	retry_after	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Rate-limit control: evidence and acceptance decision



DECISION RULE

Accept only when Load test produces 429/deny at configured thresholds. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Budget alert: architecture and control boundary

1

Token/Tool Usage

State and evidence for token/tool usage

2

Cost Estimator

State and evidence for cost estimator

3

Threshold

State and evidence for threshold

4

Circuit Breaker

State and evidence for circuit breaker

Control objective: Warn at 70%, throttle at 90%, stop at 100% in mock policy

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Budget alert: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
token/tool usage	trace_id + agent_id + policy_version	Warn at 70%, throttle at 90%, stop at 100% in mock policy	circuit breaker	Budget breach stops new calls and preserves trace

Failure to detect: Runaway planning consumes budget without operational alert

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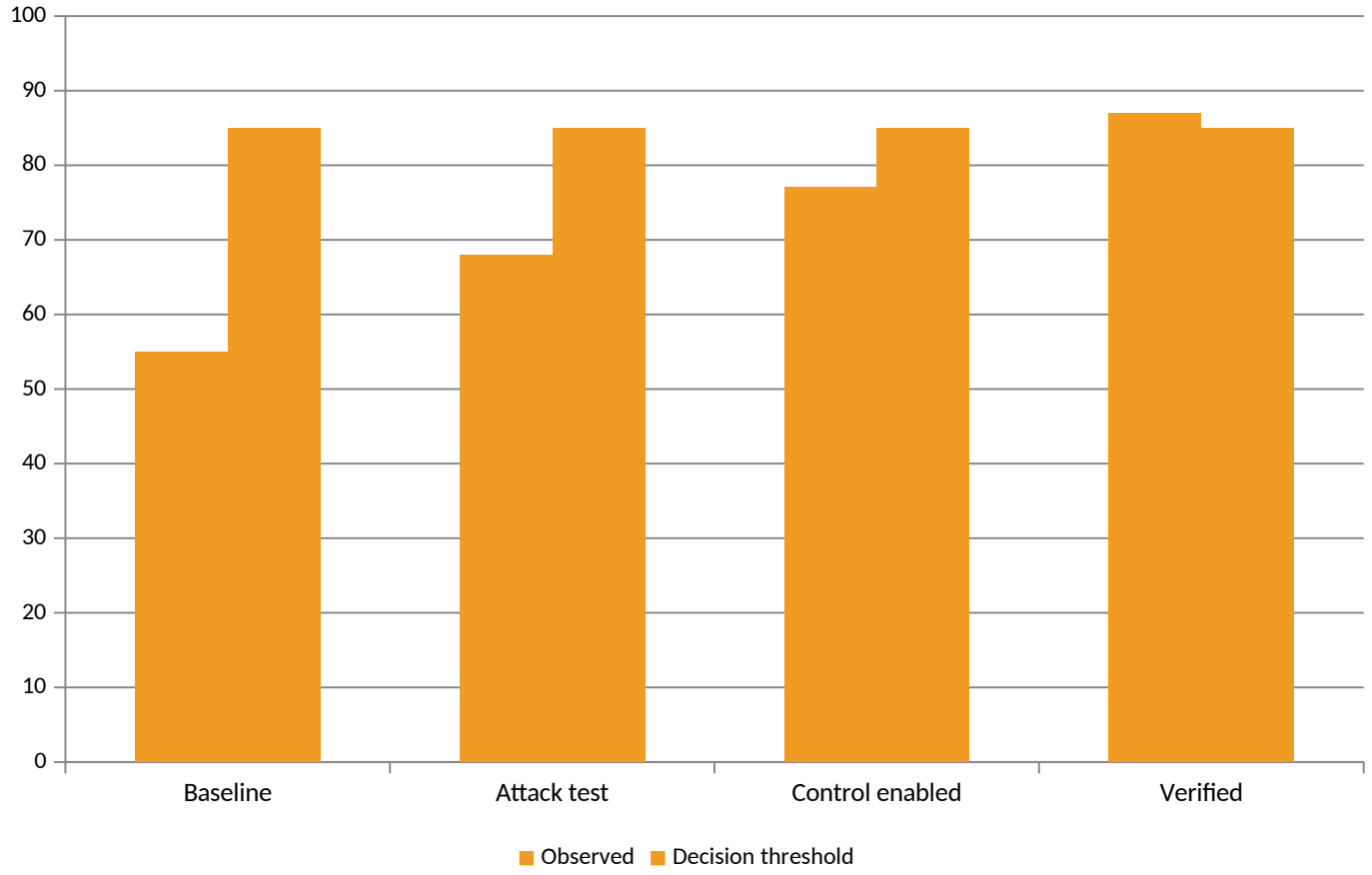
Budget alert: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	session_cost	Required in decision and audit evidence
field_2	daily_cost	Required in decision and audit evidence
field_3	warning_pct	Required in decision and audit evidence
field_4	hard_limit	Required in decision and audit evidence
field_5	action	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Budget alert: evidence and acceptance decision



DECISION RULE

Accept only when Budget breach stops new calls and preserves trace. Otherwise record a gap, owner and due date.

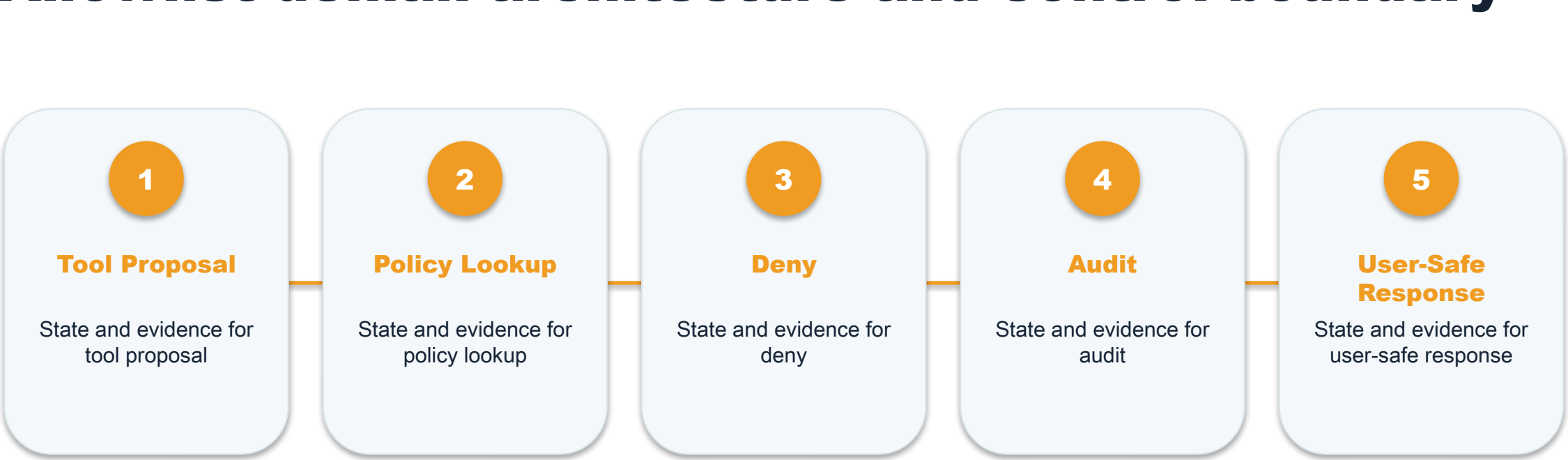
EVIDENCE STATUS

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Allowlist denial: architecture and control boundary

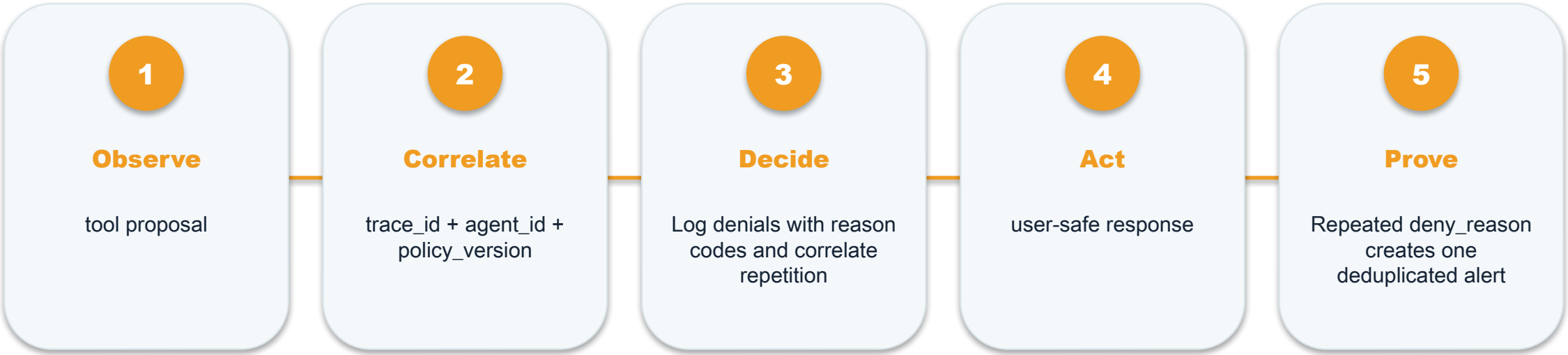


Control objective: Log denials with reason codes and correlate repetition

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Allowlist denial: runtime decision trace



Failure to detect: Blocked action fails silently and attacker keeps probing

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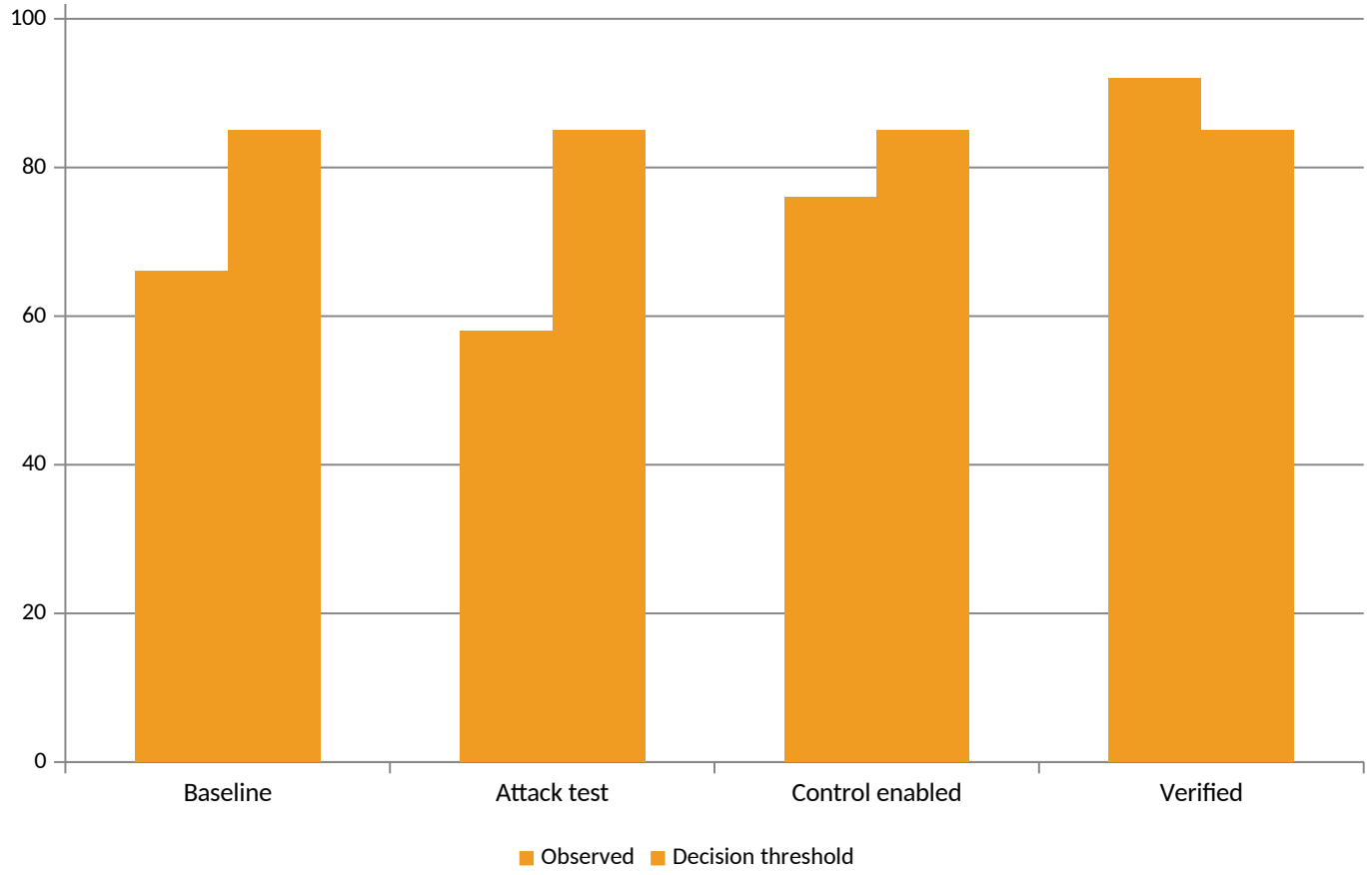
Allowlist denial: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	tool_id	Required in decision and audit evidence
field_3	action	Required in decision and audit evidence
field_4	policy_version	Required in decision and audit evidence
field_5	deny_reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Allowlist denial: evidence and acceptance decision



DECISION RULE

Accept only when Repeated deny_reason creates one deduplicated alert. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Approval timeout: architecture and control boundary

1. **Proposal**
   State and evidence for proposal
2. **Approver Queue**
   State and evidence for approver queue
3. **Timer**
   State and evidence for timer
4. **Expire**
   State and evidence for expire
5. **No Execution**
   State and evidence for no execution

Control objective: Bind approval to action hash and expiry

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Approval timeout: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
proposal	trace_id + agent_id + policy_version	Bind approval to action hash and expiry	no execution	Expired or mismatched approval_id is rejected

Failure to detect: Old approval is replayed after context changes

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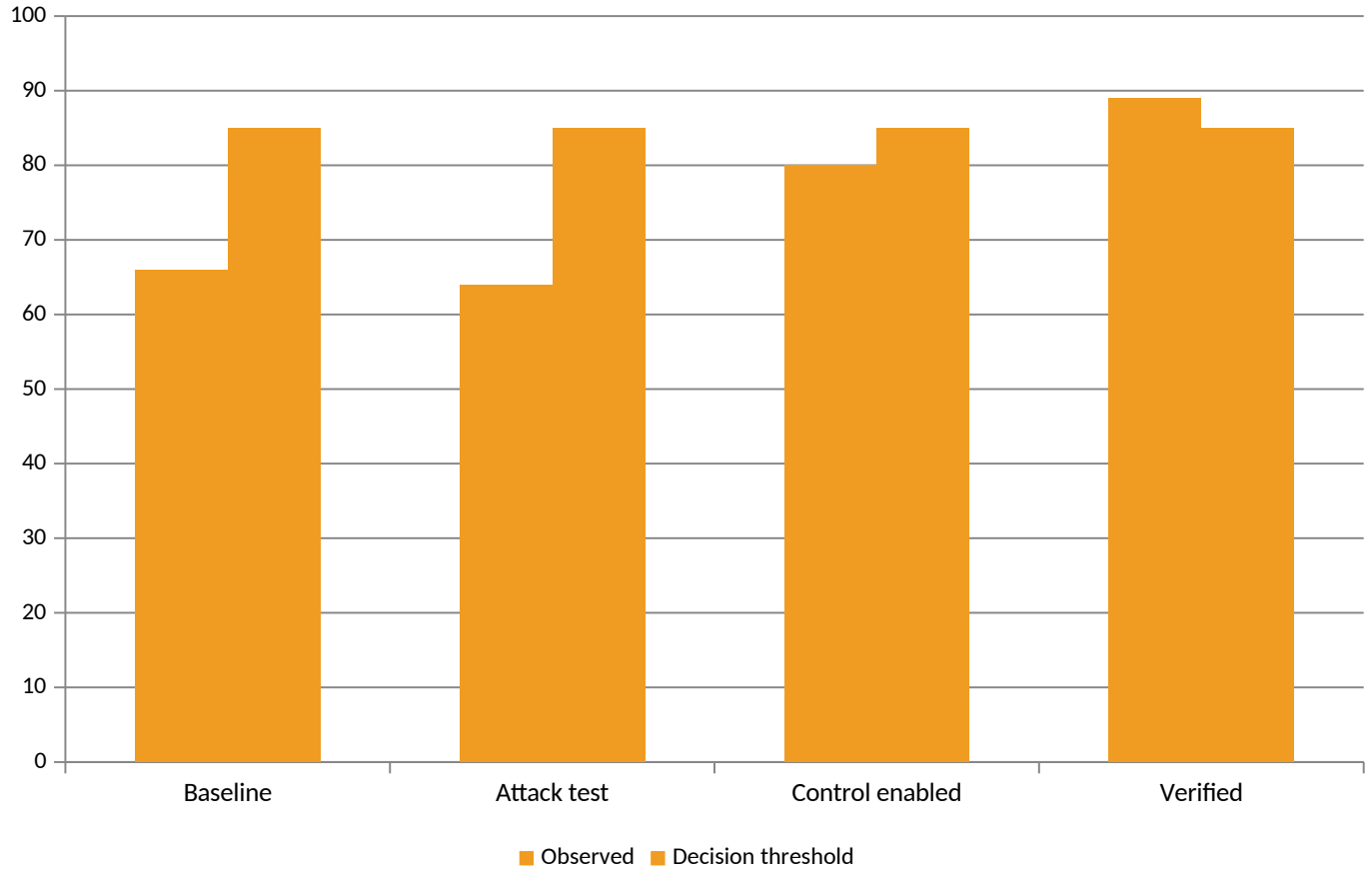
Approval timeout: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	approval_id	Required in decision and audit evidence
field_2	created_at	Required in decision and audit evidence
field_3	expires_at	Required in decision and audit evidence
field_4	reviewer	Required in decision and audit evidence
field_5	status	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Approval timeout: evidence and acceptance decision



DECISION RULE

Accept only when Expired or mismatched approval_id is rejected. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

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Alert triage: architecture and control boundary

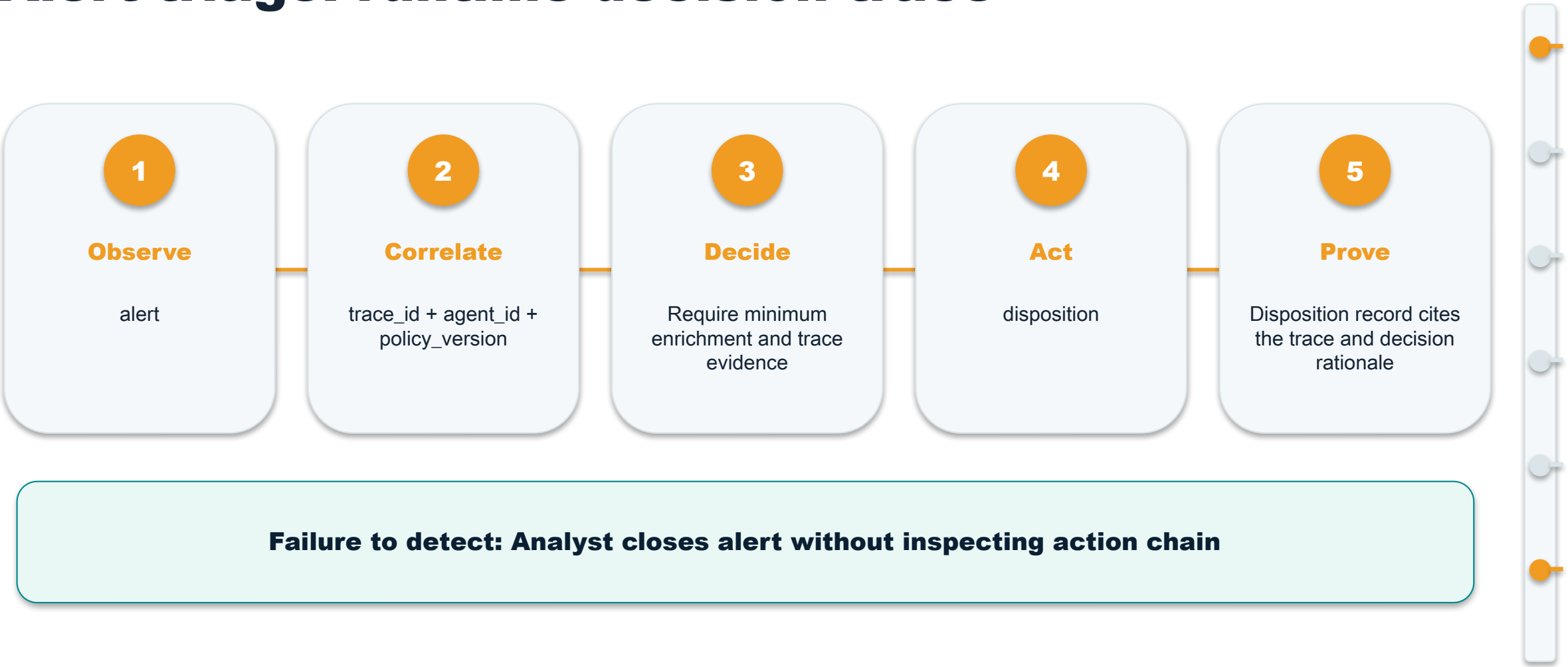
1. **Alert** — State and evidence for alert
2. **Enrichment** — State and evidence for enrichment
3. **Scope** — State and evidence for scope
4. **Confidence** — State and evidence for confidence
5. **Disposition** — State and evidence for disposition

Control objective: Require minimum enrichment and trace evidence

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Alert triage: runtime decision trace



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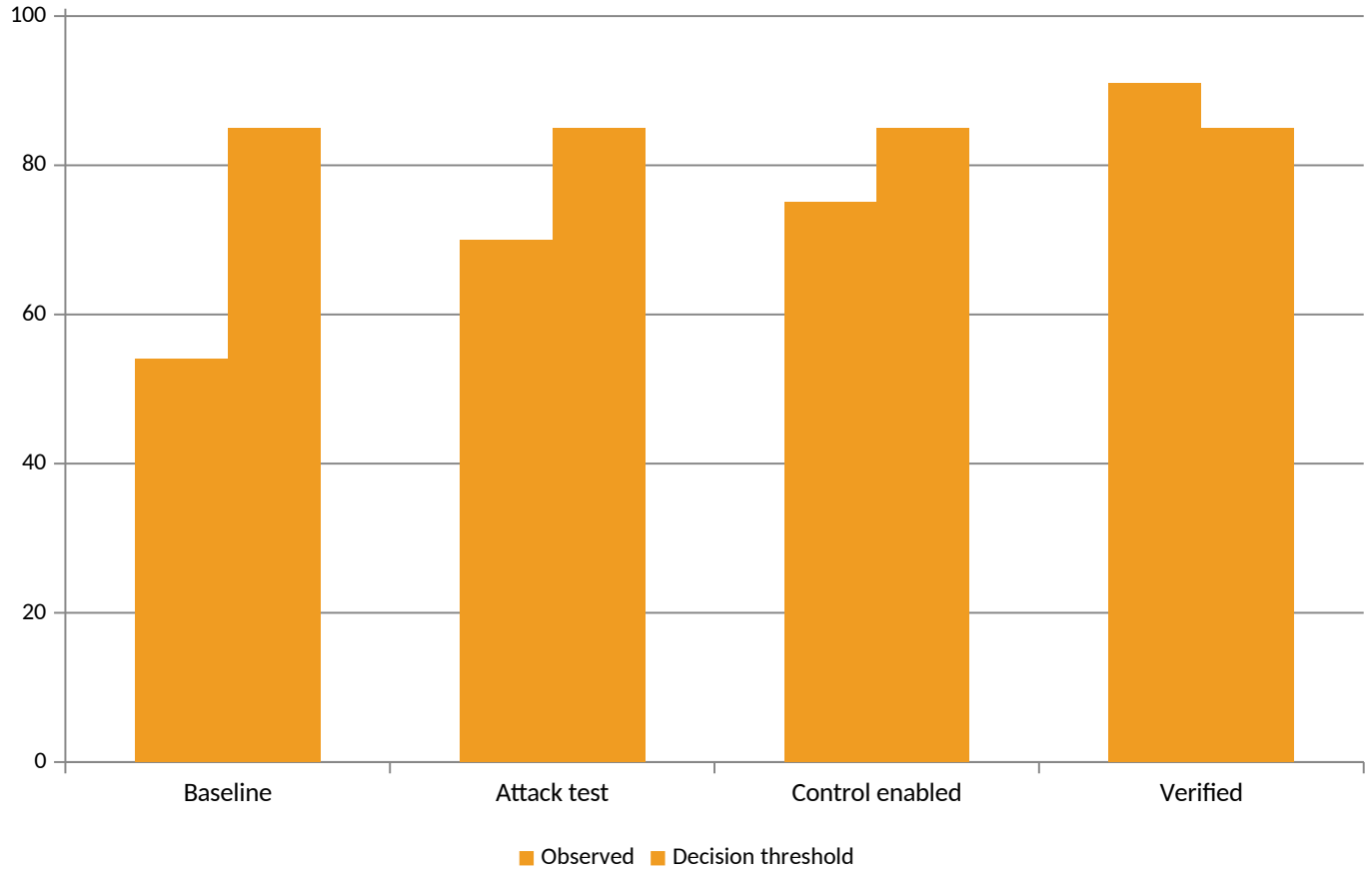
Alert triage: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	alert_id	Required in decision and audit evidence
field_2	evidence[]	Required in decision and audit evidence
field_3	affected_agents[]	Required in decision and audit evidence
field_4	severity	Required in decision and audit evidence
field_5	confidence	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

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Alert triage: evidence and acceptance decision



DECISION RULE

Accept only when Disposition record cites the trace and decision rationale. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Incident severity: architecture and control boundary

1. **Impact** — State and evidence for impact
2. **Scope** — State and evidence for scope
3. **Data Class** — State and evidence for data class
4. **Autonomy** — State and evidence for autonomy
5. **Persistence** — State and evidence for persistence
6. **Severity** — State and evidence for severity

Control objective: Account for real-world actions and ongoing autonomy

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Incident severity: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
impact	trace_id + agent_id + policy_version	Account for real-world actions and ongoing autonomy	severity	SEV-1 criteria trigger executive and containment paths

Failure to detect: Agent incident is ranked like a low-impact chatbot error

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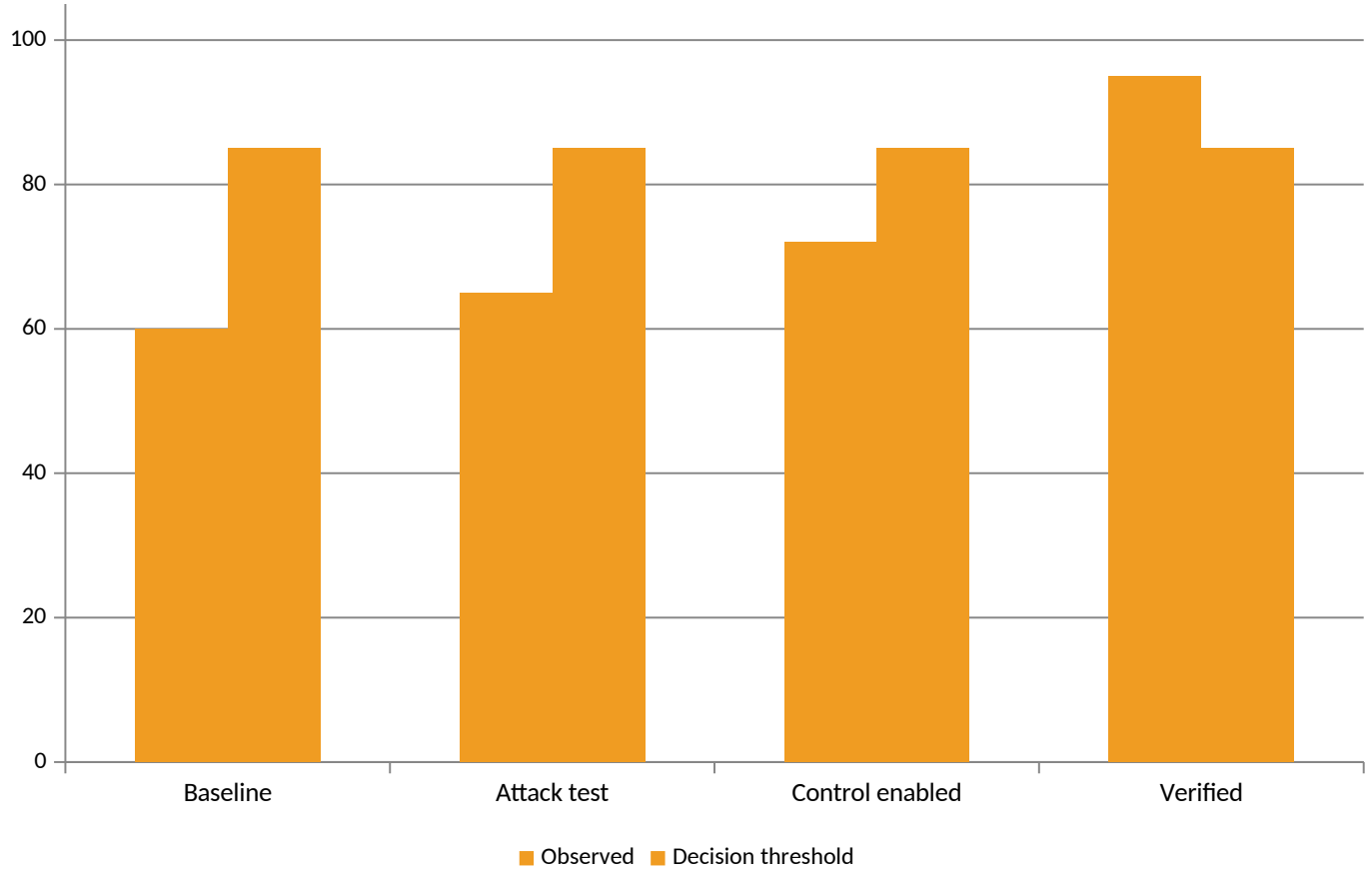
Incident severity: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agents_affected	Required in decision and audit evidence
field_2	data_class	Required in decision and audit evidence
field_3	external_effect	Required in decision and audit evidence
field_4	persistence	Required in decision and audit evidence
field_5	sev	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Incident severity: evidence and acceptance decision



DECISION RULE

Accept only when SEV-1 criteria trigger executive and containment paths. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Containment circuit breaker: architecture and control boundary

1

Soc Decision

State and evidence for SOC decision

2

Gateway Deny-All

State and evidence for gateway deny-all

3

Queue Drain

State and evidence for queue drain

4

Confirmation

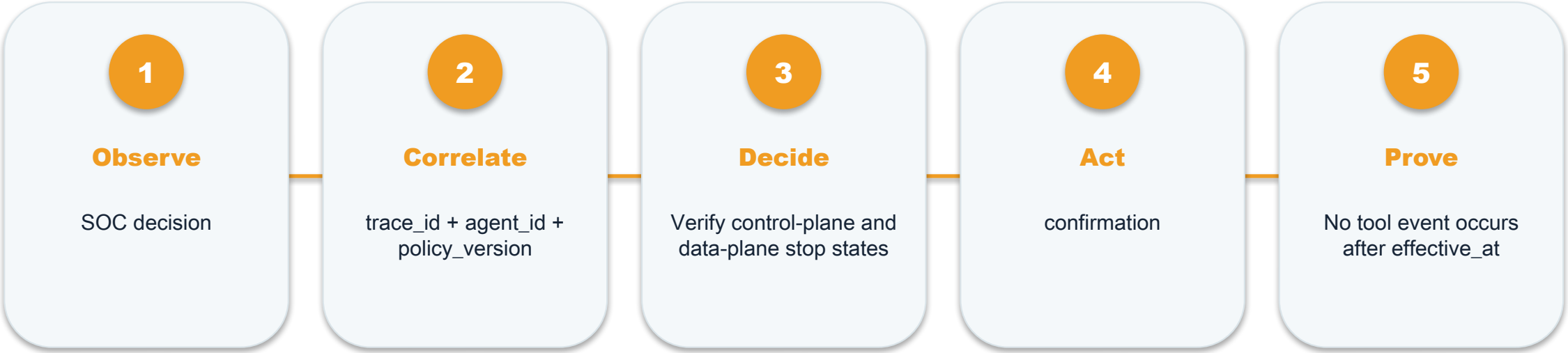
State and evidence for confirmation

Control objective: Verify control-plane and data-plane stop states

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Containment circuit breaker: runtime decision trace



Failure to detect: Containment request succeeds in UI but actions continue

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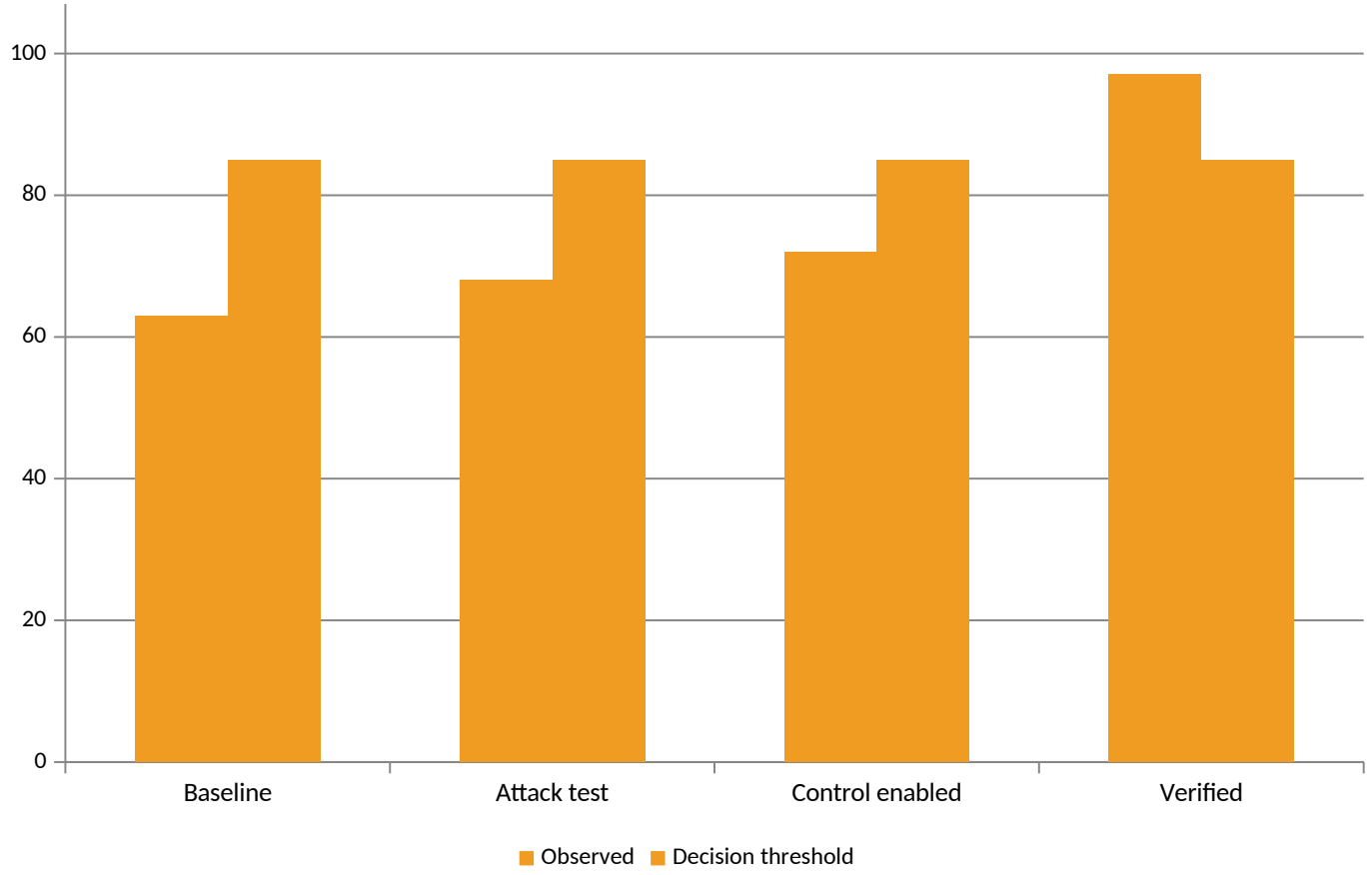
Containment circuit breaker: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	incident_id	Required in decision and audit evidence
field_2	scope	Required in decision and audit evidence
field_3	initiated_by	Required in decision and audit evidence
field_4	effective_at	Required in decision and audit evidence
field_5	agents_stopped	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Containment circuit breaker: evidence and acceptance decision



DECISION RULE

Accept only when No tool event occurs after effective_at. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Token revocation: architecture and control boundary

1

Incident

State and evidence for incident

2

Identity Provider

State and evidence for identity provider

3

Revoke Session/Tokens

State and evidence for revoke session/tokens

4

Resource Deny

State and evidence for resource deny

Control objective: Revoke tokens and enforce short TTL at resources

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Token revocation: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
incident	trace_id + agent_id + policy_version	Revoke tokens and enforce short TTL at resources	resource deny	Post-revocation access attempt returns deny

Failure to detect: Disabled agent reuses an already-issued token

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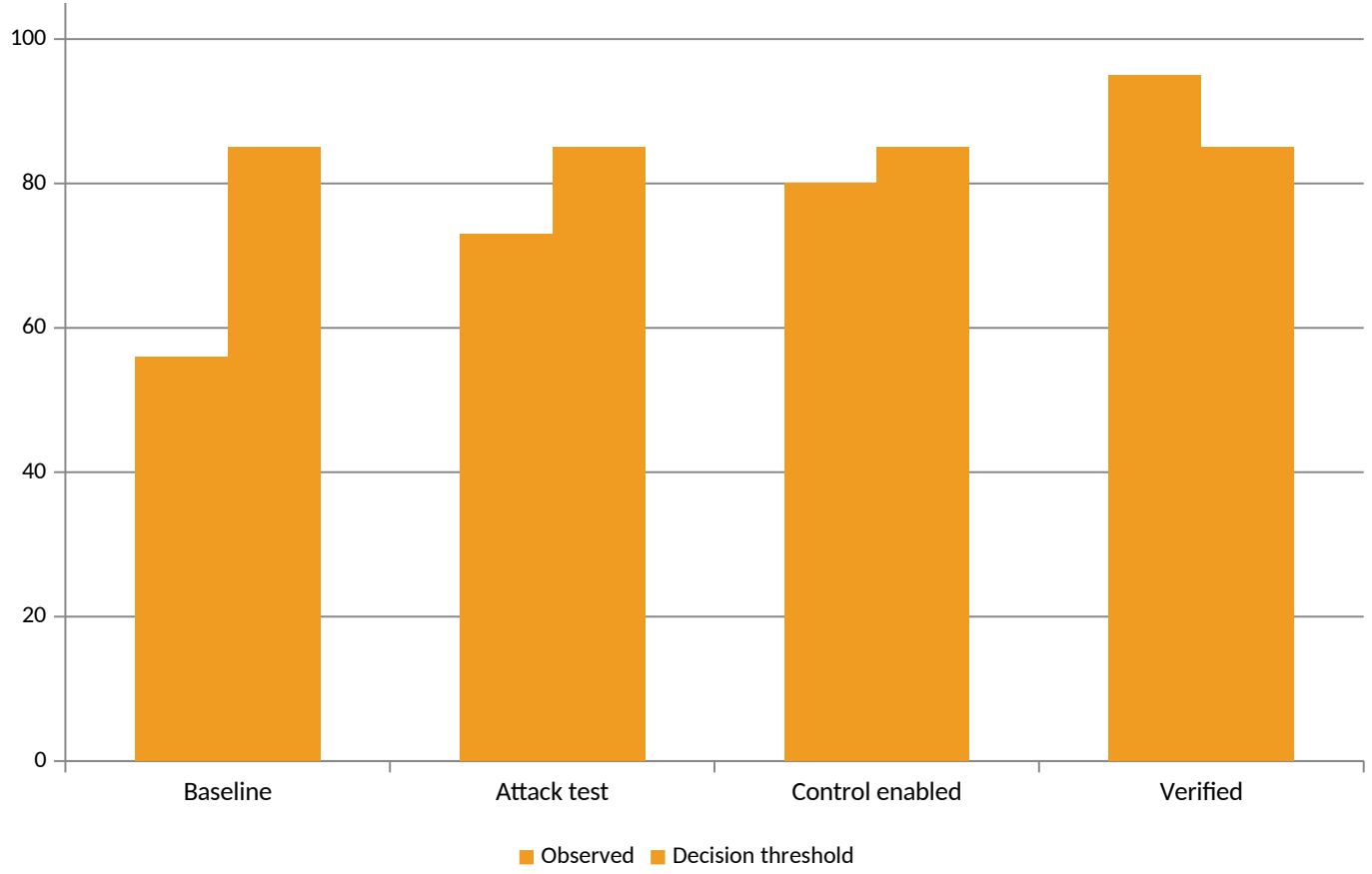
Token revocation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	token_id	Required in decision and audit evidence
field_2	subject	Required in decision and audit evidence
field_3	audience	Required in decision and audit evidence
field_4	revoked_at	Required in decision and audit evidence
field_5	reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

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Token revocation: evidence and acceptance decision



DECISION RULE

Accept only when Post-revocation access attempt returns deny. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Agent disablement: architecture and control boundary

1

Registry Status

State and evidence for registry status

2

Scheduler

State and evidence for scheduler

3

Runtime

State and evidence for runtime

4

Tool Gateway

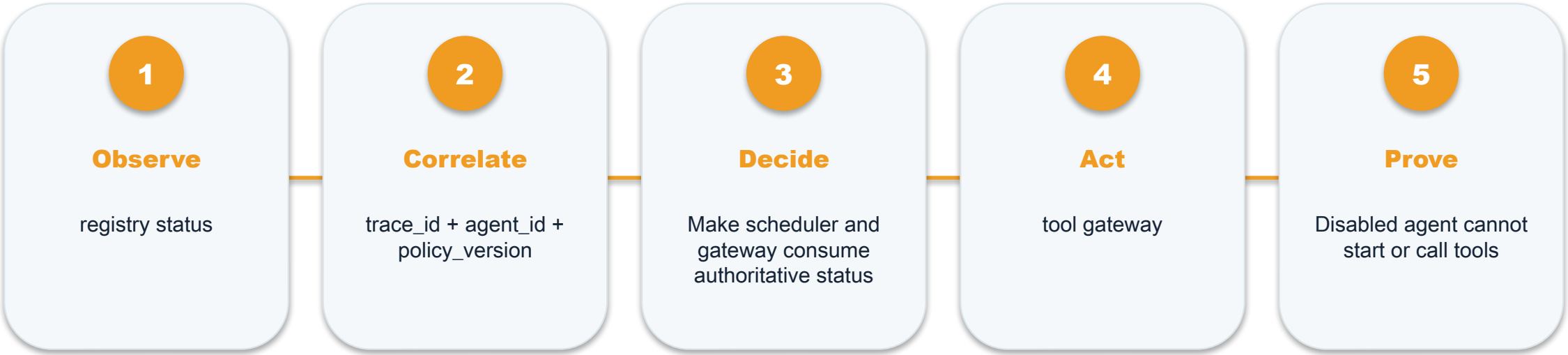
State and evidence for tool gateway

Control objective: Make scheduler and gateway consume authoritative status

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Agent disablement: runtime decision trace



Failure to detect: New worker starts after the incident

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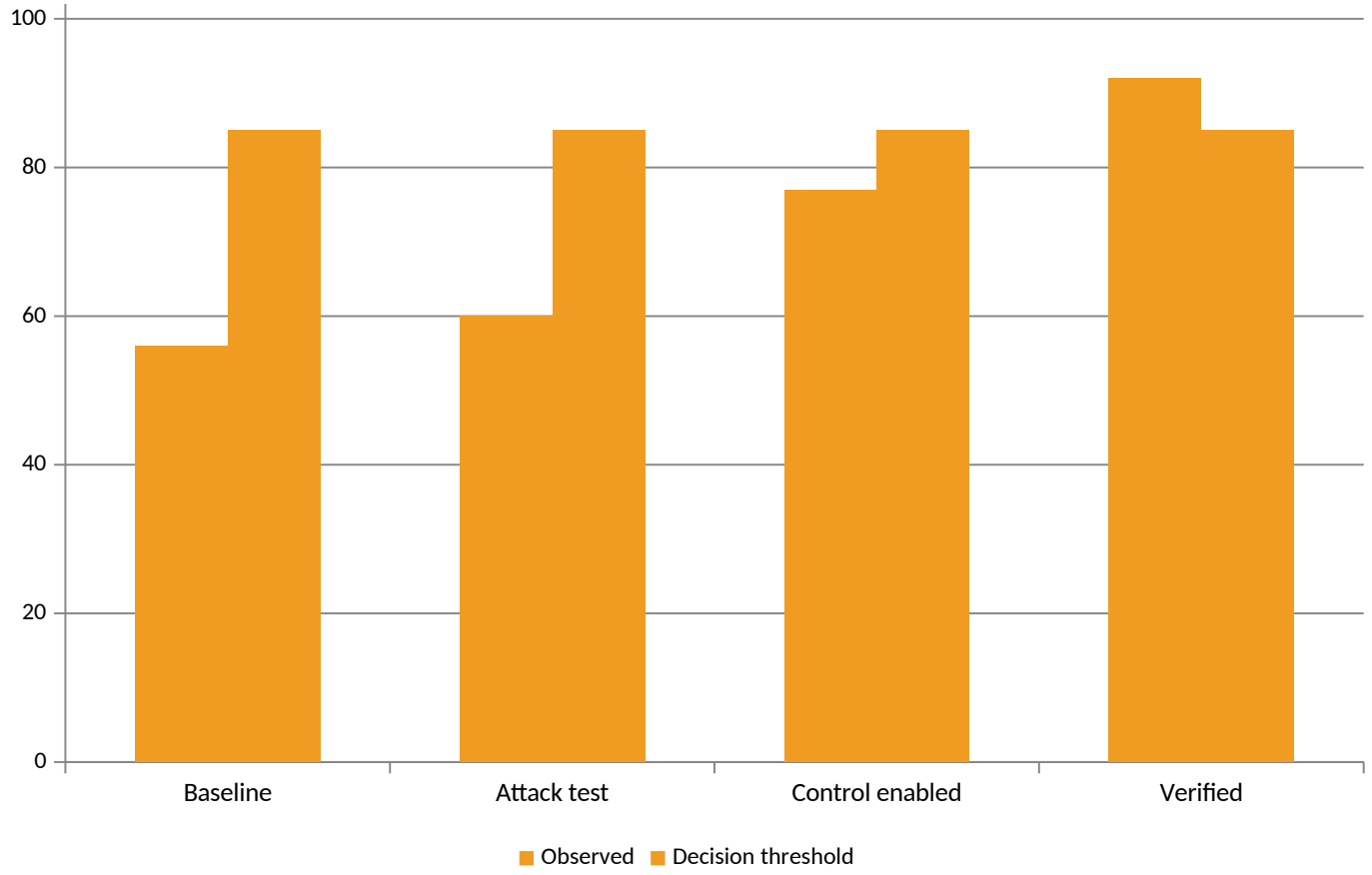
Agent disablement: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
status	disabled	Required in decision and audit evidence
field_3	version	Required in decision and audit evidence
field_4	reason	Required in decision and audit evidence
field_5	changed_by	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Agent disablement: evidence and acceptance decision



DECISION RULE

Accept only when Disabled agent cannot start or call tools. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

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Evidence preservation: architecture and control boundary

1

Volatile Session

State and evidence for volatile session

2

Snapshot

State and evidence for snapshot

3

Immutable Store

State and evidence for immutable store

4

Chain Of Custody

State and evidence for chain of custody

Control objective: Snapshot prompts, memory, policies, logs and tool outputs

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Evidence preservation: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
volatile session	trace_id + agent_id + policy_version	Snapshot prompts, memory, policies, logs and tool outputs	chain of custody	Stored hash matches acquisition hash

Failure to detect: Investigation changes or loses ephemeral context

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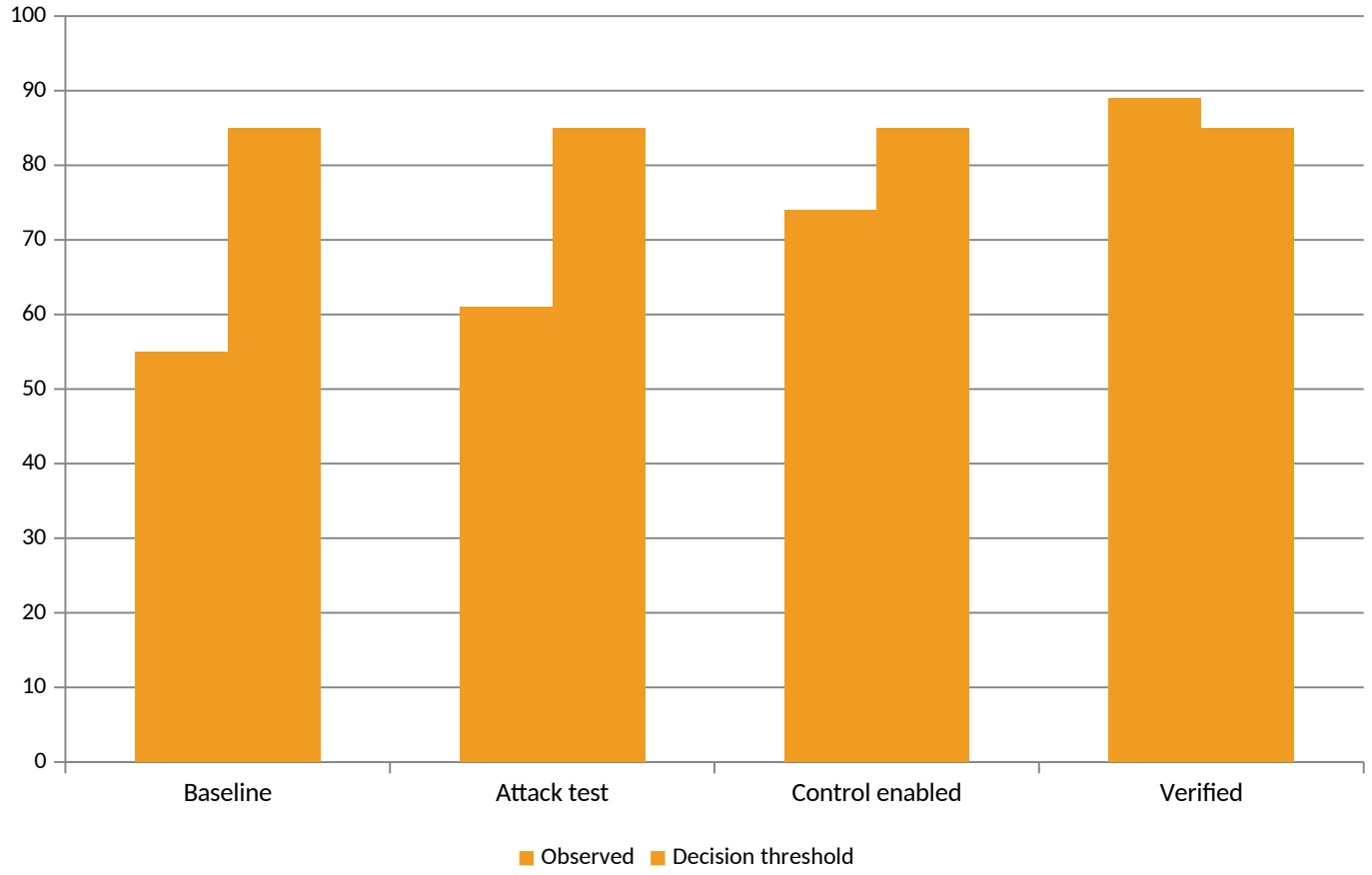
Evidence preservation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	evidence_id	Required in decision and audit evidence
field_2	source	Required in decision and audit evidence
field_3	collected_at	Required in decision and audit evidence
field_4	hash	Required in decision and audit evidence
field_5	collector	Required in decision and audit evidence
field_6	location	Required in decision and audit evidence

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Evidence preservation: evidence and acceptance decision



DECISION RULE

Accept only when Stored hash matches acquisition hash. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Session replay: architecture and control boundary

1

Ordered Events

State and evidence for ordered events

2

Versions

State and evidence for versions

3

Mock Tools

State and evidence for mock tools

4

Deterministic Replay

State and evidence for deterministic replay

Control objective: Replay with fixed artifacts and recorded tool outputs

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Session replay: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
ordered events	trace_id + agent_id + policy_version	Replay with fixed artifacts and recorded tool outputs	deterministic replay	Replay identifies the first policy divergence

Failure to detect: Team cannot reproduce the malicious decision path

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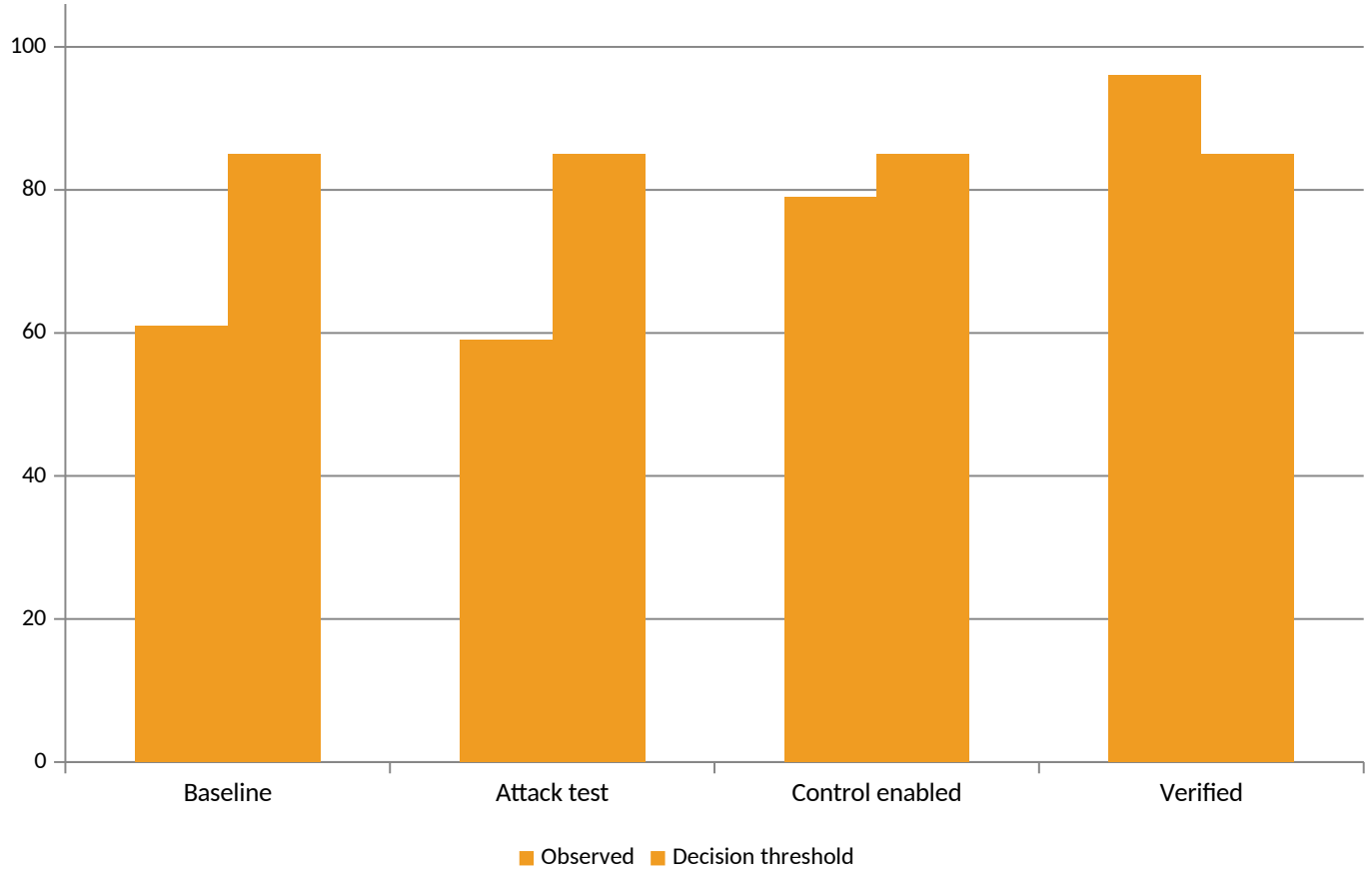
Session replay: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	trace_id	Required in decision and audit evidence
field_2	model_version	Required in decision and audit evidence
field_3	prompt_hash	Required in decision and audit evidence
field_4	tool_fixtures	Required in decision and audit evidence
field_5	outcome_diff	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Session replay: evidence and acceptance decision



DECISION RULE

Accept only when Replay identifies the first policy divergence. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Eradication: architecture and control boundary

1

Root Cause

State and evidence for root cause

2

Prompt/Policy/Tool Fix

State and evidence for prompt/policy/tool fix

3

Regression Tests

State and evidence for regression tests

4

Redeploy

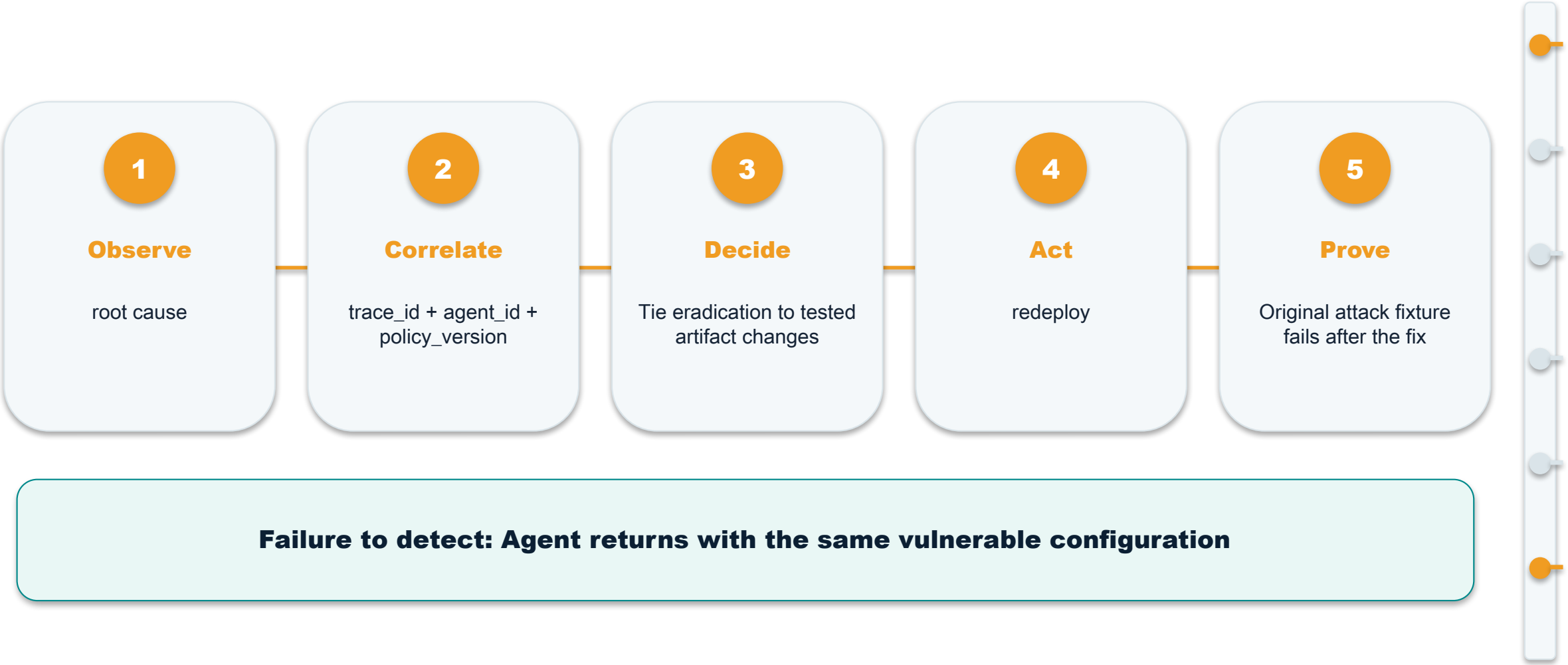
State and evidence for redeploy

Control objective: Tie eradication to tested artifact changes

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Eradication: runtime decision trace



Failure to detect: Agent returns with the same vulnerable configuration

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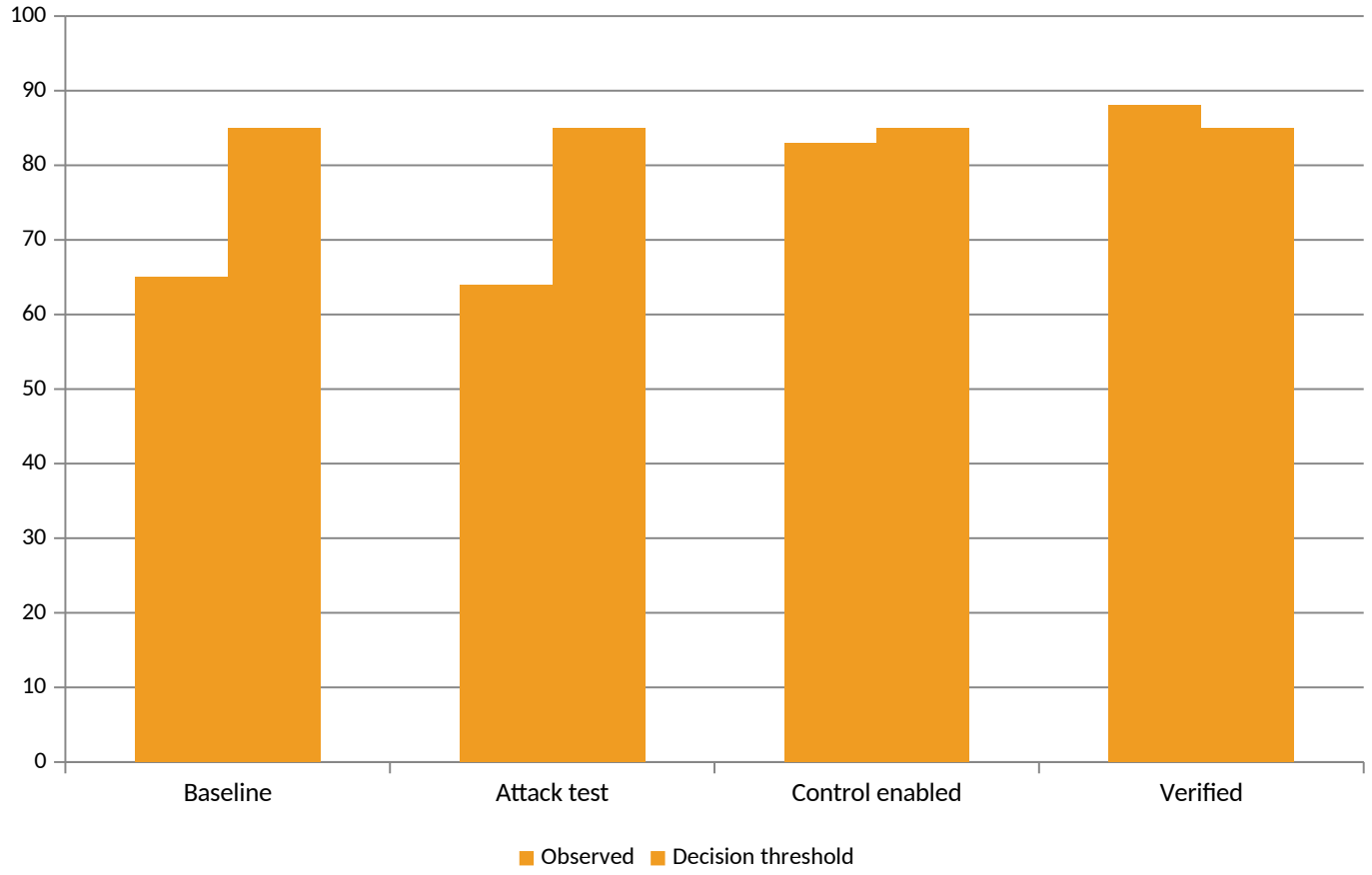
Eradication: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	root_cause	Required in decision and audit evidence
field_2	changed_artifacts[]	Required in decision and audit evidence
field_3	tests[]	Required in decision and audit evidence
field_4	approver	Required in decision and audit evidence
field_5	release_id	Required in decision and audit evidence
control_6	explicit value	Reject if missing

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Eradication: evidence and acceptance decision



DECISION RULE

Accept only when Original attack fixture fails after the fix. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Recovery: architecture and control boundary

1

Clean Release

State and evidence for clean release

2

Staged Enablement

State and evidence for staged enablement

3

Monitoring

State and evidence for monitoring

4

Normal Service

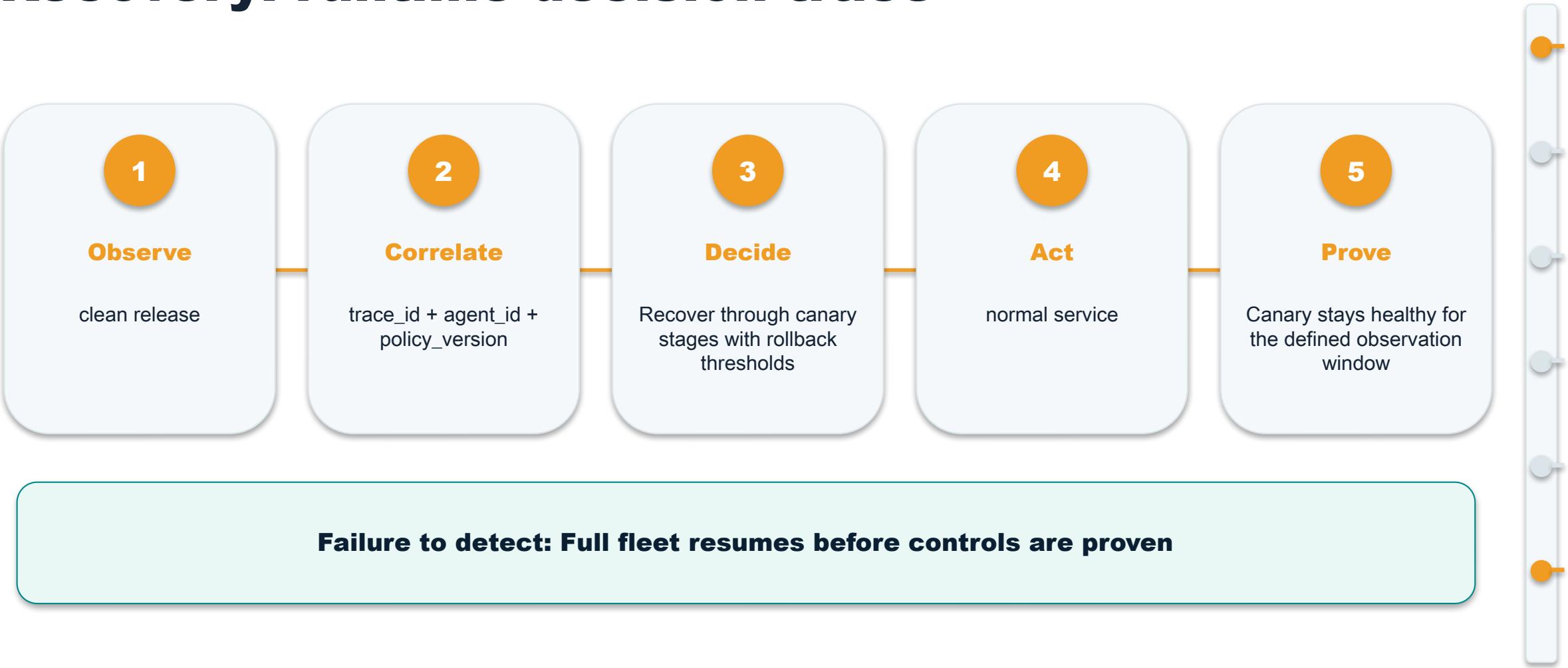
State and evidence for normal service

Control objective: Recover through canary stages with rollback thresholds

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Recovery: runtime decision trace



Failure to detect: Full fleet resumes before controls are proven

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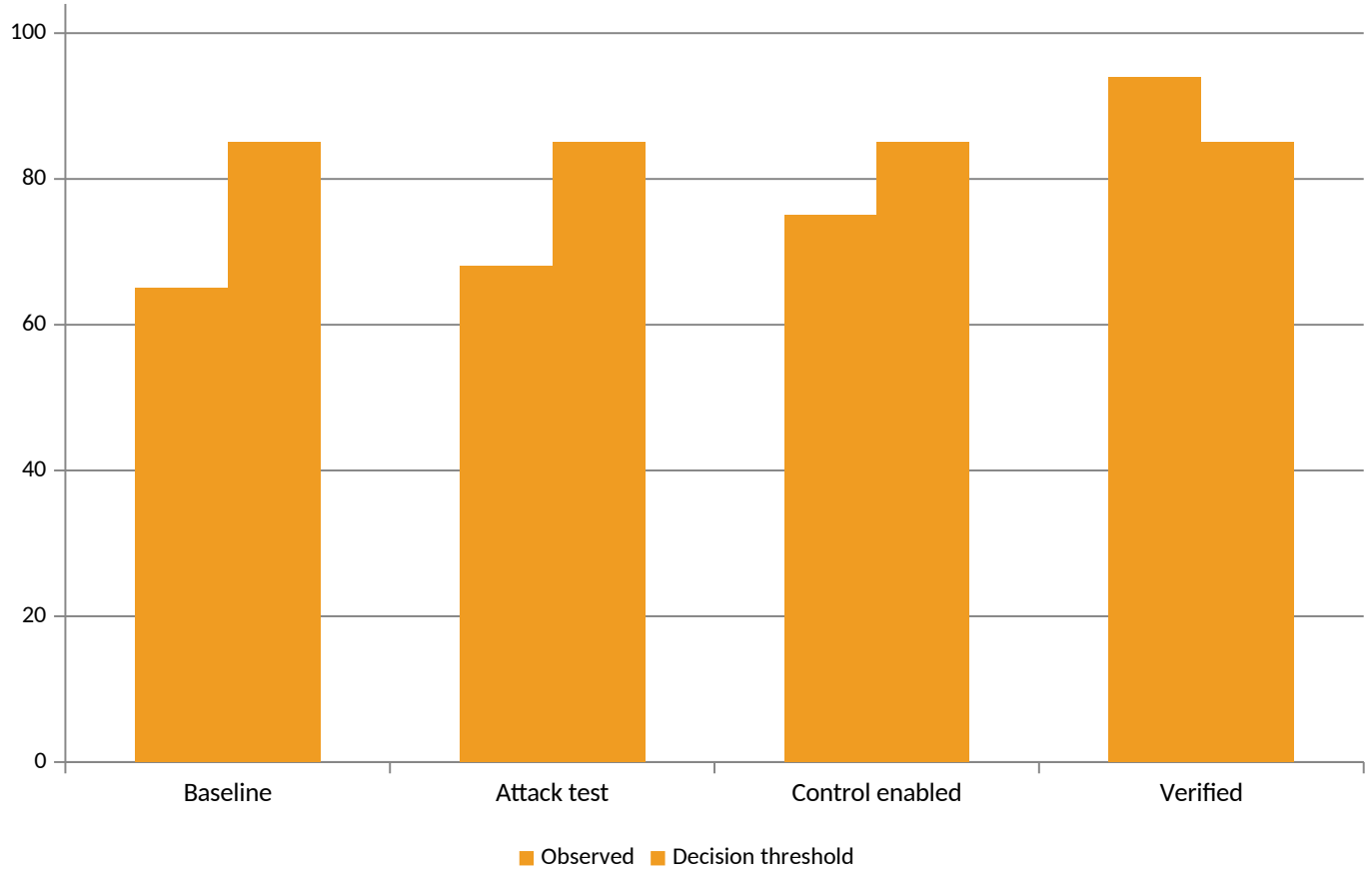
Recovery: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	release_id	Required in decision and audit evidence
field_2	canary_pct	Required in decision and audit evidence
field_3	health_metrics[]	Required in decision and audit evidence
field_4	rollback_threshold	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

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Recovery: evidence and acceptance decision



DECISION RULE

Accept only when Canary stays healthy for the defined observation window. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Lessons and metrics: architecture and control boundary

1. **Incident Timeline**
   State and evidence for incident timeline
2. **Control Gaps**
   State and evidence for control gaps
3. **Actions**
   State and evidence for actions
4. **Verification**
   State and evidence for verification
5. **Trend**
   State and evidence for trend

Control objective: Require evidence-based closure and recurrence metrics

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Lessons and metrics: runtime decision trace

1	2	3	4	5
Observe	**Correlate**	**Decide**	**Act**	**Prove**
incident timeline	trace_id + agent_id + policy_version	Require evidence-based closure and recurrence metrics	trend	Action is closed only after the acceptance test passes

Failure to detect: Post-incident actions close administratively but not technically

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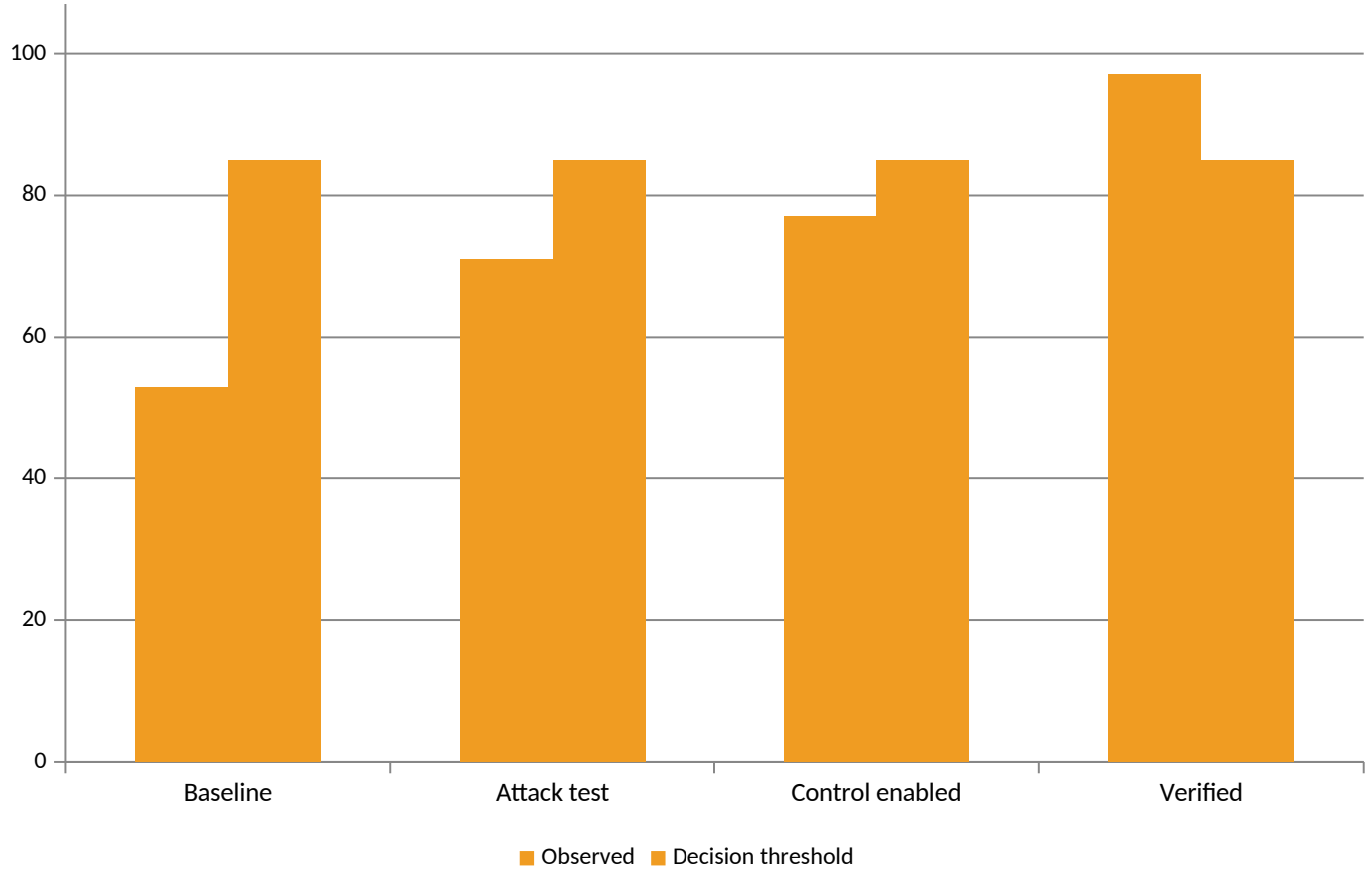
Lessons and metrics: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	lesson_id	Required in decision and audit evidence
field_2	control_gap	Required in decision and audit evidence
field_3	action_owner	Required in decision and audit evidence
field_4	due_date	Required in decision and audit evidence
field_5	validation	Required in decision and audit evidence
field_6	metric	Required in decision and audit evidence

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Lessons and metrics: evidence and acceptance decision



DECISION RULE

Accept only when Action is closed only after the acceptance test passes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

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Activity 08 — Detect Suspicious Agent Behaviour

1 **Scenario**

Use synthetic data in activity-08-agent-detection to make a security-operations decision.

2 **Input contract**

mock_events.jsonl

3 **Runnable evidence**

activity.py

4 **Output artifact**

detections.json

5 **Acceptance**

The detector finds abnormal tool rates, new destinations, blocked actions and sensitive-read/egress chains.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 08: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_events.jsonl	required fields and types	control and decision rules	detections.json	The detector finds abnormal tool rates, new destinations, blocked actions and sensitive-read/egress chains.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 08: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_events.jsonl	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	detections.json	Generated locally and reproducible
Criterion	A7	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete detections.json only	Retain script and mock data

Aligned activity files in labs/

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Activity 09 — Contain a Prompt-Injection Incident

1 **Scenario**

Use synthetic data in activity-09-prompt-injection-response to make a security-operations decision.

2 **Input contract**

mock_incident.json

3 **Runnable evidence**

activity.py

4 **Output artifact**

containment_record.json

5 **Acceptance**

The record proves agent disablement, token revocation, egress block and evidence preservation in the correct order.

6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 09: mechanism model

1	2	3	4	5
Load	**Validate**	**Evaluate**	**Produce**	**Verify**
mock_incident.json	required fields and types	control and decision rules	containment_record.json	The record proves agent disablement, token revocation, egress block and evidence preservation in the correct order.

The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 09: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_incident.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	containment_record.json	Generated locally and reproducible
Criterion	A7	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete containment_record.json only	Retain script and mock data

Aligned activity files in labs/

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Activity 10 — Run Recovery and Post-Incident Review

1 **Scenario**

Use synthetic data in activity-10-recovery-tabletop to make a security-operations decision.

2 **Input contract**

mock_recovery.json

3 **Runnable evidence**

activity.py

4 **Output artifact**

after_action_report.md

5 **Acceptance**

Recovery gates, rollback criteria, lessons, owners and acceptance tests are complete.

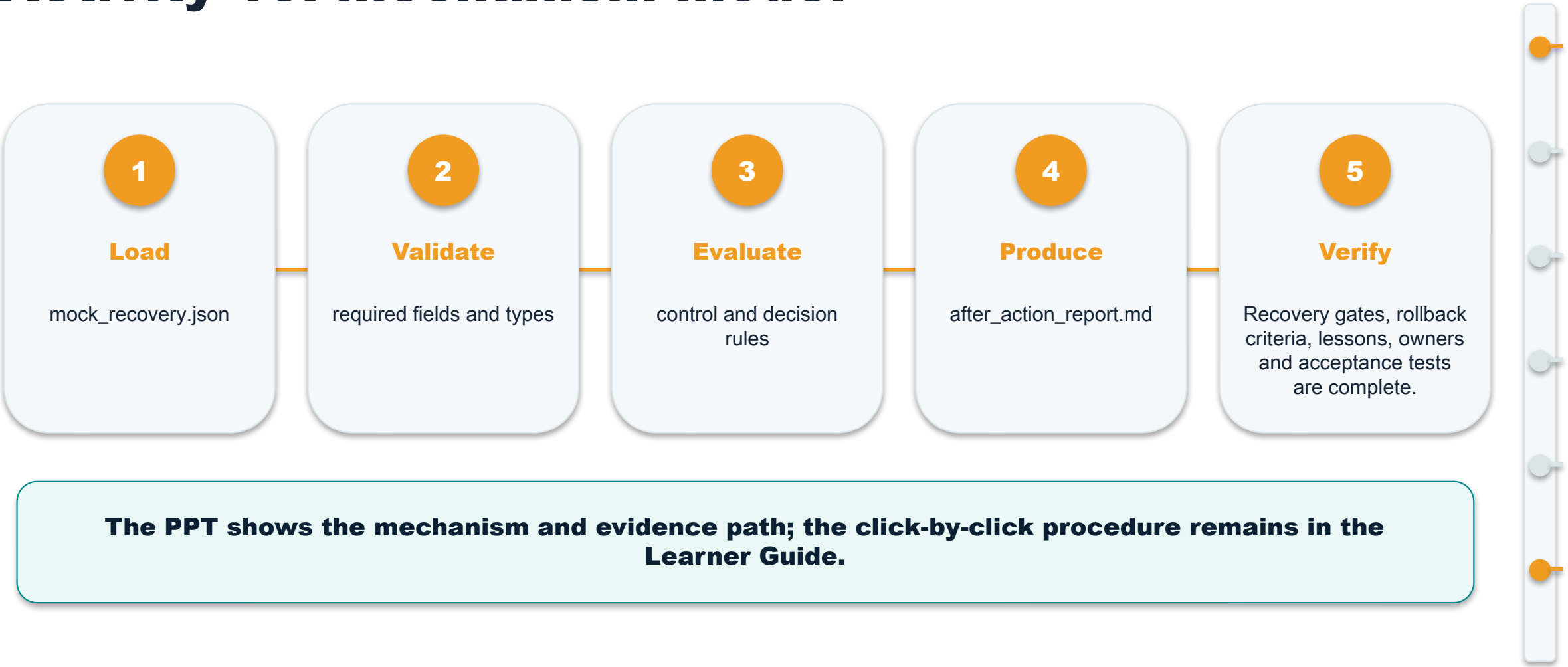
6 **Safety boundary**

Offline simulation only; no live credentials, systems or external services.

Aligned activity folder and synthetic mock data

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Activity 10: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Aligned activity design; complete procedure is in the Learner Guide

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Activity 10: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_recovery.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	after_action_report.md	Generated locally and reproducible
Criterion	A7	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete after_action_report.md only	Retain script and mock data

Aligned activity files in labs/

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Topic 04 technical recap

1 **Agent disablement**

Disabled agent cannot start or call tools

2 **Evidence preservation**

Stored hash matches acquisition hash

3 **Session replay**

Replay identifies the first policy divergence

4 **Eradication**

Original attack fixture fails after the fix

5 **Recovery**

Canary stays healthy for the defined observation window

6 **Lessons and metrics**

Action is closed only after the acceptance test passes

Topic technical-anchor inventory

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WRAP-UP

From strategy to resilient operations

Goals → policy → risk → telemetry → containment → recovery



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What you can now do

1 LO1

Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.

2 LO2

Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.

3 LO3

Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.

4 LO4

Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.



Official current course outcomes

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Assessment

1 **WA**

7 open-ended questions · K1–K7 · 1 hour.

2 **PP**

6 practical tasks · A1–A7 · 3 hours.

3 **Submission**

Upload completed answers to lms-tms.tertiaryinfotech.com.

4 **Materials**

Use the approved slides, Learner Guide and activity files.

5 **Evidence**

Outputs must reconcile with the supplied mock data.

6 **Privacy**

Answer keys remain trainer-only.

Verified legacy assessment mapping

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Assessment flow



Complete the full sequence and retain submission evidence.

WSQ assessment workflow

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Digital attendance

1 **AM**

Recorded

2 **PM**

Recorded

3 **Assessment**

Recorded

4 **Minimum**

75%

5 **Failure**

Tell trainer immediately

6 **Evidence**

Successful confirmation screen

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Thank you

Secure autonomy is an operating capability—designed, measured, rehearsed and improved.

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